

Tesis Doctoral

Ranks over pro- p groups

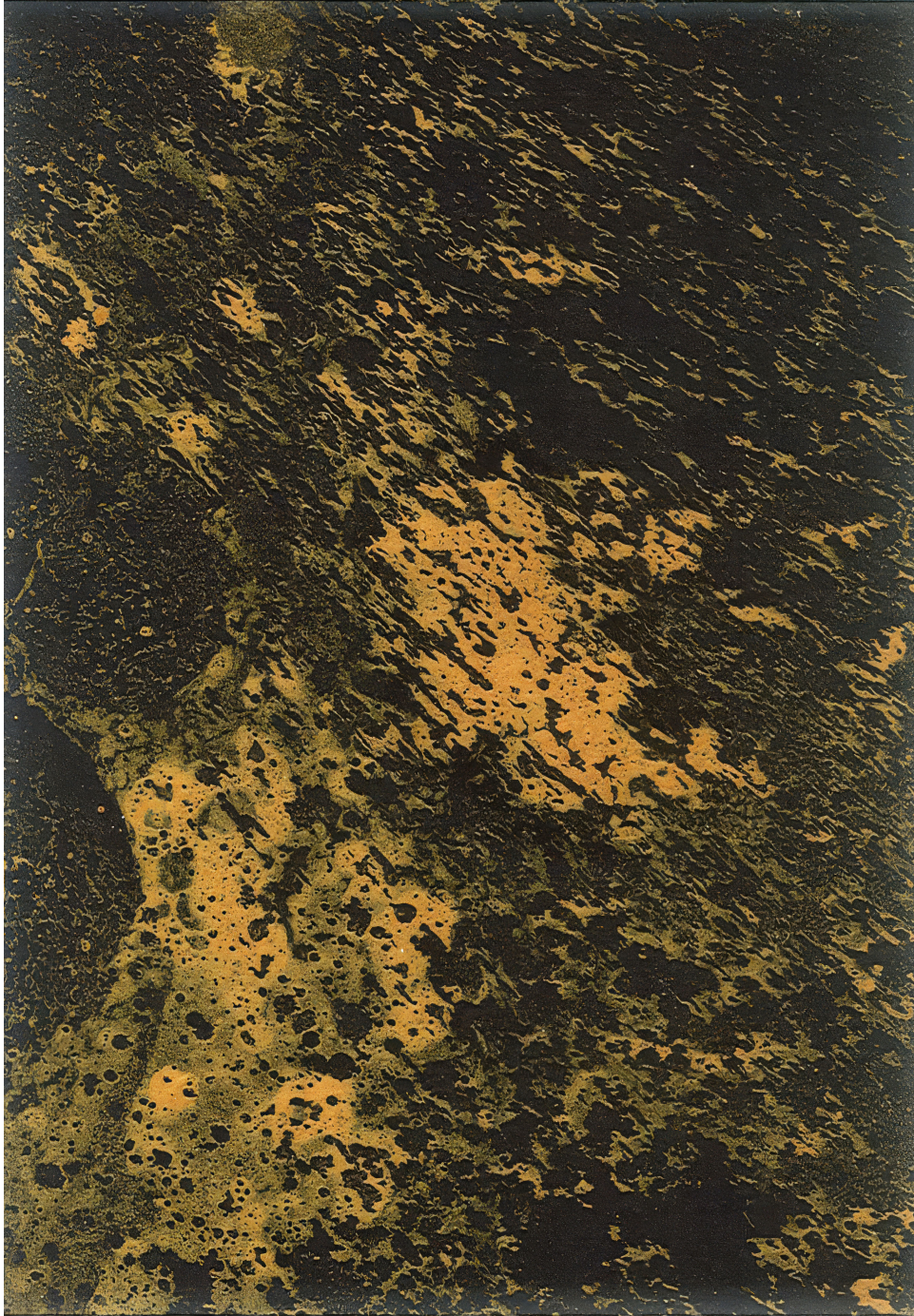
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Resumen

Esta tesis investiga el espacio de funciones de rango de matrices sobre álgebras de grupo completadas de grupos profinitos, teniendo en mente aplicaciones a la topología, la teoría de representaciones y la aritmética. El concepto de función de rango de Sylvester es una de varias generalizaciones de la noción de “rango” desde matrices sobre cuerpos a matrices sobre anillos arbitrarios. Muchos aspectos combinatorios y asintóticos de la teoría geométrica de grupos son en realidad fenómenos encubiertos en el espacio de funciones de rango sobre anillos adecuados. Baste mencionar, como ejemplos, el teorema de aproximación de Lück, la conjetura de Hanna Neumann y la coherencia de los grupos con una sola relación.

El trabajo se divide de forma natural en dos partes. El primer capítulo desarrolla la álgebra topológica y homológica necesaria para anillos profinitos, filtrados y graduados que se utilizará a lo largo de esta memoria. El segundo capítulo explica las distintas operaciones de localización que se aplicarán a los anillos bajo estudio, incluyendo la localización de Ore, la microlocalización algebraica y la localización matricial de Cohn. En dicho capítulo ofrecemos también la primera de tres definiciones de dominio de Sylvester y mostramos cómo topologizar la localización matricial de un anillo profinito. El tercer capítulo está dedicado al concepto de funciones de rango matricial de Sylvester, donde damos la segunda definición de dominio de Sylvester. Además probamos un resultado de aproximación en el espacio de funciones de rango de un álgebra de grupo completada $\mathbb{F}_p[[G]]$ de un grupo pro- p para rangos inducidos por una inmersión topológica en un cuerpo de división valuado.

La segunda parte contiene la mayor parte del material original. En el cuarto capítulo damos la tercera y última definición de dominio de Sylvester. También extendemos la construcción de A. Jaikin-Zapirain del anillo de división universal para un álgebra pro- p libre de rango numerable infinito, y demostramos que las álgebras de grupo completadas sobre $\mathbb{F}_q$ de grupos pro- p sin torsión virtualmente libres-por- $\mathbb{Z}_p$ son dominios de Sylvester. Asimismo probamos que el hecho de que su álgebra de grupo completada sea un dominio de Sylvester es una propiedad invariante por conmensurabilidad entre grupos pro- p sin torsión. El quinto capítulo estudia funciones de rango en la $\mathbb{Z}_p$ -álgebra completada $\mathbb{Z}_p[[G_1]]$ de un subgrupo de congruencia G_1 de un grupo de Lie p -ádico G semisimple escindido y simplemente conexo. Planteamos dos conjeturas sobre el comportamiento asintótico de las funciones de rango inducidas por representaciones racionales de G , que confirmamos cuando G es un producto de copias de SL_2 . El capítulo final está dedicado a aplicaciones de los resultados de los capítulos 4 y 5 a la teoría

de grupos abstractos, a subgrupos de grupos de Lie complejos semisimples, a la geometría hiperbólica y a las formas automorfas.

Palabras clave: grupos profinitos, álgebras de grupo completadas, funciones de rango de Sylvester, dominios de Sylvester, grupos de Lie p -ádicos.

Abstract

This thesis investigates the space of rank functions on matrices over completed group algebras of profinite groups, with applications to topology, representation theory and arithmetic in mind. The concept of a Sylvester rank function is one of several generalizations of the notion of “rank” from matrices over fields to matrices over general rings. Many combinatorial and asymptotic aspects of geometric group theory are actually phenomena on the space of rank functions over suitable rings in disguise, and we cite the Lück approximation theorem, the Hanna Neumann conjecture and the coherence of one-relator groups to name a few examples.

The work is naturally divided in two parts. The first chapter develops the necessary topological and homological algebra for profinite, filtered and graded rings that will be used throughout this work. The second chapter explains the various localization operations that will be applied to the rings under study, including Ore localization, algebraic microlocalization and Cohn’s matrix localization. We also offer the first of three definitions of a Sylvester domain in said chapter and show how to topologize the matrix localization of a profinite ring. The third chapter is devoted to the concept of Sylvester matrix rank functions, where we give the second definition of a Sylvester domain. We also prove an approximation result on the space of rank functions of a completed group algebra $\mathbb{F}_p[[G]]$ of a pro- p group for ranks induced from a topological embedding into a valuation division ring.

The second part contains most of the original material. In the fourth chapter we give the third and final definition of a Sylvester domain. We also extend A. Jaikin-Zapirain’s construction of the universal division ring of fractions for a free pro- p algebra of infinite countable rank, and we prove that completed $\mathbb{F}_q$ -group algebras of torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups are Sylvester domains. We also prove the fact that its completed group algebra being Sylvester domain is a commensurability invariant among torsion-free pro- p groups. The fifth chapter studies rank functions on the completed $\mathbb{Z}_p$ -algebra $\mathbb{Z}_p[[G_1]]$ of a congruence subgroup G_1 of a simply connected split semisimple p -adic Lie group G . We pose two conjectures about the asymptotic behaviour of the rank functions induced from rational representations of G , which we confirm when G is a product of copies of SL_2 . The final chapter is devoted to applications of the results in chapters 4 and 5 to the theory of abstract groups, subgroups of semisimple complex Lie groups, hyperbolic geometry and automorphic forms.

Keywords: profinite groups, completed group algebras, Sylvester rank functions, Sylvester domains, p -adic Lie groups.

Introduction

In 2020, my then M.Sc. advisor committed a grave mistake: he told me to write as much as I wanted in my dissertation. History would come to repeat itself five years later.

The concept of rank of a real matrix is standard knowledge for any first year undergraduate STEM student. Indeed, most linear algebra courses beeline to an explanation of the algorithm of Gaussian elimination, which associates to each matrix its canonical row echelon form. The rank is then simply defined as the number of non-zero rows in the row echelon form of the matrix. Usual properties of the rank are then derived from this definition, such as additivity for block diagonal matrices and the rank-nullity theorem.

Less standard is the knowledge of how to generalize the notion of rank to a ring that is not a field like the reals. Indeed, *many* such generalizations exist. To cite one, we could work with the *inner rank* $\text{irk}(A)$ of an $n \times m$ matrix A : it is the smallest integer k such that A factors as a product BC of an $n \times k$ matrix B with a $k \times m$ matrix C . It is an enlightening exercise in basic linear algebra to check that this definition recovers the usual notion of rank over a field. Is it the “correct” one, though?

A different generalization is given by the notion of a *Sylvester matrix rank function*, named after James Joseph Sylvester. This is a real valued function rk defined on the set of all matrices over a given ring which satisfies a list of axioms similar to the properties of the usual rank over a field: the rank of the zero matrix is null, the rank of an $n \times n$ identity matrix is n , rank is additive under block diagonal constructions and so on. On one hand, it has an advantage over the inner rank: it behaves very nicely under change of rings, that is, if $\varphi: R \rightarrow S$ is a ring homomorphism and rk is a Sylvester matrix rank function on S , we can use φ to define a Sylvester matrix rank function on R . Ring homomorphisms that preserve the inner rank are very special and not often occur. On the other hand, we lose uniqueness: a ring often has infinitely many distinct Sylvester matrix rank functions, at least as many as it has homomorphisms to fields!

The plethora of Sylvester rank functions on a given ring can be both a blessing and a curse. Indeed, a fruitful direction of research from the past decades has been the study of

approximations of rank functions. That is, given a Sylvester rank function rk on a ring R that I am interested in, can I find a sequence of perhaps simpler rank functions rk_i on R such that for every matrix A over R we have

$$\lim_{i \rightarrow \infty} \text{rk}_i(A) = \text{rk}(A) ?$$

Another good warm-up exercise for the reader is to consider the following sequence of ranks on the ring $\mathbb{Z}$: if you enumerate the prime numbers $p_1, p_2, \dots$, what is the limit of the ranks rk_i given by reducing modulo p_i and computing the rank over the finite field $\mathbb{F}_{p_i}$?

This has been fruitful in the following sense: many combinatorial aspects of geometric group theory have come to be explained through the lens of rank functions defined on appropriate group algebras. Let us illustrate this with some examples. If M is a finite CW-complex and

$$\dots \rightarrow M_2 \rightarrow M_1 \rightarrow M$$

is a sequence of finite normal covering spaces with the property that $\bigcap \pi_1(M_i)$ is trivial as a subgroup of $\pi_1(M)$, then a conjecture of Mikhael Gromov later confirmed by Wolfgang Lück predicted that the Betti numbers of the M_i approximate the ℓ^2 -Betti numbers of M :

$$\lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{Q}} H_j(M_i; \mathbb{Q})}{\deg(M_i \rightarrow M)} = \beta_j^{(2)}(M) .$$

This theorem was immensely generalized in the following decades, with its most “recent” iteration being due to Andrei Jaikin-Zapiran: it says that over the complex group algebra of a sofic group G one can approximate its *von Neumann rank* by the Sylvester rank functions obtained from any sofic approximation of G . The original question of Gromov can be recovered by taking $G = \pi_1(M)$ and $X_i = \pi_1(M)/\pi_1(M_i)$ the corresponding sofic approximation, and realizing that Betti numbers can be uniformly described as ranks of some fixed matrices (depending only on G).

Another example comes from the theory of modular forms. If F is a number field and Γ is a congruence subgroup of $\text{SL}_2(\mathcal{O}_F)$, where $\mathcal{O}_F$ is the ring of integers of F , one can associate to it a complex vector space $S(\Gamma)$ of SL_2 -cuspidal forms of level Γ . Those are complex-valued functions defined on a locally-symmetric orbifold with fundamental group Γ that satisfy various invariance and growth conditions – but that is not important right now. The important part is that the functions on $S(\Gamma)$ encode critical arithmetical information about the number field F . This vector space is infinite-dimensional, but it can be graded by a tuple of integers $k = (k_1, \dots, k_n)$ such that each graded part $S_k(\Gamma)$ is finite dimensional. A major open problem in number theory is to describe the growth of $S_k(\Gamma)$ as k grows. The Eichler-Shimura isomorphism relates the spaces $S_k(\Gamma)$ to the homology groups $H_n(\Gamma, V_k)$ for some representation V_k of $\text{SL}_2(\mathbb{C})$. The best currently known upper bound for S_k can be recovered as an approximation result for Sylvester rank functions on the complex group algebra $\mathbb{C}[\Gamma]$, bounding the error as a function of k .

In both of these examples, there are natural profinite groups lying in the background. Inspired by these and other problems in profinite and geometric group theory, we can describe the questions at the center of this thesis as follows:

(A) Given a profinite group G and a ring R , are there “nice” Sylvester matrix rank functions on the completed group algebra $R[[G]]$?

(B) Is the inner rank a Sylvester matrix rank function on $R[[G]]$?

(C) If the answer to either (A) or (B) is “yes”, can we approximate this rank function by even nicer ones?

In my mission to study these three questions, I was led down two intertwined but distinguishable threads, which we will describe below.

Thread A: free-by-cyclic groups and the inner rank

We have remarked before that the inner rank does not behave well under ring homomorphisms. The situation is in fact more dire: the inner rank fails to satisfy the axioms of a Sylvester matrix rank function for most rings. The special rings for which the inner rank *does* satisfy said axioms are called *Sylvester domains*. In particular, question (B) can be rephrased as: “Which completed group algebras $R[[G]]$ are Sylvester domains?”

The concept of a Sylvester domain was introduced explicitly in a paper from 1978 by Warren Dicks and Eduardo Sontag. Its defining property, however, was previously studied by Paul Moritz Cohn in the context of *free ideal rings*. Throughout the 60’s and 70’s, Cohn was interested in the following question: given a free K -algebra A , that is, the tensor algebra of a K -vector space, is there a suitable notion of the “freest division K -algebra” into which A embeds?

This naturally led him to introduce the concepts of matrix localization, universal localization and that of a free ideal ring or *fir* — a ring in which every ideal is free of unique rank. In particular, he showed that for firs and semifirs (rings such that every finitely generated ideal is free of unique rank) (1) under any such embedding, any matrix that becomes invertible has maximal inner rank and (2) there exists an embedding into a division ring such that *all* matrices of maximal inner rank become invertible, and this embedding is “universal” among all embeddings. What Dicks and Sontag did was to recognize that a necessary and sufficient condition for (2) to hold is that the inner rank be a Sylvester matrix rank function and that Cohn’s argument was actually a proof that semifirs are Sylvester domains.

A corollary of Cohn’s results is that the group algebra $K[F]$ of a free group F is a Sylvester domain. Through their work, it became explicit that the Sylvester domain condition was very restrictive: for instance, if $K[G]$ is to be a Sylvester domain for some group G , then G must have cohomological dimension at most 2 over the field K . Another example is that while the polynomial algebras $K[x]$ and $K[x, y]$ are Sylvester domains, the polynomial algebra in three

variables $K[x, y, z]$ isn't anymore. A consequence that is implicit in the name but not obvious from the definition is that Sylvester domains cannot contain zero divisors either. One can also use their techniques to show, for instance, that $K[\mathbb{Z}^2]$ is a Sylvester domain.

Decades would pass before more group algebras were shown to be Sylvester domains. In this setting, a breakthrough came in a 2020 paper by Fabian Henneke and Diego López-Álvarez, where they showed that if G is a torsion-free free-by-cyclic group, that is, if G contains a normal free subgroup N such that $G/N \simeq \mathbb{Z}$, then $K[G]$ was a Sylvester domain if and only if $K[G]$ admitted *stably free cancellation*. This condition is known to hold for some G such as $\mathbb{Z}^2$ and $F \times \mathbb{Z}$, and also known to fail for others such as the fundamental group of the Klein bottle or of the complement of the trefoil knot. It is still open whether $\mathbb{C}[\pi_1(S)]$ is a Sylvester domain or not for S a closed hyperbolic surface.

It is in this setting that Jaikin proposed I started my PhD in 2021 by studying the following profinite analogue: if G is a free-by- $\mathbb{Z}_p$ pro- p group, that is, a pro- p group G containing a closed normal free pro- p subgroup N such that $G/N \simeq \mathbb{Z}_p$, is the completed $\mathbb{F}_p$ -group algebra $\mathbb{F}_p[[G]]$ a Sylvester domain?

When F is a finitely generated free pro- p group, the Magnus isomorphism due to Michel Lazard identifies the completed group algebra $\mathbb{F}_p[[F]]$ with the algebra $\mathbb{F}_p\langle\langle X \rangle\rangle$ of noncommutative power series over $\mathbb{F}_p$ in a finite set of variables X . Those rings were among the rings studied by Cohn, who also proved that they are semifirs. In particular, $\mathbb{F}_p[[F]]$ is a Sylvester domain. However, Cohn's proof is not exactly constructive: while he shows the existence of a universal embedding of $\mathbb{F}_p[[F]]$ into a division ring, his description of said division ring is too abstract to be of much use. An explicit construction of the embedding was later given by Jaikin in 2020, where he also gave a homological criterion to detect whether an embedding into a division ring was universal and witnessed the Sylvester property. His suggestion was that such methods could be combined with the techniques of Henneke and López-Álvarez to study free-by- $\mathbb{Z}_p$ pro- p groups.

And that is what we did. For the surprise of all involved, we could actually combine these ideas with some extra profinite inputs to prove something even stronger: $\mathbb{F}_p[[G]]$ is a Sylvester domain for any torsion-free pro- p group G that is *virtually* free-by- $\mathbb{Z}_p$. This includes, for example, the pro- p completions $\widehat{\pi_1(S)}_p$ of the fundamental groups of closed orientable surfaces as well as the pro-2 completion of the fundamental group of the Klein bottle. Another by-product of our proof was the following theorem: if G is a torsion-free pro- p group and U is an open subgroup of G , then $\mathbb{F}_p[[G]]$ is a Sylvester domain if and only if $\mathbb{F}_p[[U]]$ is a Sylvester domain. No analogue of this theorem for abstract group algebras has been found yet.

Our argument splits into three separate directions. The first one harnesses the power of John Labute's notion of a *mild* pro- p group G to derive strong structural consequences for the group algebra $\mathbb{F}_p[[G]]$ and its filtrations. We also prove that any free-by- $\mathbb{Z}_p$ pro- p is virtually mild. The second one is a methodical study of the filtration by powers of the augmentation ideal on $\mathbb{F}_p[[G]]$. Inspired by the methods for $\mathbb{F}_p\langle\langle X \rangle\rangle$, we construct a series of embeddings of $\mathbb{F}_p[[G]]$ into larger and larger filtered rings until we arrive at a filtered division ring with enough

control on the associated graded counterparts to ensure we can apply Jaikin's homological criteria.

The third part concerns question (C). Once we know that the inner rank is a Sylvester matrix rank function on $\mathbb{F}_p[[G]]$, can we approximate it by other natural rank functions? Whenever $R \rightarrow \text{End}_K(V)$ is a representation of a ring R as endomorphisms of a finite-dimensional K -vector space V , one can use the natural notion of rank on $\text{End}_K(V) \simeq \text{Mat}_n(K)$ to define a Sylvester matrix rank function on R . This in particular applies to case where $R = \mathbb{F}_p[[G]]$ and $V = \mathbb{F}_p[G/U]$ for some open subgroup U of G . We prove that the ranks coming from finite quotients of G approximate the inner rank in $\mathbb{F}_p[[G]]$ for G torsion-free virtually free-by- $\mathbb{Z}_p$.

Thread B: asymptotic representation theory

An infinite profinite group admits, almost by definition, *many* representations into finite dimensional vector spaces. Indeed, if G is profinite and U is an open subgroup of G , then for any field K we can represent the completed group algebra $K[[G]]$ as endomorphisms of the finite-dimensional vector space $K[G/U]$ by the regular representation. Not all representations, however, have to be of this kind.

A family of profinite groups which admit representations of a completely different nature comes from Lie groups. If G is a Lie group that “makes sense” over $\mathbb{Q}_p$, for example, $G = \text{SL}_2$ or $G = \text{PGL}_3$, then its set of $\mathbb{Z}_p$ points $G = G(\mathbb{Z}_p)$ is a (usually infinite) profinite group. In particular, it will have many representations over $\mathbb{Q}_p$ coming from its finite quotients. However, if $G \rightarrow \text{GL}_k$ is a Lie group representation that also “makes sense” over $\mathbb{Q}_p$, we can take $\mathbb{Q}_p$ points to get a representation $G \rightarrow \text{GL}_k(\mathbb{Q}_p)$ that behaves very differently to all the representations $\mathbb{Q}_p[G/U]$.

For example, consider the p -adic Lie group $\text{SL}_2(\mathbb{Q}_p)$ of 2×2 matrices over the p -adic field $\mathbb{Q}_p$ with determinant one. If $f(X, Y)$ is a polynomial in two variables with coefficients in $\mathbb{Q}_p$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a matrix in $\text{SL}_2(\mathbb{Q}_p)$, then I can define an action of A on the polynomial f by using A as a linear change of variables:

$$(fA)(X, Y) = f(aX + bY, cX + dY).$$

Another good exercise in linear algebra is to check that (1) this is a well defined left group action and (2) if f is homogeneous of degree d , then (fA) is again homogeneous of degree d . Since the space V_d of homogeneous polynomials in degree d is a $\mathbb{Q}_p$ -subspace of dimension $d + 1$ with basis

$$X^d, X^{d-1}Y, \dots, XY^{d-1}, Y,$$

we get a representation $\text{SL}_2(\mathbb{Q}_p) \rightarrow \text{GL}_d(\mathbb{Q}_p)$.

How is this representation, when restricted to $G = \mathrm{SL}_2(\mathbb{Z}_p)$, different from the representations coming from the finite quotients $\mathbb{Q}_p[G/U]$? For instance, the latter representations always have an open kernel, while one can easily check that the representations on V_d have a finite kernel for every $d > 1$. If v is any vector on $\mathbb{Q}_p[G/U]$, then its stabilizer in G is an open subgroup. On the other hand, non-zero vectors in V_d tend to have stabilizers which are closed but not open. To distinguish them from one another, we will call representations such as these V_d *Lie-theoretic* representations, whose precise meaning will be made clear in Chapter 5.

If the Lie group G is “simply-connected split semisimple” and p is odd, one can show that the first congruence kernel $G_1 = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$ is an example of a *uniform* pro- p group. Such groups have a very transparent structure: they are torsion-free, their completed group algebras $\mathbb{Z}_p[[G_1]]$ over the p -adic integers are Noetherian domains and they embed naturally into an Ore division ring of fractions $\mathcal{Q}$. If $G_i = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{Z}/p^i\mathbb{Z}))$ denotes the lower congruence kernels, a classic theorem of M. Harris relates the ranks rk_{G_1/G_i} on $\mathbb{Z}_p[[G_1]]$ coming from the representations $\mathbb{Z}_p[[G_1]] \rightarrow \mathbb{Q}_p[G_1/G_i]$ with the rank $\mathrm{rk}_{\mathcal{Q}}$ coming from the division ring $\mathcal{Q}$: for any matrix A over $\mathbb{Z}_p[[G_1]]$, we have

$$\left| \mathrm{rk}_{G_1/G_i}(A) - \mathrm{rk}_{\mathcal{Q}}(A) \right| \leq \frac{M}{p^{di}}$$

for some constants M and d depending only on G . In some sense, this addresses points (A), (B) and (C) for the group rings $\mathbb{Z}_p[[G_1]]$.

However, under these hypothesis, G also comes with infinitely many irreducible Lie-theoretic representations $G \rightarrow \mathrm{GL}_k$. These are usually parametrized by a unique tuple of non-negative integers $\lambda = (\lambda_1, \dots, \lambda_k)$ called its highest weight. Let us denote by V_λ the irreducible Lie-theoretic $\mathbb{Q}_p$ representation of $G(\mathbb{Q}_p)$ with highest weight λ . As we have discussed, the representations V_λ and $\mathbb{Q}_p[G_1/G_n]$ have little in common.

It may come as a surprise, then, our conjecture that from afar they are probably very similar. What we mean is that *asymptotically* they describe the same rank function: if rk_λ denotes the rank function on $\mathbb{Z}_p[[G_1]]$ induced from the representation $G_1 \rightarrow \mathrm{GL}(V_\lambda)$, we expect that

$$\left| \mathrm{rk}_\lambda(A) - \mathrm{rk}_{\mathcal{Q}}(A) \right| \leq \frac{M}{\min\{\lambda_1, \dots, \lambda_n\}}$$

for sufficiently large λ and any matrix A over $\mathbb{Z}_p[[G_1]]$, where M is a constant depending only on G . In particular, we conjecture that the ranks rk_λ also approximate the Ore rank $\mathrm{rk}_{\mathcal{Q}}$. As evidence, we present a proof that this conjecture holds for almost all primes p if $G = \prod_{i=1}^k \mathrm{SL}_2$ is a product of copies of SL_2 . The argument is due to Weibo Fu, but reinterpreted in the language of rank functions.

A particularly intriguing consequence of this result is the following original contribution by me and Lander Guerrero-Sánchez: if Γ is torsion-free subgroup of $\prod_{i=1}^k \mathrm{SL}_2(\mathbb{C})$ of type

$FP_{n+1}(\mathbb{C})$, then the cohomology of Γ with coefficients in the various Lie-theoretic representations W_λ of $\prod_{i=1}^k \mathrm{SL}_2(\mathbb{C})$ over $\mathbb{C}$ approximate ℓ^2 -Betti numbers:

$$\lim_{\min \lambda_i \rightarrow \infty} \frac{\dim_{\mathbb{C}} H_n(\Gamma, W_\lambda)}{\dim_{\mathbb{C}} W_\lambda} = \beta_n^{(2)}(\Gamma).$$

This has far reaching consequences: it yields new proofs of John Lott and Wolfgang Lück's computation of the ℓ^2 -Betti numbers of hyperbolic 3-manifolds of finite volume and extends this to infinite volume topologically finite hyperbolic 3-manifolds. It can also be used to give a new proof of the best known upper bound on the space $S_k(\Gamma)$ of cusp forms via the Eichler-Shimura isomorphism directly, without resorting to the machinery of completed cohomology as the original proof by Fu does. We also show how, modulo a question on the asymptotic representation theory of completed universal enveloping algebras we currently can only answer for $\mathfrak{sl}_2$, this could be extended to arbitrary complex semisimple Lie groups.

Again, the main idea of the proof is divided into two independent steps. One is handling the passage from $\mathbb{C}$ to $\mathbb{Q}_p$: it is a fairly straightforward consequence of Cassels' embedding theorem that any finitely generated subgroup Γ of $G(\mathbb{C})$ embeds in $G(\mathbb{Z}_p)$ for infinitely many primes p . Less obvious is how to give enough room to manoeuvre over finite index subgroups and over-groups. Since we want to work with subgroups of the first congruence subgroup G_1 , we must prove that convergence of the relevant rank functions over $\Gamma \cap G_1$ implies convergence over Γ .

Once this is dealt with, we can forget about the abstract groups entirely and work solely with the congruence subgroups $G_1 = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$. Now, the theory of algebraic microlocalization enters into play. If $\mathfrak{g}$ denotes the Lie $\mathbb{Q}_p$ -algebra of G_1 , a theorem of Konstantin Ardakov and Simon Wadsley that dates back to Lazard embeds the completed group algebra $\mathbb{Z}_p[[G_1]]$ into a completion $\widehat{U(\mathfrak{g})}$ of the universal enveloping $\mathbb{Q}_p$ -algebra $U(\mathfrak{g})$ of $\mathfrak{g}$ under a suitable norm. Moreover, this ring map $\mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g})}$ is flat.

The rank function $\mathrm{rk}_{\mathbb{Q}}$ on $\mathbb{Z}_p[[G_1]]$ can also be described as the unique rank function satisfying $\mathrm{rk}_{\mathbb{Q}}(x) = 1$ for any non-zero element $x \in \mathbb{Z}_p[[G_1]]$. Since we can see the rank functions rk_λ as also induced from the change of rings $\mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g})}$, the question of bounding the difference between rk_λ and $\mathrm{rk}_{\mathbb{Q}}$ on $\mathbb{Z}_p[[G_1]]$ becomes a question of giving a lower bound on $\mathrm{rk}_\lambda(x)$ for $x \in \widehat{U(\mathfrak{g})}$. Fu's proof is an argument that obtains a lower bound whenever x is *generic* and $G = \prod_{i=1}^k \mathrm{SL}_2$, that is, satisfies some list of conditions that Ardakov and Wadsley already proved to hold whenever x lies in the image of $\mathbb{Z}_p[[G_1]]$. Giving such lower bounds to other Lie groups G remains the main obstruction to extending this results to higher rank.

Main results

Before we move on to a breakdown of the structure of this thesis, let me briefly state and describe the main original results of this work. For a precise explanation of all the definitions involved, we refer the reader to the next section, the index at the end or the relevant chapter.

We recall that a ring R is a Sylvester domain if the inner rank irk_R is a Sylvester matrix rank function on R . If $G = \varprojlim G_i$ is a profinite group and R is a ring, the completed group ring of G with coefficients in R is the inverse limit $R[[G]] = \varprojlim R[G_i]$. An embedding $\varphi: R \rightarrow \mathcal{Q}$ from a Sylvester domain R into a division ring $\mathcal{Q}$ is universal if $\varphi(A)$ is invertible for any $n \times n$ matrix A over R such that $\text{irk}_R(A) = n$. The first main original contribution of this work is the following theorem.

Theorem 4.3.7. *Let G be a torsion-free finitely generated pro- p group. Then, the following are equivalent:*

- (A) $\mathbb{F}_q[[G]]$ is a Sylvester domain for any finite field $\mathbb{F}_q$ of characteristic p .
- (B) $\mathbb{F}_q[[U]]$ is a Sylvester domain for every open subgroup U of G and any finite field $\mathbb{F}_q$ of characteristic p .
- (C) $\mathbb{F}_q[[U]]$ is a Sylvester domain for some open subgroup U of G and some finite field $\mathbb{F}_q$ of characteristic p .

Moreover, in case one of the above holds and U is an open normal subgroup of G with a universal embedding $\mathbb{F}_p[[U]] \rightarrow \mathcal{Q}$, then $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is a division ring and the induced map $\mathbb{F}_q[[G]] \rightarrow (\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is also a universal embedding.

We apply this result to G a torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p group and U a particular mild subgroup of G . If $G > G_1 > G_2 > \dots$ is a chain of open normal subgroups of G with trivial intersection, then for any $n \times m$ matrix A over $\mathbb{F}_q[[G]]$ the limit

$$\lim_{i \rightarrow \infty} \text{rk}_{G/G_i}(A) = \frac{\dim_{\mathbb{F}_q} A \mathbb{F}_q[G/G_i]^m}{n|G: G_i|}$$

exists and does not depend on the chain G_i . We denote this limit Sylvester matrix rank function by $\text{rk}_G(A)$. We explicitly construct a universal embedding for $\mathbb{F}_p[[U]]$ to prove:.

Theorem 4.6.6. *Let G be a finitely generated torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p group. Then $\mathbb{F}_q[[G]]$ is a Sylvester domain and $\text{rk}_G = \text{irk}_{\mathbb{F}_q[[G]]}$.*

We fix $G(\mathbb{C})$ a complex semisimple Lie group with Lie algebra $\mathfrak{g}$. A central character of $G(\mathbb{C})$ is a homomorphism $\chi: Z(G(\mathbb{C})) \rightarrow \mathbb{C}^\times$. We denote by $P(\mathfrak{g})$ the set of integral dominant weights of $\mathfrak{g}$, that is, the set of weights λ of $\mathfrak{g}$ that are highest weights of some

irreducible representation V_λ of $G(\mathbb{C})$. If $\omega_1, \dots, \omega_n$ are the fundamental weights of $\mathfrak{g}$, then any $\lambda \in P(\mathfrak{g})$ can be uniquely written as

$$\lambda = \lambda_1 \omega_1 + \dots + \lambda_n \omega_n$$

where $\lambda_1, \dots, \lambda_n$ are non-negative integers. We set $\min \lambda = \min\{\lambda_1, \dots, \lambda_n\}$.

We denote by $P(\mathfrak{g})_\chi$ the set of integral dominant weights λ such that $Z(G(\mathbb{C}))$ acts on V_λ through χ . We let S_0 be a subring of $\mathbb{C}$ and Γ a subgroup of $G(\mathbb{C})$. The rank functions rk_{V_λ} and rk_Γ^χ are the Sylvester matrix rank function induced by the representation V_λ and the twisted von Neumann rank function on $S_0[\Gamma]$ respectively.

We pose the following two conjectures:

Conjecture 6.2.1. *For any $\lambda \in P(\mathfrak{g})_\chi$ and any matrix A over $S_0[\Gamma]$, if V_λ is the corresponding irreducible representation of G we have that*

$$\left| \text{rk}_{V_\lambda}(A) - \text{rk}_\Gamma^\chi(A) \right| = O\left(\frac{1}{\min \lambda}\right).$$

Conjecture 6.2.2. *As rank functions on $S_0[\Gamma]$, we have*

$$\lim_{\min \lambda \rightarrow \infty} \text{rk}_{V_\lambda} = \text{rk}_\Gamma^\chi,$$

where the limit is taken over $P(\mathfrak{g})_\chi$ and rk_Γ^χ is the twisted von Neumann rank of Γ by the central character χ restricted to Γ .

When the irreducible factors of the root system of $\mathfrak{g}$ are of type A_1 , we give positive results for these conjectures.

Corollary 6.2.7. *Let $G(\mathbb{C})$ be a connected semisimple complex Lie group such that its Lie algebra $\mathfrak{g}$ is isomorphic to $\prod_{i=1}^k \mathfrak{sl}_2(\mathbb{C})$ and Γ be a finitely generated subgroup of $G(\mathbb{C})$. Then, for any central character χ and each finitely generated subring S_0 of $\mathbb{C}$, Conjecture 6.2.1 holds over S_0 for some finite index subgroup of Γ and Conjecture 6.2.2 holds over $\mathbb{C}$ for Γ .*

Structure

This dissertation is naturally split into two parts plus one appendix. The first three chapters are devoted mostly to collecting results from the literature that will be needed for our proofs, as well as setting up general notation and nomenclature. With some exceptions, most of the results in these chapters are not original. The exposition, however, is focused mostly on profinite rings and completed group algebras of profinite groups. In this sense, some originality may be claimed in the choice of guiding examples, although most of their computations are certainly well known to the specialist.

The last three chapters contain most of the original material. They cover the results presented by the author in two research papers, one already published¹ and another submitted². In some parts, significantly more detail is given in the arguments in comparison with the published version. The exposition is also more extensive, with examples illustrating the general picture of each section. The appendix provides the definition and some basic results about the twisted von Neumann rank. While we could not find a reference for these facts, they follow standard arguments about group von Neumann algebras with the necessary adaptations to the twisted setting. Hence, their proofs are laid out for the convenience of the reader. It is presented as an appendix by virtue of its techniques being noticeably different, but it is not less important nor less carefully presented.

Chapter 1 deals with the basic theory of topological rings that will be fundamental to the development of the thesis. It sets up the notation and states, mostly without proofs but with references to the literature, the main results about profinite, filtered and graded rings that will be needed. Several examples relevant to the later chapters are also discussed.

The theme of Chapter 2 is the localization of non-commutative rings: how to build from a ring R a universal over-ring S in which certain elements of R become invertible. Many flavours of localization are presented: the Ore localization, which generalizes the commutative notion of fractions, the microlocalization for filtered rings which preserves completeness and Cohn's theory of matrix localization. All three localizations will play some role in the main results. In this chapter we also show how to topologize a matrix localization of a profinite ring.

Chapter 3 presents the main object we will study on profinite rings: Sylvester rank functions. It also connects the concept of an integral and of an ϵ -separated rank function with that of a map to a division ring and to a finite dimensional representation via Peter Malcomson's and Aidan Schofield's theorems. This chapter is also rich with examples of rank functions on completed group rings, and how they interplay.

In Chapter 4 we start a systematic study of which completed group algebras of profinite groups can be Sylvester domains. It contains: (1) an extension of Jaikin's construction of an universal embedding for free pro- p algebras of finite rank to infinite countable rank, (2) the proof that the Sylvester property is a commensurability invariant for torsion-free pro- p groups and (3) the proof that completed group $\mathbb{F}_q$ -algebras of torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups are Sylvester domains, and moreover their universal rank function can be approximated by the normalized rank functions from finite quotients.

In Chapter 5 we change directions to focus on the completed group rings $\mathbb{Z}_p[[G_1]]$ over $\mathbb{Z}_p$ of the first congruence subgroup of a simply-connected split semisimple p -adic Lie group for an odd prime p . The notion of "simply-connected split semisimple" is formalized through Chevalley's universal group construction, which allows one to easily move between split semisimple algebraic groups over various rings. In this chapter we present Michael Harris'

¹Andrei Jaikin-Zapirain and Henrique Souza. "Sylvester domains and pro- p groups". In: *Documenta Mathematica* (Sept. 2025). ISSN: 1431-0643. DOI: 10.4171/dm/1034.

²Lander Guerrero Sánchez and Henrique Souza. *Asymptotics of rational representations for algebraic groups*. 2024. DOI: 10.48550/ARXIV.2405.17360.

proof that the Ore rank on the Ore domain $\mathbb{Z}_p[[G_1]]$ can be approximated by ranks coming from finite quotients of G_1 . We also explain Ardakov's and Wadsley's proof that $\mathbb{Z}_p[[G_1]]$ embeds in a completion of the p -adic universal enveloping algebra $\widehat{U(\mathfrak{g})}$ of the lie algebra $\mathfrak{g}$ of G_1 . It is also in this chapter that we state a conjecture about the asymptotic behaviour of the irreducible finite dimensional representations of $\widehat{U(\mathfrak{g})}$ that we prove for $\mathfrak{g} = \prod_{i=1}^k \mathfrak{sl}_2$.

Chapter 6 ties both problems back to their origins in the theory of abstract groups. We give applications of the results in Chapters 4 and 5 to the Lück approximation conjecture in characteristic p , the growth of the homology of subgroups of semisimple complex Lie groups, the approximation of ℓ^2 -Betti numbers of hyperbolic manifolds and upper bounds for the space of cusp forms on SL_2 .

This thesis ends with a summary of the theory of the twisted von Neumann ranks in Appendix A. References to the literature are given in footnotes³, which are used extensively. A full list of the references cited is also available at the end, together with an index of mathematical terms.

³For it is better to look at the end of a page than to skip all the way to the end of a book.

Chapter 1

Ring theory

Use drugs.

S.F. on doing a PhD.

There is a famous¹ quote attributed to Italian mathematician Gian-Carlo Rota:

“Every mathematician has only a few tricks.”

If the present author may be permitted a provocation, I would amend this to “every mathematician has only one trick, and it is *linear algebra*”.

The idea of reducing non-linear problems to linear ones is as old as mathematics itself. In the context of groups and their modules, the natural objects to consider are their associated group rings. Their effectiveness is illustrated by the extent to which the modern theory of finite groups can be explained through the structure of their group algebras over various fields. In this setting, the usual notion of finite generation of modules comes with the stronger and *very* useful property of finite *dimensionality*. The very core of linear algebra depends on the elementary but crucial fact that vector spaces have well-defined ranks.

For infinite groups, the condition that a module over a group algebra be finite dimensional becomes far more restrictive. There are, however, other notions of “rank” that have proven meaningful.² The common theme of this work is the embedding completed group algebras of finitely generated infinite pro- p groups into topological algebras equipped with well-behaved rank functions, and the study of the induced ranks on matrices over the group. Hence, we start our investigation by collecting basic facts about topological rings in general, and rings whose topology comes from a filtration more specifically.

¹or, perhaps, infamous? Gian-Carlo Rota. “Ten Lessons I Wish I Had Been Taught”. In: *Notices of the American Mathematical Society* 44.1 (1997), pp. 22–25. ISSN: 0002-9920

²See Chapter 3

Most of the material in this chapter is well known, and with minor exceptions can be found in the works of Nicolas Bourbaki³, Vladimir Arnautov, Sergei Glavatsky and Aleksandr Mikhalev⁴ and Constantin Năstăsescu and Fred van Oystaeyen⁵. We also assume familiarity with the basic theory of profinite groups, as exposed in the early chapters of Luis Ribes' and Pavel Zalesskii's book⁶. Given its introductory nature, we have chosen to state the results in this chapter mostly without proofs⁷, and the reader may be encouraged to start reading this thesis from Chapter 2 and refer back to this chapter as needed.

In Section 1.1, we state basic facts and set up the notation and nomenclature we will use for topological rings and modules. We specialize in the class of profinite rings in Section 1.2, where we show how a profinite ring without zero divisors must be a pro- p local ring. In Section 1.3, we consider the class of topological filtered rings. While there is a substantial theory of filtered rings already in place, it deals mostly with ascending filtrations, and for our purposes we need to consider descending filtrations. The section with most detail in its arguments is Section 1.4, dealing with graded rings associated to filtered rings, where we conclude this chapter showing how one may carry information about Tor groups between a filtered ring, its completion and its associated graded counterpart.

All rings are assumed to be associative and unital (but not necessarily commutative) unless stated otherwise – Lie rings and algebras will appear throughout this work, but their non-associative and non-unital nature will always be explicitly characterized with the suggestive word “Lie”.

1.1 Topological rings and modules

A *topological group* is a group endowed with a topology for which multiplication and inversion are continuous functions. Analogously, a *topological ring* is a ring endowed with a topology for which addition, subtraction, and multiplication are continuous functions.

In any topological ring R , the family $\mathcal{B}$ of neighborhoods of 0 satisfies⁸:

(RT1) For every $U \in \mathcal{B}$, $0 \in U$.

(RT2) For every $U, V \in \mathcal{B}$, there is $W \in \mathcal{B}$ such that $W \subseteq U \cap V$.

³Nicolas Bourbaki. *Elements of Mathematics: General Topology, Pt. I*. Addison-Wesley Educational Publishers Inc, 1967.

⁴Sergei Timofeevich Glavatsky, Aleksandr Vasil'evich Mikhalev, and Vladimir Ivanovich Arnautov. *Introduction to the Theory of Topological Rings and Modules*. Monographs and Textbooks in Pure and Applied Mathematics 197. M. Dekker, CRC Press, 1996.

⁵Constantin Năstăsescu and Fred van Oystaeyen. *Graded Ring Theory*. North-Holland Mathematical Library 28. North-Holland Pub. Co., 1982.

⁶Luis Ribes and Pavel Zalesskii. *Profinite Groups*. 2nd ed. A Series of Modern Surveys in Mathematics. Springer-Verlag Berlin Heidelberg, 2010.

⁷A major exception being Section 1.4

⁸Glavatsky, Mikhalev, and Arnautov, *Introduction to the Theory of Topological Rings and Modules*, Prop. 1.1.2.

(RT₃) For every $U \in \mathcal{B}$ there is $V \in \mathcal{B}$ such that $V + V \subseteq U$.

(RT₄) For every $U \in \mathcal{B}$ there is $V \in \mathcal{B}$ such that $-V \subseteq U$.

(RT₅) For every $U \in \mathcal{B}$ there is $V \in \mathcal{B}$ such that $V \cdot V \subseteq U$.

(RT₆) For every $U \in \mathcal{B}$ and $x \in R$ there is $V \in \mathcal{B}$ such that $xV, Vx \subseteq U$.

Conversely⁹, any non-empty family $\mathcal{B}$ of subsets of a ring R satisfying (RT₁)-(RT₆) determines a unique ring topology on R for which $\mathcal{B}$ is a basis of neighborhoods of 0. This description is useful to obtain ring topologies described algebraically. For example, there is a unique ring topology $\tau_{\mathbb{Z}}$ on $\mathbb{Q}$ for which the family of subsets $x\mathbb{Z}$, $x \in \mathbb{Q}^{\times}$, forms a basis of neighborhoods of 0.¹⁰ When there is a basis $\mathcal{B}$ of neighborhoods of 0 consisting of open left (resp. right, two-sided) ideals of R , we say that the topology on R is *left linear* (resp. *right linear*, *linear*).

If $(R, +, \cdot)$ is a topological ring, then $(R, +)$ is also a topological group, whereas $(R^{\times}, \cdot)$ may fail to be a topological group, as there is no assumption of continuity of the multiplicative inverse. A topological ring R for which $x \mapsto x^{-1}$ is continuous on $R^{\times}$ is said to have *continuous inverses*. Any subring of a topological ring is again a topological ring with the subspace topology, and it inherits the property of continuous inverses.

Example 1.1.1. Trivially, any discrete ring is a topological ring with continuous inverses. The usual rings $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}_p$ of rationals, reals, complex and p -adic numbers are also topological rings with continuous inverses. The ring topology $\tau_{\mathbb{Z}}$ on $\mathbb{Q}$ defined previously does not have continuous inverses. An arbitrary product of topological rings with the product topology is once again a topological ring, and it has continuous inverses if and only if each factor does so.¹¹ It follows that any inverse limit of topological rings is again a topological ring, and the limit inherits the continuity of inversion. ■

Since $(R, +)$ is a topological group whenever $(R, +, \cdot)$ is a topological ring, to check if a topological ring is Hausdorff, it suffices¹² to check that the singleton $\{0\}$ is closed in R . The closure of $\{0\}$ is a two-sided ideal O of R , and the quotient ring R/O becomes a Hausdorff topological ring under the quotient topology. All our rings shall be Hausdorff. Observe that a Hausdorff ring is automatically completely regular¹³: given a closed subset S of R and a point $x \notin S$, there is a continuous function $f: R \rightarrow [0, 1]$ such that $f(x) = 1$ and $f(y) = 0$ for every $y \in S$.

⁹Ibid., Thm. 1.2.5.

¹⁰This topology also comes from the embedding of $\mathbb{Q}$ in its ring of finite adeles.

¹¹The group of units in $\prod R_i$ is $\prod R_i^{\times}$, and inversion is given by taking the product of the continuous projection from $\prod R_i^{\times}$ onto $R_i^{\times}$ and then inverting.

¹²Bourbaki, *Elements of Mathematics*, Sec. III.1.2.

¹³Or T_{3½}, or Tychonoff. See Thm. 1.3.2 in Glavatsky, Mikhalev, and Arnautov, *Introduction to the Theory of Topological Rings and Modules*

For a separable topological space X , Urysohn's metrization theorem states that X is metrizable if and only if X is Hausdorff, regular (or T_3) and second countable.¹⁴ Hence, a separable topological ring R is metrizable if and only if it has a countable family $\mathcal{B}$ of neighborhoods of 0 satisfying (RT1)-(RT6). All of the rings we will consider in the future chapters are separable and metrizable.

A *topological module* over R is simply an R -module endowed with an abelian group topology for which the multiplication map $R \times M \rightarrow M$ is continuous. Given $\mathcal{B}$, the set of neighborhoods of 0 of R , one can describe uniquely the topology of a left (resp. right) R -module in terms of a basis of neighborhoods $\mathcal{C}$ of 0 in M satisfying¹⁵

(MT1) For every $U \in \mathcal{C}$, $0 \in U$.

(MT2) For every $U, V \in \mathcal{C}$, there is $W \in \mathcal{B}$ such that $W \subseteq U \cap V$.

(MT3) For every $U \in \mathcal{C}$ there is $V \in \mathcal{B}$ such that $V + V \subseteq U$.

(MT4) For every $U \in \mathcal{C}$ there is $V \in \mathcal{B}$ such that $-V \subseteq U$.

(MT5) For every $U \in \mathcal{C}$ there is $V \in \mathcal{B}$ and $V' \in \mathcal{C}$ such that $V \cdot V' \subseteq U$ (resp. $V' \cdot V \subseteq U$).

(MT6) For every $U \in \mathcal{C}$ and $x \in M$ there is $V \in \mathcal{B}$ such that $Vx \subseteq U$ (resp. $xV \subseteq U$).

If a module has a basis of neighborhoods of 0 consisting of open submodules, we say its topology is *linear*.

For any submodule N of M , both N and the quotient M/N are topological modules under the subspace and quotient topology respectively. As with any topological group, a submodule is open if and only if it contains an interior point, and any open submodule is also closed. Any discrete submodule of a Hausdorff module is closed, and a submodule is discrete if and only if it has an isolated point. The quotient module M/N is Hausdorff if and only if N is closed, and it is discrete if and only if N is open. The closure O of $\{0\}$ on any topological R -module M is a submodule, and M/O is Hausdorff. If M is Hausdorff (resp. discrete), the annihilator of any point in M is closed (resp. open) in R .

The connected component C of 0 in a topological ring R is a closed two-sided ideal, and the quotient ring R/I is a totally disconnected ring (which is automatically Hausdorff). In a topological R -module M , the connected component C of 0 is a closed submodule and M/C is a totally disconnected R -module, and again, automatically Hausdorff.

We say that a topological R -module M is *topologically generated* by a set $X \subseteq M$ if the smallest closed submodule of M containing X is M itself. A *simple* topological R -module is a non-zero R -module M that is topologically generated by any of its non-zero elements. Equivalently, a non-trivial topological R -module is simple if its closed submodules are $\{0\}$ and

¹⁴A topological space is separable if it contains a dense countable subset, and second countable if it contains a countable basis.

¹⁵Glavatsky, Mikhalev, and Arnautov, *Introduction to the Theory of Topological Rings and Modules*, Thm. 1.2.6.

M . Hence, a topological simple R -module is of the form R/I for some maximal left or right ideal I of R (not necessarily closed).

A subset X of R is *topologically nil* if for any $x \in X$ we have $\lim_{n \rightarrow \infty} x^n = 0$, and *topologically nilpotent* if for any sequence $(x_i)_{i \geq 0}$ in X the products $x_1 \cdots x_n$ converge to 0 as $n \rightarrow \infty$. Any topologically nilpotent subset is necessarily topologically nil. We shall later need the following observation in Chapter 4.

Lemma 1.1.2. *Let R be a discrete ring and $X = \{x_1, \dots, x_n\}$ a topologically nilpotent subset of X . Then, the set of non-commutative monomials in the alphabet X that do not map to 0 in R is finite.*

Proof. The condition that X is topologically nilpotent in a discrete ring R means that no infinite chain

$$y_1, y_1 y_2, \dots, y_1 y_2 \cdots y_k, \dots$$

of non-zero products exists with each $y_i \in X$. Let C be set of all non-commutative monomials in the alphabet X that do not map to 0 in R , including the “trivial” monomial 1. We define a graph structure on C by adjoining an edge between m and mx_i for any monomial m . Observe that this graph structure has no loops, since we are considering its elements as “free” monomials in the alphabet X , not as in elements of R . Under this structure, C becomes a locally finite rooted tree. If it were infinite, by König’s lemma¹⁶ it would contain an infinite ray, contradicting the topological nilpotency of X . Hence, C must be finite. $\square$

A ring R with a linear topology is called *complete*¹⁷ if the map

$$R \rightarrow \varprojlim R/U$$

is surjective, where the inverse limit is taken over all open two-sided ideals U of R . Observe that this map is injective if and only if R is Hausdorff. We also say that a topological R -module M with a linear topology is *complete* if the map

$$M \rightarrow \varprojlim M/N$$

is surjective, where the inverse limit is taken over all open submodules N of M . Once again, this map is injective if and only if M is Hausdorff.

An arbitrary product or sum of topological (Hausdorff, linear, totally disconnected) rings or modules is again a topological (Hausdorff, linear, totally disconnected) ring or module under the product and sum topology respectively. An arbitrary product of complete rings is complete. An inverse limit of topological (Hausdorff, linear, totally disconnected, complete)

¹⁶Reinhard Diestel. *Graph Theory*. Springer Berlin Heidelberg, 2025. ISBN: 9783662701072. DOI: 10.1007/978-3-662-70107-2, Lem. 9.1.3.

¹⁷Our definition of complete ring will always mean complete with respect to a linear topology. In particular, $\mathbb{R}$ is a topological ring and a complete metric space, but not a complete ring!

rings or modules is topological (Hausdorff, linear, totally disconnected, complete) under the subspace topology of the product.

We say that a Hausdorff complete ring R is *pseudocompact* if R/U is Artinian for every open two-sided ideal U of R . We say that a Hausdorff complete module M is *pseudocompact* if M/N is of finite length for every open submodule N of M . Every discrete Artinian ring is pseudocompact. Any direct product or inverse limit of pseudocompact rings and modules is again pseudocompact. Any compact, Hausdorff, and complete ring is pseudocompact, as for example the ring $\mathbb{Z}_p$ of p -adic integers.

The category of topological R -modules with continuous homomorphisms as its morphisms is quasi-abelian but not abelian in general, as bijections may fail to be isomorphisms.¹⁸ The following lemma shows that this obstruction vanishes for pseudocompact modules, so that the category of pseudocompact R -modules with continuous homomorphisms is abelian.¹⁹

Lemma 1.1.3. *Let R be a topological ring and $\varphi: M \rightarrow N$ a continuous bijection between pseudocompact R -modules M and N . Then, $\varphi^{-1}: N \rightarrow M$ is also continuous.*

Proof. It suffices to prove that the subspace and the quotient topologies on $\varphi(M)$ coincide. Let $\{M_i\}_{i \in I}$ and $\{N_j\}_{j \in J}$ be two linear bases of neighborhoods of 0 on M and N respectively such that each M/M_i and N/N_j has finite length for each $i \in I$ and $j \in J$. Since φ is continuous, for each N_j there is an M_i for which $\varphi(M_i) \subseteq N_j \cap \varphi(M)$, and it remains to find N_j for which $N_j \cap \varphi(M) \subseteq \varphi(M_i)$.

A standard diagram chase on

$$\begin{array}{ccc}
 \varphi(M_i) & \xrightarrow{\gamma} & \varprojlim_{j \in J} \varphi(M_i) / (\varphi(M_i) \cap N_j) \\
 \downarrow & & \downarrow \\
 \varphi(M) & \xrightarrow{\psi} & \varprojlim_{j \in J} \varphi(M) / (\varphi(M) \cap N_j) \\
 \downarrow & & \downarrow \\
 \varphi(M) / \varphi(M_i) & \xrightarrow{\theta} & \varprojlim_{j \in J} \varphi(M) / (\varphi(M_i) + (\varphi(M) \cap N_j))
 \end{array}$$

shows that γ and ψ are bijective, and hence so is θ . By writing down its kernel, we get:

$$\bigcap_{j \in J} (\varphi(M_i) + (\varphi(M) \cap N_j)) = \varphi(M_i).$$

¹⁸Let M be any non-discrete $\mathbb{Z}$ -module for the discrete topology on $\mathbb{Z}$ and $N \simeq M$ as abstract $\mathbb{Z}$ -modules but with the discrete topology. Then, the identity map $N \rightarrow M$ is a continuous homomorphism that is not an isomorphism.

¹⁹Pierre Gabriel. “Des Catégories Abéliennes”. In: *Bulletin de la Société Mathématique de France* 90 (1962), pp. 323–448, Thm. 3.

Since $\varphi(M)/\varphi(M_i)$ is of finite length, there exists some $j' \in J$ such that $\varphi(M_i) + (\varphi(M) \cap N_{j'})$ is minimal among all such sums. Therefore,

$$\varphi(M_i) + (\varphi(M) \cap N_{j'}) = \varphi(M_i)$$

and hence

$$\varphi(M) \cap N_{j'} \subseteq \varphi(M_i)$$

as desired. $\square$

In a Hausdorff complete ring, a series $\sum x_i$ converges if and only if $x_i \rightarrow 0$.²⁰ Hence, if X is a topologically nil subset of a complete ring R , then $1 - x$ is a unit in R for every x in X with inverse $(1 - x)^{-1} = \sum_{i \geq 0} x^i$.

Example 1.1.4. If R is a topological ring, then we can make $S = R[[t]]$ into a topological ring as well by taking the sets $U \cdot S + t^k S$ to be a basis of neighborhoods of 0 as k ranges through the positive integers and U through a basis of neighborhoods of 0 in R . With this topology, S is Hausdorff. This topology is linear if the topology on R is linear, and in this case it is complete if R is complete. The set tS is topologically nilpotent, and hence $S^\times = R^\times \times (1 + tS)$. If moreover R also has continuous inverses, then so does S . If R is pseudocompact, then so is $R[[t]]$. $\blacksquare$

This example can be generalized to the non-commutative case as follows²¹:

Example 1.1.5. Let σ be a continuous automorphism of a Hausdorff complete ring R and assume that $\delta = \sigma - \text{id}$ is a topologically nilpotent operator, that is, for every $x \in R$ the sequence $\delta^k(x)$ converges to 0. Then, the *right skew power series ring* $S = R[[s; \sigma, \delta]]$ is the right R -module of all formal power series $\sum_{i \geq 0} s^i a_i$ with coefficients in R with multiplication defined by

$$\left(\sum_{i \geq 0} s^i a_i \right) \left(\sum_{j \geq 0} s^j b_j \right) = \sum_{m \geq 0} s^m \left(\sum_{n=0}^m \sum_{k \geq n} \binom{k}{n} \delta^{k-n}(\sigma^n(a_{m-n})) b_k \right).$$

Since δ is topologically nilpotent and R is complete and Hausdorff, the series on the right always converges to a unique limit for any choices of a_i and b_j , and this endows S with a ring structure. In S we have the relation $xs = s\sigma(x) + \delta(x)$ for every $x \in R$.

There is an analogous notion of a *left skew power series ring*. It consists of the left R -module of all formal power series $\sum_{i \geq 0} a_i s^i$ with coefficients in R and multiplication defined by

$$\left(\sum_{i \geq 0} a_i s^i \right) \left(\sum_{j \geq 0} b_j s^j \right) = \sum_{m \geq 0} \left(\sum_{n=0}^m \sum_{k \geq n} a_k \binom{k}{n} \delta^{k-n}(\sigma^n(b_{m-n})) \right) s^m.$$

²⁰Bourbaki, *Elements of Mathematics*, Thm. III.5.2.1.

²¹Peter Schneider and Otmar Venjakob. "On the Codimension of Modules over Skew Power Series Rings with Applications to Iwasawa Algebras". en. In: *Journal of Pure and Applied Algebra* 204.2 (Feb. 2006), pp. 349–367. ISSN: 0022-4049. DOI: 10.1016/j.jpaa.2005.05.007.

If $\sigma' = \sigma^{-1}$ is also continuous and $\delta' = \sigma' - \text{id} = -\delta \circ \sigma'$ is again topologically nilpotent, then the ring of left skew power series $R[[s; \sigma, \delta]]$ and the ring of right skew power series $R[[s; \sigma', \delta']]$ are naturally isomorphic under the maps

$$\begin{aligned} \sum_{i \geq 0} s^i a_i &\mapsto \sum_{j \geq 0} \left(\sum_{i \geq j} \binom{i}{j} \delta^{i-j}(\sigma^j(a_i)) \right) s^j, \\ \sum_{i \geq 0} a_i s^i &\mapsto \sum_{j \geq 0} s^j \left(\sum_{i \geq j} \binom{i}{j} (\delta')^{i-j}((\sigma')^j(a_i)) \right). \end{aligned}$$

Since we have a natural left or right R -module isomorphism between S and $\prod_{i \geq 0} R$, the product topology on the latter induces a natural left or right linear Hausdorff R -module topology on the skew power series ring. If R has a basis $\{U_i\}_{i \in I}$ of neighborhoods of 0 which are right (resp. left) ideals, then the sets $S \cdot U_i + s^k S$ (resp $U_i \cdot S + S s^k$) form a basis of neighborhoods of 0 of this topology on S .²² The multiplication map $S \times S \rightarrow S$ is continuous under this topology, making S into a topological ring. ■

1.2 Profinite rings

A *profinite ring* is a topological ring R that is isomorphic to an inverse limit of finite discrete rings. The prototypical examples of profinite rings are the ring of p -adic integers $\mathbb{Z}_p = \varprojlim \mathbb{Z} / p^i \mathbb{Z}$, the profinite completion

$$\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z} / n \mathbb{Z} \simeq \prod_{p \text{ prime}} \mathbb{Z}_p$$

of the integers and the ring of formal power series $R[[t]]$ whenever R is profinite. Note that by definition every finite discrete ring is a profinite ring. Another important class of profinite rings are the completed group rings: if R is a profinite ring and G is a profinite group, then we can form the completed group ring

$$R[[G]] = \varprojlim_{I \leq_o R, U \trianglelefteq_o G} (R/I)[G/U]$$

as the inverse limit of the finite group rings $(R/I)[G/U]$ as I runs through the open two-sided ideals of R , and U through the open normal subgroups of G .

Since the product of finite discrete rings is compact, Hausdorff and totally disconnected, then so is any profinite ring. In contrast to profinite groups, a statement even stronger than the converse holds:²³

²²If the U_i are also σ -invariant, those sets are left and right ideals of S respectively.

²³Ribes and Zalesskii, *Profinite Groups*, Prop. 5.1.2.

Proposition 1.2.1. *Let R be a topological ring. Then, the following are equivalent:*

1. R is profinite.
2. R is isomorphic to an inverse limit of profinite rings.
3. R is compact, Hausdorff and totally disconnected.
4. R is compact and Hausdorff.

In particular, a profinite ring is the same as a compact pseudocompact ring. It also shows that if R is a profinite ring and $\sigma: R \rightarrow R$ is a continuous ring automorphism, then the rings of skew power series $R[[s; \sigma, \delta]]$ are also profinite rings, since they are topologically homeomorphic to $\prod_{i \geq 0} R$ which is compact and Hausdorff.

Several basic results about profinite groups generalize immediately to profinite rings:²⁴

- a. Every profinite ring admits a fundamental system of neighborhoods of 0 consisting of open two-sided ideals, that is, any profinite topology on a ring is linear and complete.
- b. If I is a closed ideal of R , then R/I is profinite.
- c. Any continuous bijective homomorphism between profinite rings has a continuous inverse.

Some properties of profinite rings follow directly from these basic results. For example, we say that a ring R has *invariant basis number*, or IBN for short, if $R^n \simeq R^m$ as R -modules implies $n = m$. Clearly any finite ring has IBN, and therefore so does any profinite ring by passing to a finite quotients.

The continuity of inversion also follows from the finite discrete case by taking the limit. The group $R^\times$ of units is closed in R , since it is the inverse limit of the units in the finite quotients. In particular, it is also a profinite group. If R is a profinite ring, then so is the ring $\text{Mat}_n(R)$ of $n \times n$ matrices over R . Therefore, its group of units $\text{GL}_n(R) = (\text{Mat}_n(R))^\times$ is again a profinite group.

Let R be a profinite ring. A topological R -module M is *profinite* if it is isomorphic to an inverse limit of finite discrete R -modules. Equivalently, M is profinite if and only if M is compact, Hausdorff, and totally disconnected. Every profinite R -module admits a fundamental system of neighborhoods of 0 consisting of open submodules, that is, any profinite module topology is linear and complete. A continuous bijective homomorphism between profinite modules has a continuous inverse. The category of profinite modules with continuous homomorphisms is an abelian category.

Another important family of topological R -modules are the *discrete modules*. A topological R -module is discrete if and only if it is isomorphic to the direct limit of finite discrete R -modules,

²⁴Ibid., Sec. 5.1.

so in particular it must be a torsion abelian group. A discrete R -module is also the direct union of its submodules annihilated by I as I runs through the open two-sided ideals of R . The category of discrete modules and their homomorphisms is an abelian category. The Pontryagin dual of a left (resp. right) profinite R -module is a discrete right (resp. left) R -module and vice-versa, yielding a perfect duality between the subcategories of profinite R -modules and discrete R -modules.²⁵

Let X be a topological generating set for M . When X is finite, M is Hausdorff and R is profinite, this agrees with the abstract notion of generation by a set: if $R[X]$ denotes the abstract free R -module on X , then $R[X]$ is profinite and the natural homomorphism $\varphi: R[X] \rightarrow M$ has dense image in M . By the compactness of $R[X]$, we have $\varphi(R[X]) = M$. In particular, if M is Hausdorff, every finitely generated abstract R -submodule of M is profinite and closed in M . If a left (resp. right) R -module M is *finitely presented*, then we have an exact sequence

$$R^n \xrightarrow{\phi} R^m \rightarrow M \rightarrow 0$$

where the map ϕ is represented by right (resp. left) multiplication by some matrix $A \in \text{Mat}_{n \times m}(R)$ (resp. $\text{Mat}_{m \times n}(R)$). A finitely presented R -module is automatically profinite, since the image of ϕ is compact, and hence closed. Any pseudocompact R -module is also profinite, as a Hausdorff R -module of finite length must be finite and discrete.

The simple topological R -modules M don't need to be profinite in general.²⁶ A simple R -module $M \simeq R/I$ is profinite if and only if the ideal I is closed, in which case M is finite.

If X is a profinite space, say $X = \varprojlim X_i$, then one can form the profinite R -module

$$R[[X]] = \varprojlim R[X_i]$$

which satisfies the following property in the category of all profinite R -modules: any continuous function $f: X \rightarrow M$ where M is a profinite R -module extends uniquely to a continuous homomorphism $\varphi: R[[X]] \rightarrow M$. This is called the *free profinite* R -module on X . It follows that the category of profinite R -modules has enough projectives. Moreover, any finitely generated projective profinite R -module is also projective as an abstract R -module.

If M and N are two profinite R -modules, their tensor product $M \otimes N$ is, in general, not a profinite abelian group. The tensor product in the category of profinite R -modules is given by

$$M \hat{\otimes}_R N = \varprojlim M/M_i \otimes_R N/N_j$$

²⁵Ribes and Zalesskii, *Profinite Groups*, Thm. 2.9.6.

²⁶For $R = \hat{\mathbb{Z}} \simeq \prod_p \mathbb{Z}_p$, the ideal $I = \bigoplus_p \mathbb{Z}_p$ is dense and hence contained in a proper, not closed, maximal ideal. It is known that for a profinite ring R , the following statements are equivalent: (a) every left ideal of R is closed, (b) R is Noetherian, (c) R satisfies the ACC on left closed ideals, (d) R is Noetherian and the topology on R is the $J(R)$ -adic topology, where $J(R)$ is the Jacobson radical of R , $R/J(R)$ is finite and this topology is complete and Hausdorff. See Seth Warner. "Compact Rings". In: *Mathematische Annalen* 145.1 (Feb. 1962), pp. 52–63, Thm. 2

where M_i and N_j run through the open submodules of M and N respectively. This module is called the *completed tensor product* of M and N . Note that if X and Y are profinite spaces, we have $R[[X \times Y]] \simeq R[[X]] \hat{\otimes}_R R[[Y]]$. We denote by $\mathfrak{Tor}_i^R(-, -)$ the i -th left derived bifunctor of $-\hat{\otimes}_R-$ in the category of pairs of profinite R -modules.²⁷

In general, the category of profinite R -modules doesn't have enough injectives.²⁸ However, the category of discrete R -modules has enough injectives. Hence, denoting by $\mathfrak{Hom}_R(M, N)$ the abelian group of all continuous homomorphisms between topological R -modules M and N , one can form its i -th right derived functors $\mathfrak{Ext}_R^i(M, N)$ whenever M is profinite and N is discrete. Since $-\hat{\otimes}_R-$ is a profinite group and $\mathfrak{Hom}_R(-, -)$ is a discrete group (in the compact-open topology), the same is true for their respective derived functors $\mathfrak{Tor}_i^R(-, -)$ and $\mathfrak{Ext}_R^i(-, -)$.

The (abstract) *Jacobson radical* $J(R)$ of a ring R is the intersection of all maximal left (or, equivalently, right) ideals of R . This is a two-sided ideal of R . Another well known characterization of the Jacobson radical is as follows: an element x of R belongs to $J(R)$ if and only if $1 + yx$ is a unit for every $y \in R$, which is also the case if and only if $1 + xy$ is a unit for every $y \in R$.

In view of the existence of maximal ideals which are not closed, one might expect modifications to the definition in order to adapt the Jacobson ideal for the profinite world. It is, however, unnecessary:²⁹

Proposition 1.2.2. *If R is profinite, then $J(R)$ is the intersection of all maximal closed³⁰ left (or right) ideals of R . In particular, $J(R)$ is a closed two-sided ideal of R .*

Proof. Let $\tilde{J}$ denote the intersection of all maximal closed left ideals of R . All we need to show is that $\tilde{J}$ is contained in $J(R)$. Since $\tilde{J}$ is a left ideal, by the characterization of the Jacobson radical it suffices to prove that $1 + x$ is a unit whenever $x \in \tilde{J}$. If not, by compactness there would be an open ideal I_0 such that $1 \notin R(1 + x) + I_0$. Let I_1 be a maximal open left ideal containing $R(1 + x) + I_0$. We would have both x and $1 + x$ belonging to I_1 , from which we obtain that $1 \in I_1$ and therefore $I_1 = R$, a contradiction. $\square$

Since the Jacobson radical of any finite ring is nilpotent, it follows by taking the limit that the Jacobson radical of any profinite ring is topologically nilpotent. In particular, $1 + J(R)$ is a

²⁷Ribes and Zalesskii, *Profinite Groups*, Sec. 6.1.

²⁸By Pontryagin duality, there are enough injective profinite modules if and only if there are enough projective discrete R -modules. For example, if G is an infinite profinite group, then the only projective discrete $\widehat{\mathbb{Z}}[[G]]$ -module is the trivial module, while there are enough projective discrete modules for a countable product of finite fields. See the discussion by Alexandru Chirvasitu and Ryo Kanda. "Projective Discrete Modules over Profinite Groups". In: *Journal of Pure and Applied Algebra* 227.4 (Apr. 2023), p. 107260

²⁹Jan Dobrowolski and Krzysztof Krupiński. "Locally Finite Profinite Rings". In: *Journal of Algebra* 401 (Mar. 2014), pp. 161–178, Prop. 2.1.

³⁰By the previous observations, such an ideal is open in R . Hence, there is no difference between a maximal closed ideal and a closed maximal ideal.

closed subgroup of $R^\times$ for every profinite ring R , and it's precisely the kernel of the induced map $R^\times \rightarrow (R/J(R))^\times$.

We say that a ring R is *local*³¹ if $R/J(R)$ is a division ring, and *semisimple* if $J(R) = 0$. A ring is local if and only if there is a unique maximal ideal (automatically two-sided), and if R is profinite this ideal is automatically open. Hence, for a local profinite ring, Wedderburn's little theorem says that $R/J(R)$ is a finite discrete field. In this case, we call $\text{char}(R/J(R))$ the *residual characteristic* of R . Moreover, since $J(R)$ is topologically nilpotent, if $J(R)^2$ is open then the powers $J(R)^n$ form a basis of neighborhoods of 0 in R .

Proposition 1.2.3. *If R is a local profinite ring, then*

$$R^\times \simeq (1 + J(R)) \rtimes (R/J(R))^\times.$$

Proof. Let p be the residual characteristic of R . Since $(1 + J(R)^n)/(1 + J(R)^{n+1})$ is isomorphic to $J(R)^n/J(R)^{n+1}$ and the latter is a module over $R/J(R)$, it is an $\mathbb{F}_p$ -vector space. In particular, $1 + J(R)$ is a pro- p group as it admits a descending series of open normal subgroups with trivial intersection and pro- p quotients. Since the order of $(R/J(R))^\times$ is $p^k - 1$ for some $k \geq 1$, the Schur-Zassenhaus theorem³² tells us that the short exact sequence

$$1 \rightarrow 1 + J(R) \rightarrow R^\times \rightarrow (R/J(R))^\times \rightarrow 1$$

splits.³³ □

The ring $R[[G]]$ when the order of G is invertible³⁴ in R is an example of a semisimple profinite ring. On the other extreme, the rings $\mathbb{Z}_p$ and $R[[G]]$ when G is pro- p and R is local with residual characteristic p are examples of profinite local rings.³⁵ It is also relevant to note that $R[[t]]$ is a profinite local ring whenever R is a profinite local ring, with the maximal ideal being generated by $J(R)$ and t .

There is an extension of Wedderburn's theorem for artinian rings to profinite rings due to Irving Kaplansky:³⁶

³¹Some authors ask only that $R/J(R)$ is simple Artinian, and reserve the name *scalar local* for when $R/J(R)$ is a division ring.

³²Ribes and Zalesskii, *Profinite Groups*, Thm. 2.3.15.

³³The semi-direct product is not direct in general. This can be seen in the 4-dimensional $\mathbb{F}_2$ -algebra R generated by two elements a and b satisfying $a^2 = a + 1$, $ba = ab + b$ and $b^2 = 0$. We have $J(R) = Rb$, $R/Rb \simeq \mathbb{F}_4$, $J(R)^2 = 0$ but R is not commutative.

³⁴The order of a profinite group is a supernatural number, that is, a formal product $\prod_{p \text{ prime}} p^{a_p}$ where $a_p \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. To say that this order is a unit in R amounts to saying that p is a unit in R whenever $a_p > 0$.

³⁵It suffices to see this when both R and G are finite. If R is local with residual characteristic p , then the order of R/I is a power of p for every open two-sided ideal I of R . Hence, we may assume that R is a finite local p -ring and G is a finite p -group. We claim that the only simple $R[G]$ -module M is $R/J(R)$ with the trivial G -action. Since $J(R)M$ is a proper submodule, it must be annihilated by $J(R)$, so we reduce to the well known case of the group algebra of a finite p -group over a field of characteristic p . Observe that the converse is also true: if R and $R[[G]]$ are local with residual characteristic p , then G is pro- p .

³⁶Irving Kaplansky. "Topological Rings". In: *American Journal of Mathematics* 69.1 (1947), pp. 153–183, Thm. 16.

Theorem 1.2.4. *A profinite semisimple ring is isomorphic to a possibly infinite direct product of discrete matrix rings over finite fields.*

There are two key steps on the proof of this theorem. The first one is observing that $J(R)$ contains the intersection of all maximal two-sided open ideals of R , that is, the kernels of all continuous surjections of R to non-zero finite rings. The second one is showing that one has an analogue of the Chinese Remainder Theorem for distinct maximal two-sided open ideals, showing that the injective ring map $R \rightarrow \prod R/I$ where I ranges through such ideals has dense image.

We recall the non-commutative version of Nakayama's Lemma: if M is any non-zero left (resp. right) finitely generated R -module, then $J(R)M$ (resp. $MJ(R)$) is a proper submodule of M .³⁷ From this, it follows that any finitely generated profinite projective R -module over a profinite local ring R is actually free of finite rank.³⁸ In fact, one can drop the finitely generated hypothesis:³⁹

Proposition 1.2.5. *Let R be a profinite ring and M a profinite non-zero left (resp. right) R -module. Then, $J(R)M$ (resp. $MJ(R)$) is a proper submodule of M . In particular, a subset $X \subseteq M$ topologically generates M if and only if it topologically generates $M/J(R)M$ (resp. $M/MJ(R)$).*

Proof. We write out a proof for left modules, as the case for right modules is completely analogous. The statement is equivalent to: if $J(R)M = M$ for some profinite left R -module M , then $M = \{0\}$.

Let N be an open submodule of M . For every element x of M , one can find an open two-sided ideal I_x of R and an open neighborhood V_x of x such that $I_x V_x \subseteq N$. Since the module M/N is finite, it is finitely generated, say by $x_1, \dots, x_n$. If we take $I = I_{x_1} \cap \dots \cap I_{x_n}$, we have $IM \subseteq N$.

Since the ring R/I is finite and $J(R/I)$ is nilpotent, there are maximal open ideals $J_1 \dots, J_k$ of R and a positive integer m such that $(J_1 \dots J_k)^m \subseteq I$. Hence:

$$M = J(R)^m M \subseteq (J_1 \dots J_k)^m M \subseteq IM = N.$$

This shows that M is the only open submodule of itself, and therefore $M = \{0\}$. $\square$

Another construction relating the rings R and $J(R)$ is the lifting of idempotents. Recall that a non-zero element $e \in R$ is an *idempotent* if $e^2 = e$, and two idempotents e and f are *orthogonal* if $ef = fe = 0$. Lastly, an idempotent is *primitive* if it cannot be written as a

³⁷The more well known statement of $J(R)M + N = M$ implies $N = M$ for any submodule N of M follows from this formulation: indeed, we have $J(R)(M/N) = (J(R)M + N)/N = M/N$...

³⁸Another well known result of Kaplansky states that any *abstract* projective module over a local ring is an *abstract* free module.

³⁹Armand Brumer. "Pseudocompact Algebras, Profinite Groups and Class Formations". In: *Journal of Algebra* 4:3 (Nov. 1966), pp. 442–470, Lem. 1.4.

sum of two orthogonal idempotents. A key fact about profinite groups relates the primitive orthogonal idempotents of R and those of $R/J(R)$:⁴⁰

Proposition 1.2.6. *Let R be a profinite ring and $(\bar{e}_i)_{i \in I}$ a summable⁴¹ family of orthogonal primitive idempotents in $R/J(R)$. Then:*

1. *There exists a summable family $(e_i)_{i \in I}$ of orthogonal primitive idempotents in R such that $e_i + J(R) = \bar{e}_i$ for every $i \in I$. Moreover, if $\sum \bar{e}_i = 1$ in $R/J(R)$, then $\sum e_i = 1$ in R .*
2. *Let $e = \sum_{i \in I} e_i$. Then the product of the inclusion maps $Re_i \rightarrow R$ induces an isomorphism of left profinite R -modules $\prod_{i \in I} Re_i \simeq Re$, and analogously for right R -modules.*

For a proof of the above proposition, we refer the reader to the work⁴² of Leonid Positselski and Jan Štoviček. The proof adapts the well known argument for when $J(R)$ is a nilpotent ideal to the topologically nilpotent setting.

A *pro- p ring* is a inverse limit of finite discrete rings whose order is a power of p . As a commutative profinite abelian group, $(R, +)$ decomposes as the direct product of its Sylow p -subgroups R_p . Since they are invariant under every group endomorphism of R , each R_p is a closed two-sided ideal of R and hence we get a decomposition $R = \prod_{p \text{ prime}} R_p$ of any profinite ring R as an infinite direct product of pro- p subrings.

Hence, any profinite local ring is pro- p , as well as any profinite ring without zero divisors. If R doesn't have zero divisors, then it must also be local in view of Proposition 1.2.6 since one could lift the non-identity orthogonal idempotents of the semisimple profinite ring $R/J(R)$. In particular, if a completed group algebra $R[[G]]$ has no zero divisors, then R and G are both pro- p .

We finish this introductory section with the some remarks about commutative profinite rings.

Proposition 1.2.7. *Any commutative profinite ring has a decomposition as a direct product of profinite commutative local rings.*⁴³

Proof. The semisimple commutative ring $R/J(R)$ is a (possibly infinite) direct product $\prod_{i \in I} F_i$ of finite fields by Theorem 1.2.4, and so by Proposition 1.2.6 we can lift the orthogonal primitive idempotents $\bar{e}_i$ corresponding to the unit in each factor F_i of $R/J(R)$ to orthogonal primitive idempotents e_i of R such that $\sum_{i \in I} e_i = 1$. We then have an isomorphism of ring $R \simeq \prod_{i \in I} Re_i$, where each Re_i is a ring with unit e_i . We claim that each Re_i is a local ring. Since the Jacobson radical of Re_i is $J(R)e_i = Re_i \cap J(R)$, we have

$$Re_i/J(Re_i) = Re_i/J(R)e_i = Re_i/(Re_i \cap J(R)) = (R/J(R))\bar{e}_i.$$

⁴⁰This proposition was shown in much greater generality for pseudocompact rings in Gabriel, “Des Catégories Abéliennes”, pp. 393–394

⁴¹Summable means that the the partial finite sums $e_{i_1} + \dots + e_{i_k}$ converge under the filter of cofinite subsets of I .

⁴²Leonid Positselski and Jan Štoviček. “Topologically Semisimple and Topologically Perfect Topological Rings”. In: *Publicacions Matemàtiques* 66 (July 2022), pp. 457–540, Cor. 8.6.

⁴³Ribes and Zalesskii, *Profinite Groups*, Exercise 5.1.3.

This is a finite field by hypothesis, so $J(Re_i)$ is maximal and Re_i is local as desired. $\square$

We recall how the global dimension $\text{gl.dim}(R)$ of a profinite ring R is defined:⁴⁴ it is the supremum of the projective dimensions of the profinite R -modules, where a profinite R -module M has projective dimension at most n if there exists a resolution

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \dots \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

of M by profinite projective R -modules P_i . If the ring R is Noetherian, then any module of finite projective dimension admits a finite free resolution and therefore this notion coincides with the classical notion of global dimension. The following is an application of Cohen's structure theorem:

Theorem 1.2.8. *Let R be a Noetherian commutative local profinite domain with residue field $k = R/J(R)$. If $\text{gl.dim}(R) = n$ is finite, then either $R \simeq k[[x_1, \dots, x_n]]$ is a power series ring in n variables over k or R is an Eisenstein extension of the power series ring $\mathcal{O}_K[[x_2, \dots, x_n]]$ in $n - 1$ variables where $\mathcal{O}_K$ is the ring of integers of a finite extension K of $\mathbb{Q}_p$ with residue field k .*

Proof. A classical theorem of Auslander-Buchsbaum and Serre^{45,46} states that, under these hypothesis, R is a regular local ring of Krull dimension n . Observe that the residue field k is finite of characteristic p . Then, Cohen's structure theorem⁴⁷ says that either $R \simeq k[[x_1, \dots, x_n]]$, or there exists a discrete valuation ring $W \leq R$ of characteristic 0 with residue field k with $S = W[[x_1, \dots, x_n]]$ such that $R \simeq S/(u)$, where u is either x_1 or an element of $S \setminus J(S)$ ² according to whether $p \notin J(R)^2$ or $p \in J(R)^2$. If $p \notin J(R)^2$, R is said to be unramified, and R is said to be ramified otherwise. Since W is a discrete valuation ring with a finite field of characteristic p as its residue field, it is well known that $W = \mathcal{O}_K$ for some finite extension K of $\mathbb{Q}_p$: by continuity, $\mathbb{Z}_p$ embeds into W , and $W/pW \simeq W/\pi^e W$ where π is a uniformizer for W is an $\mathbb{F}_p$ -vector space of dimension $e \cdot \dim_{\mathbb{F}_p} k$. Hence, $K = \text{Frac}(W)$ is a finite extension of $\mathbb{Q}_p$, with $W = \mathcal{O}_K$ being its valuation ring. This already covers the unramified case. In the ramified case, it can be further shown⁴⁸ that R is an extension of $S' = W'[[x_2, \dots, x_n]]$ with W' another discrete valuation ring with residue field of characteristic p by an Eisenstein polynomial

$$P(x_1) = x_1^l + a_1 x_1^{l-1} + \dots + a_l$$

⁴⁴Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations".

⁴⁵Maurice Auslander and David A. Buchsbaum. "Homological dimension in local rings". In: *Transactions of the American Mathematical Society* 85.2 (1957), pp. 390–405. ISSN: 1088-6850. DOI: 10.1090/S0002-9947-1957-0086822-7.

⁴⁶Stanislaw Balcerzyk and Tadeusz Józefiak. *Commutative Rings: Dimension, Multiplicity and Homological Methods*. 1st ed. Ellis Hollywood Limited, 1989, Thm. 2.1.5.

⁴⁷Ibid., Thm. 2.3.20.

⁴⁸Ibid., Thm. 2.3.22.

where $a_i \in J(S')$ for all i and $a_l \notin J(S')^2$. Observe that the residue field doesn't change under an Eisenstein extension: the polynomial we're adding decomposes completely modulo $J(S')$, and so the residue field of W' must also be k . $\square$

1.3 Filtered rings

Let R be a ring. A *filtration* on R is a family $F_i R$ of additive subgroups of R indexed by the integers and satisfying

$$(FR_1) \quad R = \bigcup_{i \in \mathbb{Z}} F_i R;$$

$$(FR_2) \quad 1 \in F_0 R;$$

$$(FR_3) \quad \text{For every } i, j \in \mathbb{Z} \text{ we have } F_i R \cdot F_j R \subseteq F_{i+j} R;$$

$$(FR_4) \quad \text{If } i \leq j, \text{ then } F_j R \subseteq F_i R. \quad ^{49}$$

Each set $F_i R$ is called a *level* of the filtration, the i -th level. A *filtered ring* is a ring with a choice of a filtration. The axioms (FR₁)-(FR₄) imply that the family $\{F_i R\}$ satisfies all the axioms (RT₁)-(RT₆) of a basis of neighborhoods of 0 for a ring topology on R . We say that this is the topology *induced* by the filtration.

A filtration on a left (resp. right) R -module M over a filtered ring R is a family $F_i M$ of additive subgroups of M indexed by the integers satisfying

$$(FM_1) \quad M = \bigcup_{i \in \mathbb{Z}} F_i M;$$

$$(FM_2) \quad \text{For every } i, j \in \mathbb{Z} \text{ we have } F_i R \cdot F_j M \subseteq F_{i+j} M \text{ (resp. } F_i M \cdot F_j R \subseteq F_{i+j} M);$$

$$(FM_3) \quad \text{If } i \leq j, \text{ then } F_j M \subseteq F_i M.$$

Once again, each set $F_i M$ is called a *level* of the filtration. A *filtered module* is a module with a choice of a filtration. The axioms (FM₁)-(FM₃) imply that the family $\{F_i M\}$ satisfies the axioms (MT₁)-(MT₆) for being a basis of neighborhoods of 0 of an R -module topology on M , the induced topology.

We say the filtration is *separated* if $\bigcap_{i \in \mathbb{Z}} F_i M = \{0\}$. The filtration is separated if and only if the induced topology is Hausdorff. The intersection O of all the $F_i M$ is a submodule of M for which M/O inherits a separated filtration.

If $F_0 M = M$, we say that the filtration is *positive*. In a ring with a positive filtration, each set $F_i R$ is a two-sided ideal of R , so the induced topology is linear. Moreover, if the filtration on R is positive, then the level sets $F_i M$ are submodules in every filtered R -module M , so that the

⁴⁹For us, a filtration will always mean a descending filtration, except in Section 5.4. An ascending filtration would be one in which $i \leq j$ implies $F_i R \subseteq F_j R$. Some authors also only require (FR₂)-(FR₄), and call the filtrations satisfying (FR₁) to be exhaustive. We will only need to consider exhaustive filtered modules.

induced topology on filtered R -modules is also linear. Assuming the filtration on R is positive, we say a filtered module M is *complete* if the induced linear topology is complete, that is, if the natural map

$$M \rightarrow \varprojlim_{i \in \mathbb{Z}} M/F_i M$$

is surjective. We say a filtered ring is separated, positive or complete if it is so as a filtered module over itself, the definition being left/right symmetric.

If R is filtered and separated, then we can metricize⁵⁰ the topology induced by the filtration by means of

$$d(x, y) = 2^{-\sup\{i: x-y \in F_i R\}}.$$

When R is positive, this metric is complete if and only if R is complete.

Example 1.3.1. Any ring R admits the trivial filtration by means of $F_i R = R$ if $i \leq 0$ and $F_i R = 0$ otherwise, and this filtration is positive, complete and separated.⁵¹ If I is a two-sided ideal in a ring R , then R admits a positive filtration whose n -th level is I^n for $n \geq 1$. We call this filtration the I -adic filtration, and the corresponding topology is the I -adic topology. In particular, any profinite local ring with $J(R)$ finitely generated admits a complete separated positive filtration by powers of its Jacobson radical,⁵² and this filtration induces the profinite topology. More generally, the topology on a positive filtered ring is profinite if and only if it is complete, separated and each quotient $R/F_i R$ is finite. ■

Example 1.3.2. If R is a commutative ring with identity and $\mathfrak{g}$ is a Lie algebra over R , then the *universal enveloping algebra* $U(\mathfrak{g})$ of $\mathfrak{g}$ over R is the quotient of the tensor algebra⁵³ of $\mathfrak{g}$ over R by the relations $xy - yx = [x, y]$ for every $x, y \in \mathfrak{g}$. The tensor algebra admits a natural *ascending* filtration by declaring each element in $\mathfrak{g}$ to have degree 1, and this passes as a quotient filtration on $U(\mathfrak{g})$ called the *Poincaré-Birkhoff-Witt filtration* (or PBW filtration for short).⁵⁴

We say $\mathfrak{g}$ is a *filtered Lie ring* over a filtered ring R if it is a filtered R -module and moreover it satisfies $[F_i \mathfrak{g}, F_j \mathfrak{g}] \subseteq F_{i+j} \mathfrak{g}$ for every $i, j \in \mathbb{Z}$. When $\mathfrak{g}$ is a filtered Lie ring, we can endow $U(\mathfrak{g})$ with a *descending filtration* as follows: the level $F_m U(\mathfrak{g})$ is generated by $F_m R$ and all the products $x_1 \cdots x_n$ of elements $x_j \in \mathfrak{g}$ such that there are integers $i_1, \dots, i_n$ with $x_j \in F_{i_j} \mathfrak{g}$ and $i_1 + \cdots + i_n = m$. If the filtrations on R and $\mathfrak{g}$ are positive, then so is the filtration on $U(\mathfrak{g})$.

⁵⁰Even if the filtration is not separated, the formula below still defines a pseudo-metric that induces the filtration topology on R .

⁵¹It induces the discrete topology on R .

⁵²If $J(R)$ is finitely generated, then $J(R)^2$ is open since $J(R)/J(R)^2$ is a finitely generated vector space over the finite field $R/J(R)$. By induction, since the map $\bigotimes_{R/J(R)}^n J(R)/J(R)^2 \rightarrow J(R)^n/J(R)^{n+1}$ is surjective, all the $J(R)^i$ are finitely generated and hence open. Since the radical of finite rings is nilpotent, we have that for every open ideal I of R there exists a power of $J(R)$ contained in it, so that $\bigcap_{i \geq 0} J(R)^i = \{0\}$. This establishes the isomorphism $R \simeq \varprojlim R/J(R)^i$.

⁵³if $\mathfrak{g}$ is a free R -module with basis X , the tensor algebra is simply the free non-commutative R -algebra $R\langle X \rangle$ on X .

⁵⁴Nicolas Bourbaki. *Lie Groups and Lie Algebras: Chapters 1-3*. en. Springer Science & Business Media, 1989, Sec. I.2.6.

If R is separated and $\mathfrak{g}$ is a separated and free R -module, then the filtration on $U(\mathfrak{g})$ is also separated.⁵⁵ However, it is almost never complete.⁵⁶ ■

Example 1.3.3. For any ring R , the ring of power series $R[[t]]$ admits the positive I -adic filtration by powers of $I = (t)$. This is a complete and separated filtration. If R is filtered and positive, one can modify this filtration to take into account the filtration of R by declaring $F_i(R[[t]])$ to be the subgroup generated by all the additive subgroups $t^k(F_j R)[[t]]$ where $k + j = i$. Equivalently:

$$F_i(R[[t]]) = \left\{ \sum_{j \geq 0} a_j t^j : a_j \in F_{i-j} R \right\}.$$

This filtration is positive and separated, and it is complete if R is complete. ■

Any R -submodule N of a filtered R -module M is also filtered under the filtration $F_i N = N \cap F_i M$, called the *submodule filtration*. If M is positive or separated, then so is N . If M is complete and N is closed, then N is also complete. Under the submodule filtration topology, the embedding $N \rightarrow M$ is continuous.

For any submodule N of M , the quotient $Q = M/N$ also admits a filtration given by $F_i Q = (F_i M + N)/N$, called the *quotient filtration*. This filtration is positive if the filtration on M is so, and it is separated if and only if N is closed in the induced topology of M . If M is complete, then so is Q .⁵⁷ Under the quotient filtration topology, the natural map $M \rightarrow Q$ is continuous.

An R -homomorphism $\varphi: M \rightarrow N$ of filtered R -modules is said to have degree $d \in \mathbb{Z}$ if $\varphi(F_i M) \subseteq F_{i+d} N$. If φ has degree d , then it also has degree d' for any $d' \leq d$. An R -homomorphism is filtered if it has degree 0. It is said to be a *strict* homomorphism if it has degree 0 and the quotient and subspace filtrations on $\varphi(M)$ agree, that is, $\varphi(F_i M) = \varphi(M) \cap F_i N$. Any homomorphism of R -modules of any degree d is continuous in the induced topologies, and it is strict if and only if the canonical isomorphism $M / \ker(\varphi) \simeq \varphi(M)$ is a filtered map with filtered inverse, where $M / \ker(\varphi)$ has the quotient filtration and $\varphi(M)$ the subspace filtration.⁵⁸ The category of filtered modules over a filtered ring with filtered homomorphism is a quasi-abelian category.

⁵⁵This will follow from the Poincaré-Birkhoff-Witt theorem on the PBW filtration, see Theorem 1.4.6

⁵⁶It is never complete if $\mathfrak{g}$ is a non-trivial free R -module with the direct sum filtration.

⁵⁷For each non-negative i , we obtain a short exact sequence

$$N / (N \cap F_i N) \hookrightarrow M / F_i M \twoheadrightarrow Q / F_i Q.$$

Since the inverse system of the $N / (N \cap F_i M)$ consists of surjective maps, it satisfies the Mittag-Leffler condition, so the inverse limit preserves exactness.

⁵⁸In particular, in the category of filtered modules with filtered homomorphisms, the isomorphism theorem only holds true for strict homomorphisms. It follows that the category of filtered modules with filtered homomorphisms is not abelian in general. Moreover, since the composition of strict homomorphisms is also not strict in general, one cannot restrict to strict homomorphisms in order to obtain an abelian category. This will be the main source of trouble when trying to do homological algebra over filtered modules.

If $\{M_i : i \in I\}$ is a family of filtered R -modules, then their direct sum $S = \bigoplus_{i \in I} M_i$ admits a filtration $F_j S = \bigoplus_{i \in I} F_j M_i$ satisfying the usual universal properties with respect to filtered homomorphisms $M_i \rightarrow N$. Under the quotient filtration, the direct limit also admits a filtration and becomes a direct limit in the filtered category.

For the direct product $P = \prod_{i \in I} M_i$, the collection $F_j P = \prod_{i \in I} F_j M_i$ is *not* a filtration in general, as it can fail to satisfy (FM1). If (M_i, φ_{ij}) is an inverse system of filtered modules and each $\varphi_{ij} : M_j \rightarrow M_i$ ($i \geq j$) is filtered, then the $(F_k M_i, \varphi_{ij})$ form an inverse system for each $k \in \mathbb{Z}$. Denoting by M (resp. $F_k M$) the inverse limit $\varprojlim M_i$ (resp. $\varprojlim F_k M_i$), each $F_k M$ is an additive subgroup of M . However, the family $\{F_k M\}$ may still fail to satisfy (FM1). However, if the filtration on each M_i is positive⁵⁹, then $F_0 P = P$ and $F_0 M = M$, so (FM1) trivially holds and we obtain a filtration on both the direct product and the inverse limit.

If the filtration on R is positive, for any filtered R -module M we call

$$\widehat{M} = \varprojlim_{i \in \mathbb{Z}} M/F_i M$$

the *completion* of M . If the filtration on M is positive, then it is again a positive filtered R -module with the filtration

$$F_j \widehat{M} = \varprojlim_{i \geq j} F_j M/F_i M,$$

which is now complete and separated. The completion $\widehat{R}$ of R is also a positive, separated and complete filtered ring, and the action of R on $\widehat{M}$ extends naturally to a filtered action of $\widehat{R}$. For every i , the natural map $M \rightarrow \widehat{M}$ induces isomorphisms $M/F_i M \simeq \widehat{M}/F_i \widehat{M}$. If $\varphi : M \rightarrow N$ has degree $d \geq 0$, then so does the induced map $\widehat{\varphi} : \widehat{M} \rightarrow \widehat{N}$.

Lemma 1.3.4. *If $\varphi : M \rightarrow N$ is strict, then so is $\widehat{\varphi} : \widehat{M} \rightarrow \widehat{N}$.*

Proof. Denote the elements of $\widehat{M}$ by a sequence $(x_j + F_j M)_{j \geq 0}$ where each $x_j + F_j M$ is an element of $M/F_j M$ and $x_{j'} + F_{j'} M = x_j + F_j M$ for every $j' \geq j$. Hence, the elements of $F_i \widehat{M}$ are the sequences $(x_j + F_j M)$ for which each $x_j + F_j M$ is an element of $F_i M/F_j M$ whenever $j > i$, that is, x_j can be chosen to lie in $F_i M$.

Assume that $\widehat{\varphi}(x_j + F_j M) = (\varphi(x_j) + F_j N)$ is an element of $F_i \widehat{N}$. We can assume that $\varphi(x_j) \in F_i N$ for every $j > i$. We construct an element $(z_j + F_j M)$ of $F_i \widehat{M}$ satisfying $\widehat{\varphi}(z_j + F_j M) = \widehat{\varphi}(x_j + F_j M)$ inductively as follows: for every $j \leq i$, we take $z_j = x_j$. For each $j > i$, we have a commutative diagram

$$\begin{array}{ccccc} (\ker \varphi \cap F_i M) / (\ker \varphi \cap F_j M) & \longrightarrow & F_i M / F_j M & \longrightarrow & F_i N / F_j N \\ \downarrow & & \downarrow & & \downarrow \\ (\ker \varphi \cap F_i M) / (\ker \varphi \cap F_{j-1} M) & \longrightarrow & F_i M / F_{j-1} M & \longrightarrow & F_i N / F_{j-1} N \end{array}$$

⁵⁹Or, more generally, uniformly bounded from below.

We assume that z_{j-1} is already defined as satisfies the required properties, so that $\varphi(z_{j-1}) + F_{j-1}N = \varphi(x_{j-1}) + F_{j-1}N$.

Since φ is strict and $\varphi(x_j) \in F_iN$, we know that there is an element $a_j \in F_iM$ such that $\varphi(a_j) = \varphi(x_j)$. Moreover, the difference $z_{j-1} - a_j + F_{j-1}M$ belongs to $(\ker \varphi \cap F_iM) / (\ker \varphi \cap F_{j-1}M)$. Therefore, we can lift this difference to an element $c_j + F_jM$ of $(\ker \varphi \cap F_iM) / (\ker \varphi \cap F_jM)$. Then, the element $z_j = a_j + c_j + F_jM$ has the desired properties. $\square$

Hence, taking the completion is a functor from the category of positive filtered R -modules to the category of positive, separated, complete filtered $\widehat{R}$ -modules that preserves degree and strictness. Since $N = \widehat{N}$ for any separated and complete module N , we've also shown the following universal property: any R -homomorphism $\varphi: M \rightarrow N$ of degree d (resp. strict) where M is positive and N is complete and separated induces a unique homomorphism $\widehat{\varphi}: \widehat{M} \rightarrow N$ of degree d (resp. strict).

Another operation on filtered R -modules is the *suspension* functor. For any $d \in \mathbb{Z}$ and any filtered module M , let T_dM be isomorphic to M as an abstract R -module but with a new filtration defined by $F_iT_dM = F_{i+d}M$. It has the following property: an R -homomorphism $M \rightarrow N$ of filtered modules has degree d if and only if the induced homomorphism $T_{-d}M \rightarrow N$ (or equivalently $M \rightarrow T_dN$) has degree 0.

If $\varphi: M \rightarrow N$ is a bijective homomorphism of degree d , then its inverse $\varphi^{-1}: N \rightarrow M$ doesn't need to have any degree.⁶⁰ We say a filtered bijective homomorphism $\varphi: M \rightarrow N$ is an isomorphism if its inverse is also filtered. Equivalently, an isomorphism of filtered modules is a strict bijection.

Any filtered map has a natural filtered kernel and cokernel: the kernel of $\varphi: M \rightarrow N$ is the R -module $\ker(M)$ with the submodule filtration, and the cokernel is $N/\varphi(M)$ with the quotient filtration. Unless φ is strict, there is ambiguity if the image $\varphi(M)$ should be filtered with the submodule or quotient filtrations. Because of this, we consider the following distinction on the definition of exact sequences.

A sequence

$$\dots \rightarrow M_{i+1} \xrightarrow{\varphi_{i+1}} M_i \xrightarrow{\varphi_i} M_{i-1} \rightarrow \dots$$

of filtered maps is said to be *exact at M_i* if $\ker(\varphi_i) = \varphi_{i+1}(M_{i+1})$ as abstract R -modules, and it is exact if it is exact at every M_i . An exact sequence of strict maps is called a *strict exact sequence*. Hence, a *short strict exact sequence* of filtered R -modules is a sequence

$$0 \rightarrow N \xrightarrow{\alpha} M \xrightarrow{\beta} Q \rightarrow 0$$

of filtered R -modules and strict maps α and β . In the strict exact case, the sequence

$$0 \rightarrow \widehat{N} \rightarrow \widehat{M} \rightarrow \widehat{Q}$$

⁶⁰Let M be any non-trivial module with a separated filtration, and let $N = M$ as an abstract module but with the filtration $F_iN = N$ for all $i \in \mathbb{Z}$. Then the identity map $M \rightarrow N$ has degree 0, but its inverse is not even continuous.

is also strict exact whenever the filtration on R and M are positive.⁶¹

Let X be a set and $d: X \rightarrow \mathbb{Z}$ a function. In this case, we say that X is *weighted*. The *free filtered R -module on X* is the filtered R -module $R(X) = \bigoplus_{x \in X} T_{-d(x)} R[\{x\}]$.⁶² It satisfies the following universal property: any function $f: X \rightarrow M$ from X to a filtered R -module M such that $f(x) \subseteq F_{d(x)} M$ for every $x \in X$ extends uniquely to a filtered homomorphism $R(X) \rightarrow M$ (not necessarily strict).

In general, for any weighted set X and any filtered R -module M , we define the *weighted direct sum* of M with respect to X to be the filtered R -module $M(X) = \bigoplus_{x \in X} T_{-d(x)} M$. The module $M(X)$ comes with a distinguished copy M_x of $T_{-d(x)} M$ for each $x \in X$.

Each module $R(X)$ is *strictly projective*, that is, it is projective in the category of filtered R -modules with respect to strict surjections. Formally, a filtered R -module P is strictly projective if the lifting problem

$$\begin{array}{ccc} & P & \\ \psi \swarrow & \downarrow \pi & \\ M & \xrightarrow{\varphi} N & \longrightarrow 0 \end{array}$$

has a filtered solution ψ whenever π is filtered and φ is strict.⁶³ Any strictly projective module is projective as an abstract module, since we can endow a pair of abstract modules M and N with the filtrations $F_i M = M$ and $F_i N = N$ for all i .

Proposition 1.3.5. *There are enough strictly projective R -modules. More precisely, for every filtered R -module M , there is a strictly projective R -module P and a strict surjection $P \rightarrow M$. One can also take P to be a free filtered R -module.*

Proof. Let X be the disjoint union of all the $F_i M$, $i \in \mathbb{Z}$, and $d: X \rightarrow \mathbb{Z}$ be the indicator function of this disjoint union. Take $P = R(X)$. Then, P is strictly projective, and by the universal property of the free filtered module we obtain a filtered homomorphism $\varphi: P \rightarrow M$ extending the inclusion maps $F_i M \rightarrow M$. It is immediate that φ is surjective, and we claim that φ is strict.

Indeed, since it is filtered, we already have the inclusion $\varphi(F_i P) \subseteq F_i M$. For the other inclusion, identify an element $x \in F_i M$ with its copy $y \in X$ such that $d(y) = i$. Then, $1 \cdot y \in F_i T_{-i} R[\{y\}] \subseteq F_i P$ and $\varphi(1 \cdot y) = 1 \cdot x = x$, showing that $\varphi(F_i P) = F_i M$ as desired. $\square$

If M is filtered and finitely generated, then one has a surjective filtered map $R(X) \rightarrow M$ for some finite weighted set X .⁶⁴ We say that the filtration on M is *good* if there is a finite

⁶¹The inverse limit is not an exact functor in general, as it can fail to be right exact without extra hypothesis.

⁶²Here, $R[\{x\}]$ denotes the free R -module on the singleton $\{x\}$ with the filtration obtained by $F_i R[\{x\}] = (F_i R)[\{x\}]$.

⁶³Analogously, one can define the notion of a strictly injective module. On one hand, I don't know if there are always enough strictly injective R -modules. On the other hand, we will not need them.

⁶⁴If R is in addition positive, we also have $F_d M = M$ for some $d \leq 0$.

weighted set X and a strict surjection $\varphi: R(X) \rightarrow M$. It is a straightforward exercise from the definition to show that any two good filtrations on a module are equivalent, that is, if $F_i M$ and $F'_i M$ are two good filtrations, then there exists an integer c such that $F'_{i-c} M \subseteq F_i M \subseteq F'_{i+c} M$ for all $i \in \mathbb{Z}$. Observe that for any weighted set X and any strict surjection $\varphi: R(X) \rightarrow M$ of left (resp. right) modules we have $F_i M = \sum_{x \in X} F_{i-d(x)} R \cdot \varphi(x)$ (resp. $F_i M = \sum_{x \in X} \varphi(x) \cdot F_{i-d(x)} R$).

We call a weighted set X *semicompact* if for each $i \in \mathbb{Z}$ there are only finitely many $x \in X$ for which $d(x) \leq i$.⁶⁵ If there exists a strict surjection $R(X) \rightarrow M$ where X is a semicompact weighted set, we say M is *semicompactly generated*.

If R is a positive filtered ring, we say a filtration $F_i M$ on an R -module M is *bounded* if each quotient $F_{i-1} M / F_i M$ is an $R/F_1 R$ -module of finite length.⁶⁶ The ring is *bounded* if each quotient ring $R/F_i R$ is Artinian. Observe that if R is positive, then for any filtered R -module M and any submodule N of M we have that M is bounded if and only if both N and M/N are bounded under the induced filtrations. Since the direct product of two bounded filtered modules is bounded, the category of bounded filtered modules with filtered homomorphisms is a quasi-abelian subcategory of the category of all filtered modules.

Any module with a good filtration is semicompactly generated, and if R is bounded then any semicompactly generated module is also bounded. In general, none of the converse implications are true.⁶⁷ We record the following useful description of $\widehat{M(X)}$ for later use:

Lemma 1.3.6. *Let M be a complete and separated module and X a semicompact weighted set. Then any element of $\widehat{M(X)}$ can be written uniquely as $\sum_{x \in X} m_x$ with m_x in the image of M_x inside $\widehat{M(X)}$, and conversely any such sum converges to a unique element.*⁶⁸

Proof. The map $M_x \rightarrow M(X)$ is injective, and since the filtration on $M(X)$ is also separated the natural map $M(X) \rightarrow \widehat{M(X)}$ is injective. Hence, the composition $M_x \rightarrow \widehat{M(X)}$ is injective. Moreover, for any sequence $(m_x) \in M(X)$ we have $m_x \rightarrow 0$ along the filter of cofinite subsets inside $\widehat{M(X)}$ and therefore the series $\sum_{x \in X} m_x$ is convergent.

For each $x \in X$, we have a natural projection $\varphi_x: \widehat{M(X)} \rightarrow M_x$, and since M_x is complete this extends to a surjective map $\varphi_x: \widehat{M(X)} \rightarrow M_x$. We now claim that $m = \sum_{x \in X} \varphi_x(m)$ for every $m \in \widehat{M(X)}$. Write $m = (m_i + F_i M(X))$ as a coherent sequence of elements in $M(X)/F_i M(X)$. For each m_i , we clearly have $m_i = \sum_{x \in X} \varphi_x(m_i)$. Moreover, the congruence

⁶⁵A weighted set is semicompact if and only if it is compact in the topology obtained by declaring a non-empty set to be open if it contains a subset of the form $\{x \in X: d(x) \geq i\}$.

⁶⁶However, we allow the possibility of $F_d M \neq M$ for every d .

⁶⁷Let $R = k[[t]]$ with the (t) -adic filtration over some field k , so that R is separated, positive, bounded and complete. The module $M = \sum_{i \geq 0} T_{-i} R$ with the direct sum filtration is semicompactly generated but not good since it is not abstractly finitely generated. Let $Q = R/tR$ with the quotient filtration. The module $N = \sum_{i \in \mathbb{Z}} T_{-i} Q$ again with the direct sum filtration is bounded but not semicompactly generated, since $F_d N \neq N$ for every $d \in \mathbb{Z}$.

⁶⁸Any semicompact weighted set is at most countable. Algebraically, this lemma means that $\widehat{M(X)}$ behaves like the Cartesian product $\prod_X M$.

$\sum_{x \in X} \varphi_x(m_i) \equiv \sum_{x \in X} \varphi_x(m_j) \pmod{F_i M(X)}$ implies $\varphi_x(m_i) \equiv \varphi_x(m_j) \pmod{F_i M_x}$ for every $x \in X$, as $M(X)/F_i M(X)$ decomposes as $\bigoplus_{x \in X} M_x/F_i M_x$. Since $\varphi_x: \widehat{M(X)} \rightarrow M_x$ is the inverse limit of all the induced maps $M(X)/F_i M(X) \rightarrow M_x/F_i M_x$, we also obtain that $\varphi_x(m) \equiv \varphi_x(m_i) \pmod{F_i M_x}$. Hence, the element $m - \sum_{x \in X} \varphi_x(m)$ must lie in $F_i \widehat{M(X)}$ for all i and therefore be trivial. $\square$

The topology on a filtered module is pseudocompact if and only if the filtration is complete, separated and bounded. If M is a bounded R -module, then $\widehat{M}$ is a pseudocompact $\widehat{R}$ -module, and if R is bounded as a ring, then $\widehat{R}$ is a pseudocompact ring.

If M and N are right and left filtered R -modules respectively, then the tensor product $M \otimes_R N$ is a filtered $\mathbb{Z}$ -module⁶⁹ by taking $F_i(M \otimes_R N)$ to be the subgroup of $M \otimes_R N$ generated by all the elements of the form $x \otimes y$ where $x \in F_j M$ and $y \in F_k N$ with $j + k = i$. Hence, we can see the tensor product as a functor from filtered R -modules to filtered $\mathbb{Z}$ -modules. If R, S and T are filtered rings and M (resp. N) is a filtered S - R bimodule⁷⁰ (resp. R - T bimodule), then $M \otimes_R N$ is a filtered S - T bimodule with that same filtration. If $\varphi: M_1 \rightarrow M_2$ is a filtered homomorphism of S - R bimodules, then the induced map $M_1 \otimes_R N \rightarrow M_2 \otimes_R N$ is a filtered homomorphism of S - T bimodules (and analogously for filtered left R - T bimodules). Observe that we can see left (resp. right) filtered R -modules as R - $\mathbb{Z}$ (resp. $\mathbb{Z}$ - R) filtered bimodules in a natural way.

Let $M \hat{\otimes}_R N$ be a shorthand for $\widehat{M \otimes_R N}$. If the filtrations on R, M and N are positive, we have isomorphisms

$$M \hat{\otimes}_R N \simeq \widehat{M} \hat{\otimes}_{\widehat{R}} \widehat{N} \simeq \varprojlim_{i \in \mathbb{Z}} M/F_i M \otimes_R N/F_i N.$$

of filtered $\mathbb{Z}$ -modules. This isomorphism is compatible with any filtered bimodule structure on M and N .⁷¹ This also shows that there is no ambiguity about what $M \hat{\otimes}_R N$ means when R, M and N are filtered with a profinite induced topology. We call $M \hat{\otimes}_R N$ the *completed tensor product* of the R -modules M and N . In particular, we record the following:

Proposition 1.3.7. *If R is a filtered ring with a profinite induced topology, X is a semicompact weighted set and $M \rightarrow N$ is a filtered injective map between filtered right R -modules with profinite induced topology, then the induced map*

$$M \hat{\otimes}_R \widehat{R(X)} \rightarrow N \hat{\otimes}_R \widehat{R(X)}$$

is injective, and analogously for left modules.

⁶⁹We see $\mathbb{Z}$ as filtered by $F_i \mathbb{Z} = \mathbb{Z}$ if $i \leq 0$ and $F_i \mathbb{Z} = 0$ otherwise.

⁷⁰That is, an abstract S - R bimodule which is filtered with the same filtration both as a left S -module and a right R -module.

⁷¹It suffices to check that either one of the modules on the right hand side satisfy the universal property of $M \hat{\otimes}_R N$ with respect to R -balanced S - T bilinear maps $\varphi: M \times N \rightarrow Q$ where M is a S - R filtered bimodule, N is an R - T filtered bimodule, Q is a complete S - T bimodule and $\varphi(F_i M, F_j N) \subseteq F_{i+j} Q$.

Proof. The module $\widehat{R(X)}$ is topologically isomorphic to the pointed free profinite R -module $R[[\overline{X}, *]]$ on the one-point compactification $\overline{X} = X \cup \{*\}$ of X . In particular, it is projective on the category of all profinite R -modules, and hence the functors $-\hat{\otimes}_R \widehat{R(X)}$ and $\widehat{R(X)} \hat{\otimes}_R -$ preserve monomorphisms. $\square$

Since the category of filtered $\mathbb{Z}$ -modules is not abelian, it is not immediate how to define the derived functors of $-\hat{\otimes}_R -$ as filtered modules.⁷² If the filtered modules M and N are bounded, then the filtration on $M \hat{\otimes}_R N$ is also bounded and hence the topology on $M \hat{\otimes}_R N$ is pseudocompact.

We finish with some comments about skew power series.

Example 1.3.8. If R is a positive, complete and separated ring, and σ is a strict⁷³ ring automorphism of R satisfying $\delta(F_i R) \subseteq F_{i+1} R$ for $\delta = \sigma - \text{id}$, then δ is a topologically nilpotent operator and we can form the right skew power series ring $S = R[[s; \sigma, \delta]]$, which is also isomorphic to the left skew power series ring $R[[s; \sigma^{-1}, -\delta\sigma^{-1}]]$. In this case, we can filter S by $F_i S = \sum_{j+k=i} s^k \cdot S \cdot F_j R$, and this induces the natural topology⁷⁴ on S defined at the end of Section 1.1. Another description of this filtration is:

$$F_i S = \left\{ \sum_{j \geq 0} s^j a_j : a_j \in F_{i-j} R \right\}.$$

We see that in the filtered case this topology is actually linear, so S is a positive, complete and separated filtered ring. Moreover, the inclusion $R \rightarrow S$ is strict. If R is bounded and hence pseudocompact, then so is S . $\blacksquare$

1.4 Graded rings

To any filtered ring R we can associate a new ring

$$\text{Gr } R = \bigoplus_{i \in \mathbb{Z}} F_i R / F_{i+1} R$$

called the *graded ring* associated to the filtration $\{F_i R\}$. This is naturally a $\mathbb{Z}$ -graded abelian group, and the multiplication map is defined for each homogeneous element by the map

$$F_i R / F_{i+1} R \times F_j R / F_{j+1} R \rightarrow F_{i+j} R / F_{i+j+1} R$$

⁷²One can take their derived functors inside the category of abelian groups, but this simply means forgetting the whole filtered structure and computing the usual Tor groups. The problem of trying to force a filtered definition of Tor is that tensoring in general does not preserve strictness.

⁷³That is, a ring automorphism that is also a strict bijection as a left and as a right module homomorphism. Its inverse is automatically strict, and $\delta = \sigma - \text{id}$ is a topologically nilpotent operator if and only if $\delta' = -\delta\sigma^{-1}$ is so.

⁷⁴We have $F_{2i} S \subseteq S \cdot F_i R + s^i S \subseteq F_i S$.

induced from multiplication in R and then extended linearly to all of $\text{Gr } R$. If M is a filtered R -module, then we also define the $\mathbb{Z}$ -graded abelian group

$$\text{Gr } M = \bigoplus_{i \in \mathbb{Z}} F_i M / F_{i+1} M.$$

It comes with a natural graded action of $\text{Gr } R$, and we call it the *graded module* associated to the filtration $\{F_i M\}$.

Translating information from the filtered category to the associated graded category⁷⁵ is a subtle (albeit very useful) endeavour. For example, if M is a finitely generated and filtered R -module, the associated graded module $\text{Gr } M$ doesn't need to be finitely generated over $\text{Gr } R$.⁷⁶ Some properties, however, are immediate from the definitions:⁷⁷

Proposition 1.4.1. *Let M be a filtered R -module.*

1. *If the filtration on M is separated, then $\text{Gr } M = 0$ if and only if $M = 0$.*
2. *If $M = \bigoplus_{i \in I} M_i$ with the direct sum filtration, then $\text{Gr } M = \bigoplus_{i \in I} \text{Gr } M_i$ as $\text{Gr } R$ -modules.*
3. *If $M = \varinjlim_{i \in I} M_i$ with the direct limit filtration, then $\text{Gr } M = \varinjlim_{i \in I} \text{Gr } M_i$ as $\text{Gr } R$ -modules.*
4. *If $M = \prod_{i \in I} M_i$ with each M_i a positive filtered R -module and the product filtration on M , then $\text{Gr } M = \prod_{i \in I} \text{Gr } M_i$ as $\text{Gr } R$ -modules.*

That passing to the associated graded module does not commute with inverse limits is illustrated by the following proposition:⁷⁸

Proposition 1.4.2. *If M is a positive filtered module, then $\text{Gr } M = \text{Gr } \widehat{M}$.*

If R is a filtered ring and $x \in F_i R$ but not in $F_{i+1} R$, we call $\bar{x} \in F_i R / F_{i+1} R \subseteq \text{Gr } R$ the *principal symbol* or *homogeneous part* of x , and we say x is a representative of $\bar{x}$.⁷⁹ The following proposition and its corollary are immediate.

⁷⁵The category of graded modules over a graded ring has the *graded homomorphisms* as its morphisms, that is, a graded module homomorphism sends homogeneous elements of degree d to homogeneous elements of degree d . A graded, or homogeneous, submodule N of a graded module $M = \bigoplus_{i \in \mathbb{Z}} M_i$ is one that satisfies $N = \bigoplus_{i \in \mathbb{Z}} N \cap M_i$.

⁷⁶Let $R = \mathbb{Z}$ with $F_i R = \mathbb{Z}$ for $i \leq 0$ and $F_i R = 0$ for $i > 0$. Take $M = \mathbb{Z}$ with $F_i M = p^i M$ for all $i \geq 0$. We have that M is finitely generated as an R -module but $\text{Gr } M = \bigoplus_{i \geq 0} \mathbb{F}_p$ is not finitely generated as a $\text{Gr } R$ -module.

⁷⁷Năstăsescu and Oystaeyen, *Graded Ring Theory*, Prop. D.3.1.

⁷⁸Ibid., Prop. D.3.1.

⁷⁹Whenever $\{F_i R\}$ is separated, taking the homogeneous part defines a function $R \rightarrow \text{Gr } R$ if we postulate $\bar{0} = 0$. One should be very careful with this notation: this function is neither multiplicative nor additive in general. When “taking” the homogeneous part of a product or sum, one should always pay attention to the degrees of the elements involved. For example, if $R = \mathbb{Q}[x, y] / (xy - y^3)$ with the I -adic filtration for $I = (x, y)$, then $\bar{x} \cdot \bar{y} = 0$ but $\overline{xy} = \bar{y}^3 \neq 0$.

Proposition 1.4.3. *Assume that the filtration $\{F_i R\}$ is separated. If x is a zero divisor, then $\bar{x}$ is a zero divisor.*

Proof. Let $x, y \in R$ be non-zero such that $xy = 0$ in R . Since the filtration is separated, we can find $i, j \in \mathbb{Z}$ such that $x \in F_i R$ but not in $F_{i+1} R$ and $y \in F_j R$ but not in $F_{j+1} R$. In particular, $\bar{x}$ and $\bar{y}$ are both non-zero while $\bar{x} \cdot \bar{y} \equiv \overline{xy} = 0 \pmod{F_{i+j} R}$. $\square$

Corollary 1.4.4. *If $\{F_i R\}$ is separated and $\text{Gr } R$ is a domain, then R is a domain.*

Before we continue exploring properties of associated graded rings, let us see some examples. For any ring R with the trivial filtration

$$F_i R = \begin{cases} R, & \text{if } i \leq 0 \\ 0, & \text{if } i > 0 \end{cases}$$

we have $\text{Gr } R = R$. For $R = \mathbb{Z}$ with the positive p -adic filtration $F_i R = p^i \mathbb{Z}$, we have $\text{Gr } R = \mathbb{F}_p[t]$ where t is the image of p in $p\mathbb{Z}/p^2\mathbb{Z}$.⁸⁰ If $S = R[[t]]$ with the positive filtration $F_i S = t^i S$, then $\text{Gr } S = R[\bar{t}]$ where $\bar{t}$ is the image of t in tS/t^2S . This generalizes to the setting where R is also filtered and positive and $F_i S = \sum_{k+j=i} t^k (F_j R)[[t]]$: we get $\text{Gr } S = (\text{Gr } R)[\bar{t}]$ where $\bar{t}$ is the image of t in $F_1 S/F_2 S$. Even more generally:

Proposition 1.4.5. *Let R be a positive, complete and separated ring and σ a strict ring automorphism of R satisfying $\delta(F_i R) \subseteq F_{i+1} R$ for $\delta = \sigma - \text{id}$. Let $S = R[[s; \sigma, \delta]]$ with the filtration $F_i S = \sum_{j+k=i} s^k \cdot S \cdot F_j R$. Then the graded ring $\text{Gr } S$ is naturally isomorphic to the skew polynomial ring $(\text{Gr } R)[\bar{s}; \text{id}, \bar{\delta}]$ where $\bar{\delta}: F_i R/F_{i+1} R \rightarrow F_{i+1} R/F_{i+2} R$ is induced by δ and $\bar{s}$ is the image of s in $F_1 S/F_2 S$.*

Proof. Since $R \rightarrow S$ is strict, we have a natural injection $\text{Gr } R \rightarrow \text{Gr } S$. Moreover, $\text{Gr } S$ is generated as a ring by $\text{Gr } R$ and $\bar{s}$. For the claim, we must show that the powers of $\bar{s}$ are linearly independent over $\text{Gr } R$ and that the relation

$$\bar{x} \cdot \bar{s} - \bar{s} \cdot \bar{x} = \bar{\delta}(\bar{x})$$

holds.

Suppose we have a relation of the form $\sum_{i=0}^k \bar{s}^i \cdot \bar{x}_i = 0$ with each $\bar{x}_i \in \text{Gr } R$. By breaking down $\bar{x}_i = \sum_{j \geq 0} \bar{x}_{i,j}$ with each $\bar{x}_{i,j} \in F_j R/F_{j+1} R$, we get

$$\sum_{i=0}^k \sum_{j \geq 0} \bar{s}^i \cdot \bar{x}_{i,j} = 0.$$

⁸⁰This also shows that Gr does not commute with inverse limits, for we have $\widehat{R} = \varprojlim \mathbb{Z}/p^i \mathbb{Z}$ where each $\mathbb{Z}/p^i \mathbb{Z}$ has the induced filtration. Hence, for each quotient we get $\text{Gr } \mathbb{Z}/p^i \mathbb{Z} = \mathbb{F}_p[t]/(t^i)$, and thus $\varprojlim \text{Gr } \mathbb{Z}/p^i \mathbb{Z} = \mathbb{F}_p[[t]]$. However, we have $\text{Gr } \widehat{R} = \text{Gr } R = \mathbb{F}_p[t]$.

Isolating each homogeneous component, we conclude for each $n \geq 0$ the relations

$$\sum_{i+j=n} \bar{s}^i \cdot \bar{x}_{i,j} = 0.$$

Lift each $\bar{x}_{i,j}$ to an element $x_{i,j} \in F_j R$. Those relations imply

$$\sum_{i+j=n} s^i x_{i,j} \in F_{n+1} S.$$

We recall that an element $\sum_{i \geq 0} s^i a_i$ belongs to $F_{n+1} S$ if and only if whenever $i < n + 1$ we have $a_i \in F_{n+1-i} R$, showing thus that $x_{i,j} \in F_{j+1} R$ and hence $\bar{x}_{i,j} = 0$, as desired.

Now let $x \in F_i R$ represent $\bar{x}$. We have $xs = s\sigma(x) + \partial(x)$ as elements of $F_{i+1} R$. Moreover, since $\sigma(x) \equiv x \pmod{F_{i+1} R}$, we have $s\sigma(x) \equiv sx \pmod{F_{i+2} R}$ and hence

$$xs \equiv sx + \partial(x) \pmod{F_{i+2} R}.$$

This establishes the last relation we were searching for. $\square$

We recall that if R is a commutative ring with identity and $\mathfrak{g}$ is a Lie ring over R , then the universal enveloping algebra $U(\mathfrak{g})$ has an *ascending* filtration called the PBW filtration. For ascending filtered rings one defines the associated graded ring analogously, and the computation of this graded ring for $U(\mathfrak{g})$ is the celebrated Poincaré-Birkhoff-Witt theorem:⁸¹

Theorem 1.4.6. *If $\mathfrak{g}$ is a free R -module, then graded ring associated to the PBW filtration on $U(\mathfrak{g})$ is isomorphic as a graded R -algebra to the symmetric algebra on $\mathfrak{g}$.*

It follows from this theorem that if R is a free R -module with basis $\{X_i\}_{i \in I}$, then fixing a total order on I we have that $U(\mathfrak{g})$ is a free R -module with basis given by the monomials $X_{i_1} \cdots X_{i_n}$ with $n \geq 0$ and $i_1 \leq \cdots \leq i_n$. Hence, if R is separated and $\mathfrak{g}$ is a separated filtered Lie ring which is free as an R -module, the induced *descending* filtration on $F_i U(\mathfrak{g})$ is also separated. With respect to this filtration we have:

Proposition 1.4.7. *If R is a filtered ring and $\mathfrak{g}$ is a filtered Lie ring that is free filtered as a filtered R -module, then $\text{Gr } \mathfrak{g}$ is a Lie ring over $\text{Gr } R$, free as a $\text{Gr } R$ -module, and moreover $\text{Gr } U(\mathfrak{g})$ is isomorphic to $U(\text{Gr } \mathfrak{g})$.*

Proof. The Lie bracket on $\text{Gr } \mathfrak{g}$ is defined on homogeneous elements $\bar{x} \in F_i \mathfrak{g} / F_{i+1} \mathfrak{g}$ and $\bar{y} \in F_j \mathfrak{g} / F_{j+1} \mathfrak{g}$ by the relation

$$[\bar{x}, \bar{y}] \equiv [x, y] \pmod{F_{i+j} \mathfrak{g}},$$

and extended linearly to all of $\text{Gr } \mathfrak{g}$. One readily checks that this defines a Lie ring structure over $\text{Gr } R$, and freeness follows from the fact that Gr commutes with direct sums.

⁸¹Bourbaki, *Lie Groups and Lie Algebras*, Thm. I.2.7.1.

Since $\mathfrak{g}$ is a free filtered R -module, it is isomorphic as an R -module to $R(X)$ where X is some weighted set. Fix a total order on X and let

$$X_n = \{x_{i_1} \dots x_{i_n} \in U(\mathfrak{g}) : x_{i_1} \leq x_{i_2} \leq \dots \leq x_{i_n}\}$$

denote the set of all ordered monomials of length n in X . We can extend the degree function $d: X \rightarrow \mathbb{Z}$ to a degree function $d: X_n \rightarrow \mathbb{Z}$ by taking $d(x_{i_1} \dots x_{i_n}) = d(x_{i_1}) + \dots + d(x_{i_n})$, making the disjoint union $Y = \bigcup_{n \geq 0} X_n$ into a weighted set as well. The Poincaré-Birkhoff-Witt theorem now tells us that $U(\mathfrak{g})$ is isomorphic to $R(Y)$ as a filtered R -module.⁸²

To see that $\text{Gr } U(\mathfrak{g})$ is isomorphic to $U(\text{Gr } \mathfrak{g})$, it suffices to check the following universal property: if A is any unital associative $\text{Gr } R$ -algebra and $\varphi: \text{Gr } \mathfrak{g} \rightarrow A$ is a $\text{Gr } R$ -module homomorphism such that

$$\varphi([\bar{x}, \bar{y}]) = \varphi(\bar{x})\varphi(\bar{y}) - \varphi(\bar{y})\varphi(\bar{x})$$

for any $\bar{x}, \bar{y} \in \text{Gr } \mathfrak{g}$, then there is a unique algebra homomorphism $\tilde{\varphi}: \text{Gr } U(\mathfrak{g}) \rightarrow A$ extending φ .

Since $\text{Gr } U(\mathfrak{g}) \simeq (\text{Gr } R)[\bar{Y}]$, the uniqueness of $\tilde{\varphi}$ is automatic as it must satisfy

$$\tilde{\varphi}(\overline{x_{i_1} \dots x_{i_n}}) = \tilde{\varphi}(\overline{x_{i_1}} \dots \overline{x_{i_n}}) = \varphi(\overline{x_{i_1}}) \dots \varphi(\overline{x_{i_n}})$$

for any ordered set $x_{i_1} \leq x_{i_2} \leq \dots \leq x_{i_n}$. To show that the formula above defines an algebra homomorphism, observe that

$$\tilde{\varphi}(\overline{x_{i_1} \dots x_{i_n} \cdot x_{j_1} \dots x_{j_m}}) = \tilde{\varphi}(\overline{x_{i_1} \dots x_{i_n}}) \tilde{\varphi}(\overline{x_{j_1} \dots x_{j_m}})$$

whenever $x_{i_1} \leq \dots \leq x_{i_n}$ and $x_{j_1} \leq \dots \leq x_{j_m}$ holds by induction on n and m , where the base case $n = m = 1$ follows from the identity $\varphi([\bar{x}, \bar{y}]) = \varphi(\bar{x})\varphi(\bar{y}) - \varphi(\bar{y})\varphi(\bar{x})$. $\square$

If $\varphi: M \rightarrow N$ is a filtered homomorphism of filtered R -modules, it induces a graded homomorphism $\text{Gr } \varphi: \text{Gr } M \rightarrow \text{Gr } N$ between their associated $\text{Gr } R$ -modules. If $\varphi: R \rightarrow S$ is a filtered homomorphism of rings, then $\text{Gr } \varphi: \text{Gr } R \rightarrow \text{Gr } S$ is a graded ring homomorphism. Hence, Gr defines a functor from the category of filtered R -modules to the category of graded $\text{Gr } R$ -modules. Unless we assume some form of strictness, then φ and $\text{Gr } \varphi$ can behave quite differently.⁸³ The strictness condition is what allows us to transport information between the filtered and graded worlds:⁸⁴

⁸²If an *unordered* monomial $x_1 \dots x_n$ has degree m , since $\mathfrak{g}$ is filtered as a Lie ring any expression in $U(\mathfrak{g})$ obtained by reordering the monomial will also consist of a sum of R -multiples of monomials, each multiple of degree at least m . Hence, a sum of multiples of monomials in $U(\mathfrak{g})$ has degree at least m if and only if each multiple has degree at least m .

⁸³if $N = T_{-1}M$ and φ is the identity, then $\text{Gr } \varphi$ is the zero map.

⁸⁴Năstăsescu and Oystaeyen, *Graded Ring Theory*, cf. Thm. III.3.

Proposition 1.4.8. *Let*

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} Q \quad (\dagger)$$

be a sequence of filtered R -modules and

$$\mathrm{Gr} N \xrightarrow{\mathrm{Gr} \varphi} \mathrm{Gr} M \xrightarrow{\mathrm{Gr} \psi} \mathrm{Gr} Q \quad (\mathrm{Gr} \dagger)$$

be the associated graded sequence. Then:

1. *If $(\dagger)$ is strict exact then $(\mathrm{Gr} \dagger)$ is exact.*
2. *If $(\mathrm{Gr} \dagger)$ is exact then ψ is strict.*
3. *If $(\mathrm{Gr} \dagger)$ is exact, N is complete and M is separated then $(\dagger)$ is strict exact.*

Proof. While the proof in Constantin Năstăsescu and Fred van Oystaeyen's book is given for increasing filtrations, the argument is easily adapted to our decreasing filtrations, and we do it below.

1. Let $\bar{x} \in \mathrm{Gr} M$ be such that $(\mathrm{Gr} \psi)(\bar{x}) = 0$. If $x \in F_i M$ but not in $F_{i+1} M$, this tells us that $\psi(x) \in F_{i+1} Q$. Hence, since ψ is strict, it means that there is some $y \in F_{i+1} M$ such that $\psi(x) = \psi(y)$. In particular, $x - y \in \ker \psi \cap F_i M$, so one can find $z \in F_i N$ such that $\varphi(z) = x - y$ by the strictness of φ . Hence $\bar{x} = \overline{\varphi(z) + y} = \overline{\varphi(z)} = (\mathrm{Gr} \varphi)(\bar{z})$ as desired.

2. Let $x \in \psi(M) \cap F_i Q$ be such that $x \notin F_{i+1} Q$. By our hypothesis, $x = \psi(y)$ for some $y \in F_{i-s} M$ and some $s \geq 0$, and we only need to show that $s = 0$. If $s > 0$, then $(\mathrm{Gr} \psi)(\bar{y}) = 0$ and therefore we would find some $\bar{z} \in \mathrm{Gr} N$, $z \in F_{i-s} N$, for which $(\mathrm{Gr} \varphi)(\bar{z}) = y$. Take $y' = y - \varphi(z)$. By construction, we have that $y' \in F_{i-s+1} M$ and $\psi(y') = \psi(y) = x$, so we have reduced s by one. Iterating this process gives us the desired element in $F_i M$.

3. Let $x \in \varphi(N) \cap F_i M$ be such that $x \notin F_{i+1} M$. Since $(\mathrm{Gr} \dagger)$ is exact, we know that $\bar{x} = \overline{\varphi(y_i)}$ for some $y_i \in F_i N$, that is, $x - \varphi(y_i) \in F_{i+1} M$. Applying the same construction now for $x - \varphi(y_i) \in \varphi(N) \cap F_{i+1} M$, we can obtain for every $n \geq 0$ a sequence of elements $y_i, \dots, y_{i+n}$ with $y_{i+j} \in F_{i+j} N$ such that

$$x - \varphi(y_i) - \varphi(y_{i+1}) - \dots - \varphi(y_{i+n}) \in \varphi(N) \cap F_{i+n+1} M.$$

Since N is complete, we can define $y = \sum_{n \geq 0} y_{i+n}$, and since M is separated we have $x = \varphi(y)$. The claim now follows since $y \in F_i N$. $\square$

The following consequences are immediate:⁸⁵

Corollary 1.4.9. *Let $\varphi: N \rightarrow M$ be a filtered homomorphism of R -modules.*

1. *If φ is a strict injection (resp. surjection), then $\mathrm{Gr} \varphi$ is injective (resp. surjective).*

⁸⁵Ibid., Sec. D.III.

2. If $\text{Gr } \varphi$ is injective, then φ is strict. If moreover N is separated, then φ is a strict injection.

3. If $\text{Gr } \varphi$ is surjective, N is complete and M is separated then φ is a strict surjection.

Proof. 1. The strict exact sequences

$$0 \rightarrow N \xrightarrow{\varphi} M \text{ or } N \xrightarrow{\varphi} M \rightarrow 0$$

would give rise to exact sequences

$$0 \rightarrow \text{Gr } N \xrightarrow{\text{Gr } \varphi} M \text{ or } \text{Gr } N \xrightarrow{\text{Gr } \varphi} \text{Gr } M \rightarrow 0.$$

2. The sequence

$$0 \rightarrow N \xrightarrow{\varphi} M$$

is such that

$$0 \rightarrow \text{Gr } N \xrightarrow{\text{Gr } \varphi} \text{Gr } M$$

is exact, so φ is strict. If in addition N is separated, then the first sequence must be also strict exact.

3. The sequence

$$N \xrightarrow{\varphi} M \rightarrow 0$$

is such that

$$\text{Gr } N \xrightarrow{\text{Gr } \varphi} \text{Gr } M \rightarrow 0$$

is exact, so the hypothesis guarantee us that the former is strict exact. $\square$

Let us now turn to the problem of lifting the ascending chain condition from $\text{Gr } R$ over to R . To this end, we must understand how to transport finite generation between the filtered and graded worlds, and this is where good filtrations play an important role. First, we remark that to check the Noetherian property on a graded ring it suffices to do so on graded ideals:

Lemma 1.4.10. *Let $S = \bigoplus_{i \in \mathbb{Z}} S_i$ be a graded ring. Then, S is left (resp. right) Noetherian if and only if every left (resp. right) graded ideal $I \leq S$ is finitely generated.*

Proof. We prove the lemma only for left ideals. If S is Noetherian, any ideal is finitely generated, and in particular so is any graded ideal. Conversely, assume that every left graded ideal is finitely generated and let $I \leq S$ be a left ideal of S . Given an element $x \in S$, we can write $x = \sum_{i \in \mathbb{Z}} x_i$ with $x_i \in S_i$ almost all zero. Consider the following notation:

$$\text{supp}(x) = \{i \in \mathbb{Z} : x_i \neq 0\}, \quad \text{init}(x) = x_{\min \text{supp}(x)}, \quad \text{fin}(x) = x_{\max \text{supp}(x)}.$$

We consider the graded subring $S^+ = \bigoplus_{i \geq 0} S_i$ of S , and we define the S^+ -module $I^+ = I \cap S^+$.

By hypothesis, the left graded ideal of S generated by $\{\text{init}(x) : x \in I\}$ is finitely generated, say by $\text{init}(y_1), \dots, \text{init}(y_n)$. Therefore, given any non-zero element $x \in I$, we can find homogeneous elements $z_1, \dots, z_n \in S$ such that

$$\min \text{supp}(x) < \min \text{supp}(x - z_1 y_1 - \dots - z_n y_n).$$

Hence, after finitely many steps, we produce elements $z_1, \dots, z_n \in S$ such that $x - z_1 y_1 - \dots - z_n y_n$ belongs to I^+ . This reduces the claim to proving that I^+ is finitely generated as an S^+ -module.

First, observe that every left graded ideal J of S^+ is also finitely generated: since SJ is finitely generated as a graded ideal of S , one can find homogeneous elements $x_1, \dots, x_k \in J$ that generate it. If $m = \max\{\text{supp}(x_1), \dots, \text{supp}(x_k)\}$, then any homogeneous element in J of degree greater than or equal to m lies in the ideal $\sum_{i=1}^k S^+ x_i \leq J$. On the other hand, decomposing $SJ = \bigoplus_{i \in \mathbb{Z}} J_i$ with $J_i = SJ \cap S_i$, each J_i is again a Noetherian S_0 -module. Hence, there are elements $y_1, \dots, y_l \in J$ that generate $J \cap (J_0 \oplus \dots \oplus J_{m-1})$ over S_0 . Therefore, the left ideal J is generated as an S^+ -graded ideal by $x_1, \dots, x_k, y_1, \dots, y_l$.

Consider now the left graded ideal generated by $\{\text{fin}(x) : x \in I^+\}$ of S^+ . By our previous observation, it is finitely generated, say by $\text{fin}(y'_1), \dots, \text{fin}(y'_m)$. Therefore, given any non-zero element $x' \in I^+$, we can find homogeneous elements $z'_1, \dots, z'_m \in S^+$ such that

$$\sup \text{supp}(x' - z'_1 y'_1 - \dots - z'_m y'_m) < \max \text{supp}(x).$$

After finitely many steps, we obtain that I^+ is generated as an S^+ -ideal by $y'_1, \dots, y'_m$. $\square$

Observe that for any weighted set X we have that $\text{Gr}(R(X))$ is isomorphic to the free $\text{Gr } R$ -module on the set $\overline{X}$ of homogeneous components of X . From this, we obtain:⁸⁶

Proposition 1.4.II. *Let M be a filtered R -module.*

- (a) *If $F_i M$ is a good filtration, then $\text{Gr } M$ is a finitely generated $\text{Gr } R$ -module.*
- (b) *If R is complete and M is separated, then $F_i M$ is good if and only if $\text{Gr } M$ is finitely generated.*

Proof. If $\varphi: R(X) \rightarrow M$ is a strict surjection where X is a finite weighted set, then the induced map $\text{Gr } \varphi: (\text{Gr } R)[\overline{X}] \rightarrow \text{Gr } M$ is surjective by part 1 of Corollary 1.4.9, so $\text{Gr } M$ is finitely generated.

Assuming now that R is complete and M is separated, let $\overline{X} \subset \text{Gr } M$ be a finite generating set, and choose for each $\overline{x} \in \overline{X}$ some representative $x \in M$. Then, the subset $X \subseteq M$ of such a choice of representatives is naturally a finite weighted set by declaring $d(x) = i$ if $x \in F_i M$ but not in $F_{i+1} M$. We then obtain a filtered homomorphism $\varphi: R(X) \rightarrow M$ induced by the natural inclusion. Since $\text{Gr } \varphi$ is surjective by construction, φ is a strict surjection by part 3 of Corollary 1.4.9, and hence the filtration $F_i M$ is good. $\square$

⁸⁶Li Huishi and Fred van Oystaeyen. *Zariskian Filtrations*. 1st ed. K-Monographs in Mathematics 2. Springer Netherlands, 1996. ISBN: 978-90-481-4738-0, Lem. 5.4 and Thm. 5.7.

We are now ready to show:

Proposition 1.4.12. *Let R be a separated and complete filtered ring. If $\text{Gr } R$ is (left, right) Noetherian, then R is (left, right) Noetherian.*

Proof. Let I be a left or right ideal of R , and endow it with the induced submodule filtration. Since the inclusion map $I \rightarrow R$ is a strict injection, then $\text{Gr } I \rightarrow \text{Gr } R$ is injective by Corollary 1.4.9. Given that $\text{Gr } R$ is Noetherian, we know that $\text{Gr } I$ is finitely generated, and so the induced submodule filtration on I is good by Proposition 1.4.11. In particular, I is finitely generated. $\square$

Example 1.4.13. An important class of filtered rings for which we will apply Proposition 1.4.12 consists of p -adic Iwasawa algebras. Those are the completed group rings $\mathbb{Z}_p[[G]]$ of p -adic analytic profinite groups G . If G is a uniform pro- p group, the computation of $\text{Gr } \mathbb{Z}_p[[G]]$ with respect to the filtration induced by powers of the maximal ideal is a well-know result that can be traced back to Michel Lazard.⁸⁷

Any profinite p -adic Lie group G admits an open *uniform*⁸⁸ subgroup U . Moreover, since $\mathbb{Z}_p[[G]]$ is finitely generated over $\mathbb{Z}_p[[U]]$, if $\mathbb{Z}_p[[U]]$ is Noetherian then so is $\mathbb{Z}_p[[G]]$. Henceforth, we assume that G is a uniform pro- p group.

Since $\mathbb{Z}_p[[G]]$ is profinite and G is finitely generated, it is complete and separated with respect to the J -adic filtration, where $J = J(\mathbb{Z}_p[[G]])$ is its Jacobson radical, or maximal ideal. Let $G_i = G_{i-1}^p[G, G]$ denote the lower p -central series of G with $G_1 = G$. As G is uniform, the graded Lie algebra $\text{Gr } G = \bigoplus_{i \geq 0} G_i/G_{i+1}$ is free and finitely generated as a module over $\mathbb{F}_p[\pi]$, where π acts on homogeneous elements by raising to the p -th power.⁸⁹ The subgroups G_i can also be described as the filtration induced by the J -adic filtration on $\mathbb{Z}_p[[G]]$:⁹⁰

$$G_i = \{g \in G : g - 1 \in J^i\}.$$

Moreover, the results of Lazard⁹¹ show that $\text{Gr } \mathbb{Z}_p[[G]]$ is isomorphic to the universal enveloping algebra of $\text{Gr } G$ over $\mathbb{F}_p[\pi]$. Hence, by the Poincaré-Birkhoff-Witt Theorem 1.4.6, $\text{Gr } \mathbb{Z}_p[[G]]$ is a Noetherian domain (as well as a torsion-free $\mathbb{F}_p[\pi]$ -algebra). Hence, $\mathbb{Z}_p[[G]]$ is also a Noetherian domain.

The case of the mod- p Iwasawa algebras $\mathbb{F}_p[[G]]$ for a uniform pro- p group is even simpler: if G is a p -adic manifold of dimension n , then $\text{Gr } \mathbb{F}_p[[G]]$ with respect to powers of the

⁸⁷Michel Lazard. “Groupes analytiques p -adiques”. In: *Publications Mathématiques de l’IHÉS* 26 (1965), pp. 5–219.

⁸⁸A finitely generated pro- p group is uniform if it satisfies that $[G, G] \leq G^p$ if $p \neq 2$ and $[G, G] \leq G^4$ if $p = 2$ and moreover all quotients G_i/G_{i+1} have the same size, where G_i denotes the lower p -central series of G .

⁸⁹If G is uniform, then raising to the p -th power induces isomorphisms between G_i/G_{i+1} and G_{i+1}/G_{i+2} for all i by Lem. 4.10 of John D. Dixon et al. *Analytic Pro- P Groups*. 2nd ed. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press, 1999. ISBN: 978-0-521-54218-0. DOI: 10.1017/CB09780511470882

⁹⁰Ibid., Exercise 7.2.

⁹¹Lazard, “Groupes analytiques p -adiques”, Thm. III.2.3.3.

augmentation ideal (which coincides with the radical) is a polynomial algebra over $\mathbb{F}_p$ in d variables.⁹² Hence, $\mathbb{F}_p[[G]]$ is also a Noetherian domain. More generally, it can also be shown that if G is a torsion-free p -adic Lie group, then $\mathbb{Z}_p[[G]]$ and $\mathbb{F}_p[[G]]$ have no zero divisors.⁹³ ■

An important remark to be made is that a graded module M that is also a free $\text{Gr } R$ -module when we forget the graded structure doesn't need to be free in the category of graded modules: it may not have a basis consisting of homogeneous elements. However, the following particular case will come in handy later:⁹⁴

Lemma 1.4.14. *Suppose that $S = \bigoplus_{i \in \mathbb{Z}} S_i$ is a graded ring with $S_i = 0$ for $i < 0$ and S_0 is a field. Then, any graded S -module $M = \bigoplus_{i \in \mathbb{Z}} M_i$ that is bounded from below⁹⁵ and projective as an abstract S -module must be free with a basis consisting of homogeneous elements.*

Proof. We prove the claim for right modules M , with the proof for left modules being analogous. Let $S_+ = \bigoplus_{i > 0} S_i$ be the two-sided ideal of S generated by the homogeneous elements of positive degree. First, observe that if $N = \bigoplus_{i \in \mathbb{Z}} N_i$ is a graded S -module bounded from below, then $N \subset NS_+$ if and only if $N = 0$. Indeed, let i be the smallest integer for which $N_i \neq 0$. The condition $N \subset NS_+$ implies that for any $x \in N_i$ there exists $y \in N_j$ and some $s \in S_+$ for which $x = ys$. This is only possible if $j < i$, and therefore $y = x = 0$.

Now, M/MS_+ is a graded S/S_+ -module, and since S_0 is a field, it means that it must have a basis of homogeneous elements.⁹⁶ Lift this basis to a homogeneous set X in M and consider the graded map $\varphi: S[X] \rightarrow M$. The map φ is surjective, since upon passing to the quotient modulo S_+ we get that

$$(M/\varphi(S[X]))/(M/\varphi(S[X]))S_+ \simeq \text{coker}(S_0[\bar{X}] \rightarrow M/MS_+) = 0.$$

As M is projective, this map splits, that is, we have $S[X] \simeq M \oplus N$ as S -modules, where $N \simeq \ker \varphi$ is also graded. However, $S[X]S_+ = MS_+ \oplus NS_+$, and since the kernel of $S[X]/S[X]S_+ \rightarrow M/MS_+$ is trivial, we must have $N \subset NS_+$. Hence, $N = 0$ and $M \simeq S[X]$ as graded modules. □

The remainder of this section is devoted to the study of homological algebra over filtered and graded rings. For a right filtered R -module M and a left filtered R -module N , we would like to be able to compare the Tor groups $\text{Tor}_i^R(M, N)$ and $\text{Tor}_i^{\text{Gr } R}(\text{Gr } M, \text{Gr } N)$. While this

⁹²Dixon et al., *Analytic Pro- p Groups*, Thm. 7.24.

⁹³Konstantin Ardakov and Kenneth A. Brown. “Ring-Theoretic Properties of Iwasawa Algebras: A Survey”. en. In: *Documenta Mathematica* Extra Volume Coates (2006), pp. 7–33, Sec. 4.3.

⁹⁴Hirofumi Minamoto and Izuru Mori. “The structure of AS-Gorenstein algebras”. In: *Advances in Mathematics* 226.5 (Mar. 2011), pp. 4061–4095. ISSN: 0001-8708. DOI: 10.1016/j.aim.2010.11.004, Lem. 2.5.

⁹⁵Meaning that there exists some $n \in \mathbb{Z}$ such that $M_i = 0$ for all $i < n$.

⁹⁶We see S_0 as a graded ring with the trivial gradation, so that any “grading” for a S_0 -module is simply a direct sum decomposition.

is particularly difficult to do in general, when the ring R is separated, complete and bounded, more can be said. We start by constructing special resolutions of bounded modules:⁹⁷

Proposition 1.4.15. *Let R be a separated, complete and bounded ring, and M a separated, complete and bounded R -module. Then, there are semicompact weighted sets X_i , $i \geq 0$, and a strict exact sequence*

$$\dots \rightarrow \widehat{R(X_2)} \rightarrow \widehat{R(X_1)} \rightarrow \widehat{R(X_0)} \rightarrow M \rightarrow 0.$$

Proof. First, recall that any closed submodule of a complete, separated and bounded module is again complete, separated and bounded under the induced filtration. Given that R is bounded, $\widehat{R(X)}$ is bounded for any semicompact weighted set X , so by proceeding inductively it suffices to construct X_0 . To build X_0 , choose for each $i \geq 0$ a finite set $Y_i \subseteq F_i M$ of generators of the finite length module $F_i M / F_{i+1} M$. Then, the set X_0 given by the disjoint union of the Y_i is a semicompact weighted set with the indicator function mapping any $x \in Y_i$ to $i \in \mathbb{Z}$. By construction, the natural map $\varphi: \widehat{R(X_0)} \rightarrow M$ is such that $\text{Gr } \varphi$ is surjective. By the part 3 of Corollary 1.4.9, φ is a strict surjection. $\square$

The modules $\widehat{R(X_i)}$ are not projective in general, so the resolution of Proposition 1.4.15 would be useless for computing Tor functors if we cannot show some form of acyclicity. Recall that for any weighted set X we have defined the weighted direct sum of M to be the filtered R -module $M(X) = \bigoplus_{x \in X} T_{-d(x)} M$. The following proposition⁹⁸ gives the acyclicity we are searching for.⁹⁹

Proposition 1.4.16. *Let R be a separated, complete and bounded ring, M and N separated, complete and bounded right and left R -modules respectively and X a semicompact weighted set. Then:*

1. *If M and N are finitely generated, then the natural filtered maps*

$$M \otimes_R \widehat{R(X)} \xrightarrow{\gamma} \widehat{M(X)},$$

$$\widehat{R(X)} \otimes_R N \xrightarrow{\gamma} \widehat{N(X)}$$

are strict surjections. If moreover M and N are finitely presented, then they are strict isomorphisms.

⁹⁷ Andrei Jaikin-Zapirain. “An Explicit Construction of the Universal Division Ring of Fractions of $E<<x_1, \dots, x_d>>$ ”. In: *Journal of Combinatorial Algebra* 4.4 (Dec. 2020), pp. 369–395. ISSN: 2415-6302. DOI: 10.4171/JCA/47, Prop. 4.5.

⁹⁸ Ibid., Prop. 4.9.

⁹⁹ Recall that a module M over a ring R is of type FP_k if it admits a projective resolution $\dots P_1 \rightarrow P_0 \rightarrow M$ such that P_i is finitely generated for every $i \leq k$.

2. If M and N are of type FP_k over R for $k \geq 2$, we have

$$\mathrm{Tor}_{k-1}^R(M, \widehat{R(X)}) = \mathrm{Tor}_{k-1}^R(\widehat{R(X)}, N) = 0.$$

Proof. We prove the claims for M , as the proof for N is completely analogous. Observe that since M is finitely generated, one can get an exact sequence

$$\widehat{R(U)} \xrightarrow{\varphi} R^n \xrightarrow{\psi} M \rightarrow 0$$

where φ and ψ are filtered maps and U is a semicompact weighted subset of $\ker \psi$ by Proposition 1.4.15. This exact sequence induces the following commutative diagram:

$$\begin{array}{ccccccc} \widehat{R(U)} \otimes_R \widehat{R(X)} & \xrightarrow{\varphi'} & R^n \otimes_R \widehat{R(X)} & \xrightarrow{\psi'} & M \otimes_R \widehat{R(X)} & \longrightarrow & 0 \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \\ \widehat{R(U \times X)} & \xrightarrow{\varphi''} & \widehat{R^n(X)} & \xrightarrow{\psi''} & \widehat{M(X)} & \longrightarrow & 0 \end{array}$$

where $U \times X$ is a semicompact weighted set with the sums of the weights in U and X .

The first row is exact because tensoring is right exact. Since the direct sum of finitely many complete modules is complete, the module $R^n \otimes_R \widehat{R(X)} \simeq (\widehat{R(X)})^n$ is complete. From there, one immediately checks that β is a strict isomorphism by defining its inverse through the universal property of $\widehat{R^n(X)}$. Alternatively, one may use Lemma 1.3.6.

From the description of Lemma 1.3.6, it follows that ψ'' is surjective and its kernel is precisely $\varphi''(\widehat{R(U \times X)})$. A standard diagram chase now concludes that γ must be surjective. We now claim that $\mathrm{Gr} \gamma$ is also surjective. To prove it, let us write for each $i \in \mathbb{Z}$

$$X_{\leq i} = \{x \in X : d(x) \leq i\}, \quad X_{> i} = \{x \in X : d(x) > i\}.$$

We have that X is the disjoint union of the weighted sets $X_{\leq i}$ and $X_{> i}$, and since X is semicompact each $X_{\leq i}$ is finite. Moreover, since M is bounded, there exists $k \in \mathbb{Z}$ for which $F_k M = M$. We then have

$$F_i \widehat{M(X)} = F_i(M(X_{\leq i-k})) \oplus \widehat{M(X_{> i-k})},$$

$$F_i(M \otimes_R \widehat{R(X)}) = F_i((M \otimes_R R(X_{\leq i-k})) \oplus (M \otimes_R \widehat{X_{> i-k}})).$$

Now, for each $i \in \mathbb{Z}$, we have that γ maps $F_i(M \otimes_R R(X_{\leq i-k}))$ onto $F_i(M(X_{\leq i-k}))$, and that $\gamma(M \otimes_R \widehat{R(X_{> i-k})})$ is contained in $\widehat{M(X_{> i-k})}$. Therefore, γ is surjective on each graded piece. The fact that γ is strict now follows from part 3 of Corollary 1.4.9.

If moreover M is finitely presented, then we can take U to be finite. In this case, α is also an isomorphism, and another diagram chase shows that γ must be an isomorphism as well. Since γ is filtered,

$$\gamma \left(\bigcap_{i \geq 0} F_i(M \otimes_R \widehat{R(X)}) \right) \subseteq \bigcap_{i \geq 0} F_i \widehat{M(X)} = 0,$$

and therefore we conclude that the filtered module $M \otimes_R \widehat{R(X)}$ is separated.

Finally, assume that M is of type FP_k over R for some $k \geq 2$. When $k = 2$, we can find a short exact sequence of R -modules

$$0 \rightarrow L \rightarrow R^n \rightarrow M \rightarrow 0$$

where L is finitely presented. Taking the degree in M of any basis of R^n endows it filtration, and taking L with the submodule filtration we may assume that all homomorphisms are filtered and that R^n and L are both complete, separated and bounded R -modules.

Tensoring with $\widehat{R(X)}$, we get an exact sequence

$$0 \rightarrow \mathrm{Tor}_1^R(M, \widehat{R(X)}) \rightarrow \widehat{L(X)} \rightarrow \widehat{R^n(X)} \rightarrow \widehat{M(X)} \rightarrow 0$$

where we have used part I for M, R^n and L . However, it follows immediately from Lemma 1.3.6 that the kernel of $\widehat{R^n(X)} \rightarrow \widehat{M(X)}$ is precisely the image of $\widehat{L(X)}$, and hence $\mathrm{Tor}_1^R(M, \widehat{R(X)}) = 0$.

If $k > 2$, there exists an exact sequence of R -modules

$$0 \rightarrow L \rightarrow R^{n_{k-3}} \rightarrow \dots \rightarrow R^{n_0} \rightarrow M \rightarrow 0.$$

Once again, we inductively endow each R^{n_i} with filtrations coming from the degree of an image of a basis in $R^{n_{i-1}}$. This makes all homomorphisms filtered and all modules separated, complete and bounded. A usual dimension shift argument shows that

$$\mathrm{Tor}_{k-1}^R(M, \widehat{R(V)}) \simeq \mathrm{Tor}_1^R(L, \widehat{R(V)}) = 0$$

as desired. $\square$

With the previous result in hand, we are now able to prove one can use this resolution to compute Tor groups.¹⁰⁰

Theorem 1.4.17. *Let R be a separated, complete and bounded ring, $k \geq 2$ some integer, M a filtered right R -module of such that $\mathrm{Tor}_i^R(M, \widehat{R(X)})$ vanishes for all $i < k$ and all semicompact weighted sets X and N a separated, complete and bounded left module. If*

$$\dots \rightarrow \widehat{R(X_2)} \rightarrow \widehat{R(X_1)} \rightarrow \widehat{R(X_0)} \rightarrow N \rightarrow 0 \quad (\dagger)$$

is a strict exact sequence with each X_i a semicompact weighted set, then $\mathrm{Tor}_i^R(M, N)$ can be computed as i -th homology of the complex $M \otimes_R \dagger$. When moreover M is of type FP_k , this complex is naturally isomorphic to

$$\dots \rightarrow \widehat{M(X_2)} \rightarrow \widehat{M(X_1)} \rightarrow \widehat{M(X_0)} \rightarrow 0$$

¹⁰⁰Observe that the following theorem has a natural analogue swapping the roles of M and N as the right and the left module.

for $i \leq k - 1$, and it is obtained by applying either of the functors $M \otimes_R -$ or $M \hat{\otimes}_R -$ to $(\dagger)$. If moreover R , M and N are profinite, we then obtain isomorphisms

$$\mathrm{Tor}_i^R(M, N) \simeq \mathfrak{Tor}_i^R(M, N)$$

for all $i \leq k - 1$.

Proof. Consider F the functor $M \otimes_R -$ defined on left R -modules to abelian groups. It is well known¹⁰¹ that one can compute its left derived functors $L_*F(N)$ by taking the homology of the complex

$$\cdots \rightarrow F(P_2) \rightarrow F(P_1) \rightarrow F(P_0) \rightarrow 0$$

where

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow N \rightarrow 0$$

is any *acyclic*¹⁰² resolution of N . The resolution $(\dagger)$ is given by modules which are partially acyclic, that is, we only have the vanishing of $L_iF(P_j)$ for some values of i . However, to compute low degrees of $L_*F(N)$ we claim it is enough.

Claim. Let R be any ring, M be any right R -module and suppose that the modules on an exact sequence

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow 0$$

satisfy $\mathrm{Tor}_i^R(M, P_j) = 0$ for every j and every $1 \leq i \leq k - 2$ with $k \geq 2$. Then, the sequence

$$M \otimes_R P_k \rightarrow \cdots \rightarrow M \otimes_R P_2 \rightarrow M \otimes_R P_1 \rightarrow M \otimes_R P_0 \rightarrow 0$$

is exact.

We already have exactness up to

$$M \otimes_R P_2 \rightarrow M \otimes_R P_1 \rightarrow M \otimes_R P_0 \rightarrow 0.$$

Break the exact sequence into short exact sequences as follows:

$$\begin{array}{ccccccc}
 \cdots & \xrightarrow{\quad} & P_2 & \xrightarrow{\quad} & P_1 & \xrightarrow{\quad} & P_0 \longrightarrow 0 \\
 & & \nearrow & \searrow & \nearrow & \searrow & \nearrow \\
 & & X_3 & & X_2 & & X_1 \\
 & \nearrow & & \nearrow & & \nearrow & \\
 0 & & 0 & & 0 & &
 \end{array}$$

¹⁰¹Serge Lang. *Algebra*. 3rd ed. Graduate Texts in Mathematics 211. Springer-Verlag New York, 2002. ISBN: 978-0-387-95385-4, Sec. XX.7.

¹⁰²An object P is acyclic for the functor F if $L_iF(P) = 0$ for all $i > 0$.

The short exact sequence starting at X_2 gives us the exact sequence

$$0 = \text{Tor}_1^R(M, P_0) \rightarrow M \otimes_R X_2 \rightarrow M \otimes_R P_1 \rightarrow M \otimes_R P_0 \rightarrow 0,$$

which shows that $M \otimes_R X_2$ is the kernel of $M \otimes_R P_1 \rightarrow M \otimes_R P_0$. Since it also fits in the exact sequence

$$M \otimes_R P_3 \rightarrow M \otimes_R P_2 \rightarrow M \otimes_R X_2 \rightarrow 0$$

by right exactness, we are able to extend our desired exact sequence by one more step.

Observe that X_2 satisfies $\text{Tor}_i^R(M, X_2) = 0$ for each $1 \leq i \leq k-3$, since we have the short exact sequence

$$0 = \text{Tor}_{i+1}^R(M, P_0) \rightarrow \text{Tor}_i^R(M, X_2) \rightarrow \text{Tor}_i^R(M, P_1) = 0.$$

Hence, the claim now follows from induction on k applied to the sequence

$$\dots \rightarrow P_3 \rightarrow P_2 \rightarrow X_2 \rightarrow 0.$$

Let us now return to the proof of the theorem. We can find a projective resolution Q_* of N such that we have a commuting diagram with surjective columns:

$$\begin{array}{ccccccc} \dots & \longrightarrow & Q_2 & \longrightarrow & Q_1 & \longrightarrow & Q_0 \longrightarrow N \\ & & \downarrow & & \downarrow & & \downarrow \text{id} \\ \dots & \longrightarrow & \widehat{R(X_2)} & \longrightarrow & \widehat{R(X_1)} & \longrightarrow & \widehat{R(X_0)} \longrightarrow N \end{array}$$

For each $i \geq 0$, let P_i be the kernel of the map $Q_i \rightarrow \widehat{R(X_i)}$. From the long exact sequence

$$\dots \rightarrow \text{Tor}_{i+1}^R(M, \widehat{R(X_j)}) \rightarrow \text{Tor}_i^R(M, P_j) \rightarrow \text{Tor}_i^R(M, Q_j) \rightarrow \dots$$

and the vanishing of $\text{Tor}_{i+1}^R(M, \widehat{R(X_j)})$ we obtain $\text{Tor}_i^R(M, P_j) = 0$ for every $1 \leq i \leq k-2$. Hence, by the previous claim we know that the diagram

$$\begin{array}{ccccccc} M \otimes_R P_k & \longrightarrow & M \otimes_R P_{k-1} & \longrightarrow & \dots & \longrightarrow & M \otimes_R P_0 \longrightarrow 0 \\ \downarrow & & \downarrow & & & & \downarrow \\ M \otimes_R Q_k & \longrightarrow & M \otimes_R Q_{k-1} & \longrightarrow & \dots & \longrightarrow & M \otimes_R Q_0 \longrightarrow 0 \\ \downarrow & & \downarrow & & & & \downarrow \\ M \otimes_R \widehat{R(X_k)} & \longrightarrow & M \otimes_R \widehat{R(X_{k-1})} & \longrightarrow & \dots & \longrightarrow & M \otimes_R \widehat{R(X_0)} \longrightarrow 0 \end{array}$$

has an exact top row and each column is a short exact sequence because of the vanishing of $\text{Tor}_1(M, \widehat{R(X_i)})$. Applying the Snake Lemma to each column one finds the exact sequences of homology groups

$$H_i(M \otimes_R P_*) \rightarrow H_i(M \otimes_R Q_*) \rightarrow H_i(M \otimes_R \widehat{R(X_*)}) \rightarrow H_{i-1}(M \otimes_R P_*),$$

yielding the desired isomorphism for every $i \leq k - 1$. When M is of type FP_k , we moreover have natural isomorphisms $M \otimes_R \widehat{R(X_i)} \simeq \widehat{M(X_i)}$ by part 1 of Proposition 1.4.16. $\square$

We will finish this section with the following result¹⁰³, the promised way of lifting information from Tor groups over $\text{Gr } R$ to Tor groups over R .

Theorem 1.4.18. *Let R be a separated, complete and bounded ring and $k \geq 1$, and let M and N be separated and complete right and left R -modules respectively. Assume that:*

1. M is of type FP_{k+1} ,
2. N is bounded and
3. $\text{Tor}_k^{\text{Gr } R}(\text{Gr } M, \text{Gr } N) = 0$.

Then, $\text{Tor}_k^R(M, N) = 0$.

Proof. By Proposition 1.4.15, there exists a strict resolution

$$\cdots \rightarrow \widehat{R(X_2)} \rightarrow \widehat{R(X_1)} \rightarrow \widehat{R(X_0)} \rightarrow N \rightarrow 0$$

of N where each X_i is semicompact. Since this resolution is strict exact and $\text{Gr}(\widehat{R(X_i)}) \simeq (\text{Gr } R)[\overline{X_i}]$, by part 1 of Proposition 1.4.8 it gives us a free resolution of $\text{Gr } N$ when passing to the graded ring:

$$\cdots \rightarrow (\text{Gr } R)[\overline{X_2}] \rightarrow (\text{Gr } R)[\overline{X_1}] \rightarrow (\text{Gr } R)[\overline{X_0}] \rightarrow \text{Gr } N \rightarrow 0.$$

Since $\text{Gr}(\widehat{M(X)}) \simeq \text{Gr}(M(X))$ and Gr commutes with direct sums, we obtain that for any weighted set X we have a natural isomorphism $\text{Gr}(\widehat{M(X)}) \simeq \text{Gr } M \otimes_{\text{Gr } R} (\text{Gr } R)[\overline{X}]$. Hence, if we start from the sequence

$$\widehat{M(X_{k+1})} \rightarrow \widehat{M(X_k)} \rightarrow \widehat{M(X_{k-1})}$$

and we apply the graded functor, we are left with the sequence

$$\text{Gr } M \otimes_{\text{Gr } R} (\text{Gr } R)[\overline{X_{k+1}}] \rightarrow \text{Gr } M \otimes_{\text{Gr } R} (\text{Gr } R)[\overline{X_k}] \rightarrow \text{Gr } M \otimes_{\text{Gr } R} (\text{Gr } R)[\overline{X_{k-1}}].$$

This sequence is exact by our hypothesis of $\text{Tor}_k^{\text{Gr } R}(\text{Gr } M, \text{Gr } N) = 0$. By part 3 of Proposition 1.4.8 and Theorem 1.4.17 we obtain that $\text{Tor}_k^R(M, N) = H_k(\widehat{M(X_*)}) = 0$ as desired. $\square$

¹⁰³Jaikin-Zapirain, “An Explicit Construction of the Universal Division Ring of Fractions of $E\langle\langle x_1, \dots, x_d \rangle\rangle$ ”, Thm. 4.10.

Chapter 2

Localization

Сегодня концерта не будет, потому что
Гриша, Миша и Антон подпиздили
патефон.

J. W. and G. V. to a very confused me.

What is localization? There are two viewpoints one may adopt when discussing the term "localization" in ring theory. The internal viewpoint is probably the first that comes to mind: to "localize" a subset T of a unital ring R is to formally build a new ring $T^{-1}R$ that "contains" R and in which every element of T is now invertible. Moreover, such a localization should be universal among all rings with this property. This is often carried out intrinsically, by studying the necessary and sufficient conditions the subset T must satisfy for the ring $T^{-1}R$ to exist and behave well.

The other viewpoint is the external one: in commutative rings, localizations at prime ideals can be used to obtain epimorphisms¹ $R \rightarrow F$ where F is a field. Hence, the problem of localization in a ring R is deeply connected with the question of "what are all the homomorphisms from R to a field?". In the more general context of non-commutative algebra, this still holds true: there is a correspondence between epimorphisms from R to division rings and certain "localizations" of R , taken in a broader sense to be defined below.

This extension from the commutative to the non-commutative setting has been led first by the works of Øystein Ore, Joseph Wedderburn, Ludwig Schwarz, Dudley Littlewood, Keizō Asano and Alfred Goldie in the first half of the twentieth century, who focused on building rings of fractions as in the commutative case. While the theory developed is fairly complete in comparison to its commutative analogue, the necessary conditions for a ring of fractions to exist fail to hold in many important rings such as free algebras over a field or group rings of

¹in the category of rings, which is not the same as surjective homomorphisms! Also, not every epimorphism is a compositum of localizations and quotients.

free groups.²

By the second half of the 20th century, the “ring of fractions” theory was already very well established and widely known among ring theorists. However, the problem of localization in more general rings was being solved at a slower pace. In a series of papers throughout the late 60’s and early 70’s, Paul Cohn made breakthrough discoveries on the structure of free algebras and their embeddings in division rings. In fact, a paper of his dated from 1971³ reads:

“A well-known (and easily proved) theorem states that each integral domain can be embedded in a field. This was generalized to certain non-commutative rings in a brilliant paper by Ore in 1930; his results are essentially as simple as in the commutative case, and the proofs, though longer, are no harder. Beyond this, very little is known, so little that it can be set down in quite a brief article. I thought this was worth doing because some interesting problems on the embedding of rings in skew fields remain, and because it gives me the chance to mention some recent work which makes it seem that these embedding problems are not quite as hard as they appear at first sight.”

Just a few months later, Cohn would publish the first edition of his book “Free rings and their relations”⁴, where his work⁵ on matrix localization would give necessary and sufficient intrinsic conditions for a general ring to admit an embedding into a division ring. This approach to localization opened new doors to study heavily non-commutative rings such as free algebras, and this field of research has seen constant development ever since. Moreover, the work of Peter Malcolmson would later connect the concept of matrix localization with that of a Sylvester rank function⁶, which is the main object under study in this thesis.

This matrix localization is what paints the general picture of localization that we will describe below. The first three sections will define and recall the construction of the Ore ring of fractions, microlocalization and of matrix localization, as well as state some of its basic properties and how it all fits within the framework of this thesis. The core of this chapter is Section 2.3, where we also give the first definition of a Sylvester domain and explore some examples. Most of this material is not original, with the exception of Theorem 2.3.17 in Section 2.3 that restricts which completed group algebras can be Sylvester domains. The last section considers the problem of extending the topology of a topological ring to a topology

²For an account of the surprising early modern history of localization in ring theory, see Severino Collier Coutinho. “Quotient Rings of Noncommutative Rings in the First Half of the 20th Century”. en. In: *Archive for History of Exact Sciences* 58.3 (Mar. 2004), pp. 255–281

³Paul Moritz Cohn. “Rings of Fractions”. In: *The American Mathematical Monthly* 78.6 (June 1971), pp. 596–615.

⁴Paul Moritz Cohn. *Free Rings and Their Relations*. en. 2nd ed. London Mathematical Society Monographs 19. London: Academic Press, Sept. 1985.

⁵along with contributions by others, mainly from George Mark Bergman’s PhD thesis “Commuting elements in free algebras and related topics in ring theory”.

⁶see Chapter 3.

on its matrix localizations, recalling some of the literature and describing the main original contribution of this chapter in the form of Theorem 2.4.1.

2.1 Ore localization

We keep the following assumptions from the first chapter: all rings are assumed to be associative and unital, and ring homomorphisms must send the multiplicative identity to the multiplicative identity. The results on Ore localization stated here can be found in any standard textbook on noncommutative Noetherian rings. We choose to reference the book by Ken Goodearl and Robert Warfield Jr.⁷ for most results.

If R is a commutative ring and T is a subset of R that is closed under multiplication, contains 1 and consists of non zero divisors, then one can build a new ring $T^{-1}R$ consisting of all fractions a/b with $a \in R$ and $b \in T$ which behave like ordinary fractions in $\mathbb{Q}$. This ring admits a natural embedding $R \rightarrow T^{-1}R$ which makes it universal among all ring homomorphisms $R \rightarrow S$ in which the image of T becomes invertible in S . When R is a domain and T is the set of non-zero elements of R , the ring $T^{-1}R$ is a field, called the field of fractions of R . As an R -module, $T^{-1}R$ is always a flat extension of R . For non-commutative rings, Ore localization arises from the natural attempt to introduce fractions in order to invert elements. However, some obstructions become immediate.

First, should fractions be written on the right (i.e. a/b meaning ab^{-1}) or on the left (i.e. $b \backslash a$ meaning $b^{-1}a$)? Does it yield different structures? The answer is to drop fraction notation altogether and refer to the expressions of ab^{-1} and $b^{-1}a$ as right and left fractions, respectively. Whether it yields different structures is a subtle problem we will discuss further down the line.

Definition 2.1.1. Let R be a ring and T a subset of R consisting of non zero divisors that is closed under multiplication and contains 1. A *right ring of fractions* (resp. *left ring of fractions*) for R with respect to T is a ring S equipped with a ring homomorphism $\iota: R \rightarrow S$ such that $\iota(b)$ is a unit for every $b \in T$ and every element of S can be written as $\iota(a)\iota(b)^{-1}$ (resp. $\iota(b)^{-1}\iota(a)$) for some $a \in R$ and some $b \in T$.

If ι is injective, we can identify R with $\iota(R) \subseteq S$ and we drop it altogether and regard the embedding as implicit from the context. The second problem now appears when we try to add fractions: to compute $ab^{-1} + cd^{-1}$, one would like to bring both fractions to a common denominator in T . This is easy in the commutative setting, but in the non-commutative one it may be impossible⁸ to find a common right multiple of b and d .

⁷Kenneth Ralph Goodearl and Robert Breckinridge Warfield Jr. *An Introduction to Noncommutative Noetherian Rings*. 2nd ed. London Mathematical Society Student Texts. Cambridge University Press, 2004.

⁸Consider the free algebra $k\langle x, y \rangle$ over a field k . Then, the elements x and y have no common non-trivial right or left multiple. Indeed, if $xa = yb$ for some $a, b \in k\langle x, y \rangle$, any monomial with a non-zero coefficient on the left hand side must begin with the letter x while on the right hand side they must begin with the letter y . An analogous argument holds for left multiples.

Lastly, multiplying fractions introduces further difficulties. In order to compute $ab^{-1}cd^{-1}$, one would need to move b^{-1} past c , that is, we would like to find $b' \in T$ and $c' \in R$ such that an identity like $b^{-1}c = c'b'^{-1}$ holds. In order to fix these last two problems, we consider the following condition:

Definition 2.1.2. Let T be a set of non zero divisors in a ring R closed under multiplication and that contains 1. We say that T satisfies the *right Ore condition* if for every $b \in T$ and $c \in R$ there is a $b' \in T$ and a $c' \in R$ such that $bc' = cb'$.⁹ We say that T satisfies the *left Ore condition* if for every $b \in T$ and $c \in R$ there is a $b' \in T$ and a $c' \in R$ such that $b'c = c'b$. If it satisfies both the left and right Ore conditions, we simply say that it satisfies the *Ore condition*.

If R is commutative, any multiplicative set T of non zero divisors containing 1 automatically satisfies the Ore condition. More generally, this is also true if T consists of central elements of R .¹⁰ A large source of left and right Ore sets is given by the following easy proposition:¹¹

Proposition 2.1.3. *Let R be a right (resp. left) Noetherian domain and T be the set of all non-zero elements of R . Then T satisfies the right (resp. left) Ore condition.*

Proof. We prove the claim for right Noetherian rings, since the proof for left Noetherian rings is similar. To show the right Ore condition for T it suffices to prove that for any non zero $a, b \in R$ we have $aR \cap bR \neq 0$. Since the right ideal generated by the elements $b, ab, a^2b, \dots$ is finitely generated, there exist elements $r_0, \dots, r_{n-1}$ such that

$$a^n b = br_0 + abr_1 + \dots + a^{n-1}br_{n-1}.$$

Let k be the smallest index such that $r_k \neq 0$. Left-canceling the a^k and rearranging the terms, we obtain

$$br_k = -abr_{k+1} - \dots - a^{n-k-1}br_{n-1} + a^{n-k}b \neq 0.$$

This is a non zero element of $aR \cap bR$, as we set out to prove. □

A ring without zero divisors such that the set of all non zero elements satisfies both the left (resp. right, both) Ore condition is called a *left Ore domain* (resp. *right Ore domain*, *Ore domain*). The relevance of the Ore condition is the following theorem:¹²

⁹Formally, just think of multiplying on the left by b^{-1} and on the right by b'^{-1} to arrive at $c'b'^{-1} = b^{-1}c$. Intuitively, the right Ore condition ensures that any hypothetical left fraction can be rewritten as a right fraction. Analogously, the left Ore condition ensures that any hypothetical right fraction can be rewritten as a left fraction.

¹⁰For an example of a set T that is not central and satisfies the Ore condition, we take R to be the quotient of the free algebra $\mathbb{Q}\langle x, y \rangle$ by the relation $yx = 2xy$. Any element of R can be written as a sum $\sum_{i,j \geq 0} a_{ij}x^i y^j$ where almost all of the coefficients $a_{ij} \in \mathbb{Q}$ are zero. It's a simple exercise to show that the center of R consists of the scalar field $\mathbb{Q}$. Let T be the set of non-negative powers of x . Using the defining relation of R , one immediately checks that any left multiple of x^i is also a right multiple of x^i and vice versa, so that T is an Ore set. The left and right Ore conditions are genuinely distinct, for an example of a pair (R, T) such that T satisfies the left Ore condition but not the right Ore condition see Exercise 6E in Goodearl and Warfield Jr, *An Introduction to Noncommutative Noetherian Rings*

¹¹This proof comes from Cohn, "Rings of Fractions"

¹²Goodearl and Warfield Jr, *An Introduction to Noncommutative Noetherian Rings*, Sec. 6.1.

Theorem 2.1.4. *Let R be a ring and T be a set satisfying the right (resp. left) Ore condition. Then, a right (resp. left) ring of fractions for R with respect to T exists and is unique up to a unique isomorphism. It is denoted by RT^{-1} (resp. $T^{-1}R$). If T satisfies both the right and the left Ore conditions, then any right ring of fractions for R with respect to T is also a left ring of fractions, and vice versa.*

It follows immediately from the definitions that if R is a left or right Ore domain, then the ring of fractions with respect to all non-zero elements is a division ring called the *division ring of fractions* of R . The ring of fractions of R with respect to T also satisfies the following universal property:¹³ if $\varphi: R \rightarrow S$ is any ring homomorphism such that $\varphi(t)$ is a unit for every $t \in T$, then φ extends uniquely to a ring homomorphism $\varphi: T^{-1}R \rightarrow S$. In particular, every Noetherian domain admits such a division ring of fractions. This result will apply in particular to the completed group algebras of torsion-free p -adic Lie groups in Chapter 5. The concept of microlocalization is a special case of the construction of a ring of fractions in the setting of filtered rings and will play a key role in the study of completed universal enveloping algebras in that same chapter.

Let us state some results on the structure of the rings of fractions RT^{-1} . First, since T consists of non zero divisors, the natural map $R \rightarrow RT^{-1}$ is an embedding¹⁴, and endows RT^{-1} with an R - R bimodule structure. Moreover, any finite family $a_1b_1^{-1}, \dots, a_nb_n^{-1}$ can be brought to a common denominator. It is also clear that the map $R \rightarrow RT^{-1}$ is an epimorphism in the category of rings, since any ring homomorphism $RT^{-1} \rightarrow S$ is completely determined by its restriction to R . Clearly, analogous results hold for the left ring of fractions with respect to a left Ore set T .

The main result about the structure of the Ore localizations $T^{-1}R$ and RT^{-1} as R -modules is the following:¹⁵

Proposition 2.1.5. *For any right (resp. left) Ore set $T \subseteq R$, the map $R \rightarrow RT^{-1}$ (resp. $T^{-1}R$) is flat.*

For any right Ore set $T \subset R$ and any left (resp. right) R -module M , define $MT^{-1} = RT^{-1} \otimes_R M$ (resp. $M \otimes_R RT^{-1}$) as the left (resp. right) RT^{-1} -module called the *right Ore localization* of M . We analogously define $T^{-1}M$ for any left Ore set T . Its elements can be written as equivalence classes of the form mb^{-1} for right Ore sets or $b^{-1}m$ for left Ore sets, with $m \in M$ and $b \in T$. When T satisfies both the left and right Ore conditions, we have natural isomorphisms $MT^{-1} \simeq T^{-1}M$, compatible with the isomorphism $RT^{-1} \simeq T^{-1}R$.

We have a natural R -module map $M \rightarrow MT^{-1}$, and its kernel $\text{tor}_T(M)$ consists of all elements $m \in M$ that are annihilated by some $t \in T$. This is called the *T -torsion* of M , and it

¹³Ibid., Prop. 6.3.

¹⁴a “fraction” $a \cdot 1^{-1}$ represents the zero element $0 \cdot 1^{-1}$ if and only if there is a pair of elements $c, d \in R$ such that $ac = 0d = 0$ and $1 \cdot c = 1 \cdot d$ is in T . Since T does not contain zero divisors, a must be zero.

¹⁵Goodearl and Warfield Jr, *An Introduction to Noncommutative Noetherian Rings*, Cor. 10.13.

defines a submodule of M^{16} . We say a module M is T -torsion-free whenever $\text{tor}_T(M) = 0$. If T is the set of all non zero divisors of R , we omit the subscript and simply call $\text{tor}(M)$ the torsion submodule of M and say that M is torsion-free whenever $\text{tor}(M) = 0$. The quotient module $M / \text{tor}_T(M)$ is always T -torsion-free. Hence, we obtain a short exact sequence

$$0 \rightarrow \text{tor}_T(M) \rightarrow M \rightarrow M / \text{tor}_T(M) \rightarrow 0$$

where the quotient map

$$M \rightarrow M / \text{tor}_T(M)$$

induces an isomorphism after taking the Ore localization at T .

2.2 Microlocalization in filtered rings

In the present section, we shall briefly touch on the subject of *microlocalization*. The term originates in the works of Mikio Sato in the late 60's, who introduced microlocalization in the context of algebraic analysis to study the singularities of solutions to linear partial differential equations, not only in terms of their location, but also with respect to directions in the cotangent space: more precisely, this is achieved by analyzing the rate of decay of the Fourier transform of a distribution tested against bump functions supported in conic neighborhoods of the point in question, leading to the definition of the so called wave front set.

Algebraically, in the setting of rings of differential operators, this corresponds to passing to an overring where one inverts not only the coefficients of the differential operators, but also the directional differential operators themselves, provided their principal symbols are invertible in the given region. In the microlocal setting, one studies modules over a localization of the sheaf of differential operators along cotangent directions, and the wave front set of a distribution is then reflected in the support of the corresponding microlocal module. We refer the reader to Pierre Schapira's survey¹⁷ for a detailed account of microlocal methods in algebraic analysis.

In our applications, microlocalization will refer to the process of lifting an Ore set from the associated graded ring of a complete filtered ring to the original ring, in order to build a new, complete filtered ring in which the elements of the Ore set become invertible. This is a purely algebraic construction, and is precisely the construction that occurs in the context of differential operators in the analytic setting.

Definition 2.2.1. Let R be a filtered ring with a filtration $F.R$. The *Rees ring* of R is the graded ring

$$\tilde{R} = \bigoplus_{n \in \mathbb{Z}} F_n R.$$

¹⁶Goodearl and Warfield Jr, *An Introduction to Noncommutative Noetherian Rings*, Lem. 4.21.

¹⁷Pierre Schapira. "Microlocal Analysis and Beyond". In: *New Spaces in Mathematics*. Ed. by Mathieu Anel and Gabriel Catren. 1st ed. Cambridge University Press, Apr. 2021, pp. 117–152. DOI: 10.1017/9781108854429.005.

We can also identify $\widetilde{R}$ with a subring of the Laurent polynomial ring $R[X, X^{-1}]$ given by $\bigoplus_{n \in \mathbb{Z}} F_{-n} R X^n$ if we identify $1 \in F_{-1} R$ with X . One immediately checks that the associated graded ring $\text{Gr } R$ is isomorphic as a graded ring to the quotient $\widetilde{R}/\widetilde{R}X$. We also have a natural filtered ring isomorphism $\widetilde{R}/\widetilde{R}(X - 1) \simeq R$, where $\widetilde{R}$ is endowed with the filtration $F_n \widetilde{R} = \bigoplus_{i \geq n} F_i R$. Hence, in some sense, the Rees ring $\widetilde{R}$ interpolates between the filtered ring R and its graded counterpart $\text{Gr } R$.

Definition 2.2.2. A filtered ring R is said to have the left (resp. right) *Artin-Rees property* if for any finite subset $\{x_1, \dots, x_m\}$ of R there is an integer c such that the left (resp. right) ideal I generated by the x_i satisfies

$$F_n R \cap I \subseteq \sum_{i=1}^m F_{n+c} R \cdot x_i \quad \left(\text{resp. } \sum_{i=1}^m x_i \cdot F_{n+c} R \right)$$

for all $n \in \mathbb{Z}$.

The Artin-Rees property allows us to relate the Noetherian condition on $\widetilde{R}$ with that on R and $\text{Gr } R$.¹⁸

Lemma 2.2.3. *If R is separated, then $\widetilde{R}$ is left (resp. right) Noetherian if and only if R and $\text{Gr } R$ are left (resp. right) Noetherian and R has the left (resp. right) Artin-Rees property.*

A particular class of filtered rings that satisfy the Artin-Rees property comes from the following result:

Lemma 2.2.4. *Let R be a separated and complete filtered ring such that $\text{Gr } R$ is left (resp. right) Noetherian. Then, R satisfies the left (resp. right) Artin-Rees property.*

Proof. We prove the claim for the left Artin-Rees property, since the proof for the right Artin-Rees property is similar. Let I be a left ideal of R generated by a finite set of elements $\{x_1, \dots, x_m\}$. Since $\text{Gr } R$ is left Noetherian, the graded $\text{Gr } R$ -module $\text{Gr } I$ is finitely generated. Observe that I is separated under the induced filtration from R . By Proposition 1.4.11, since $\text{Gr } I \leq \text{Gr } R$ is finitely generated, the induced filtration on I is a good filtration. Since any two good filtrations induce the same topology, there is an integer c such that

$$F_n R \cap I \subseteq \sum_{i=1}^m F_{n+c} R \cdot x_i$$

for all $n \in \mathbb{Z}$, as we wanted to show. □

¹⁸For a proof of the following lemma using ascending filtrations, see Huishi and van Oystaeyen, *Zariskian Filtrations*, Thm. II.1.5. Note that while the authors work with ascending filtrations, they do not assume that the filtrations are positive, that is, a filtered ring $F_\bullet R$ in our definition is also a filtered ring in their definition by considering the ascending filtration $F'_i R = F_{-i} R$. This association preserves the isomorphism class of the graded ring (with the opposite grading), the Rees ring (also with the opposite grading) as well as the presence of the Artin-Rees property.

We now turn to the problem of lifting Ore sets in $\text{Gr } R$ to Ore sets in R and $\tilde{R}$. If T' is a left or right Ore set in $\text{Gr } R$ consisting of homogeneous elements, we define the sets

$$T = \{x \in R : \bar{x} \in T'\}, \quad \tilde{T} = \{x \in \tilde{R} : x \text{ homogeneous and } \bar{x} \in T'\}.$$

It is clear that T and $\tilde{T}$ are then multiplicative sets consisting of non zero divisors by Proposition 1.4.3, with $\tilde{T}$ homogeneous. We call T the *microlocal Ore set* associated to T' . The name is justified by the following result:¹⁹

Proposition 2.2.5. *If $\tilde{R}$ is left (resp. right) Noetherian and T' is a left (resp. right) Ore set in $\text{Gr } R$, then $\tilde{T}$ is a left (resp. right) Ore set in $\tilde{R}$ and T is a left (resp. right) Ore set in R . In this case, $\tilde{T}^{-1}\tilde{R}$ is a graded ring and the quotient map $\tilde{R} \rightarrow R$ induces a surjection $\tilde{T}^{-1}\tilde{R} \rightarrow T^{-1}R$. Hence, $T^{-1}R$ inherits a filtered ring structure for which the map $R \rightarrow T^{-1}R$ is a strict ring homomorphism. Moreover, $\text{Gr } T^{-1}R \simeq (T')^{-1} \text{Gr } R$ as graded rings.*

We are now ready to define microlocalization:

Definition 2.2.6. Let R be a graded ring such that the Rees ring $\tilde{R}$ is Noetherian and let T' be an Ore set in R . If T is the microlocal Ore set associated to T' , then the *microlocalization* of R at T' is the completion $\widehat{T^{-1}R}$ of the Ore localization $T^{-1}R$ with respect to the filtration induced by that of $\tilde{R}$.

The microlocalization satisfies the following universal property:^{20,21} if $\varphi: R \rightarrow S$ is a filtered ring homomorphism such that R is separated, S is complete and $\varphi(t)$ is a unit in $F_{-n}S$ for every $t \in T \cap (F_n R \setminus F_{n+1} R)$, then φ extends uniquely to a filtered ring homomorphism $\varphi: \widehat{T^{-1}R} \rightarrow S$. Observe that the property of $\varphi(t) \in F_{-n}S$ is automatically satisfied if the filtration on R is positive. Another property we shall need about microlocalization is the following:²²

Proposition 2.2.7. *Assume that R is a filtered ring for which the Rees ring $\tilde{R}$ is Noetherian and let T be the microlocal Ore set in R associated to an Ore set T' in $\text{Gr } R$. Then, the functor $\widehat{T^{-1}R} \otimes_R -$ is exact and preserves strict maps.*

Let us consider an example of microlocalization that will be relevant in Chapter 5.²³

¹⁹Li Huishi. “Lifting Ore Sets of Noetherian Filtered Rings and Applications”. en. In: *Journal of Algebra* 179.3 (Feb. 1996), pp. 686–703. ISSN: 00218693. DOI: 10.1006/jabr.1996.0031, Cor. 2.2 and Prop. 2.3.

²⁰Ibid., Cor. 2.4.

²¹Huishi and van Oystaeyen, *Zariskian Filtrations*, Prop. IV.1.3.

²²Huishi, “Lifting Ore Sets of Noetherian Filtered Rings and Applications”, Cor. 2.7.

²³See Thm. III.2.3.3 and Lem. III.2.3.6 in Lazard, “Groupes analytiques p-adiques” or Sec. 3.3 in Otmar Venjakob. “On the structure theory of the Iwasawa algebra of a p-adic Lie group”. In: *Journal of the European Mathematical Society* 4.3 (Sept. 2002), pp. 271–311. ISSN: 1435-9863. DOI: 10.1007/s100970100038

Proposition 2.2.8. *Let G be a uniform pro- p group and $J \leq \mathbb{Z}_p[[G]]$ be the radical of its completed group algebra over $\mathbb{Z}_p$. Then, with respect to the J -adic filtration, the set*

$$T = \bigcup_{i \geq 0} p^i + J^{i+1}$$

is a microlocal Ore set in $\mathbb{Z}_p[[G]]$ associated to the Ore set $T' = \{\pi^i : i \geq 0\}$ in $\text{Gr } \mathbb{Z}_p[[G]]$, where π is the principal symbol of p in $\text{Gr } \mathbb{Z}_p[[G]]$.

Proof. We've already seen in Example 1.4.13 that $\text{Gr } \mathbb{Z}_p[[G]]$ is isomorphic to the universal enveloping algebra of $\text{Gr } G$ over $\mathbb{F}_p[\pi]$, which is a Noetherian domain and a torsion-free $\mathbb{F}_p[\pi]$ -algebra of which $\pi = \bar{p}$ is a central regular element. The conclusion now follows from Lemma 2.2.4 and Proposition 2.2.5. $\square$

We shall see in Chapter 5 that the microlocalization $\widehat{T^{-1} \mathbb{Z}_p[[G]]}$ is intimately connected to the universal enveloping algebra of the $\mathbb{Q}_p$ -Lie algebra of G .

2.3 Matrix localization

The study of matrix localization is a natural generalization of the Ore condition to arbitrary associative rings with unity, which stemmed from Paul Cohn's approach to the following problem: given a ring R , what are all the possible embeddings of R into a division ring?

This question is a subtle one. First, there may not exist any such embedding at all: for instance, if R has zero divisors. Hence, it is also natural to consider the weaker question of finding ring homomorphisms from R to division rings. Observe that it is also possible that all such homomorphisms are trivial, as is the case when $R = \text{Mat}_2(K)$ is a matrix ring over a field K .

Moreover, it is also not clear how to “distinguish” two given homomorphisms $R \rightarrow \mathcal{D}$ from R to a division ring $\mathcal{D}$. For instance, the two maps $\mathbb{Z}[i] \rightarrow \mathbb{Q}(i)$ given by inclusion and complex conjugation should be considered different maps or the same map? Also observe that we can always compose $R \rightarrow \mathcal{D}$ with an extension of division rings $\mathcal{D} \rightarrow \mathcal{D}'$ to obtain a new homomorphism $R \rightarrow \mathcal{D}'$, such as for example the embedding of $\mathcal{D}$ into the division ring of formal Laurent series $\mathcal{D}((t))$.

To work around this issues, we shall concern ourselves exclusively with *epic* homomorphisms $R \rightarrow \mathcal{D}$ to division rings $\mathcal{D}$, that is, homomorphisms that are epimorphisms in the category of rings. This means that if we have two ring homomorphisms $\phi_1, \phi_2 : \mathcal{D} \rightarrow S$ such that $\phi_1 \circ \varphi = \phi_2 \circ \varphi$, then $\phi_1 = \phi_2$. In this case, we call $\mathcal{D}$ an *epic division R -ring*. There is a natural notion of isomorphism of epic division R -rings²⁴, and hence Cohn's question can be better formulated as: what are all the isomorphism classes of epic division R -rings?

²⁴Notably, $\varphi_1 : R \rightarrow \mathcal{D}_1$ and $\varphi_2 : R \rightarrow \mathcal{D}_2$ are isomorphic if there exists an isomorphism $\psi : \mathcal{D}_1 \rightarrow \mathcal{D}_2$ such that $\psi \circ \varphi_1 = \varphi_2$. In particular, the two embeddings of $\mathbb{Z}[i]$ into $\mathbb{Q}(i)$ induce isomorphic epic division $\mathbb{Z}[i]$ -ring structures on $\mathbb{Q}(i)$, so they could count as the same embedding.

Before we continue, we give a different characterization of epic ring homomorphisms:²⁵

Proposition 2.3.1. *Let $\varphi: R \rightarrow S$ be a ring homomorphism. Then, the following are equivalent:*

- (a) φ is an epimorphism in the category of rings.
- (b) $1 \otimes s = s \otimes 1$ in $S \otimes_R S$ for any $s \in S$.
- (c) The multiplication map $\mu: S \otimes_R S \rightarrow S$ given by $\mu(s_1 \otimes s_2) = s_1 s_2$ is an isomorphism of S - S bimodules.

If S is a division ring, then these properties hold if and only if S is the smallest division subring containing $\varphi(R)$.

Proof. Suppose that φ is an epimorphism and regard $S \otimes_R S$ as an S - S bimodule. The abelian group $\mathcal{A} = S \oplus (S \otimes_R S)$ can be endowed with an associative ring structure where the multiplication map is given by

$$(a, x) \cdot (b, y) = (ab, ay + xb),$$

the unit element being $(1, 0)$. We have two ring homomorphisms $\psi_1, \psi_2: S \rightarrow \mathcal{A}$ given by $\psi_1(s) = (s, 0)$ and $\psi_2(s) = (s, s \otimes 1 - 1 \otimes s)$. Since $\psi_1 \circ \varphi = \psi_2 \circ \varphi$, we have that $\psi_1 = \psi_2$, and so $s \otimes 1 = 1 \otimes s$ for any $s \in S$, showing that (a) implies (b).

Observe that the multiplication map $S \otimes_R S \rightarrow S$ is always an S -bimodule homomorphism. If (b) holds, then we have a well defined inverse bimodule homomorphism $S \rightarrow S \otimes_R S$ given by $s \mapsto 1 \otimes s$, showing that (c) holds. Finally, if (c) holds, let $\psi_1, \psi_2: S \rightarrow T$ be two ring homomorphisms that agree on $\varphi(R)$. Then, the map $S \times S \rightarrow T$ given by $(s_1, s_2) \mapsto \psi_1(s_1)\psi_2(s_2)$ is R -bilinear, and so it induces a homomorphism of S - S -bimodules $\psi: S \otimes_R S \rightarrow T$. Since the multiplication map μ is an isomorphism, we have that

$$\psi_1(s) = \psi(s \otimes 1) = \psi(\mu^{-1}(s)) = \psi(1 \otimes s) = \psi_2(s)$$

for any $s \in S$, showing that (a) holds.

Assume now that S is a division ring, φ is epic and let S' be the smallest division subring of S containing $\varphi(R)$. Tensoring the inclusion of right S' -modules $S' \rightarrow S$ with the left S' -modules S' and S/S' we obtain the commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & S & \longrightarrow & S \otimes_{S'} S \\ & & \downarrow & & \downarrow \gamma \\ 0 & \longrightarrow & S/S' & \xrightarrow{\beta} & S \otimes_{S'} S/S' \end{array}$$

²⁵Bo Stenström. *Rings of Quotients*. Springer Berlin Heidelberg, 1975. ISBN: 9783642660665. DOI: 10.1007/978-3-642-66066-5. URL: <http://dx.doi.org/10.1007/978-3-642-66066-5>, Prop. 1.1, 1.2 and 1.4.

of left S' -modules with exact rows, as every S' -module is flat. Since $S \otimes_{S'} S$ is a quotient of $S \otimes_R S$, property (b) implies that $1 \otimes s = s \otimes 1$ holds in $S \otimes_{S'} S$ for any $s \in S$ as well. Hence, for any $s \in S$, if $\bar{s}$ denotes its image in S/S' , we have

$$\beta(\bar{s}) = 1 \otimes \bar{s} = \gamma(1 \otimes s) = \gamma(s \otimes 1) = s \otimes \bar{1} = 0,$$

from which we deduce $\bar{s} = 0$, that is, $S = S'$. Conversely, given two ring homomorphisms $\psi_1, \psi_2: S \rightarrow T$ that agree on $\varphi(R)$, the equalizer

$$E = \{x \in S: \psi_1(x) = \psi_2(x)\}$$

is a division subring of S containing $\varphi(R)$. Hence, if $S = S'$, then $E = S$, showing that $\psi_1 = \psi_2$ and so φ is epic. $\square$

In particular, epimorphisms do not need to be surjective. The prototypical example of an epimorphism is the inclusion map $\mathbb{Z} \rightarrow \mathbb{Q}$. There are examples of epimorphisms $R \rightarrow S$ where the ring R is finite and S is infinite.²⁶ A particular class on which epimorphisms coincide with surjections is the class of left or right Kasch rings. A ring is said to be *left Kasch* if every simple left R -module embeds in a free left R -module, and analogously for *right Kasch* rings. Examples of rings that are both left and right Kasch include commutative Artinian rings R , Artinian local rings R and group rings $R[G]$ of finite groups G over left and right Kasch rings R .²⁷

Proposition 2.3.2. *Let R be a left or right Kasch ring and $\varphi: R \rightarrow S$ an epimorphism. Then, φ is surjective.*²⁸

Proof. We'll prove the proposition assuming that R is right Kasch, as the proof for the left Kasch condition is analogous. The condition that the multiplication map $S \otimes_R S \rightarrow S$ is an isomorphism implies that in the exact sequence of right S -modules

$$\varphi(R) \otimes_R S \xrightarrow{\alpha} S \otimes_R S \rightarrow S/\varphi(R) \otimes_R S \rightarrow 0$$

the map α is an isomorphism. Hence, the right S -module $S/\varphi(R) \otimes_R S$ is trivial. Therefore:

$$\text{Hom}_S(S/\varphi(R) \otimes_R S, S) = \text{Hom}_R(S/\varphi(R), \text{Hom}_S(S, S)) = 0.$$

But, as a right R -module, $\text{Hom}_S(S, S) \simeq S$ under the evaluation at 1 map. Hence, we get that $\text{Hom}_R(S/\varphi(R), S)$ is the trivial group. If $S/\varphi(R)$ were non-trivial, it would surject onto a

²⁶John R. Isbell. "Epimorphisms and Dominions". In: *Proceedings of the Conference on Categorical Algebra*. Springer Berlin Heidelberg, 1966, pp. 232–246. ISBN: 9783642999024. DOI: 10.1007/978-3-642-99902-4_9, Sec. 3.7.

²⁷Liang Shen. "Group rings with annihilator conditions". In: *Acta Mathematica Hungarica* 156.1 (Aug. 2018), pp. 38–46. ISSN: 1588-2632. DOI: 10.1007/s10474-018-0860-5, Thm. 2.4.

²⁸Hans H. Storrer. "Epimorphic extensions of non-commutative rings". In: *Commentarii Mathematici Helvetici* 48.1 (Dec. 1973), pp. 72–86. ISSN: 1420-8946. DOI: 10.1007/bf02566112, Prop. 4.2.

simple R -module M . However, R^k contains M as a submodule, so projecting onto a non-trivial coordinate shows that R contains M as a non-trivial submodule. This produces a non-zero element in $\text{Hom}_R(S/\varphi(R), S)$, showing that $S/\varphi(R) = 0$, that is, φ is surjective. $\square$

Another closely related notion is that of the rational closure of a homomorphism.²⁹ If $\varphi: R \rightarrow S$ is a ring homomorphism, the *rational closure* $R^\varphi(S)$ of R in S is the set of all entries of inverses of matrices over $\varphi(R)$ that are invertible over S . It can be shown that $R^\varphi(S)$ is a subring of S containing $\varphi(R)$.³⁰ Moreover, the map $R \rightarrow R^\varphi(S)$ is always epic.³¹ In general, it is not true that $R^\varphi(S) = S$ when φ is epic, but this does hold when S is a division ring.

Proposition 2.3.3. *Let $\varphi: R \rightarrow \mathcal{D}$ be a ring homomorphism to a division ring $\mathcal{D}$. Then, φ is epic if and only if $R^\varphi(\mathcal{D}) = \mathcal{D}$.*

Proof. It suffices to show that $R^\varphi(\mathcal{D})$ is closed under inverses when they exist in $\mathcal{D}$. If $x \in R^\varphi(\mathcal{D})$ is non-zero, then it can be expressed as

$$x = e_i^T \varphi(A)^{-1} e_j$$

where A is a matrix over R and e_i, e_j are standard column basis vectors of appropriate sizes.

Then, the block matrix $B = \begin{pmatrix} \varphi(A) & e_j \\ e_i^T & 0 \end{pmatrix}$ is invertible over $\mathcal{D}$ with inverse

$$B^{-1} = \begin{pmatrix} \varphi(A)^{-1} - \varphi(A)^{-1} e_j x^{-1} e_i^T \varphi(A)^{-1} & \varphi(A)^{-1} e_j x^{-1} \\ x^{-1} e_i^T \varphi(A)^{-1} & -x^{-1} \end{pmatrix},$$

so we have that $-x^{-1}$ is an entry of the inverse of a matrix over $\varphi(R)$, showing that $x^{-1} \in R^\varphi(\mathcal{D})$. $\square$

This gives a way to construct an epic division R -ring $\mathcal{D}$ from information completely intrinsic to R : namely, by specifying which matrices over R become invertible in $\mathcal{D}$. This is the idea behind matrix localization. Observe that over a division ring, every square matrix is either invertible or singular (that is, a zero divisor). Hence, the information about the matrices that become invertible in $\mathcal{D}$ is equivalent to the information about the *singular kernel* of φ : the set Σ_φ of all square matrices over R that become singular over $\mathcal{D}$.

The main insight of Cohn was to axiomatize the properties of singular kernels. To state these properties, we recall the definition of the *inner rank* of a matrix: if A is an $n \times n$ matrix over a ring R , its inner rank $\text{irk}(A)$ is the smallest integer k such that A can be factored as a product of an $n \times k$ matrix by a $k \times n$ over R . By definition, we say that $\text{irk}(A) = 0$ for any zero square matrix A . We say that an $n \times n$ matrix is *full* if $\text{irk}(A) = n$.

²⁹Cohn, *Free Rings and Their Relations*, Sec. 7.1.

³⁰Ibid., Thm. 7.1.2.

³¹If $\psi_1, \psi_2: R^\varphi(S) \rightarrow T$ are two ring homomorphisms that agree on $\varphi(R)$ and $x \in R^\varphi(S)$ is an entry in $\varphi(A)^{-1}$, then $\psi_1(x)$ and $\psi_2(x)$ are the corresponding entries in $\psi_1(\varphi(A))^{-1} = \psi_2(\varphi(A))^{-1}$, so they must be equal.

Definition 2.3.4. Let Σ be a collection of square matrices over a ring R . We say that Σ is a *matrix ideal* if the following properties hold:

(MI₁) Every non-full matrix is in Σ .

(MI₂) If A and B are two matrices in Σ of the same size that differ by a single row or column, then the matrix $A \nabla B$ obtained by summing that row or column is also in Σ .³²

(MI₃) If A is in Σ and B is any square matrix over R , then their block diagonal sum $A \oplus B$ is also in Σ .

(MI₄) If a block diagonal matrix $\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$ is in Σ , then so is A .

A matrix ideal Σ is said to be *proper* if no identity matrix lies in Σ , and *prime* if it is proper and also satisfies:

(MI₅) If A and B are two square matrices such that $A \oplus B \in \Sigma$, then either $A \in \Sigma$ or $B \in \Sigma$.

Lets check that these axioms properly describe singular kernels of epic division R -rings.

Lemma 2.3.5. Let $\varphi: R \rightarrow \mathcal{D}$ be an epic homomorphism to a division ring $\mathcal{D}$. Then, its singular kernel Σ_φ is a prime matrix ideal.

Proof. Let A be a non-full matrix and write it as $A = BC$ where B is an $n \times k$ matrix and C is a $k \times n$ matrix with $k < n$. Then, the $n \times n$ matrix $\varphi(A) = \varphi(B)\varphi(C)$ has rank at most k over the division ring $\mathcal{D}$, so it is singular. This shows (MI₁). For (MI₂), it suffices to prove the claim for column operations since Σ_φ is closed under taking transposes. Also, it is clear that Σ_φ is closed under permutations of rows and columns, so we may assume that A and B differ by their first column. Let a_1 and b_1 be the first columns of A and B respectively, and let $a_2, \dots, a_n$ be the other columns of both matrices. The image of left multiplication by A (resp. B) in $\mathcal{D}^n$ is the right $\mathcal{D}$ -module spanned by $\{\varphi(a_1), \varphi(a_2), \dots, \varphi(a_n)\}$ (resp. $\{\varphi(b_1), \varphi(a_2), \dots, \varphi(a_n)\}$). Since both A and B are in Σ_φ , these sets are linearly dependent. Hence, the set $\{\varphi(a_1) + \varphi(b_1), \varphi(a_2), \dots, \varphi(a_n)\}$ is also linearly dependent, showing that $A \nabla B$ is in Σ_φ .

For (MI₃), observe that if P is a non-zero matrix over $\mathcal{D}$ such that $P\varphi(A) = 0$, then

$$\begin{pmatrix} P & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \varphi(A) & 0 \\ 0 & \varphi(B) \end{pmatrix} = 0,$$

showing that $\varphi(A \oplus B)$ is singular as well. For (MI₄), let P be a non-zero matrix over $\mathcal{D}$ such that $P(\varphi(A) \oplus (1)) = 0$. Writing $P = \begin{pmatrix} p_1 & p_2 \\ p_3 & x \end{pmatrix}$ where p_2 and p_3 are column vectors of the

³²The matrix $A \nabla B$ is also called the determinantal sum of A and B .

size of A , we have

$$P(\varphi(A) \oplus (1)) = \begin{pmatrix} P_1 & p_2 \\ p_3^T & x \end{pmatrix} \begin{pmatrix} \varphi(A) & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} P_1 \varphi(A) & p_2 \\ p_3^T \varphi(A) & x \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

showing that $\varphi(A)$ is singular over $\mathcal{D}$, as desired. Finally, for (MI₅), let A and B be two square matrices over R . If the direct sum $\varphi(A) \oplus \varphi(B)$ is invertible in $\mathcal{D}$, its inverse must also be block diagonal of the form $P \oplus Q$ for matrices P and Q with the same sizes as A and B respectively, and a direct computation shows that $\varphi(A)P$ and $\varphi(B)Q$ are identity matrices. Hence, neither A nor B can be in Σ_φ . $\square$

We can now state Cohn's theorem classifying epic division R -rings:³³

Theorem 2.3.6. *Let R be a ring.*

(a) *If Σ is a prime matrix ideal of R , then there exists an epic local R -ring R_Σ such that its residue division ring $\mathcal{D}$ is an epic division R -ring with singular kernel Σ .*

(b) *If $\mathcal{D}$ is an epic division R -ring with singular kernel Σ , then there is an isomorphism of epic R -rings between $\mathcal{D}$ and the residue division ring of R_Σ .*

The local ring R_Σ from Theorem 2.3.6 is called the *localization* of the ring R at the matrix ideal Σ . It can also be described by the following universal property: it comes with a map $\psi: R \rightarrow R_\Sigma$ such that every matrix A over R that is not in Σ becomes invertible over R_Σ , and given any other ring homomorphism $\varphi: R \rightarrow S$ such that every matrix not in Σ becomes invertible over S , there exists a unique ring homomorphism $\tilde{\varphi}: R_\Sigma \rightarrow S$ such that $\varphi = \tilde{\varphi} \circ \psi$.³⁴

The explicit construction R_Σ by Cohn is somewhat involved, obtained by formally adjoining inverses to the matrices in the complement of Σ , so we shall not give the details here. What we shall mention is that Theorem 2.3.6 establishes more than just a bijection between the prime matrix ideals of R and the isomorphism classes of epic division R -rings: if we introduce a partial order on the prime matrix ideals by inclusion, and a partial order on the set of isomorphism classes of epic division R -rings by setting $\mathcal{D}_1 \leq \mathcal{D}_2$ if there exists a local R -subring $L \leq \mathcal{D}_2$ whose residue division ring is isomorphic to $\mathcal{D}_1$ as R -rings, then these bijections are order-reversing isomorphisms of posets. This order relation on the set of all epic division R -rings is denoted by saying that $\mathcal{D}_1$ is an *specialization* of $\mathcal{D}_2$.³⁵

We shall say that an epic division R -ring $\mathcal{D}$ is *universal* if it is the maximal element in this poset, or equivalently, if for any other epic division R -ring $\mathcal{D}'$ there exists a local R -subring of $\mathcal{D}$ whose residue division ring is isomorphic to $\mathcal{D}'$. In this case, we also say that the localization

³³Cohn, *Free Rings and Their Relations*, Thm. 7.4.8.

³⁴A ring R_Σ with this universal property can be shown to exist for any collection Σ of matrices, but without further hypotheses on Σ it can be very ill-behaved, including cases where it degenerates to the trivial ring.

³⁵Cohn, *Free Rings and Their Relations*, Sec. 7.2.

R_Σ at its singular kernel Σ is a *universal localization* of R . By Theorem 2.3.6, a universal epic division R -ring exists if and only if there is a unique minimal prime matrix ideal for R .

A first example of a ring admitting a universal division ring is $\mathbb{Z}$, with $\mathbb{Q}$ as its universal epic division ring and the various quotients $\mathbb{Q} \geq \mathbb{Z}_{(p)} \rightarrow \mathbb{F}_p$ as its specializations. In this case, the minimal prime matrix ideal is given by the set of all square matrices over $\mathbb{Z}$ whose determinant is zero. This example can be generalized to any commutative ring R admitting a unique minimal prime ideal. This is misleading evidence for the general picture, however: there are examples of Ore domains that do not admit universal division rings.³⁶

Another class of rings admitting universal division rings are called *Sylvester domains*. We shall see more characterizations of Sylvester domains in Chapter 4, but for now we give the following definition:

Definition 2.3.7. Let R be a ring. We say that R is a *Sylvester domain* if the set of all non-full square matrices over R is a prime matrix ideal.

Observe that by property (MIr), any matrix ideal must contain all non-full matrices. Hence, if the set Σ of all non-full matrices is a prime matrix ideal, it must be the unique minimal one. In this case, the universal epic division R -ring $\mathcal{D}$ associated to Σ is called the *universal division ring of fractions* of R . Sylvester domains appeared first implicitly in the work of Cohn³⁷ and were later defined explicitly by Warren Dicks and Eduardo Sontag³⁸. Also, the kernel of the map $R \rightarrow \mathcal{D}$ must be contained in Σ , and since non-zero 1×1 matrices are always full, we deduce that Sylvester domains are always domains and inject into $\mathcal{D}$, thus justifying the name.

A remarkable property of Sylvester domains is that the universal division ring is precisely the matrix localization at the set of all non-full matrices.³⁹

Proposition 2.3.8. Let R be a Sylvester domain and let Σ be the set of all non-full square matrices over R . Then, the matrix localization R_Σ is a division ring, and hence it is the universal division ring of fractions of R .

Some examples of Sylvester domains are as follows.

Example 2.3.9. Every division ring is a Sylvester domain, with itself as its universal division ring of fractions. ■

Example 2.3.10. Any ring without zero divisors where any left or right ideal is principal is a Sylvester domain, with its Ore field of fractions as its universal division ring of fractions. This follows from the non-commutative analogue of the Smith normal form for matrices over principal ideal domains, and the fact that a diagonal matrix over such a ring is full if and

³⁶D. S. Passman. “Universal fields of fractions for polycyclic group algebras”. In: *Glasgow Mathematical Journal* 23.2 (July 1982), pp. 103–113. ISSN: 1469-509X. DOI: 10.1017/S0017089500004869.

³⁷Cohn, *Free Rings and Their Relations*.

³⁸Warren Dicks and Eduardo D. Sontag. “Sylvester domains”. en. In: *Journal of Pure and Applied Algebra* 13.3 (Dec. 1978), pp. 243–275. ISSN: 0022-4049.

³⁹Cohn, *Free Rings and Their Relations*, Prop. 7.5.7.

only if it is non-zero. In particular, the polynomial ring $\mathcal{D}[X]$ is a Sylvester domain for any division ring $\mathcal{D}$, with the division ring of rational functions $\mathcal{D}(X)$ as its universal division ring of fractions. ■

Example 2.3.11. A finite ring is a Sylvester domain if and only if it is a field. Indeed, such a ring must be a division ring, since multiplication by any non-zero element is an injective and thus surjective abelian group homomorphism. Now, we apply Wedderburn's little theorem: a finite division ring is commutative. ■

Example 2.3.12. An example of an infinite profinite Sylvester domain is given by the ring $\mathbb{F}_p[[t]]$ of formal power series over the finite field $\mathbb{F}_p$, since this is a principal ideal domain. A less obvious example is the ring $\mathbb{Z}_p[[t]]$ of formal power series over the p -adic integers $\mathbb{Z}_p$. To see why it is a Sylvester domain, observe that its field of fractions is the field $\mathbb{Q}_p^*((t))$ of formal Laurent series over the p -adic numbers $\mathbb{Q}_p$ where the norm of the coefficients is bounded⁴⁰, and every non-zero element $x \in \mathbb{Q}_p^*((t))$ can be written uniquely as $x = up^l t^k$ where $u \in \mathbb{Z}_p[[t]]$ is a unit and $k, l \in \mathbb{Z}$. If A is an $n \times n$ matrix over $\mathbb{Z}_p[[t]]$, we claim that A is full if and only if its determinant is non-zero in $\mathbb{Z}_p[[t]]$. It is clear that a matrix with non-zero determinant is invertible over $\mathbb{Q}_p^*((t))$, so it must be full. For the converse, note that if $\det A = 0$, then there exists $v \in \mathbb{Q}_p^*((t))^n$ non-zero such that $Av = 0$. Multiplying v by an appropriate power of t and p , we may assume that $v \in \mathbb{Z}_p[[t]]^n$ is non-zero. Hence, the columns of A are linearly dependent over $\mathbb{Z}_p[[t]]$. Without loss of generality, we may assume that the first column $a_1 = \alpha_2 a_2 + \dots + \alpha_n a_n$ of A is a linear combination of the other columns $a_2, \dots, a_n$. Then, we have

$$A = \begin{pmatrix} a_2 & a_3 & \dots & a_n \end{pmatrix}_{n \times (n-1)} \begin{pmatrix} \alpha & e_1 & \dots & e_{n-1} \end{pmatrix}_{(n-1) \times n}$$

where α is the column vector $(\alpha_2, \dots, \alpha_n)^T$ and e_i are the standard basis vectors of $\mathbb{Z}_p[[t]]^{n-1}$. This shows that $\text{irk}(A) < n$, so A is not full. Both rings are also examples of completed group algebras, with $\mathbb{F}_p[[t]]$ isomorphic to $\mathbb{F}_p[[\mathbb{Z}_p]]$ and $\mathbb{Z}_p[[t]]$ isomorphic to $\mathbb{Z}_p[[\mathbb{Z}_p]]$. ■

A landmark theorem of Cohn, which was also the reason why the Sylvester condition was introduced in the first place, concerns *free ideal rings*, or *firs*: these are rings where every left or right ideal is free of unique rank as a module, a class which includes free algebras over division rings.⁴¹ More generally, a ring R is called a *semifir* if every finitely generated left or right ideal is free of unique rank. Cohn's theorem states the following:⁴²

Theorem 2.3.13. *Every semifir is a Sylvester domain.*

Using this theorem, we can describe another important example of a Sylvester domain that is relevant for this thesis.

⁴⁰ Every element of $\mathbb{Z}_p[[t]]$ can be written uniquely as $up^l t^k$ for some l, k non-negative integers and u a unit. Hence, it suffices to invert t and p .

⁴¹ Cohn, *Free Rings and Their Relations*, Prop. 4.2 and Cor. 4.3.

⁴² Ibid., Prop. 5.1 and Thm. 7.5.10.

Proposition 2.3.14. *Let F be a free pro- p group and $\mathbb{F}_q$ a finite field of characteristic p . Then, the completed group algebra $\mathbb{F}_q[[F]]$ is a Sylvester domain.*

Proof. Any finitely generated ideal of $\mathbb{F}_q[[F]]$ is profinite and therefore closed. By a theorem of Armand Brumer,⁴³ the global dimension⁴⁴ of the category of profinite $\mathbb{F}_q[[F]]$ -modules is 1. Hence, any finitely generated ideal I of $\mathbb{F}_q[[F]]$ is projective. Since $\mathbb{F}_q[[F]]$ is local, I must be free. As profinite rings have invariant basis number, the rank of I is also unique. This shows that $\mathbb{F}_q[[F]]$ is a semifir, so Theorem 2.3.13 implies that it is a Sylvester domain. $\square$

Later in Section 4.2 we shall see an explicit construction of the universal division ring of fractions of $\mathbb{F}_q[[F]]$ using ideas of Andrei Jaikin-Zapirain. We now record⁴⁵ a fundamental property of Sylvester domains:⁴⁶

Proposition 2.3.15. *Any Sylvester domain has weak global dimension⁴⁷ at most 2.*

This condition imposes strong restrictions on which completed group algebras can be Sylvester domains. We recall Brumer's formula about the global dimension of the category of profinite modules over completed group algebras:⁴⁸ if R is a commutative profinite ring and G is a profinite group, then

$$\text{gl.dim}(R[[G]]) = \text{gl.dim}(R) + \sup_{p \notin (R/J(R))^\times} \text{cd}_p(G).$$

Moreover, recall from the end of Section 1.2 that if $R[[G]]$ has no zero divisors, then both R and G must be pro- p , with R a commutative pro- p local ring.

Lemma 2.3.16. *Let R be a commutative pro- p local ring and G a pro- p group. If the abstract weak global dimension of $R[[G]]$ is at most 2 and the finite $R[[G]]$ -module $k = R/J(R)$ with trivial G -action is of type FP_2 , then the profinite global dimension $\text{gl.dim}(R[[G]])$ in the category of profinite $R[[G]]$ -modules is also at most 2.*

Proof. Since $R[[G]]$ is a local ring, the only simple profinite $R[[G]]$ module is the finite field k with the trivial G action. Therefore, it suffices⁴⁹ to show that $\mathfrak{Tor}_3^{R[[G]]}(k, k) = 0$. By our hypothesis, there exists a finitely generated free resolution

$$R[[G]]^{n_2} \xrightarrow{\varphi_2} R[[G]]^{n_1} \xrightarrow{\varphi_1} R[[G]]^{n_0} \xrightarrow{\varphi_0} k \rightarrow 0.$$

⁴³Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations", Thm. 4.1.

⁴⁴The maximal length of a profinite projective resolution of all profinite modules

⁴⁵See also Proposition 4.1.9

⁴⁶Dicks and Sontag, "Sylvester domains", Cor. 4.

⁴⁷The maximal length of a flat resolution of any module

⁴⁸Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations", Thm. 4.1.

⁴⁹Ibid., Thm. 3.5.

Let $K_i = \ker \varphi_i$. Since the abstract weak global dimension of $R[[G]]$ is at most 2, we have

$$0 = \mathrm{Tor}_3^{R[[G]]}(k, M) = \mathrm{Tor}_2^{R[[G]]}(K_0, M) = \mathrm{Tor}_1^{R[[G]]}(K_1, M)$$

for any $R[[G]]$ -module M , that is, the module K_1 is flat. Since K_1 is finitely generated and $R[[G]]$ is local, it is projective⁵⁰ and therefore free and profinite. Hence,

$$0 \rightarrow K_1 \rightarrow R[[G]]^{n_1} \rightarrow R[[G]]^{n_0} \rightarrow k$$

is a profinite projective resolution of k of length 2, showing that $\mathfrak{Tor}_3^{R[[G]]}(k, k) = 0$. $\square$

This implies that under suitable finiteness hypotheses on G , Brumer's formula can be used to restrict which completed group algebras can be Sylvester domains.

Theorem 2.3.17. *Let G be a finitely presented pro- p group and R a commutative pro- p local ring. Then, if $R[[G]]$ is a Sylvester domain, one of the following holds:*

(a) $\mathrm{gl.dim}(R) = 2$. Then G is trivial. If moreover R is Noetherian, then it must be either a power series ring $\mathbb{F}_{p^n}[[x, y]]$ in two variables over a finite field or an Eisenstein extension of the ring of power series $\mathcal{O}_K[[x]]$ in one variable over the ring of integers of a finite extension K of $\mathbb{Q}_p$ with residue field $R/J(R)$.

(b) $\mathrm{gl.dim}(R) = 1$. Then G is a finitely generated free pro- p group and R is either a power series ring $\mathbb{F}_{p^n}[[x]]$ over a finite field or the ring of integers $\mathcal{O}_K$ of a finite extension K of $\mathbb{Q}_p$.

(c) $\mathrm{gl.dim}(R) = 0$. Then $R = \mathbb{F}_{p^n}$ is a finite field and $\mathrm{cd}_p G \leq 2$.

Proof. First, observe that if G is finitely presented and $k = R/J(R)$ is the residue field of R , then the $R[[G]]$ -module k with trivial G -action is of type FP_2 . Indeed, choose a free profinite resolution

$$F_1 \rightarrow R[[G]] \xrightarrow{\varphi_0} k \rightarrow 0$$

of k over $R[[G]]$ where φ_0 is the composition of the augmentation map with the projection modulo $J(R)$. Since G is finitely generated, $H_1(G, k)$ is a finite dimensional k -vector space. As we have⁵¹

$$H_1(G, k) = \mathfrak{Tor}_1^{R[[G]]}(k, k) \simeq \mathfrak{Tor}_0^{R[[G]]}(\ker(\varphi_0), k) \simeq \ker(\varphi_0)/J(R[[G]])\ker(\varphi_0),$$

the module $\ker(\varphi_0)$ is finitely generated as a profinite $R[[G]]$ -module by Nakayama's lemma (Lemma 1.2.5). If $d = \dim_k H_1(G, k)$, we may now repeat the argument with a free profinite resolution

$$F_2 \rightarrow R[[G]]^d \xrightarrow{\varphi_1} \ker(\varphi_0) \rightarrow 0$$

⁵⁰T. Y. Lam. *Lectures on Modules and Rings*. 1st ed. Graduate Texts in Mathematics 189. Springer-Verlag New York, 1999. ISBN: 978-0-387-98428-5, Thm. 2.4.38.

⁵¹Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations", Lem. 4.2.

to conclude using

$$H_2(G, k) = \mathcal{T}or_2^{R[[G]]}(k, k) \simeq \mathcal{T}or_1^{R[[G]]}(\ker(\varphi_0), k) \simeq \mathcal{T}or_0(\ker(\varphi_1), k)$$

that $\ker(\varphi_1)$ is also finitely generated, which gives us the desired resolution of k .

Hence, Lemma 2.3.16 applies, and we can conclude that $\text{gl.dim}(R[[G]]) \leq 2$. We now consider the cases $\text{gl.dim}(R)$ equal to 2, 1 and 0 separately.

Case 1. ($\text{gl.dim}(R) = 2$) In this case, by Brumer's formula, $\text{cd}_p(G) = 0$, that is, G is trivial and $R[[G]] = R$. If R is Noetherian, the claims follow from Theorem 1.2.8.

Case 2. ($\text{gl.dim}(R) = 1$) By Brumer's formula, we must have $\text{cd}_p(G) \leq 1$, that is, G is a finitely generated free pro- p group. Moreover, every closed ideal in R must be projective and hence a free profinite R -module. Since R is a domain and $R[X]$ embeds in $R[[X]]$, any free profinite submodule of R must be cyclic by tensoring with the field of fractions $\text{Frac}(R)$ and computing dimensions, so every closed ideal of R is principal. In particular, R satisfies the ascending chain condition on closed ideals. By a theorem of Seth Warner, this implies that R is Noetherian.⁵² Its description now follows from Theorem 1.2.8 with the obvious remark that any Eisenstein extension of a p -adic field is again a p -adic field.

Case 3. ($\text{gl.dim}(R) = 0$) Every finitely generated R -module is free. In particular, $k = R/J(R)$ must be free, necessarily of rank 1, so that $R = k$ is a finite field. Brumer's formula then gives us $\text{cd}_p(G) \leq 2$. $\square$

With this result in hand, one may attempt a classification of the finitely presented pro- p groups G and pro- p domains R such $R[[G]]$ is a Sylvester domain. A theorem⁵³ of Dicks and Sontag says that a commutative domain R is a coherent Sylvester domain if and only if it is coherent, projective R -modules are free and the weak global dimension of R is at most two. Hence, it follows that $\mathbb{F}_q[[x, y]]$ and Eisenstein extensions of $\mathcal{O}_K[[x]]$ are Sylvester domains, and so the case where G is trivial and R is Noetherian is completely determined. The author is not aware of any profinite local domain of finite global dimension that is not Noetherian.⁵⁴ We shall see later in Chapter 4 that if $R[[G]]$ is a Sylvester domain, then $R[[G]]$ is coherent. We leave this problem as a question:

Question 2.3.18. Is there a commutative profinite coherent domain of finite global dimension that is not Noetherian?

For G a finitely generated free pro- p group, that $\mathbb{F}_p[[x]][[G]] \simeq \mathbb{F}_p[[G \times \mathbb{Z}_p]]$ is a Sylvester domain will follow from Theorem 4.6.6 in Section 4.4. A natural extension⁵⁵ of Magnus'

⁵² Warner, "Compact Rings", Thm. 2.

⁵³ Dicks and Sontag, "Sylvester domains", Thm. 10.

⁵⁴ That no examples of such rings are known is also mentioned in the remark after Cor. 3.7 of Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations"

⁵⁵ Observe that the Magnus expansion identifies $\mathbb{Z}_p[[G]]$ with $\mathbb{Z}_p[[X]]$, and the extension to $\mathcal{O}_K$ is given by taking the completed tensor product with the flat profinite $\mathbb{Z}_p$ -module $\mathcal{O}_K$. See Thm. 5.9.1 of Ribes and Zalesskii, *Profinite Groups*

expansion identifies $\mathcal{O}_K[[G]]$ topologically with the ring of non-commutative power series $\mathcal{O}_K\langle\langle X \rangle\rangle$ where X is a basis of G as a free pro- p group. The question of whether or not $\mathcal{O}_K\langle\langle X \rangle\rangle$ is a Sylvester domain is still open.⁵⁶

This leaves us with the case of $R = \mathbb{F}_q$ and G a pro- p group of cohomological dimension two, since $\mathbb{F}_q[[G]]$ for free pro- p groups G is already covered by Proposition 2.3.14. One of the main contributions of this thesis is to provide the first non-analytic examples of such groups in Section 4.4, the class of torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups. Although it is a big class containing important historical groups such as Demushkin groups, there are still many open questions regarding this case. For instance, in private discussion with the author, Jaikin-Zapirain raised the following question:

Question 2.3.19. Let G be a torsion-free one-relator pro- p group. Is $\mathbb{F}_p[[G]]$ a Sylvester domain?

It is not known whether torsion-free one-relator pro- p groups have cohomological dimension at most two, a famous open question due to Jean-Pierre Serre⁵⁷. A positive answer to Question 2.3.19 would also imply a positive answer to Serre's question.

2.4 Topology and localization

As we remarked in Section 2.3, given a prime matrix ideal Σ of a ring R , there exists a local epic R -ring R_Σ such that every matrix $A \notin \Sigma$ is invertible in R_Σ , and moreover R_Σ is universal for this property. The construction of R_Σ by Paul Cohn is completely formal: one adds the necessary generators and relations to R as a ring to ensure that any matrix $A \notin \Sigma$ is invertible. While this proves the existence of R_Σ , it sheds no light on the structure of this ring itself, nor how the map $R \rightarrow R_\Sigma$ behaves.

Observe that for *any* set Δ of square matrices, the procedure above constructs an R -ring such that any matrix in Δ becomes invertible. This ring will be denoted⁵⁸ by $\Delta^{-1}R$, henceforth called the universal Δ -inverting R -ring. In particular, for a prime matrix ideal Σ , we have $R_\Sigma = \Delta^{-1}R$ where Δ is the complement of Σ on the set of all square matrices over R . Our main goal in this section is to prove the following theorem:

Theorem 2.4.1. *If S is profinite and Δ is a collection of square matrices over S , then the universal localization $\Delta^{-1}S$ admits a Hausdorff ring topology such that the canonical map $S \rightarrow \Delta^{-1}S$ is continuous.*

⁵⁶Pere Ara and Warren Dicks. “Universal localizations embedded in power-series rings”. In: *Forum Mathematicum* 19.2 (Jan. 2007). ISSN: 1435-5337. DOI: 10.1515/forum.2007.015, Sec. 3.9.

⁵⁷D. Gildenhuys. “On pro- p -groups with a single defining relator”. In: *Inventiones Mathematicae* 5.4 (Dec. 1968), pp. 357–366. ISSN: 1432-1297. DOI: 10.1007/bf01389782.

⁵⁸This notation and others in the present section differs from the one in the published version, since in this thesis the notation R_Δ is already taken for localizations at prime matrix ideals, cf. Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”

To do so, we will use another construction of $\Delta^{-1}S$ due to John Beachy.⁵⁹ Let us denote by $\widetilde{\Delta}$ the set of all square matrices over R that become invertible in $\Delta^{-1}R$. In general, $\widetilde{\Delta}$ is greater than Δ , but due to the universal property of matrix localization we have that $\Delta^{-1}R$ and $\widetilde{\Delta}^{-1}R$ are isomorphic as R -rings. We say that Δ is complete if $\widetilde{\Delta} = \Delta$.

It is clear that a complete collection of square matrices is multiplicative, that is, it contains $\text{GL}_n(R)$ for every n , is closed under products when defined and if $A, B \in \Delta$ then $\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \in \Delta$ for any matrix C of the appropriate size. It also has the following property regarding diagonal summands: if $A \oplus B \in \Delta$, then A and B are both in Δ .

From now on, we assume that Δ is complete. Let R^n denote the rows of n elements in R , nR the columns of n elements in R and Δ_n denote the subset of $n \times n$ matrices in Δ . Consider the set T_Δ given by the disjoint union of all the products $R^n \times \Delta_n \times {}^n\Delta$, that is, the triples (a, C, x) where a is a row matrix over R of length n , $C \in \Delta_n$ and x is a column matrix over R of height n . We define two binary operations of sum and product in T_Δ as follows:⁶⁰

$$(a, C, x) + (b, D, y) = \left((a \ b), \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} \right),$$

$$(a, C, x) \cdot (b, D, y) = \left((a \ 0), \begin{pmatrix} C & -xb \\ 0 & D \end{pmatrix}, \begin{pmatrix} 0 \\ y \end{pmatrix} \right),$$

where the matrices make sense because Δ is multiplicative, x is an $n \times 1$ matrix and b is a $1 \times m$ matrix.

We define an equivalence relation $\mathcal{R}_1 \subseteq T_\Delta \times T_\Delta$ on Δ by means of $(a, C, x) \sim_{\mathcal{R}_1} (b, D, y)$ if there exists invertible matrices U, V over R such that $b = aU$, $y = Vx$ and $D = VCU$. Observe that, in particular, if $(a, C, x) \sim_{\mathcal{R}_1} (b, D, y)$, then C and D must have the same size. We denote the quotient $T_\Delta / \mathcal{R}_1$ by $T_\Delta^{(1)}$ and the equivalence class of a triple (a, C, x) in $T_\Delta^{(1)}$ by $(a : C : x)$.⁶¹

The operations of sum and product are well defined on the equivalence classes and descend to a sum and a product on the quotient $T_\Delta^{(1)}$. With the sum, it becomes a commutative semigroup. To obtain an abelian additive group, we further introduce a new equivalence relation.

Let $OT_\Delta^{(1)}$ be the subsemigroup of $T_\Delta^{(1)}$ generated by all the elements of the form $(a : C : 0)$ and $(0 : C : y)$. Using permutation matrices and induction, it is easy to show that any element of $OT_\Delta^{(1)}$ has the form $(a : C : 0)$, $(0 : C : y)$ or $\left((a \ 0) : \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} : \begin{pmatrix} 0 \\ x \end{pmatrix} \right)$ for blocks of appro-

⁵⁹John A. Beachy. "On universal localization at semiprime Goldie ideals". In: *Ring theory (Granville, OH, 1992)*. World Sci. Publ., River Edge, NJ, 1993, pp. 41–57. ISBN: 978-981-02-1385-5.

⁶⁰This definition comes from (a, C, x) being supposed to represent the element $aC^{-1}x \in R_\Sigma$, and formally developing what the operations on two such elements must look like.

⁶¹The notation with $:$ is inspired by the usual cancellation laws for fractions: $(aU : CU : x) = (a : C : x)$ and $(a : VC : Vx) = (a : C : x)$.

prate sizes. This subsemigroup allows us to define the congruence relation $\mathcal{R}_2 \subseteq T_\Delta^{(1)} \times T_\Delta^{(1)}$ on $T_\Delta^{(1)}$ where two elements $p, q \in T_\Delta^{(1)}$ are equivalent if there exists $z_1, z_2 \in OT_\Delta^{(1)}$ such that $p + z_1 = q + z_2$.

The quotient space $T_\Delta^{(1)} / \sim_{\mathcal{R}_2}$ inherits the sum and product, under which it becomes⁶² an associative ring with the zero element being the class $[(1: 1: 0)]$ and the identity element being the class $[(1: 1: 1)]$. By construction, this ring is readily seen to be R -isomorphic to $\Delta^{-1}R$, since any homomorphism $\psi: R \rightarrow S$ such that $\psi(\Delta)$ consists of invertible matrices gives a well defined homomorphism $\Delta^{-1}\psi: T_\Delta^{(1)} / \sim_{\mathcal{R}_2} \rightarrow S$ by taking $\Delta^{-1}\psi([(a: C: y)]) = \psi(a)\psi(C)^{-1}\psi(y)$. The map $R \rightarrow T_\Delta^{(1)} / \sim_{\mathcal{R}_2}$ is given by $x \mapsto [(1: 1: x)]$.

We henceforth identify $T_\Delta^{(1)} / \sim_{\mathcal{R}_2}$ with $\Delta^{-1}R$ as R -rings. If we suppose furthermore that R is a topological ring, then all the spaces nR and R^n have a natural product topology, and Δ_n becomes a topological space as it is contained in $\text{Mat}_n(R) \simeq R^{n^2}$. Hence, T_Δ becomes a topological space endowed with the direct union topology on which the operations of sum and product defined are continuous. Under the quotient topology, these operations are still continuous on $T_\Delta^{(1)}$ and $\Delta^{-1}R$, so that $\Delta^{-1}R$ becomes a topological ring. We call this topology structure on $\Delta^{-1}R$ the *induced* topological structure from R .

Proof of Theorem 2.4.1. Without loss of generality, we assume that Δ is complete. Since R is profinite, T_Δ is a Hausdorff space under the induced topology. We will first prove that the quotient topology in $T_\Delta^{(1)}$ is Hausdorff.

To do so, we must show that the equivalence relation $\mathcal{R}_1$ is a closed subspace of $T_\Delta \times T_\Delta$. Since T_Δ is given the union topology and each equivalence class under $\mathcal{R}_1$ is entirely contained in one of the subsets $T_n = R^n \times \Delta_n \times {}^nR$, it suffices to prove that $X_n = (\mathcal{R}_1 \cap (T_n \times T_n))$ is closed in $T_n \times T_n$.

Consider the continuous maps $\psi: T_n \times T_n \times \text{GL}_n(R)^2 \rightarrow R^n \times \text{Mat}_n(R) \times {}^nR$ and $\pi: T_n \times T_n \times \text{GL}_n(R)^2 \rightarrow T_n \times T_n$ given by

$$\psi(a, C, x, b, D, y, U, V) = (aU - b, VCU - D, Vx - y),$$

$$\pi(a, C, x, b, D, y, U, V) = (a, C, x, b, D, y).$$

By construction, we have $X_n = \pi(\psi^{-1}(0, 0, 0))$. Since $\text{GL}_n(R)^2$ is compact, π is a closed map, so we are done.

Now, we already know that the topology on $T_\Delta^{(1)}$ is Hausdorff. Our next step is to show that $OT_\Delta^{(1)}$ is closed in $T_\Delta^{(1)}$. It again suffices to consider $T_n^{(1)} = T_n / \sim_{\mathcal{R}_1}$ and show that $Y_n = OT_\Delta^{(1)} \cap T_n^{(1)}$ is closed in $T_n^{(1)}$, as we have a homeomorphism between $T_\Delta^{(1)}$ and the disjoint union $\bigcup_{n \geq 1} T_n^{(1)}$. Decompose $Y_n = Z_I \cup Z_{II} \cup \bigcup_{i=1}^{n-1} Z_{III,i}$ where

$$Z_I = \{(a: C: 0) \in Y_n\},$$

⁶²Beachy, "On universal localization at semiprime Goldie ideals", Prop. 2.3 and Prop. 3.2.

$$Z_{II} = \{(0 : C : x) \in Y_n\},$$

$$Z_{III,i} = \left\{ \left((a \ 0) : \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix} : \begin{pmatrix} 0 \\ x \end{pmatrix} \right) \in Y_n : a \in R^i, x \in {}^{n-i}R, C \in \Delta_i, D \in \Delta_{n-i} \right\}.$$

We only need to show that each one of the Z_I , Z_{II} and $Z_{III,i}$ is closed in $T_n^{(1)}$.

The inverse image of Z_I in T_n is the set of triples of the form $(a, C, 0)$, which immediately shows that it is closed. An analogous description holds for the inverse image of Z_{II} , hence it is also closed. The only case requiring a finer description is that of $Z_{III,i}$, whose inverse image in T_n is

$$\left\{ \left((a \ 0) U, V \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix} U, V \begin{pmatrix} 0 \\ x \end{pmatrix} \right) : U, V \in \text{GL}_n(R) \right\}.$$

Define

$$Z_i = \left\{ \left((a \ 0), \begin{pmatrix} C & 0 \\ 0 & D \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix} \right) \in T_n \right\}.$$

Then, the inverse image in question is the image of $Z_i \times \text{GL}_n(R)^2$ under the continuous right group action map $\mu: T_n \times \text{GL}_n(R)^2 \rightarrow T_n$ given by

$$\mu(a, C, y, U, V) = (aU, V^{-1}CU, V^{-1}y).$$

Once again, the map μ is closed by virtue of $\text{GL}_n(R)^2$ being compact. Since Z_i is closed, both $\mu(Z_i \times \text{GL}_n(R)^2)$ and $Z_{III,i}$ are closed.

The kernel of $T_\Delta^{(1)} \rightarrow \Delta^{-1}R$ is the subsemigroup

$$OT_\Delta^{(1)} - OT_\Delta^{(1)} = \{p \in T_\Delta^{(1)} : \text{exists } z \in OT_\Delta^{(1)} \text{ such that } p + z \in OT_\Delta^{(1)}\},$$

which contains $OT_\Delta^{(1)}$ but might be strictly larger. Since $\Delta^{-1}R$ is a topological ring, it is Hausdorff if and only if $\{0\}$ is closed, that is, if and only if $OT_\Delta^{(1)} - OT_\Delta^{(1)}$ is a closed subset of $T_\Delta^{(1)}$. To see that it is the case, observe that $(OT_\Delta^{(1)} - OT_\Delta^{(1)}) \times OT_\Delta^{(1)}$ is precisely the inverse image of the closed subset $OT_\Delta^{(1)}$ under the continuous addition map $T_\Delta^{(1)} \times OT_\Delta^{(1)} \rightarrow T_\Delta^{(1)}$. Hence, $OT_\Delta^{(1)} - OT_\Delta^{(1)}$ is closed in $T_\Delta^{(1)}$, as we set out to prove. $\square$

We record the following immediate corollary of Theorem 2.4.1:

Corollary 2.4.2. *Let R be a profinite ring and Σ a set of square matrices over R . Then, for any $r \geq 1$, all finitely generated R -submodules of $(\Sigma^{-1}R)^r$ are profinite under the subspace topology.*

Chapter 3

Rank functions

Da igual si las cosas se llaman punto,
recta y plano o silla, mesa y banquillo de
cerveza.

L.M. on a famous Hilbert quote.

In the 1957 science fiction short story “*The Ifth of the Oofth*”¹ by author Walter Tevis, the two good friends Oliver and George are sharing drinks over an open fire and discussing one of Oliver’s newest inventions:

“Well,” I said. And then, “What do you call it?”

“A pentaract.”

He sat back down again in his armchair. “Of course, that name really isn’t accurate. I suppose a pentaract should really be a four-dimensional pentagon, and this is meant to be a picture of a five-dimensional cube.”

“A *picture*?” It didn’t look like a picture to me.

“Well, it couldn’t *really* have five-dimensionality—length, width, breadth, ifth and oofth—or I don’t think it could.” His voice faltered a little at that. “But it’s supposed to illustrate what you might call the layout of an object that did have those.”

“What kind of object would that be?” I looked back at the thing in my lap and was mildly surprised to see that I had folded a good many of the cubes together.

“Suppose,” he said, “you put a lot of points in a row, touching; you have a line—a one-dimensional figure. Put four lines together at right angles and on a plane;

¹Walter S. Tevis. “The Ifth of the Oofth”. eng. In: *The Sixth Galaxy Reader*. Ed. by H. L. Gold. 1st ed. Garden City, New York: Doubleday & Company, Inc., 1962.

a square—two-dimensional. Six squares at right angles and extended into real space give you a cube—three dimensions. And eight cubes extended into four physical dimensions give you a tesseract, as it’s called—”

“And eight tesseracts make a pentaract,” I said. “Five dimensions.”

“Exactly. But naturally this is just a picture of a pentaract, in that sense. There probably isn’t any ifth and oofth at all.”

As we explained in the beginning of Chapter 1, a fundamental concept of modern² mathematics is that of dimension. Most concepts of dimension revolve around the idea of “degrees of freedom”: for example, the dimension of a vector space is the size of the maximal set of vectors one can find before dependence relations between them arise and the dimension of a manifold is the number of coordinates required to describe a point in one of its charts.

The concept is so fundamental that usually a month or more of any undergraduate course in linear algebra is dedicated solely to the proof that vector spaces have a well defined dimension that behaves well under short exact sequences: the rank-nullity theorem states that if

$$0 \rightarrow V \rightarrow W \rightarrow Q \rightarrow 0$$

is a short exact sequence of vector spaces, then $\dim W = \dim V + \dim Q$.

For modules over more general rings, many variants of “dimension” have been proposed to tackle different problems.³ The generalization that we will concern ourselves with was first introduced in the work of Peter Malcolmson⁴ and later expanded on by Aidan Schofield⁵, and is that of a Sylvester rank function. It was axiomatized to reflect how, given a ring homomorphism $R \rightarrow S$ to a simple Artinian ring S , the natural notion of dimension over S pulls back to a notion of dimension over R . However, as in Oliver’s picture of a pentaract, it is only a picture: some essential properties of dimension may be missing, such as the rank-nullity theorem or even integrality.⁶

The connection of a Sylvester rank function with the author’s work is threefold: first, it can be used to give an alternate description of the set isomorphism classes of epic division R -rings via Malcolmson’s theorem, as explained in Section 3.1. Second, it gives an alternative definition of Sylvester domains in terms of the inner rank, which is of great importance to our original results and hence delegated a chapter of its own (Chapter 4). Third, many homological approximation phenomena in group theory such as W. Lück’s approximation theorem on ℓ^2 -Betti numbers or W. Fu’s asymptotic description of the twisted cohomology of arithmetic

²or, perhaps, even ancient?

³To mention a few, we have Krull dimension, composition length, uniform dimension, von Neumann dimensions, and so on...

⁴Peter Malcolmson. “Determining homomorphisms to skew fields”. In: *Journal of Algebra* 64.2 (June 1980), pp. 399–413. ISSN: 0021-8693. DOI: 10.1016/0021-8693(80)90153-2.

⁵Aidan H. Schofield. *Representations of Rings over Skew Fields*. Cambridge University Press, Apr. 1985. ISBN: 9780511661914. DOI: 10.1017/cbo9780511661914.

⁶It may not have any ifth or oofth at all!

3-manifolds is best understood as approximation phenomena in the space of Sylvester rank functions of a suitable ring, a theme that will be thoroughly explored in Chapters 5 and 6.

The chapter is organized as follows: in Section 3.1 we present the definition of a Sylvester rank function and state various of its properties, specially regarding the “change of rings procedure”. We also describe many examples of Sylvester rank functions that are relevant to this work, as well as Malcolmson and Schofield’s theorems describing the rank functions with a discrete image. In Section 3.2 we discuss the notion of *universal* Sylvester rank functions, which is inspired by the notion of a universal epic division R -ring from Section 2.3. Here, we state a original result (Theorem 3.2.5) concerning universality of a canonical Sylvester rank function on completed $\mathbb{F}_p$ -algebras of pro- p groups.

3.1 Sylvester rank functions

Our standard assumptions still stand: rings are assumed to be associative and unital. The contents of this section are well known to specialists, and good references for the subject are Aidan Schofield’s book⁷ for an early treatment of the theory, as well as Andrei Jaikin-Zapirain’s survey⁸ and Diego López-Álvarez thesis⁹ for a more modern account.

Definition 3.1.1. Let R be a ring. A *Sylvester module rank function* on R is a non-negative real-valued function $\dim$ defined on all finitely presented right R -modules that satisfies:

$$(\text{SDim}_1) \quad \dim(0) = 0 \text{ and } \dim(R) = 1;$$

$$(\text{SDim}_2) \quad \dim(M \oplus N) = \dim(M) + \dim(N);$$

$$(\text{SDim}_3) \quad \text{If}$$

$$M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$$

is an exact sequence of finitely presented modules, then

$$\dim(M_3) \leq \dim(M_2) \leq \dim(M_1) + \dim(M_3) .$$

The dimension of vector spaces is the prototypical example of a Sylvester module rank function. More generally, if R is an Artinian ring, then any finitely generated module is finitely presented and in fact of finite length, that is, it admits a finite composition series by simple R -modules. If $l(M)$ denotes the composition length of an R -module M , then

⁷Schofield, *Representations of Rings over Skew Fields*.

⁸Andrei Jaikin-Zapirain. “L2-Betti Numbers and Their Analogues in Positive Characteristic”. In: *Groups St Andrews 2017 in Birmingham*. Ed. by C. M. Campbell et al. London Mathematical Society Lecture Note Series. Cambridge: Cambridge University Press, 2019, pp. 346–405. ISBN: 978-1-108-72874-4. DOI: 10.1017/9781108692397.015.

⁹Diego López-Álvarez. “Sylvester Rank Functions, Epic Division Rings and the Strong Atiyah Conjecture for Locally Indicible Groups”. PhD thesis. Spain: Universidad Autónoma de Madrid, 2021.

$\dim(\mathcal{M}) = l(\mathcal{M})/l(R)$ is a Sylvester module rank function on R . It also satisfies a property stronger than (SDim₃), namely it is additive under short exact sequences: if

$$0 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{M}_2 \rightarrow \mathcal{M}_3 \rightarrow 0$$

is a short exact sequence, then $\dim(\mathcal{M}_2) = \dim(\mathcal{M}_1) + \dim(\mathcal{M}_3)$. Sylvester module rank functions with this property are called *exact*. An example of a Sylvester module rank function on the ring $\mathbb{Z}$ that is not exact is given by

$$\dim_p(\mathbb{Z}^k \oplus \mathbb{Z}/p_1^{a_1}\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/p_n^{a_n}\mathbb{Z}) = k + |\{1 \leq i \leq n: p_i = p\}|,$$

where p is any prime number.

If $\dim$ is a Sylvester rank function on a ring S and $\varphi: R \rightarrow S$ is a ring homomorphism, then we can define a Sylvester rank function $\varphi^\# \dim$ on R by means of the formula

$$\varphi^\# \dim(\mathcal{M}) = \dim(\mathcal{M} \otimes_R S)$$

for any right R -module $\mathcal{M}$. The dimension function $\varphi^\# \dim$ is called the *pullback* of $\dim$ along φ . We also say that $\varphi^\# \dim$ was obtained from $\dim$ by a *change of rings* along φ , or that $(\varphi, \dim)$ is an *envelope* for $\varphi^\# \dim$. For example, the dimension function $\dim_p$ of the previous paragraph was obtained from the usual dimension over the field $\mathbb{F}_p$ by a change of rings along the projection $\mathbb{Z} \rightarrow \mathbb{F}_p$. It also follows from the change of rings that non-trivial profinite rings always admit Sylvester dimension rank functions, as any profinite ring is a projective limit of finite rings, and finite rings are Artinian.¹⁰

Definition 3.1.2. Let R be a ring. A *Sylvester matrix rank function* on R is a non-negative real-valued function rk defined on all matrices over R satisfying:

(SMat₁) $\text{rk}(A) = 0$ if A is any zero matrix and $\text{rk}(1) = 1$;

(SMat₂) $\text{rk}(AB) \leq \min\{\text{rk}(A), \text{rk}(B)\}$ for any matrices A and B that can be multiplied;

(SMat₃) $\text{rk}\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \text{rk}(A) + \text{rk}(B)$ for any matrices A and B ;

(SMat₄) $\text{rk}\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \geq \text{rk}(A) + \text{rk}(B)$ for any matrices A, B and C of appropriate sizes.

Once more, the prototypical example of a Sylvester matrix rank function is the usual rank of matrices over a field. Some intuitive properties of rank can be deduced from the axioms: for example, from (SMat₁) and (SMat₃) it follows that $\text{rk Id}_n = n$ for the identity $n \times n$ matrix Id_n ; from this and (SMat₂) it follows that if A is an $m \times n$ matrix, then $\text{rk}(A) \leq \min\{m, n\}$. If U and V are invertible matrices, then they are rk -full, that is, $\text{rk}(U) = n$ (resp. $\text{rk}(V) = m$) if U is square of size n (resp. of size m). For any rk -full matrices U, V of sizes n and m respectively, we have $\text{rk}(VAU) = \text{rk}(A)$.

Another general fact we shall need later is the following:¹¹

¹⁰There are rings which admit no Sylvester rank functions, for instance, rings without invariant basis number.

¹¹Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Lem. 2.4.

Lemma 3.1.3. *Let rk be a Sylvester matrix rank function on a ring R and A a matrix over R . If B is a submatrix of A , then $\text{rk}(B) \leq \text{rk}(A)$.*

Proof. Any submatrix B of $A \in \text{Mat}_{n \times m}(R)$ can be obtained by removing rows or columns of A . Hence, it suffices to prove the claim for $A = \begin{pmatrix} B & a \end{pmatrix}$ and for $A = \begin{pmatrix} B \\ a' \end{pmatrix}$ for arbitrary columns $a \in \text{Mat}_{n \times 1}(R)$ and rows $a' \in \text{Mat}_{1 \times m}(R)$. Since

$$B = \begin{pmatrix} B & a \end{pmatrix} \begin{pmatrix} \text{Id}_{n-1} \\ 0 \end{pmatrix} = \begin{pmatrix} \text{Id}_{m-1} & 0 \end{pmatrix} \begin{pmatrix} B \\ a' \end{pmatrix},$$

the lemma follows from property (SMat2). $\square$

If $\dim$ is a Sylvester module rank function on a ring R , then we can define a Sylvester matrix rank function $\text{rk}_{\dim}$ on R by means of the formula

$$\text{rk}_{\dim}(A) = m - \dim(R^m / AR^n)$$

for any $m \times n$ matrix A over R , where we consider A as a homomorphism $A: R^n \rightarrow R^m$ of right R -modules by left multiplication. It is straightforward to check that $\text{rk}_{\dim}$ satisfies the axioms of a Sylvester matrix rank function. Conversely, if rk is a Sylvester matrix rank function on R , then we can define a Sylvester module rank function $\dim_{\text{rk}}$ on R by means of the formula

$$\dim_{\text{rk}}(M) = m - \text{rk}(A)$$

for any finitely presented right R -module M with presentation

$$R^n \xrightarrow{A} R^m \rightarrow M \rightarrow 0.$$

Proposition 3.1.4. *The functions $\text{rk} \mapsto \dim_{\text{rk}}$ and $\dim \mapsto \text{rk}_{\dim}$ are bijections between the set of all Sylvester matrix rank functions and all Sylvester module rank functions over a given ring R .*

Proof. We will show that $\dim_{\text{rk}}$ is well-defined, that is, it does not depend on our choice of presentation for M . If

$$R^{n'} \xrightarrow{A'} R^{m'} \rightarrow M \rightarrow 0$$

is another presentation for M , then Schanuel's lemma gives us an isomorphism $\psi: R^{m+m'} \rightarrow R^{m+m'}$ of right R -modules sending $AR^n \oplus R^{m'}$ to $A'R^{n'} \oplus R^{m'}$. If W is a square matrix of size

$m + m'$ representing ψ , from the commutative diagram

$$\begin{array}{ccc}
 & R^n \oplus R^{m'} & \xrightarrow{U} R^{n'} \oplus R^m \\
 & \downarrow \text{ } & \downarrow \text{ } \\
 A \oplus \text{Id}_{m'} \cdot & AR^n \oplus R^{m'} & \xrightarrow{\psi} A'R^{n'} \oplus R^m & A' \oplus \text{Id}_m \cdot \\
 & \downarrow \text{ } & \downarrow \text{ } \\
 & R^{m+m'} & \xrightarrow{W} R^{m'+m}
 \end{array}$$

we see that there exists an $(n' + m) \times (n + m')$ matrix U and an $(n + m') \times (n' + m)$ matrix V over R such that

$$(A' \oplus \text{Id}_m)U = W \cdot A \oplus \text{Id}_{m'},$$

from which we obtain $m + \text{rk}(A') \leq \text{rk}(A) + m'$ by applying (SMat2) and (SMat3). By symmetry, we also have $m' + \text{rk}(A) \leq \text{rk}(A') + m$, so $\dim_{\text{rk}}(\mathcal{M})$ is well-defined. It is now straightforward to check that $\dim \mapsto \text{rk}_{\dim}$ and $\text{rk} \mapsto \dim_{\text{rk}}$ are inverse to each other. $\square$

From now on, we will not distinguish between Sylvester module rank functions and Sylvester matrix rank functions, and we will refer to both simply as *Sylvester rank functions*. If $\dim$ and rk correspond under the bijection set up by Proposition 3.1.4, then we shall say that they are *associated*.

If $\varphi: R \rightarrow S$ is a ring homomorphism and rk is a Sylvester matrix rank function on S , then the pullback $\varphi^\# \text{rk}$ is defined by means of the formula

$$\varphi^\# \text{rk}(A) = \text{rk}(\varphi(A))$$

for any matrix A over R , where $\varphi(A)$ is the matrix over S obtained by applying φ to each entry of A . The dimension function associated to the pullback $\varphi^\# \text{rk}$ is precisely the pullback of the dimension function associated to rk .

Observe that the set $\mathbb{P}(R)$ of Sylvester rank functions on R is closed under convex linear combinations, that is, if $\dim$ and $\dim'$ (resp. rk and rk') are Sylvester module (resp. matrix) rank functions on R , then for any $\lambda \in [0, 1]$ the function $\lambda \dim + (1 - \lambda) \dim'$ (resp. $\lambda \text{rk} + (1 - \lambda) \text{rk}'$) is again a Sylvester rank function on R . Although we will not use it, it can moreover be shown¹² that $\mathbb{P}(R)$ is a compact subset of $\mathbb{R}^{\text{Mat}(R)}$ under the topology of pointwise convergence. Let us see some examples of Sylvester rank functions that will be relevant to us.

¹² For each $n \times m$ matrix A , let $s(A) = \min\{m, n\}$. Then, we can regard $\mathbb{P}(R)$ as a subset of the compact space $\prod_{A \in \text{Mat}(R)} [0, s(A)]$. The axioms of a Sylvester matrix rank function imply it is an intersection of closed subsets, and hence it is itself closed and compact.

Example 3.1.5. Let $\mathcal{D}$ be a division ring. Then $\mathbb{P}(\mathcal{D})$ consists of a single Sylvester rank function, namely the usual dimension over $\mathcal{D}$. Indeed, if $\dim$ is any Sylvester module rank function on $\mathcal{D}$ and M is a finitely presented right $\mathcal{D}$ -module, then $\mathcal{D}$ is in fact a finite-dimensional free module over $\mathcal{D}$, so if $\{m_1, \dots, m_k\}$ is a basis of $\mathcal{D}$, we have an isomorphism of right $\mathcal{D}$ -modules $\mathcal{D}^k \rightarrow M$ sending the standard basis of $\mathcal{D}^k$ to $\{m_1, \dots, m_k\}$. Hence, by (SDim2), $\dim(M) = k = \dim_{\mathcal{D}}(M)$. More generally, if R is a simple Artinian ring, that is, $R \simeq \text{Mat}_n(\mathcal{D})$ for some division ring $\mathcal{D}$ and integer $n \geq 1$, then $\mathbb{P}(R)$ also consists of a single Sylvester rank function: we have $R \simeq (\mathcal{D}^n)^n$ as right R -modules, and so any Sylvester module rank function $\dim$ on R satisfies $\dim(\mathcal{D}^n) = 1/n$ by (SDim2). Hence, for any finitely presented right R -module M , by semisimplicity we have $\dim(M) = \frac{1}{n^2} \dim_{\mathcal{D}}(M)$. ■

Example 3.1.6. Let G be a finite group and $\mathbb{F}$ a field. If M is a finitely presented $\mathbb{F}[G]$ -module, then it is of finite $\mathbb{F}$ -dimension. The function $\dim_G(M) = \frac{\dim_{\mathbb{F}}(M)}{|G|}$ is a Sylvester module rank function on $\mathbb{F}[G]$, called the *normalized dimension* over G . The associated Sylvester matrix rank function rk_G is called the *normalized rank function over G* . The associated Sylvester matrix rank function rk_G is obtained as the pullback of the usual normalized rank over the matrix ring $\text{Mat}_{|G|}(\mathbb{F})$ along the regular representation $\mathbb{F}[G] \rightarrow \text{Mat}_{|G|}(\mathbb{F})$. Observe that $\mathbb{F}_p[G]$ is an Artinian ring, so $\dim(M) = l(M)/l(\mathbb{F}_p[G])$ is another Sylvester module rank function on $\mathbb{F}_p[G]$ where l denotes the composition length. These two rank functions are different in general: for example, if $G = C_4$ is the cyclic group of order 4 and $\mathbb{F} = \mathbb{F}_3$, then $\dim_G(\mathbb{F}_3) = 1/4$ whereas $\dim(\mathbb{F}_3) = 1/3$. However, if G is a finite p -group and $\mathbb{F}$ is a field of characteristic p then the only simple $\mathbb{F}[G]$ -module is the trivial module $\mathbb{F}$.¹³ Hence, in this case, since R is an augmented $\mathbb{F}$ -algebra, for any finitely presented $\mathbb{F}[G]$ -module M , we have $\dim_G(M) = \dim(M) = \dim_{\mathbb{F}}(M)/|G|$.

If H is a subgroup of G and $\varphi: \mathbb{F}[H] \rightarrow \mathbb{F}[G]$ is the inclusion map, then $\varphi^{\#} \dim_G = \dim_H$. Indeed, since $\mathbb{F}[G] \simeq \mathbb{F}[H]^{|G:H|}$ as left $\mathbb{F}[H]$ -modules, for any finitely presented right $\mathbb{F}[H]$ -module M , we have

$$\varphi^{\#} \dim_G(M) = \frac{\dim_{\mathbb{F}} M \otimes_{\mathbb{F}[H]} \mathbb{F}[G]}{|G|} = \frac{|G:H| \dim_{\mathbb{F}} M}{|G|} = \frac{\dim_{\mathbb{F}} M}{|H|} = \dim_H M.$$

The normalized dimension is also stable under field extensions: if $\mathbb{F} \rightarrow \mathbb{L}$ is a field extension and $\varphi: \mathbb{F}[G] \rightarrow \mathbb{L}[G]$ is the induced embedding, then for any $\mathbb{F}[G]$ -module M we have

$$M \otimes_{\mathbb{F}[G]} \mathbb{L}[G] \simeq M \otimes_{\mathbb{F}} \mathbb{L}$$

as right $\mathbb{L}[G]$ -modules, there the right G -action on the latter comes from the right G -action on M . Hence:

$$\varphi^{\#} \dim_G = \frac{\dim_{\mathbb{L}} M \otimes_{\mathbb{F}[G]} \mathbb{L}[G]}{|G|} = \frac{\dim_{\mathbb{L}} M \otimes_{\mathbb{F}} \mathbb{L}}{|G|} = \frac{\dim_{\mathbb{F}} M}{|G|} = \dim_G.$$

¹³As mention in Section 1.2, this is well known. Indeed, the augmentation ideal $\ker(\mathbb{F}[G] \rightarrow \mathbb{F})$ is nilpotent, so it must be the unique maximal ideal.

Therefore, there is no ambiguity when writing $\dim_G$ for the normalized dimension over any group algebra of a finite group. $\blacksquare$

Example 3.1.7. Let G be a countably based pro- p group and $\mathbb{F}_q$ a finite field of characteristic p . If M is a finitely presented $\mathbb{F}_q[[G]]$ -module and $G > U_1 > U_2 > \dots$ is a chain of open normal subgroups of G with trivial intersection, we¹⁴ claim that the limit

$$\dim_G(M) = \lim_{i \rightarrow \infty} \dim_{G/U_i}(M)$$

exists and is independent of the choice of the chain, where $\dim_{G/U_i}$ is the pullback of the normalized dimension over the finite group G/U_i along the natural projection $\mathbb{F}_q[[G]] \rightarrow \mathbb{F}_q[G/U_i]$. To see this, observe that it suffices to show that whenever $G > U > V$ are open normal subgroups of G , we have

$$\dim_{G/V}(M) \leq \dim_{G/U}(M),$$

so that

$$\dim_G(M) = \inf_{U \trianglelefteq_o G} \dim_{G/U}(M).$$

By Nakayama's lemma, if G' is a finite p -group and N is a finitely generated right $\mathbb{F}_q[G']$ -module, then $\dim_{\mathbb{F}_q} N \leq |G'| \cdot \dim_{\mathbb{F}_q} N \otimes_{\mathbb{F}_q[G']} \mathbb{F}_q$. Hence, for any open normal subgroups $G > U > V$, we have

$$\begin{aligned} \dim_{\mathbb{F}_q} M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/V] &\leq |U:V| \dim_{\mathbb{F}_q} (M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/V]) \otimes_{\mathbb{F}_q[U/V]} \mathbb{F}_q \\ &= |U:V| \dim_{\mathbb{F}_q} (M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/U]) \end{aligned}$$

We then obtain:

$$\begin{aligned} \dim_{G/U} M &= \frac{\dim_{\mathbb{F}_q} M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/U]}{|G:U|} \\ &= \frac{|U:V| \cdot \dim_{\mathbb{F}_q} M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/U]}{|G:V|} \\ &\geq \frac{\dim_{\mathbb{F}_q} M \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q[G/V]}{|G:V|} \\ &= \dim_{G/V} M. \end{aligned}$$

This rank function $\dim_G$ is called the *normalized dimension* over G . The associated Sylvester matrix rank function is denoted by rk_G , and shall be called the *normalized rank function over G* . As in the Example 3.1.6, after reducing to the finite case one sees that if H is a closed subgroup of G and $\mathbb{F}_q \rightarrow \mathbb{F}_{q'}$ is a finite field extension, then under the change of rings map $\varphi: \mathbb{F}_q[[H]] \rightarrow \mathbb{F}_{q'}[[G]]$ we have $\varphi^\# \dim_G = \dim_H$. $\blacksquare$

¹⁴Nicolas Bergeron et al. "On the growth of Betti numbers in p -adic analytic towers". In: *Groups, Geometry, and Dynamics* 8.2 (July 2014), pp. 311–329. ISSN: 1661-7207. DOI: 10.4171/ggd/227, Lem. 4.1.

Example 3.1.8. If $R \rightarrow \text{End}_{\mathbb{F}}(V)$ is a representation of a ring R as endomorphisms of a finite dimensional vector space V over a field $\mathbb{F}$, seen as a left R -module, then the function

$$\dim_V(\mathcal{M}) = \frac{\dim_{\mathbb{F}} \mathcal{M} \otimes_R V}{\dim_{\mathbb{F}} V}$$

is a Sylvester module rank function on R , called the V -multiplicity of $\mathcal{M}$. The associated Sylvester matrix rank function rk_V is called the rank function associated to the representation V . If A is an $m \times n$ matrix, then

$$\text{rk}_V(A) = \frac{\dim_{\mathbb{F}} AV^n}{\dim_{\mathbb{F}} V} = m - \frac{\dim_{\mathbb{F}} V^m / AV^n}{\dim_{\mathbb{F}} V}.$$

We can also see that rk_V is obtained as the pullback of the unique rank function over the matrix ring $\text{Mat}_{\dim_{\mathbb{F}} V}(\mathbb{F})$ along the representation $R \rightarrow \text{End}_{\mathbb{F}}(V) \simeq \text{Mat}_{\dim_{\mathbb{F}} V}(\mathbb{F})$. The rank functions $\dim_{G/U}$ of the previous example are particular cases of this construction when $V = \mathbb{F}_q[G/U]$. ■

Example 3.1.9. Let Γ be a countable discrete group and $\mathbb{C}[\Gamma]$ its group algebra over the complex numbers. The group von Neumann algebra $\mathcal{N}(\Gamma)$ of all bounded linear operators that commute with the right action of $\mathbb{C}[\Gamma]$ on the space $\ell^2(\Gamma)$ is a finite von Neumann algebra endowed with a trace $\text{tr}: \mathcal{N}(\Gamma) \rightarrow \mathbb{C}$. If A is an $m \times n$ matrix over $\mathbb{C}[\Gamma]$, we can consider it as a bounded linear operator $A: \ell^2(\Gamma)^n \rightarrow \ell^2(\Gamma)^m$ by left multiplication. The *trace* of A is then defined as

$$\text{tr}_{\mathcal{N}(\Gamma)}(A) = \sum_{i=1}^m \text{tr}(P_{ii}),$$

where $P = (P_{ij})$ is the orthogonal projection onto the closure of the image of A in $\ell^2(\Gamma)^m$. The function rk_{Γ} defined by $\text{rk}_{\Gamma}(A) = \text{tr}_{\mathcal{N}(\Gamma)}(A)$ is a Sylvester matrix rank function on $\mathbb{C}[\Gamma]$, called the *von Neumann rank* over Γ .¹⁵ The associated dimension function $\dim_{\Gamma}$ is called the *von Neumann dimension* over Γ . When Γ is finite, we have $\ell^2(\Gamma) = \mathbb{C}[\Gamma]$ and this is simply the normalized dimension $\dim_{\Gamma} = \frac{1}{|\Gamma|} \dim_{\mathbb{C}}$ of the group algebra $\mathbb{C}[\Gamma]$. This rank function plays a central role in the theory of ℓ^2 -Betti numbers and their approximation properties, see Chapter 6. As with the normalized dimension, if Δ is a subgroup of Γ and $\varphi: \mathbb{C}[\Delta] \rightarrow \mathbb{C}[\Gamma]$ is the induced embedding, then $\varphi^{\#} \dim_{\Gamma} = \dim_{\Delta}$. ■

Example 3.1.10. We've already seen how the pullback along a ring map $\varphi: R \rightarrow S$ induces a map $\varphi^{\#}: \mathbb{P}(S) \rightarrow \mathbb{P}(R)$. In particular, if T is an Ore set in R and $S = T^{-1}R$, we must have $\varphi^{\#} \text{rk}(x) = 1$ for any $x \in T$ and any Sylvester rank function rk on $T^{-1}R$. It turns out that this property characterizes the image of the pullback map:

$$\varphi^{\#}(\mathbb{P})(T^{-1}R) = \{\text{rk} \in \mathbb{P}(R): \text{rk}(x) = 1 \text{ for all } x \in T\}.$$

¹⁵Wolfgang Lück. *L2-invariants: Theory and applications to geometry and K-theory*. 1st ed. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge A Series of Modern Surveys in Mathematics v. 44. Springer, 2002. ISBN: 978-3-540-43566-2, Thm. 6.5.

To see this, it suffices to show that any rank function in the latter set extends to a rank function on $T^{-1}R$. Since any finite set of elements in $T^{-1}R$ can be written with a common denominator in T , any $m \times n$ matrix A over $T^{-1}R$ can be written as $A = (x^{-1} \text{Id}_m)B$ for some matrix B over R and some $x \in T$. We then define $\text{rk}(A) = \text{rk}(B)$. This definition is independent of the choice of B and x because if $A = (y^{-1} \text{Id}_m)C$ is another such expression, we can find $u \in T$ and $r \in R$ such that $ux = ry$. We have

$$(u \text{Id}_m)B = (uxy^{-1} \text{Id}_m)C = (r \text{Id}_m)C.$$

Hence, by (SMat₂) and our assumption that $\text{rk}(u) = 1$ for all $u \in T$, we obtain $\text{rk}(B) \leq \text{rk}(C)$. A symmetric argument shows the reverse inequality, so $\text{rk}(B) = \text{rk}(C)$.

Proving that rk satisfies the axioms (SMat₁), (SMat₃) and (SMat₄) of a Sylvester rank function is straightforward. For (SMat₂), if A and B are $m \times n$ and $n \times k$ matrices over $T^{-1}R$ respectively, we can write $A = (x^{-1} \text{Id}_m)C$ and $B = (y^{-1} \text{Id}_n)D$ for some matrices C and D over R and some $x, y \in T$. We once again use the Ore condition to get a common denominator u in T and some matrix C' over R such that $(u \text{Id}_m)C = C'(y \text{Id}_n)$. Since both u and y are in T , we have $\text{rk}(C) = \text{rk}(C')$. We then obtain:

$$AB = (x^{-1} \text{Id}_m)C(y^{-1} \text{Id}_n)D = ((ux)^{-1} \text{Id}_m)C'D$$

and therefore

$$\text{rk}(AB) = \text{rk}(C'D) \leq \min\{\text{rk}(C'), \text{rk}(D)\} = \min\{\text{rk}(A), \text{rk}(B)\},$$

as desired. ■

In light of Example 3.1.10 one might wonder how to characterize the image of the pullback map $\varphi^\#: \mathbb{P}(\Delta^{-1}R) \rightarrow \mathbb{P}(R)$ induced by a matrix localization $\varphi: R \rightarrow \Delta^{-1}R$. The answer is given by a deep theorem due to Schofield¹⁶:

Theorem 3.1.11. *Let Δ be a set of square matrices over a ring R and $\varphi: R \rightarrow \Delta^{-1}R$ the corresponding matrix localization. Then, the image of the pullback map $\varphi^\#: \mathbb{P}(\Delta^{-1}R) \rightarrow \mathbb{P}(R)$ is the set of all Sylvester rank functions rk on R such that $\text{rk}(A) = n$ for all $n \geq 1$ and all $n \times n$ matrices $A \in \Delta$.*

We now move on to describe the relation between Sylvester rank functions and prime matrix ideals. The following result is due to Malcolmson, which motivated the original definition of a Sylvester matrix rank function.¹⁷

Theorem 3.1.12. *Let rk be an integer valued Sylvester matrix rank function on R . Then, the complement of the set of all rk -full matrices over R is a prime matrix ideal.*

¹⁶Schofield, *Representations of Rings over Skew Fields*, Thm. 7.4.

¹⁷Malcolmson, "Determining homomorphisms to skew fields", Thm. 2.

Hence, we obtain:

Corollary 3.1.13. *Let R be a ring. Then, the correspondences*

$$\begin{array}{ccc}
 & \varphi: R \rightarrow \mathcal{D} & \\
 \text{matrix localization} \nearrow & & \searrow \varphi^\# \text{rk}_{\mathcal{D}} \\
 \Sigma & & \text{rk} \\
 & \xleftarrow{\text{singular matrices}} &
 \end{array}$$

induce bijections between the sets of prime matrix ideals of R , epic division R -rings and integer valued Sylvester matrix rank functions on R .

Proof. We already know from Cohn's result (Theorem 2.3.6) that there is a bijection between prime matrix ideals and epic division R -rings. By Theorem 3.1.12, there is a map from integer valued Sylvester matrix rank functions to prime matrix ideals. To see that this map is bijective, let $\varphi: R \rightarrow \mathcal{D}$ be an epic division R -ring and $\text{rk} = \varphi^\# \text{rk}_{\mathcal{D}}$ the associated Sylvester matrix rank function. If A is an $n \times n$ square matrix over R , then $\text{rk}(A) < n$ if and only if $\varphi(A)$ is not invertible in $\text{Mat}_n(\mathcal{D})$, which happens if and only if $\varphi(A)$ is singular in $\text{Mat}_n(\mathcal{D})$, since $\mathcal{D}$ is a division ring. Hence, the set of singular matrices for rk coincides with the set of singular matrices for φ . That the other composition induces the identity follows, from example, from Theorem 3.1.11. $\square$

This result illustrates that the study of integer valued Sylvester matrix rank functions is essentially equivalent to the study of epic division rings. However, in many situations, it is very difficult to determine whether a given Sylvester rank function is integer valued or not. For instance, the following two conjectures are still open:

Conjecture 3.1.14 (Atiyah conjecture). *Let Γ be a countable discrete group such that*

$$n = \text{lcm}\{|\Delta| : \Delta \text{ is a finite subgroup of } \Gamma\}$$

is finite. Then, the von Neumann rank function rk_Γ on $\mathbb{C}[\Gamma]$ takes values in $\frac{1}{n} \mathbb{Z}$.

Conjecture 3.1.15 (Pro- p Atiyah conjecture). *Let G be a countably based pro- p group such that*

$$n = \text{lcm}\{|H| : H \text{ is a finite subgroup of } G\}$$

is finite. Then, the normalized rank function rk_G on $\mathbb{F}_p[[G]]$ takes values in $\frac{1}{n} \mathbb{Z}$.

These conjectures are known as the strong Atiyah conjecture¹⁸ and its pro- p analogue respectively. The original question of Michael Atiyah concerns the ℓ^2 -Betti numbers of aspherical closed manifolds with fundamental group Γ , which are known to be given by the von

¹⁸Quite controversially, since Atiyah himself actually asked for counterexamples to the original version of its statement.

Neumann dimension over Γ of certain homology groups.¹⁹ Some classes of groups for which Conjecture 3.1.14 is known to hold include free groups, surface groups, cocompact special groups, elementary amenable groups and subgroups of torsion-free p -adic Lie groups.²⁰ Up to our knowledge, the following list contains all groups for which we know that Conjecture 3.1.15 holds: torsion-free p -adic analytic pro- p groups,²¹ residually torsion-free p -adic analytic pro- p groups²²—a class which includes free pro- p and Demushkin groups—and torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups, which we will prove ahead in Theorem 4.6.6.²³

A priori, these two conjectures seem unrelated. Indeed, rk_Γ is defined analytically on the group algebra $\mathbb{C}[\Gamma]$, whereas rk_G is defined by approximation through finite quotients on $\mathbb{F}_p[[G]]$. However, the reason for this profinite version of the Atiyah conjecture to be considered alongside the original one stems from the following result, known as the Lück approximation theorem²⁴:

Theorem 3.1.16. *Let Γ be a residually finite group and $\Gamma \geq \Gamma_1 \geq \Gamma_2 \geq \dots$ a chain of finite index normal subgroups with trivial intersection. For any matrix A over $\mathbb{C}[\Gamma]$, we have*

$$\text{rk}_\Gamma(A) = \lim_{i \rightarrow \infty} \text{rk}_{\Gamma/\Gamma_i}(A_i),$$

where $\text{rk}_{\Gamma/\Gamma_i}$ is the normalized rank function over the finite group algebras $\mathbb{C}[\Gamma/\Gamma_i]$ and A_i denotes the image of A modulo Γ_i .

Hence, the von Neumann rank function over Γ can be approximated by rank functions over finite groups, in a way that is reminiscent of the definition of the normalized rank function over pro- p groups. We finish this section with some remarks on discrete Sylvester matrix rank functions, that is, functions $\text{rk} \in \mathbb{P}(R)$ whose image in $\mathbb{R}_{\geq 0}$ is discrete.

It is well known that a discrete subgroup G of the real numbers is isomorphic to $\varepsilon\mathbb{Z}$ for some $\varepsilon > 0$. If moreover $1 \in G$, then ε is rational. In light of this, one might hope that Sylvester rank functions with discrete image are necessarily rational valued. However, this is not the case, as the following proposition shows:²⁵

¹⁹Michael Atiyah. “Elliptic operators, discrete groups and Von Neumann algebras”. In: *Colloque analyse et topologie*. Astérisque 32-33. Société mathématique de France, 1976, pp. 43–72, Quest. 6.4(iii).

²⁰Jaikin-Zapirain, “L2-Betti Numbers and Their Analogues in Positive Characteristic”, Sec. 12.

²¹Michael Harris. “p-adic representations arising from descent on abelian varieties”. fr. In: *Compositio Mathematica* 39.2 (1979), pp. 177–245; Michael Harris. “Correction to p-adic Representations Arising from Descent on Abelian Varieties”. en. In: *Compositio Mathematica* 121.1 (Mar. 2000). Publisher: London Mathematical Society, pp. 105–108. ISSN: 1570-5846, 0010-437X. DOI: 10.1023/A:1001730616194, Thm. 1.10.

²²Andrei Jaikin-Zapirain. “Approximation by subgroups of finite index and the Hanna Neumann conjecture”. In: *Duke Mathematical Journal* 166.10 (July 2017). Publisher: Duke University Press, pp. 1955–1987. ISSN: 0012-7094, 1547-7398. DOI: 10.1215/00127094-0000015X, Thm. 2.3.

²³Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Cor. 1.2.

²⁴Andrei Jaikin-Zapirain. “The base change in the Atiyah and the Lück approximation conjectures”. en. In: *Geometric and Functional Analysis* 29.2 (Apr. 2019), pp. 464–538. ISSN: 1420-8970. DOI: 10.1007/s00039-019-00487-3, Thm. 1.3.

²⁵The savvy reader will immediately notice that underlying this proposition is the following elementary fact: any two generated submonoid of $\mathbb{R}_{\geq 0}$ is discrete.

Proposition 3.1.17. *There exists a discrete Sylvester matrix rank function on $\mathbb{Z}$ which has an irrational value.*

Proof. Let λ be an irrational number in $(0, 1)$ and consider the rank function $\text{rk} = \lambda \text{rk}_p + (1 - \lambda) \text{rk}_q$ on $\mathbb{Z}$, where p and q are distinct primes and rk_p and rk_q are the Sylvester matrix rank functions obtained as the pullbacks of the ranks over $\mathbb{F}_p$ and $\mathbb{F}_q$ respectively. We have that $\text{rk}(q) = \lambda$ which is irrational. To see that rk is discrete, observe that its value set is contained in the set X of numbers of the form $\lambda m + (1 - \lambda)n$ for integers $m, n \geq 0$. This forces each intersection $X \cap [0, k]$ to be finite for any positive real number $k > 0$, so rk is discrete. $\square$

If instead of discrete we assume that a Sylvester matrix rank function is ϵ -separated for some $\epsilon > 0$, that is, the distance between any two distinct values in its image is at least ϵ , then we do obtain integrality up to a scalar factor:

Lemma 3.1.18. *Let rk be a Sylvester matrix rank function on a ring R that is ϵ -separated for some $\epsilon > 0$. Then, there exists an integer $d \geq 1$ such that $d \cdot \text{rk}$ is integer valued.*

Proof. Let A be a matrix over R such that $\text{rk}(A) = \alpha$. Then, choosing a natural number $N > \frac{1}{\epsilon}$, the Dirichlet approximation theorem²⁶ implies that there exists positive integers p, q such that

$$|\text{rk}(\oplus_q A) - \text{rk}(\text{Id}_p)| = |q\alpha - p| < \frac{1}{N} < \epsilon.$$

Hence, this difference must be zero, so $\alpha = p/q$ is rational. This already shows that the image of rk is contained in $\mathbb{Q}$. Let X be the set of denominators of all values of rk written in lowest terms.

If X were infinite, then we could find a sequence of matrices A_i over R such that the denominators of $\text{rk}(A_i) = p_i/q_i$ expressed in lowest form are all distinct. Observe that the fractional part of $\text{rk}(A_i)$ is also of the form p'_i/q_i for some p'_i coprime with q_i . Let N_0 be a positive integer bigger than $\frac{1}{\epsilon}$ and N another positive integer that is bigger than $\max\{N_0, \text{rk}(A_1), \dots, \text{rk}(A_{N_0+1})\}$. For each $1 \leq i \leq N_0 + 1$, let $s_i = N - [\text{rk}(A_i)] \in \mathbb{Z}_{\geq 0}$. Hence, for each $1 \leq i \leq N_0 + 1$, we have

$$\text{rk}(A_i \oplus \text{Id}_{s_i}) = \text{rk}(A_i) + s_i \in [N, N + 1),$$

with the fractional part of $\text{rk}(A_i \oplus \text{Id}_{s_i})$ being equal to the fractional part of $\text{rk}(A_i)$. If we partition the interval $[N, N + 1)$ into N_0 subintervals of length $1/N_0 < \epsilon$, by the pigeonhole principle there must exist two distinct indices $1 \leq i < j \leq N_0 + 1$ such that $|\text{rk}(A_i \oplus \text{Id}_{s_i}) - \text{rk}(A_j \oplus \text{Id}_{s_j})| < \epsilon$. Hence, those ranks must coincide. However, this is impossible since their fractional parts have distinct denominators in lowest form. This contradiction shows that X is finite, so if we let d be the least common multiple of all elements in X , then $d \cdot \text{rk}$ is integer valued. $\square$

²⁶Serge Lang. *Introduction to Diophantine Approximations*. Springer New York, 1995. ISBN: 9781461242208. DOI: 10.1007/978-1-4612-4220-8, Thm. II.1.

Observe that the unique rank function on $\text{Mat}_n(\mathcal{D})$ for a division ring $\mathcal{D}$ has values in $\frac{1}{n}\mathbb{Z}$. An extension of Malcolmson's theorem, due to Schofield,²⁷ shows that any ϵ -separated Sylvester matrix rank function rk on a ring R arises as the pullback of a rank function over a simple Artinian R -ring.²⁸

Theorem 3.1.19. *Let R be a ring and $\text{rk} \in \mathbb{P}(R)$ be a Sylvester matrix rank function that assumes values in $\frac{1}{d}\mathbb{Z}$ for some integer $d \geq 1$. Then, there exists a homomorphism $\varphi: R \rightarrow \text{Mat}_d(\mathcal{D})$ for some division ring $\mathcal{D}$ such that $\text{rk} = \varphi^\# \text{rk}_{\text{Mat}_d(\mathcal{D})}$.²⁹*

3.2 Universality and topological localizations

Recall the bijections established by Malcolmson's and Cohn's work:

$$\begin{array}{ccc}
 & \varphi: R \rightarrow \mathcal{D} & \\
 \text{matrix localization} \nearrow & & \searrow \varphi^\# \text{rk}_{\mathcal{D}} \\
 \Sigma & \xleftarrow{\text{singular matrices}} & \text{rk}
 \end{array}$$

In Section 2.3, we showed how the sets of matrix localizations and epic division R -rings are both endowed with a natural partial order, and that the bijection between them is an order-reversing isomorphism of posets. Similarly, the set of Sylvester matrix rank functions on R is also a poset with respect to the pointwise order: $\text{rk}_1 \leq \text{rk}_2$ if and only if $\text{rk}_1(A) \leq \text{rk}_2(A)$ for all matrices A over R . It is then natural to ask whether the bijection above is also an isomorphism of posets.

The answer is affirmative, since if $\text{rk}_1 \leq \text{rk}_2$, then any rk_1 -full matrix is also rk_2 -full, so the set of singular matrices for rk_2 is contained in that of rk_1 . By Cohn's theorem, this implies that the epic division ring associated to rk_2 specializes to the epic division ring associated to rk_1 . This motivates the following definition:

Definition 3.2.1. A Sylvester matrix rank function rk on a ring R is said to be *universal* if it is the maximum element in the poset $\mathbb{P}(R)$. Equivalently, the associated Sylvester module rank function $\dim$ is said to be *universal* if it is the minimum element in the poset $\mathbb{P}(R)$.

Observe that R has a universal epic division R -ring if and only if it admits an integer valued universal rank function. In this case, we say that the map $R \rightarrow \mathcal{D}$ is a *universal epimorphism* for R , or that $\mathcal{D}$ is a *universal envelope* for R . From (SMat2), it follows that any Sylvester matrix rank function $\text{rk} \in \mathbb{P}(R)$ satisfies $\text{rk}(A) \leq \text{irk}(A)$ for any matrix A over R . However, irk itself

²⁷Schofield, *Representations of Rings over Skew Fields*, Thm. 8.3.

²⁸Ibid., Thm. 7.11 and 7.12.

²⁹This establishes a bijection between ϵ -separated Sylvester rank functions on R and equivalence classes of homomorphisms $\varphi: R \rightarrow S$ where S is a simple Artinian ring where the equivalence relation states that $\varphi_1: R \rightarrow S_1$ and $\varphi_2: R \rightarrow S_2$ are equivalent if there exists another simple Artinian ring S and ring homomorphisms $\psi_1: S_1 \rightarrow S$ and $\psi_2: S_2 \rightarrow S$ such that $\psi_1 \circ \varphi_1 = \psi_2 \circ \varphi_2$.

may fail to be a Sylvester matrix rank function.³⁰ We now arrive at the second characterization of Sylvester domains:³¹

Proposition 3.2.2. *Let R be a ring. Then, R is a Sylvester domain if and only if irk is a Sylvester matrix rank function on R .*

In this case, since irk is integer valued and necessarily the maximum rank function on R , it is also universal. Moreover, there exists an epic division R -ring $\mathcal{D}$ such that pullback of the rank function over $\mathcal{D}$ coincides with irk .

It is, in general, difficult to obtain criteria for a given Sylvester matrix rank function to be universal. Although it may be feasible to check whether $\text{rk}_1 \leq \text{rk}_2$ for two given rank functions rk_1 and rk_2 on a ring R , in general the space $\mathbb{P}(R)$ is too unwieldy to obtain a full description of its elements. We illustrate this with the following example:

Example 3.2.3. Let G be a finite p group and $R = \mathbb{F}_q[G]$ its group algebra over $\mathbb{F}_q$. As we saw in Example 3.1.6, the normalized rank function $\dim_G$ is a Sylvester module rank function on R . However, it is not known in general whether $\dim_G$ is universal or not. We can, however, compare it with any ε -separated Sylvester module rank function $\dim$ on R .

Indeed, such a rank function $\dim$ assumes values in $\frac{1}{d}\mathbb{Z}$ for some positive integer d by Lemma 3.1.18. Hence, by Schofield's Theorem 3.1.19, it arises as the pullback of the rank function over a homomorphism $\varphi: R \rightarrow \text{Mat}_d(\mathcal{D})$ for some division ring $\mathcal{D}$. Let $N = \mathcal{D}^d$. On one hand, this abelian group has an $R - \mathcal{D}$ -bimodule structure coming from the left action of R via φ and the right action of $\mathcal{D}$ by multiplication. It is with this bimodule structure that we compute $\dim$ for some finitely presented right R -module M :

$$\dim M = \frac{\dim_{\mathcal{D}} M \otimes_R N}{d}.$$

On the other hand, we can endow this abelian group with an $\mathbb{F}_q - \mathcal{D}[G]$ bimodule structure by letting $g \in G$ act on the right through multiplication by $\varphi(g^{-1})$ on the left. This makes the tensor product $Q = M \otimes_{\mathbb{F}_q} N$ into a finitely generated right $\mathcal{D}[G]$ -module with the diagonal action

$$(n \otimes m)g = ng \otimes \varphi(g^{-1})m.$$

Since $\mathcal{D}$ has characteristic p , as it admits a homomorphism from $\mathbb{F}_q$, the group algebra $\mathcal{D}[G]$ is local³² with its unique maximal ideal given by the kernel of the augmentation map $\mathcal{D}[G] \rightarrow \mathcal{D}$. By Nakayama's lemma, the number of generators of Q as a right $\mathcal{D}[G]$ -module is precisely

$$\mu(Q) = \dim_{\mathcal{D}} Q \otimes_{\mathcal{D}[G]} \mathcal{D} = \dim_{\mathcal{D}} M \otimes_R N.$$

³⁰The inner rank irk is not a matrix rank function over $\mathbb{Z}/4\mathbb{Z}$, for example. Property (SMat₄) fails for the matrix $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. There are also examples of Ore domains which do not admit a universal rank function, see Passman, "Universal fields of fractions for polycyclic group algebras"

³¹Dicks and Sontag, "Sylvester domains", Prop. 2.

³²The augmentation ideal is nilpotent, which can be proven by an easy induction on $|G|$ and quotienting out a central element of order p .

From a surjection $\mathcal{D}[G]^{\mu(Q)} \rightarrow Q$ we obtain the bound

$$d \cdot \dim_{\mathbb{F}_q} M = \dim_{\mathcal{D}} M \otimes_{\mathbb{F}_q} N = \dim_{\mathcal{D}} Q \leq |G| \cdot \dim_{\mathcal{D}} M \otimes_R N.$$

This proves

$$\dim_G M = \frac{\dim_{\mathbb{F}_q} M}{|G|} \leq \frac{\dim_{\mathcal{D}} M \otimes_R N}{d} = \dim(M),$$

that is, $\dim_G \leq \dim$. ■

The argument in the previous example used in a fundamental way the finiteness of the group G and the ε -separability of $\dim$. The following questions remain open:

Question 3.2.4. Let G be a pro- p group and $\dim_G$ the normalized rank function on $\mathbb{F}_p[[G]]$. Is $\dim_G$ universal among Sylvester module rank functions on $\mathbb{F}_p[[G]]$? What about among ε -separated Sylvester module rank functions on $\mathbb{F}_p[[G]]$?

In the case of completed $\mathbb{F}_p$ group algebras of pro- p groups, we have the following useful approximation result, which is the main result of this chapter:³³

Theorem 3.2.5. *Let G be a pro- p group, $\mathcal{Q}$ a division ring with a non-archimedean discrete valuation³⁴ $w: \mathcal{Q} \rightarrow \{p^k: k \in \mathbb{Z}\} \cup \{0\}$ and $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathcal{Q}$ a continuous embedding with respect to the topology on $\mathcal{Q}$ induced by w . Then*

$$\varphi^\# \text{rk}_{\mathcal{Q}} \leq \text{rk}_G$$

as rank functions on $\mathbb{F}_p[[G]]$. In particular, if $\varphi^\# \text{rk}_{\mathcal{Q}} = \text{irk}_{\mathbb{F}_p[[G]]}$, we have $\text{irk}_{\mathbb{F}_p[[G]]} = \text{rk}_G$.

We will apply this theorem in Section 4.4 to prove that torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups satisfy the pro- p Atiyah conjecture (Conjecture 3.1.15). The proof of Theorem 3.2.5 will be broken down into a series of lemmas.

We assume now that G , $(\mathcal{Q}, w)$ and $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathcal{Q}$ are as in the statement of Theorem 3.2.5. For each $i \in \mathbb{Z}$, we define the following subsets of G :

$$G_i = \{g \in G: w(\varphi(g - 1)) \leq p^{-i}\}.$$

Each G_i is a clopen and non-empty subset of G , as $1 \in G_i$ for every i . Moreover, since G is compact, there exists a minimum

$$k_0 = \min \{k \in \mathbb{Z}: w(\varphi(g - 1)) \leq p^{-k} \text{ for all } g \in G\}.$$

This implies that $G_i = G$ for every $i \leq k_0$.

³³Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Thm. 7.1.

³⁴This means that $w(ab) = w(a)w(b)$, $w(a + b) \leq \max\{w(a), w(b)\}$ with equality if $w(a) \neq w(b)$, $w(1) = 1$ and $w(a) = 0$ if and only if $a = 0$.

Lemma 3.2.6. *The integer k_0 is non-negative. In particular, $G_0 = G$. Moreover, $w(\varphi(g)) = 1$ for every $g \in G$.*

Proof. Suppose by contradiction that $k_0 < 0$ and let $g \in G$ be such that $w(\varphi(g-1)) = p^{-k_0}$. Observe that since $\mathbb{F}_p[[G]]$ embeds into a division ring, G is torsion-free³⁵, and in particular $g^p \neq 1$. Since $\mathbb{Q}$ has characteristic p , we get

$$w(\varphi(g^p-1)) = w(\varphi(g-1)^p) = p^{-pk_0} > p^{-k_0},$$

contradicting the definition of k_0 .

Suppose now that $w(\varphi(g)) > 1$ for some $g \in G$. Then, since $w(\varphi(g)) \neq w(-1) = 1$, we have

$$w(\varphi(g-1)) = \max\{w(\varphi(g)), w(-1)\} = w(\varphi(g)) > 1,$$

another contradiction. On the other hand, if $w(\varphi(g)) < 1$, then the same argument shows that $w(\varphi(g-1)) = 1$. We get a contradiction by considering:

$$w(\varphi(g^{-1}-1)) = w(\varphi(g^{-1}(1-g))) = w(\varphi(g^{-1})) \cdot w(\varphi(g-1)) = w(\varphi(g^{-1})) > 1. \quad \square$$

By the continuity of w it follows that the G_i form a chain of normal open subgroups of G , for it is easily verified that each G_i contains 1 and is closed under products, inverses and conjugation. Since $w(q) = 0$ if and only if $q = 0$, we can also conclude that $\bigcap_{i \geq 0} G_i = \{1\}$, that is, the chain G_i is residual in G . Hence, for any finitely presented right $\mathbb{F}_p[[G]]$ -module M , we have

$$\dim_G M = \lim_{i \rightarrow \infty} \dim_{G/G_i} M = \lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{F}_p} M \otimes_{\mathbb{F}_p[[G]]} \mathbb{F}_p[G/G_i]}{|G : G_i|}.$$

Let $\mathcal{Q}_i$ be the set of elements $q \in \mathbb{Q}$ with $w(q) \leq p^{-i}$. In particular, $\mathcal{Q}_0$ is an $\mathbb{F}_p$ -algebra containing $\varphi(\mathbb{F}_p[[G]])$ with ideals $\mathcal{Q}_i$ for each $i \geq 0$.

Lemma 3.2.7. *There exists a division ring extension $\mathbb{Q} \leq \mathbb{Q}'$ and an extension w' of the valuation w to $\mathbb{Q}'$ such that $\mathbb{Q}'$ contains a central element z with $w'(z) = p^{-1}$. In particular, $\mathcal{Q}'_i = z^i \mathcal{Q}'_0$ for every $i \in \mathbb{Z}$.*

Proof. Consider the polynomial ring $\mathbb{Q}[z]$ and define $w' : \mathbb{Q}[z] \rightarrow \{p^k : k \in \mathbb{Z}\} \cup \{0\}$ to be

$$w' \left(\sum_{i=0}^n q_i z^i \right) = \max \{w(q_i) p^{-i}\}.$$

We check that w' is an additive non-archimedean valuation on $\mathbb{Q}[z]$, for it is immediate that it is discrete.

³⁵If $g \in G$ has order k , then $(g-1)(1+g+\dots+g^{k-1}) = 0$ in any group algebra of G .

Let $r = \sum_{i=1}^n q_i z^i$ and $r' = \sum_{i=1}^n q'_i z^i$ be two arbitrary elements in $\mathcal{Q}[z]$. We have $w'(r) = 0$ if and only if all the q_i have valuation 0, that is, if and only if $q_i = 0$ for all i . Hence, $w'(r) = 0$ implies $r = 0$. Suppose the maximal w' valuation on r and r' is attained at the terms $q_m z^m$ and $q'_{m'} z^{m'}$ with m and m' minimal for this properties. Then, for all $0 \leq i, j \leq n$, we have:

$$w(q_m) p^{-m} \geq w(q_i) p^{-i}, \quad w(q'_{m'}) p^{-m'} \geq w(q'_j) p^{-j}.$$

Multiplying both inequalities implies:

$$w(q_m q'_{m'}) p^{-m-m'} \geq w(q_i q'_j) p^{-i-j}$$

for all $1 \leq i, j \leq n$. In particular, for every $1 \leq k \leq 2n$ we have

$$w(q_m q'_{m'}) p^{-m-m'} \geq \max \{w(q_i q'_j) : i + j = k\} p^{-k} \geq w \left(\sum_{i+j=k} q_i q'_j \right) p^{-k}. \quad (3.1)$$

Suppose now, by contradiction, that there exists $i \neq m$ and $j \neq m'$ with $i + j = m + m'$ such that $w(q_i q'_j) \geq w(q_m q'_{m'})$. Without loss of generality, we may suppose that $i < m$, so that $m' < j$. By the minimality of m and m' , we must have $w(q_i) p^{-i} < w(q_m) p^{-m}$ and $w(q'_j) p^{-j} < w(q'_{m'}) p^{-m'}$. This implies:

$$\frac{w(q_i)}{w(q_m)} < p^{i-m} = p^{n-j} \leq \frac{w(q'_{m'})}{w(q'_j)},$$

and thus $w(q_i q'_j) < w(q_m q'_{m'})$, a contradiction. Therefore, if $i + j = m + n$, we have $w(q_i q'_j) < w(q_m q'_{m'})$. Hence

$$w(q_m q'_{m'}) p^{-m-m'} = w \left(\sum_{i+j=m+n} q_i q'_j \right) p^{-m-n}. \quad (3.2)$$

We have

$$rr' = \sum_{k=0}^{2n} \left(\sum_{i+j=k} q_i q'_j \right) z^k.$$

Since

$$w \left(\sum_{i+j=k} q_i q'_j \right) p^{-k} \leq w(q_m q'_{m'}) p^{-m-m'} = w \left(\sum_{i+j=m+n} q_i q'_j \right) p^{-m-m'}$$

by (3.1) and (3.2), the maximal w' valuation on rr' must be attained in the z -degree $m + m'$, with value exactly $w'(r)w'(r')$.

At last, we must show that $w'(r + r') \leq \max\{w'(r), w'(r')\}$. We may suppose, without loss of generality, that $w(q_m) p^{-m} = w'(r) \leq w'(r') = w(q'_{m'}) p^{-m'}$. Observe that in this case

$$w(q_i + q'_i) p^{-i} \leq \max\{w(q_i), w(q'_i)\} p^{-i} \leq w(q'_{m'}) p^{-m'} = w'(r').$$

Since

$$r + r' = \sum_{i=1}^n (q_i + q'_i) z^i,$$

we are done. Now, $\mathbb{Q}[z]$ is a Noetherian domain and thus we can construct the division ring of rational functions $\mathbb{Q}' = \mathbb{Q}(z)$ by localizing at the non-zero elements. Hence, the valuation w' also extends to $\mathbb{Q}'$, which concludes the proof. $\square$

In virtue of Lemma 3.2.7, we may assume from now on that $\mathbb{Q}$ contains a central element z with $w(z) = p^{-1}$, so that $\mathbb{Q}_i = z^i \mathbb{Q}_0$ for every $i \in \mathbb{Z}$. Consider now $\mathbb{Q}_0$, the ring of elements with valuation less than or equal to 1. We note that $\varphi(\mathbb{F}_p[[G]]) \leq \mathbb{Q}_0$ by Lemma 3.2.6 and the non-zero left or right ideals of $\mathbb{Q}_0$ are precisely the sets $\mathbb{Q}_i = z^i \mathbb{Q}_0$ for $i \geq 0$. This implies that $\mathbb{Q}_0/z\mathbb{Q}_0$ is the only simple $\mathbb{Q}_0$ -module, and the length of the $\mathbb{Q}_0$ -modules $\mathbb{Q}_0/z^i \mathbb{Q}_0$ is equal to i .

We will now prove the inequality $\varphi^\# \text{rk}_{\mathbb{Q}} \leq \text{rk}_G$ for $\varphi^\# \text{rk}_{\mathbb{Q}}$ -full matrices over $\mathbb{F}_p[[G]]$, that is, we will show that if A is an $n \times n$ matrix over R such that $\text{rk}_{\mathbb{Q}} \varphi(A) = n$, then $\text{rk}_G(A) = n$. For any such matrix A , we define the right $\mathbb{F}_p[[G]]$ -module $M = \mathbb{F}_p[[G]]^n / A \mathbb{F}_p[[G]]^n$. By the correspondence between Sylvester matrix rank functions and Sylvester module rank functions of Proposition 3.1.4, we have the equalities

$$\dim_G(M) = n - \text{rk}_G(A), \quad \varphi^\# \dim_{\mathbb{Q}}(M) = n - \varphi^\# \text{rk}_{\mathbb{Q}}(A).$$

Therefore, we must show that if $\varphi^\# \text{rk}_{\mathbb{Q}}(A) = n$, then $\dim_G(M) = 0$.

For any open subgroup U of G , we define its $\mathbb{Q}_0$ length over U as

$$l_U(M) = l_{\mathbb{Q}_0}(M \otimes_{\mathbb{F}_p[[U]]} \mathbb{Q}_0),$$

the length of the induced $\mathbb{Q}_0$ -module.

Lemma 3.2.8. *The $\mathbb{Q}_0$ length $l_G(M)$ is finite.*

Proof. Since A is invertible over φ , we have

$$\varphi^\# \dim_{\mathbb{Q}}(M) = \dim_{\mathbb{Q}} M \otimes_{\mathbb{F}_p[[G]]} \mathbb{Q} = 0.$$

Observing that

$$M \otimes_{\mathbb{F}_p[[G]]} \mathbb{Q} = (M \otimes_{\mathbb{F}_p[[G]]} \mathbb{Q}_0) \otimes_{\mathbb{Q}_0} \mathbb{Q},$$

this implies that the $\mathbb{Q}_0$ -module $M \otimes_{\mathbb{F}_p[[G]]} \mathbb{Q}_0$ is torsion. Given that M is finitely generated and every proper quotient of $\mathbb{Q}_0$ is of the form $\mathbb{Q}_0/z^i \mathbb{Q}_0$ and thus has finite length, that module must have finite length as well. $\square$

Lemma 3.2.9. *Let $U \leq V$ be open subgroups of G with $|V:U| = p$. Then, $l_U(M) \leq pl_V(M)$. In particular, $l_U(M) \leq |G:U|l_G(M)$.*

Proof. Let $L = M \otimes_{\mathbb{F}_p[[U]]} \mathbb{Q}_0$ and take any element $g \in V$ but not in U . Define the function $\tau: M \times \mathbb{Q}_0 \rightarrow L$ by

$$\tau(m, q) = mg \otimes \phi(g^{-1})q - m \otimes q.$$

Since U must be normal in V , we have that $\tau(mu, q) = \tau(m, uq)$ for every $u \in U$, that is, τ induces a homomorphism of right $\mathbb{Q}_0$ -modules $\psi: L \rightarrow L$ such that

$$\psi(m \otimes q) = mg \otimes g^{-1}q - m \otimes q.$$

It is immediate to verify that $\psi^i(m \otimes q) = \sum_{j=0}^i (-1)^j \binom{i}{j} mg^{i-j} \otimes g^{j-i}q$ and therefore $\psi^p = 0$. Moreover, $L/\psi(L) \simeq M \otimes_{\mathbb{F}_p[[V]]} \mathbb{Q}_0$, and by applying ψ we find that this surjects onto $\psi^i(L)/\psi^{i+1}(L)$ for every i . Hence:

$$l_U(M) = l_{\mathbb{Q}_0} L \leq p \cdot l_{\mathbb{Q}_0} (M \otimes_{\mathbb{F}_p[[V]]} \mathbb{Q}_0) = pl_V(M). \quad \square$$

Recall that $\overline{\mathbb{Q}} = \mathbb{Q}_0/\mathbb{Q}_1$ is a division ring.

Lemma 3.2.10. *We have that*

$$\dim_{\overline{\mathbb{Q}}} M \otimes_{\mathbb{F}_p[[G_i]]} \overline{\mathbb{Q}} \leq \frac{l_{G_i}(M)}{i}$$

for every $i \geq 1$

Proof. When we view M as a right $\mathbb{F}_p[[G_i]]$ -module it has a presentation of the form

$$M = \mathbb{F}_p[[G_i]]^{n|G: G_i|} / B \mathbb{F}_p[[G_i]]^{n|G: G_i|}$$

for some matrix $B \in \text{Mat}_{n|G: G_i|}(\mathbb{F}_p[[G_i]])$. By projecting the matrix onto $\mathbb{F}_p$ and applying elementary matrix operations, we can find one such matrix B satisfying

$$B = \begin{pmatrix} \text{Id}_a + B_1 & B_2 \\ B_3 & B_4 \end{pmatrix},$$

where the B_i are matrices with entries in the augmentation ideal I_{G_i} of $\mathbb{F}_p[[G_i]]$ and $n|G: G_i| - a = \dim_{\mathbb{F}_p} M/MI_{G_i}$. Since the B_i vanish when acting on $\overline{\mathbb{Q}}$, one gets that

$$\dim_{\overline{\mathbb{Q}}} M \otimes_{\mathbb{F}_p[[G_i]]} \overline{\mathbb{Q}} = \dim_{\mathbb{F}_p} M \otimes_{\mathbb{F}_p[[G_i]]} \mathbb{F}_p = \dim_{\mathbb{F}_p} M/MI_{G_i} = n|G: G_i| - a.$$

By definition, we have that $w(\phi(x)) \leq p^{-i}$ for any $x \in I_{G_i}$, so that $\phi(I_{G_i}) \leq z^i \mathbb{Q}_0$. Hence, there are matrices B'_2 and B'_4 over $\mathbb{Q}_0$ such that we can express $\phi(B)$ as

$$\phi(B) = \underbrace{\begin{pmatrix} \text{Id}_a + \phi(B_1) & B'_2 \\ \phi(B_3) & B'_4 \end{pmatrix}}_{C_1} \cdot \underbrace{\begin{pmatrix} \text{Id}_a & 0 \\ 0 & z^i \text{Id}_{n|G: G_i| - a} \end{pmatrix}}_{C_2}.$$

This implies that

$$(\mathbb{Q}_0/z^i\mathbb{Q}_0)^{n|G:G_i|-a} = \mathbb{Q}_0^{n|G:G_i|} / \mathbb{Q}_0^{n|G:G_i|} C_2$$

is a quotient of $M \otimes_{\mathbb{F}_p[[G_i]]} \mathbb{Q}_0 \simeq \mathbb{Q}_0^{n|G:G_i|} / \mathbb{Q}_0^{n|G:G_i|} \varphi(B)$. Therefore, we get

$$\begin{aligned} l_{G_i}(M) &= l_{\mathbb{Q}_0}(M \otimes_{\mathbb{F}_p[[G_i]]} \mathbb{Q}_0) \\ &\geq l_{\mathbb{Q}_0}(\mathbb{Q}_0/z^i\mathbb{Q}_0)^{n|G:G_i|-a} \\ &= i(n|G:G_i| - a) \\ &= i \dim_{\overline{\mathbb{Q}}} M \otimes_{\mathbb{F}_p[[G_i]]} \overline{\mathbb{Q}}. \end{aligned}$$

□

We are now ready to write:

Proof of Theorem 3.2.5. Note that, by Lemma 3.1.3, to show that $\varphi^\# \text{rk}_{\mathbb{Q}} \leq \text{rk}_G$, it suffices to show that they coincide over square matrices invertible under φ . Indeed, any matrix $A \in \text{Mat}_{n \times n}(\mathbb{F}_p[[G]])$ contains a maximal submatrix $B \in \text{Mat}_k(\mathbb{F}_p[[G]])$ such that $\varphi(B)$ is invertible and

$$\varphi^\# \text{rk}_{\mathbb{Q}}(B) = \varphi^\# \text{rk}_{\mathbb{Q}}(A) = k.$$

In this case, if $\text{rk}_G(B) = k$, then Lemma 3.1.3 shows us that $k \leq \text{rk}_G(A)$.

Thus, let $M = \mathbb{F}_p[[G]]^n / A \mathbb{F}_p[[G]]^n$ be the right R -module presented by a matrix $A \in \text{Mat}_n(\mathbb{F}_p[[G]])$ that is invertible under φ . We want to show that $\dim_G(M) = 0$. We have that

$$\begin{aligned} \frac{\dim_{\mathbb{F}_p} M \otimes_{\mathbb{F}_p[[G_i]]} \mathbb{F}_p}{|G:G_i|} &= \frac{\dim_{\overline{\mathbb{Q}}} M \otimes_{\mathbb{F}_p[[G_i]]} \overline{\mathbb{Q}}}{|G:G_i|} \\ &\leq \frac{l_{G_i}(M)}{i|G:G_i|}, \text{ by Lemma 3.2.10} \\ &\leq \frac{l_G(M)}{i}, \text{ by Lemma 3.2.9.} \end{aligned}$$

Since $l_G(M)$ is finite by Lemma 3.2.8, we have

$$\lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{F}_p} M \otimes_{\mathbb{F}_p[[G_i]]} \mathbb{F}_p}{|G:G_i|} \leq \lim_{i \rightarrow \infty} \frac{l_G(M)}{i} = 0,$$

which concludes the proof. □

Given Theorem 2.4.1, which says that any matrix localization of a profinite ring admits a Hausdorff topology such that the embedding is continuous, and Theorem 3.2.5, as well as Question 3.2.4, one is tempted to ask whether any epic division R -rings for R a profinite domain admit such valuations. Indeed, this is the case for all the examples that we can decide whether rk_G is universal³⁶.

³⁶Namely, only torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups.

The question of classifying the valuated epic division R -rings has already been studied before in the literature, and for completeness of exposition we summarize its main results below. Following Mohammad Hezavehi³⁷, if R is a ring and Γ is a totally ordered abelian group, written additively³⁸, a *semi-valuation* on R with values in Γ is a function $v: R \rightarrow \Gamma \cup \{\infty\}$ such that for all $a, b \in R$:

$$(V_1) \quad v(ab) = v(a) + v(b);$$

$$(V_2) \quad v(a + b) \geq \min\{v(a), v(b)\};$$

$$(V_3) \quad v(0) = \infty.$$

We say that v is a *valuation* if $v(a) = \infty$ implies $a = 0$. In general, $P = v^{-1}(\infty)$ is a two-sided ideal of R such that R/P has no zero-divisors and inherits a valuation $\bar{v}$ for every semi-valuation v on R .

Analogously, a *matrix valuation* on a ring R is a function V defined on all square matrices over R with values in $\Gamma \cup \{\infty\}$ such that:

$$(MV_1) \quad V(A \oplus B) = V(A) + V(B) \text{ for all square matrices } A \text{ and } B.$$

(MV₂) If A and B are two matrices of the same size that differ by a single row or column and $A \nabla B$ is the matrix obtained by summing that row or column, then $V(A \nabla B) \geq \min\{V(A), V(B)\}$.

$$(MV_3) \quad V(A) \text{ is unchanged if we multiply a row or a column of } A \text{ by } -1.$$

$$(MV_4) \quad V(1) = 0.$$

$$(MV_5) \quad V(A) = \infty \text{ for every non-full matrix } A \text{ over } R.$$

It is straightforward to check that for any matrix valuation V on a ring R the set $V^{-1}(\infty)$ is a prime matrix ideal of R . If $\varphi: R \rightarrow S$ is a ring homomorphism and V is a matrix valuation on S , then the pullback

$$\varphi^\# V(A) = V(\varphi(A))$$

is a matrix valuation on R . Then, Hezavehi's result is:³⁹

Theorem 3.2.11. *Every valuation v on a division ring extends uniquely to a matrix valuation by the rule*

$$V(A) = \begin{cases} v(\det A), & \text{if } A \text{ is invertible,} \\ \infty, & \text{otherwise,} \end{cases}$$

³⁷M. Mahdavi Hezavehi. "Matrix valuations and their associated skew fields". In: *Results in Mathematics* 5.1-2 (Mar. 1982), pp. 149-156. ISSN: 1420-9012. DOI: 10.1007/bf03323311.

³⁸By taking $w = p^{-v}$, one can pass between additive valuations with values in $\mathbb{Z}$ in Hezavehi's sense and multiplicative valuations in our sense.

³⁹Hezavehi, "Matrix valuations and their associated skew fields", Thm. 2.1 and 3.1.

where $\det A$ is the Dieudonné determinant⁴⁰, and this extension gives a bijection between valuations and matrix valuations on division rings. Each matrix valuation V on an arbitrary ring R determines an associated epic division R -ring $\mathcal{D}$ endowed with a valuation v . Conversely, every epic division R -ring with a valuation v arises in this way. This correspondence gives a bijection between the sets of matrix valuations on R and valuated epic division R -rings.

We are particularly interested in the following open problem:

Question 3.2.12. If $\mathbb{F}_p[[G]]$ is a Sylvester domain, is there a continuous⁴¹ matrix valuation V over $\mathbb{F}_p[[G]]$ for which $V^{-1}(\infty)$ consists precisely of the non-full matrices?

Should Question 3.2.12 have an affirmative answer, it would imply by Theorem 3.2.5 that the pro- p Atiyah conjecture (Conjecture 3.1.15) has a positive solution among pro- p groups G for which $\mathbb{F}_p[[G]]$ is a Sylvester domain. Moreover, it would imply that in this case rk_G is universal (Question 3.2.4). The evidence so far suggests that this is the case.

⁴⁰This is the universal homomorphism $\mathrm{GL}_n(\mathcal{D}) \rightarrow \mathcal{D}^\times / [\mathcal{D}^\times, \mathcal{D}^\times]$ given by the group-theoretic abelianization.

⁴¹Meaning that the restriction of V to 1×1 -matrices is a continuous valuation $v: \mathbb{F}_p[[G]] \rightarrow \Gamma \cup \{\infty\}$.

Chapter 4

Sylvester domains

Es mejor pedir perdón que pedir
permiso.

J.B. on jumping over walls.

The English lawyer and mathematician James Joseph Sylvester (1814 – 1897) played a central role in the early development of the mathematics in this thesis. He coined the term “matrix” in 1850 as an “oblong arrangement of terms”.¹ To modern day mathematicians, he is probably best known by the following result that bears his name:

Theorem 4.0.1 (Sylvester’s law of inertia). *If D is a real symmetric matrix, the number of positive, negative and zero entries in any orthogonal diagonalization of D is invariant under matrix congruences.*

James Sylvester was an avid reader of poetry and studied multiple languages, such as French, German, Italian and Greek, and his style of mathematical writing is notably verbose.² He is known to have coined many terms in mathematics, giving him a certain notoriety among his

¹James Joseph Sylvester. “XLVII. Additions to the articles in the September number of this journal, “On a new class of theorems,” and on Pascal’s theorem”. In: *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 37.251 (Nov. 1850), pp. 363–370. ISSN: 1941-5974. DOI: 10.1080/14786445008646629, p. 369.

²John A. Garraty and Mark C. Carnes, eds. *American National Biography*. 21: Stratton-Tunney. eng. Vol. 21. New York: Oxford University Press, 1999. ISBN: 978-0-19-512780-5.

peers.³ In a footnote to the title (!) of a paper published in *Nature* in 1887, he states:⁴

“Perhaps I may without immodesty lay claim to the appellation of Mathematical Adam, as I believe that I have given more names (passed into general circulation) of the creatures of mathematical reason than all other mathematicians of the age combined.”

It was perhaps his prolific nature in naming mathematical objects, as well as his inclinations towards legal subjects, that led to the accident of the current author being more interested in Sylvester’s other, a bit less known, law. It was first published in a paper from 1888:⁵

Theorem 4.0.2 (Sylvester’s other, a bit less known, law of nullity). *If A and B are two real matrices of sizes $n \times k$ and $k \times m$ respectively, then*

$$\text{rk}(AB) \geq \text{rk}(A) + \text{rk}(B) - n .$$

This “elementary” inequality can be proved with the modern tools of linear algebra in one line:

$$\text{rk}(AB) = \text{rk}(B) - \dim(\ker(A) \cap \text{im}(B)) \geq \text{rk}(B) - \dim(\ker(A)) = \text{rk}(B) + \text{rk}(A) - n .$$

In particular, his law is valid for matrices over any field (or even a division ring). At the time, however, those tools were not available to him, with this law being proven in the same paper where the concept of “nullity”⁶ was introduced. The reader is encouraged to skim the algebraic works of Sylvester to appreciate his skill in the manipulation of matrices, as well how much of the theory of matrices is best and more easily understood under the lens of linear algebra. It is through his law of nullity that we will see our third definition of a Sylvester domain.

This chapter is organized as follows. In Section 4.1, we recap the two definitions of Sylvester domains that we have seen so far as well as describing a new one in terms of Sylvester’s law of nullity. We also recall previous examples of Sylvester domains as well as giving new ones. We

³Some notable examples include: allotropic factor, anallagmatic, Bézoutiant, canonical form, catalecticant, meiocatalecticizant, combinant, covariant, cumulant, cyclotomic, dialytic, discriminant, latent roots, Fermat’s last theorem, graph (as in graph theory), Hessian, invariant (as in invariant theory), isomorphic, Jacobian, linear combination, linearly independent, minor (of a matrix), plagiograph, quintic, sequence (of natural numbers), syzygy, totient (Euler’s ϕ function), tree (supposedly, as in connected circuitless graph), umbral calculus, universal algebra, $x/y/z$ -coordinates (in $\mathbb{R}^3$) and zetaic multiplication. See Jeff Miller. *Earliest Uses of Some Words of Mathematics*. en. URL: <https://mathshistory.st-andrews.ac.uk/Miller/mathword/> (visited on 11/11/2025)

⁴James Joseph Sylvester. “Note on a Proposed Addition to the Vocabulary of Ordinary Arithmetic”. In: *Nature* 37.946 (Dec. 1887), pp. 152–153. ISSN: 1476-4687. DOI: 10.1038/037152b0.

⁵James Joseph Sylvester. “On Involutants and Other Allied Species of Invariants to Matrix Systems.” eng. In: *The Collected Mathematical Papers of James Joseph Sylvester*. Ed. by H. F. Baker. Vol. 4. Providence, R.I.: AMS Chelsea Pub., 2006. ISBN: 978-0-8218-4725-1.

⁶The smallest i such that any minor of size of $n - i + 1$ an $n \times n$ matrix vanishes.

finish this section by improving Theorem 2.3.17 to the class of finitely generated pro- p groups (See Corollary 4.1.13) and stating Jaikin-Zapirain's criterion for an embedding into a division ring to be universal and induce the inner rank in Theorem 4.1.14.

In Section 4.2, we explicitly construct the universal division ring of fractions of a free pro- p algebra in countably many generators, extending the methods of the finite case as done by Jaikin-Zapirain.⁷ In Section 4.3, we explore how the properties of $\mathbb{F}_p[[G]]$ being a Sylvester domain and of $\text{rk}_G = \text{irk}$ passes to overgroups and subgroups of finite index. Here, we prove a surprising original fact: both properties are commensurability invariants in the class of torsion-free pro- p groups (Theorem 4.3.7 and Corollary 4.3.8). No abstract analogue of this result is known.

In Section 4.4, we start our structural study of free-by- $\mathbb{Z}_p$ pro- p groups. We prove that such groups admit a special type of presentation called a *flag* presentation, and moreover are virtually *mild* in the sense of John Labute. A free-by- $\mathbb{Z}_p$ pro- p group with a mild flag presentation will be called a *mild flag pro- p group*. The main goal of this section is to describe the structure of $\text{Gr } \mathbb{F}_p[[G]]$ for G a mild flag pro- p group and the filtration by powers of the augmentation ideal.

In Section 4.5, we use the decomposition of $\mathbb{F}_p[[G]]$ for a mild flag pro- p group into a skew power series ring over a free pro- p algebra and the results of Section 4.2 to embed $\mathbb{F}_p[[G]]$ into a division ring $\mathcal{D}$ endowed with a valuation w that extends the valuation on $\mathbb{F}_p[[G]]$ induced by the powers of augmentation ideal. This embedding is shown to be universal in Section 4.6, where we prove the main original theorem of this chapter: completed group $\mathbb{F}_q$ -algebras of torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups G are Sylvester domains, and moreover satisfy $\text{irk} = \text{rk}_G$ (Theorem 4.6.6).

4.1 Definitions, examples, properties and criteria

In Section 2.3, we defined a *Sylvester domain* as a ring R where the set Σ of non-full square matrices, that is, those $n \times n$ matrices A that factor as $A = BC$ for some $n \times k$ matrix B and some $k \times n$ matrix C with $k < n$, is a *prime matrix ideal*. We later saw in Section 3.2 that this is the case if and only if the inner rank irk , that is, the least k for which such a factorization exists, is an example of a *Sylvester matrix rank function*.

Our goal in this section is to give one final characterization of Sylvester domains and furthermore explore its properties. We start by observing that any Sylvester matrix rank function satisfies Sylvester's law of nullity:⁸

Proposition 4.1.1. *Let R be a ring and $\text{rk} \in \mathbb{P}(R)$ be a Sylvester matrix rank function on R .*

⁷Jaikin-Zapirain, "An Explicit Construction of the Universal Division Ring of Fractions of $E\langle\langle x_1, \dots, x_d \rangle\rangle$ ".

⁸López-Álvarez, "Sylvester Rank Functions, Epic Division Rings and the Strong Atiyah Conjecture for Locally Indicible Groups", Prop. 1.2.2(5).

Then, for any matrices A and B of sizes $m \times n$ and $n \times k$ respectively, we have

$$\text{rk}(AB) \geq \text{rk}(A) + \text{rk}(B) - n.$$

Proof. We have:

$$\begin{aligned} \text{rk}(AB) + n &= \text{rk} \begin{pmatrix} AB & 0 \\ 0 & \text{Id}_n \end{pmatrix}, \text{ by (SMat}_3\text{)} \\ &\geq \text{rk} \left(\begin{pmatrix} 0 & -\text{Id}_n \\ \text{Id}_m & A \end{pmatrix} \begin{pmatrix} AB & 0 \\ 0 & \text{Id}_n \end{pmatrix} \begin{pmatrix} \text{Id}_k & 0 \\ -B & \text{Id}_n \end{pmatrix} \right), \text{ by (SMat}_2\text{)} \\ &= \text{rk} \begin{pmatrix} B & -\text{Id}_n \\ 0 & A \end{pmatrix} \\ &\geq \text{rk}(A) + \text{rk}(B), \text{ by (SMat}_4\text{)} \end{aligned}$$

and therefore $\text{rk}(AB) \geq \text{rk}(A) + \text{rk}(B) - n$. □

We are now ready to state:⁹

Proposition 4.1.2. *Let R be a ring. The following are equivalent:*

- (1) *The set of all non-full square matrices over R is a prime matrix ideal.*
- (2) *The inner rank irk is a Sylvester matrix rank function on R .*
- (3) *The inner rank irk satisfies Sylvester's law of nullity:*

$$\text{irk}(AB) \geq \text{irk}(A) + \text{irk}(B) - n$$

for any matrices $A \in \text{Mat}_{m \times n}(R)$ and $B \in \text{Mat}_{n \times k}(R)$.

A ring satisfying any of these equivalent properties is called a Sylvester domain.

The main source of examples of Sylvester domains in the literature comes from *firs* and *semifirs*, that is, rings R in which every (finitely generated, for semifirs) left and right ideal is free of a unique rank.¹⁰

Example 4.1.3. Any principal ideal domain is a fir. Any polynomial ring $\mathcal{D}[x]$ over a division ring is also a fir by usual Euclidean algorithm of polynomial division, so they are also Sylvester domains. It is a theorem of Paul Cohn¹¹ that free associative algebras $\mathcal{D}\langle X \rangle$ over division rings are firs, and hence they are also Sylvester domains. If $\mathbb{F}$ is a field, then the group algebra $\mathbb{F}[F]$ of any abstract free group F is also a fir¹², so it is a Sylvester domain. We have also shown in Proposition 2.3.14 that if F is a free pro- p group and $\mathbb{F}_q$ is a finite field of characteristic p , then the completed group algebra $\mathbb{F}_q[[F]]$ is a semifir and therefore a Sylvester domain. ■

⁹Dicks and Sontag, "Sylvester domains", Thm. 3.

¹⁰Paul Moritz Cohn. *Free Ideal Rings and Localization in General Rings*. New Mathematical Monographs 3. Cambridge University Press, 2006, Prop. 5.5.1.

¹¹Ibid., Cor. 2.5.2.

¹²Ibid., Cor. 7.11.8.

Example 4.1.4. An example of a Sylvester domain that is not a semifir is given by the free R -algebra $R\langle X \rangle$ when R is a commutative principal ideal domain.¹³ For example, if $\mathbb{F}$ is a field, then we can take $R = \mathbb{F}[x]$ and $X = \{y\}$ to conclude that the polynomial ring $\mathbb{F}[x, y]$ in two variables over any field is a Sylvester domain. For three variables $\mathbb{F}[x, y, z]$ this is no longer true, for there exists a full 3×3 -matrix that has a non-trivial kernel:¹⁴

$$\begin{pmatrix} 0 & z & -y \\ -z & 0 & x \\ y & -x & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0.$$

Hence, there can be no homomorphism from $\mathbb{F}[x, y, z]$ to a division ring that inverts all full matrices. ■

Example 4.1.5. As we saw in Example 2.3.12, a non-obvious example of a profinite Sylvester domain is the ring $\mathbb{Z}_p[[t]]$ of formal power series over $\mathbb{Z}_p$. This is not a semifir but nevertheless admits a homomorphism to a field which inverts all full matrices. This can also be seen as the group algebra $\mathbb{Z}_p[[\mathbb{Z}_p]]$ of the infinite procyclic pro- p group $\mathbb{Z}_p$. ■

Before we move on to general properties of Sylvester domains, there is a final example that is crucial to our later results, so we shall develop it now. If X is any set and $\mathbb{F}_q$ is a finite field of characteristic p , we can form the *free pro- p $\mathbb{F}_q$ -algebra* $\mathbb{F}_q\langle\langle X \rangle\rangle$ as the inverse limit of all finite algebra quotients of the free algebra $\mathbb{F}_q\langle X \rangle$ in which the image of X is nilpotent and almost all elements of X go to zero.¹⁵ This free pro- p $\mathbb{F}_q$ -algebra has the following universal property: if R is another profinite $\mathbb{F}_q$ -algebra and $f: X \rightarrow R$ is a function with the properties that $f(X)$ is topologically nilpotent and that any open ideal of R contains almost all elements of $f(X)$, then there exists a unique continuous algebra homomorphism $\varphi: \mathbb{F}_q\langle\langle X \rangle\rangle \rightarrow R$ extending f . This is by construction a profinite $\mathbb{F}_q$ -algebra, and in particular a complete Hausdorff compact ring. One can also check that it is local, since it is augmented and any element in the augmentation ideal $\ker(\mathbb{F}_q\langle\langle X \rangle\rangle \rightarrow \mathbb{F}_q)$ is topologically nil.

When X is finite, this ring is nothing but the $\mathbb{F}_q$ -algebra of non-commutative power series on X endowed with a suitable topology, which can be directly checked to satisfy the desired universal property. When X is infinite, its structure is more intricate. We are able to describe it more explicitly when X is countable:

Proposition 4.1.6. *Let $X = \{x_1, \dots, x_n, \dots\}$ be a countable set and $v: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 1}$ a function with finite fibers.¹⁶ Then, the free pro- p $\mathbb{F}_q$ -algebra $\mathbb{F}_q\langle\langle X \rangle\rangle$ is the completion of $\mathbb{F}_q\langle X \rangle$ under the*

¹³Ibid., Thm. 5.5.4.

¹⁴This matrix comes from a map $\mathbb{F}[x, y, z]^3 \rightarrow \mathbb{F}[x, y, z]^3$ appearing in the Koszul resolution of $\mathbb{F}$ as a $\mathbb{F}[x, y, z]$ -module, which is a minimal resolution. If it were not full, this resolution would not be minimal. One can also directly check that no factorization over $\mathbb{F}[x, y, z]$ exists.

¹⁵It is the inverse limit of the quotients of $\mathbb{F}_q\langle X \rangle$ by finite index ideals I containing some power of the augmentation ideal as well as almost all elements of X .

¹⁶For simplicity, one could take $v = \text{id}$. However, it will later be useful to have the proposition stated in this higher level of generality. Observe that this condition implies that $v^{-1}([1, k])$ is finite for every $k \in \mathbb{Z}$ and that v is unbounded.

positive, separated filtration where $F_i \mathbb{F}_q \langle X \rangle$ is the two-sided ideal of $\mathbb{F}_q \langle X \rangle$ generated by the set $\{x_{k_1} x_{k_2} \cdots x_{k_j} : \sum_{l=1}^j v(k_l) \geq i \text{ and } k_1, \dots, k_j \geq 1\}$ for every $i \geq 1$.

Proof. Let $R = \mathbb{F}_q \langle X \rangle$ with $F_i R$ the given filtration and $\tilde{R}$ its completion. As an $\mathbb{F}_q$ -vector space, $R/F_i R$ is spanned by all the j -fold monomials $x_{k_1} \cdots x_{k_j}$ where $\sum_{l=1}^j v(k_l)$ does not exceed i . There are finitely many such monomials, showing that $R/F_i R$ is finite and therefore $\hat{R}$ is profinite. It suffices then to prove that $\hat{R}$ satisfies the universal property of $\mathbb{F}_q \langle\langle X \rangle\rangle$. Hence, let $f: X \rightarrow S$ be a function to a profinite $\mathbb{F}_q$ -algebra such that $f(X)$ is topologically nilpotent and that any open ideal of S contains almost all elements of $f(X)$. By passing to the inverse limit, it suffices to consider the case of S finite and discrete. Observe that, by the universal property of R , we get the existence of a unique $\mathbb{F}_q$ -algebra morphism $\psi: R \rightarrow S$. Let $I = \ker(\psi)$.

In this case, there exists an $N > 0$ such that $f(x_i) = 0$ for all $i > N$, that is, $x_i \in I$ for all $i > N$. Since the non-zero elements of $f(X)$ are contained in $Y = \{f(x_1), \dots, f(x_N)\}$, all but finitely many monomials in the alphabet Y are not contained in I by Lemma 1.1.2. Hence, there exists some j for which $F_j R \leq I$. This shows that $\psi: R \rightarrow S$ factors through $\bar{\psi}: R/F_j R \rightarrow S$, and therefore we get an extension $\varphi: \hat{R} \rightarrow S$ of f by composing the natural projection $\hat{R} \rightarrow R/F_j R$ with $\bar{\psi}$. This extension φ is necessarily unique, since ψ is unique and the image of R is dense in $\hat{R}$. Hence, $\hat{R} \simeq \mathbb{F}_q \langle\langle X \rangle\rangle$. $\square$

The following result is well known, and we give a proof for completeness:¹⁷

Theorem 4.1.7 (Magnus isomorphism). *If F denotes the free pro- p group on the set X converging¹⁸ to 1, then the map*

$$\psi: \mathbb{F}_q \langle\langle X \rangle\rangle \rightarrow \mathbb{F}_q \llbracket F \rrbracket$$

extending $x \mapsto x - 1$ is a continuous isomorphism.

Proof. Observe that the completed group algebra $\mathbb{F}_q \llbracket F \rrbracket$ also has a universal property: any continuous group homomorphism $f: F \rightarrow R^\times$ from F to the units of a profinite $\mathbb{F}_q$ -algebra R extends uniquely to a continuous $\mathbb{F}_q$ -algebra homomorphism $\mathbb{F}_q \llbracket F \rrbracket \rightarrow R$. Since the elements x in $\mathbb{F}_q \langle\langle X \rangle\rangle$ are topologically nil, $1 + x$ is a unit in $\mathbb{F}_q \langle\langle X \rangle\rangle$ for every $x \in X$. Moreover, the sets $1 + U$ where U is an open two-sided ideal of $\mathbb{F}_q \langle\langle X \rangle\rangle$ contained in its augmentation ideal form a basis of neighborhoods of 1 in the pro- p group $1 + J(\mathbb{F}_q \langle\langle X \rangle\rangle) \leq (\mathbb{F}_q \langle\langle X \rangle\rangle)^\times$, so that the map $x \mapsto 1 + x$ converges to 1. Hence, this map extends to a continuous homomorphism $F \rightarrow (\mathbb{F}_q \langle\langle X \rangle\rangle)^\times$ and therefore to a continuous algebra homomorphism $\varphi: \mathbb{F}_q \llbracket F \rrbracket \rightarrow \mathbb{F}_q \langle\langle X \rangle\rangle$. It is now immediate to check that $\psi \circ \varphi$ and $\varphi \circ \psi$ are the identity maps when restricted to X , so by their universal properties they must be the identity maps themselves. $\square$

Corollary 4.1.8. *The free pro- p $\mathbb{F}_q$ -algebra $\mathbb{F}_q \langle\langle X \rangle\rangle$ is a Sylvester domain for any set X .*

¹⁷Lazard, “Groupes analytiques p -adiques”.

¹⁸This means that any open subgroup U of F contains almost all elements of X .

Therefore, there exists a universal epic division $\mathbb{F}_q\langle\langle X \rangle\rangle$ -ring of fractions $\mathcal{D}$. The explicit construction of $\mathcal{D}$ by Jaikin-Zapirain is the topic of Section 4.2. Let us now list some properties of Sylvester domains that were established in the foundational paper of Dicks and Sontag:

Proposition 4.1.9. *Let R be a ring.*

(1) *If R is a Sylvester domain, then R has weak global dimension¹⁹ at most two, every projective R -module is free and every flat R -module is a direct union of finitely generated free submodules.²⁰*

(2) *If R is an Ore domain, then R is a Sylvester domain if and only if it has weak global dimension at most 2 and every flat R -module is a direct union of finitely generated free submodules.²¹*

(3) *If R is a right or left coherent Ore domain, then R is a Sylvester domain if and only if it has weak global dimension at most 2 and every projective R -module is free. In particular, a coherent local commutative ring is a Sylvester domain if and only if it has weak global dimension at most 2.²²*

(4) *If R has weak global dimension at most 1, then R is a Sylvester domain if and only if R is a semifir. If moreover R is commutative of weak global dimension at most 1, then R is a Sylvester domain if and only if it is an integral domain and every projective R -module is free.²³*

The fact that Sylvester domains have weak global dimension at most 2 is another explanation for the fact that a polynomial algebra $\mathbb{F}[x, y, z]$ in three variables is not a Sylvester domain. It also allows us to classify completely which Iwasawa algebras are Sylvester domains.

Proposition 4.1.10. *Let G be a non-trivial torsion-free p -adic analytic profinite group.*

1. $\mathbb{Z}_p[[G]]$ is a Sylvester domain if and only if $G \simeq \mathbb{Z}_p$.

2. $\mathbb{F}_p[[G]]$ is a Sylvester domain if and only if $G \simeq \mathbb{Z}_p$ or $G \simeq \mathbb{Z}_p \rtimes \mathbb{Z}_p$.

Proof. Recall from Example 1.4.13 that $\mathbb{Z}_p[[G]]$ and $\mathbb{F}_p[[G]]$ are Noetherian domains. Moreover, since G is pro- p , they are local rings, so every projective module is free. Hence, by part (3) of Proposition 4.1.9, they are Sylvester domains if and only if they have weak global dimension at most two.

¹⁹Recall: this means that $\mathrm{Tor}_3^R(M, N) = 0$ for any R -modules M and N . Equivalently, it's the maximal length of a flat resolution of any module.

²⁰Dicks and Sontag, "Sylvester domains", Thm. 6.

²¹Ibid., Thm. 7.

²²Ibid., Thm. 10.

²³Ibid., Prop. 11.

For a Noetherian profinite ring, there is no distinction between its weak global dimension as an abstract ring and its global dimension in the category of profinite modules, since any module admits a resolution by finitely generated free modules. Hence, by Brumer's formula²⁴

$$\text{gl. dim}(R[[G]]) = \text{gl. dim}(R) + \sup_{p \in (R/J(R))^\times} \text{cd}_p(G),$$

we must have $\text{gl. dim}(\mathbb{Z}_p[[G]]) = 1 + \text{cd}(G)$ and $\text{gl. dim}(\mathbb{F}_p[[G]]) = \text{cd}(G)$ both bounded by two.

For the p -adic Iwasawa algebra $\mathbb{Z}_p[[G]]$, this means that $\text{cd}(G) = 1$, that is, G is a free pro- p group. The only free pro- p group that is also p -adic analytic is the infinite procyclic pro- p group $\mathbb{Z}_p$. For the mod- p Iwasawa algebra $\mathbb{F}_p[[G]]$, we can have either $\text{cd}(G) = 1$, in which case $G = \mathbb{Z}_p$, or $\text{cd}(G) = 2$, in which case G is a solvable Demushkin group²⁵. Those are known²⁶ to be the pro- p subgroups of $\text{Aut}(\mathbb{Z}_p) \simeq \mathbb{Z}_p \rtimes \mathbb{Z}_p^\times$, and are precisely the semidirect products $\mathbb{Z}_p \rtimes \mathbb{Z}_p$. $\square$

If R is a Sylvester domain, then there exists a *universal epimorphism* $\varphi: R \rightarrow \mathcal{D}$ to a division R -ring, that is, $\text{irk} = \varphi^\# \text{rk}_{\mathcal{D}}$. In other words, φ is an *universal envelope* for the inner rank irk in R . The homological properties of $\mathcal{D}$ as an R -module is also expressed in the following result:²⁷

Proposition 4.1.11. *Let R be a Sylvester domain and $\mathcal{D}$ its universal division R -ring of fractions. Then:*

(1) *For any left (resp. right) R -submodule M (resp. N) of a left (resp. right) $\mathcal{D}$ -module, we have that*

$$\text{Tor}_1^R(N, M) = 0.$$

(2) *For any finitely generated left or right R -submodule M of a left (resp. right) $\mathcal{D}$ -module and any exact sequence*

$$0 \rightarrow I \rightarrow R^n \rightarrow M \rightarrow 0,$$

I is a (set-theoretic) union of submodules isomorphic to R^k where

$$k = n - \dim M = \dim I,$$

where $\dim$ is the Sylvester module rank function associated to the inner rank in R .

²⁴Brumer, "Pseudocompact Algebras, Profinite Groups and Class Formations", Thm. 4.1.

²⁵A Demushkin group is a pro- p group whose $\mathbb{F}_p$ -cohomology satisfies Poincaré duality in dimension two. In other words, they are finitely presented one-relator groups G for which the cup product $H^1(G, \mathbb{F}_p) \times H^1(G, \mathbb{F}_p) \rightarrow H^2(G, \mathbb{F}_p)$ is a non-degenerate bilinear form.

²⁶Jean-Pierre Serre. *Galois Cohomology*. en. Springer Monographs in Mathematics. Berlin Heidelberg: Springer-Verlag, 1997. ISBN: 978-3-540-42192-4. DOI: 10.1007/978-3-642-59141-9, Exercises of Section I.4.5.

²⁷Jaikin-Zapirain, "An Explicit Construction of the Universal Division Ring of Fractions of $E\langle x_1, \dots, x_d \rangle$ ", Prop. 2.2.

When R is profinite, this can be slightly improved:²⁸

Proposition 4.1.12. *Let R be a profinite Sylvester domain and $\mathcal{D}$ its universal division R -ring of fractions. Then, for any finitely generated left or right R -submodule M of a $\mathcal{D}$ -module and any short exact sequence*

$$0 \rightarrow I \rightarrow R^n \rightarrow M \rightarrow 0,$$

the R -module I is free of finite rank. In particular, R is coherent.

Proof. We prove the claim for right modules, with the proof for left modules being equal *mutatis mutandis*. By Proposition 2.3.8 we can identify $\mathcal{D}$ with the matrix localization R_Σ of R at the prime matrix ideal of all non-full matrices. Since M is finitely generated, we can assume it is contained in a finitely generated $\mathcal{D}$ -module. By Corollary 2.4.2, M is a profinite R -module. Hence, the module I is closed and therefore profinite itself. By Proposition 4.1.11, the R -module I is a direct union of submodules I_i isomorphic to R^k . We claim that I itself is isomorphic to R^k .

Since R is a profinite domain, it must be a pro- p local ring for some prime p with maximal ideal $J = J(R)$ and finite residue field $R/J(R) = \mathbb{F}_q$ of characteristic p . As I is the direct union of submodules isomorphic to R^k , the $\mathbb{F}_q$ -vector space I/IJ must be isomorphic to the direct union of subspaces of dimension at most k . Therefore, $\dim_{\mathbb{F}_q} I/IJ \leq k$ and there exists i such that $I = I_i + IJ$. By Nakayama's lemma, $I = I_i$ is a free module of rank k . $\square$

Corollary 4.1.13. *Let $\mathbb{F}_q$ be a finite field of characteristic p and G a finitely generated pro- p group. If $\mathbb{F}_q[[G]]$ is a Sylvester domain, then G is finitely presented and $\text{cd}(G) \leq 2$. If $\mathcal{O}_K$ is the ring of integers of a finite extension K of $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$ and $\mathcal{O}_K[[G]]$ is a Sylvester domain then the group G is also a finitely presented pro- p group.*

Proof. For the first part, apply the Proposition 4.1.12 with M being the augmentation ideal of $\mathbb{F}_q[[G]]$ to obtain a finite free resolution

$$0 \rightarrow \mathbb{F}_q[[G]]^k \rightarrow \mathbb{F}_q[[G]]^n \rightarrow \mathbb{F}_q[[G]] \rightarrow \mathbb{F}_q \rightarrow 0$$

of the trivial G -module $\mathbb{F}_q$ of length two. Since $\mathbb{F}_q[[G]]$ is a finitely generated free $\mathbb{F}_p[[G]]$ -module, this is also a finite free resolution by $\mathbb{F}_p[[G]]$ -modules. Therefore, the cohomology groups

$$H^i(G, \mathbb{F}_p) = \text{Ext}_{\mathbb{F}_p[[G]]}^i(\mathbb{F}_p, \mathbb{F}_p)$$

are finite, showing that G is finitely presented, and moreover

$$H^i(G, M) = \text{Ext}_{\mathbb{F}_p[[G]]}^i(\mathbb{F}_p, M) = 0$$

for all discrete $\mathbb{F}_p[[G]]$ -modules M and all $i \geq 3$, showing that G has cohomological dimension at most 2.

²⁸Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Lem. 3.1.

For the second part, apply Proposition 4.1.12 with M being the kernel of the map $\mathcal{O}_K[[G]] \rightarrow \mathbb{F}_q$ to obtain a finite free resolution

$$0 \rightarrow \mathcal{O}_K[[G]]^k \rightarrow \mathcal{O}_K[[G]]^n \rightarrow \mathcal{O}_K[[G]] \rightarrow \mathbb{F}_q \rightarrow 0.$$

Since $\mathcal{O}_K[[G]]$ is a finitely generated free $\mathbb{Z}_p[[G]]$ -module and $\mathbb{F}_q \simeq \mathbb{F}_p^l$ as $\mathbb{Z}_p[[G]]$ -modules for $l = |\mathbb{F}_q : \mathbb{F}_p|$, we also obtain that the cohomology groups

$$H^i(G, \mathbb{F}_p^l) = \mathfrak{E}xt_{\mathbb{Z}_p[[G]]}^i(\mathbb{F}_p^l, \mathbb{F}_p)$$

are finite, showing that once again G is finitely presented. $\square$

Hence, if we are interested in the question of which completed group algebras $R[[G]]$ of finitely generated pro- p groups G are Sylvester domains, it suffices to consider the group algebras listed by Theorem 2.3.17. As of the writing of this thesis, the following three cases remain undecided:

1. G is the trivial group and R is a coherent but non-Noetherian profinite local domain of global dimension 2. It is not known if such rings R exist, see Question 2.3.18
2. G is a finitely generated free pro- p group and $R = \mathcal{O}_K$ is the ring of integers of some finite extension K of $\mathbb{Q}_p$. An argument similar to the proof of Theorem 4.1.7 shows that $R[[G]]$ is isomorphic to the ring of non-commutative power series $\mathcal{O}_K\langle\langle X \rangle\rangle$ where X is any basis of G .
3. $R = \mathbb{F}_q$ and G is a finitely presented pro- p group of cohomological dimension 2.

We will finish this section by stating the following theorem of Jaikin-Zapirain, which is a partial converse to Proposition 4.1.9 and serves as a homological criterion to test whether or not the pullback of a rank function induces the inner rank.²⁹

Theorem 4.1.14. *Let R be a ring and rk by a Sylvester matrix rank function with an envelope $\varphi: R \rightarrow \mathcal{U}$ which is a von Neumann regular ring.³⁰ Assume that:*

1. $\text{Tor}_1^R(\mathcal{U}, \mathcal{U}) = 0$, and
2. *for any finitely generated left or right R -submodule M of $\mathcal{U}$ and any exact sequence*

$$0 \rightarrow I \rightarrow R^n \rightarrow M \rightarrow 0,$$

I is a free R -module.

Then, $\text{irk}_R = \text{rk}$. In particular, R is a Sylvester domain and the rational closure of $\varphi(R)$ in $\mathcal{U}$ is R -isomorphic to the universal division ring of fractions of R .

²⁹Jaikin-Zapirain, “An Explicit Construction of the Universal Division Ring of Fractions of $E\langle\langle x_1, \dots, x_d \rangle\rangle$ ”, Cor. 2.5.

³⁰A ring is von Neumann regular if for every non-zero x there exists y such that $xyx = x$. In particular, every division ring is von Neumann regular, with $y = x^{-1}$.

4.2 A universal embedding for free pro- p $\mathbb{F}_q$ -algebras

Although we already know that the free pro- p $\mathbb{F}_q$ -algebra $R = \mathbb{F}_q\langle\langle X \rangle\rangle$ is a Sylvester domain, we shall later need an explicit embedding of R into its universal epic division R -ring of fractions. We shall follow the arguments of Jaikin-Zapirain,³¹ who proved it for finite X .

We fix the following notation for the remainder of this section: $X = \{x_1, x_2, \dots\}$ is a countable set, $v: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 1}$ is a function with finite fibers, $R = \mathbb{F}_q\langle X \rangle$ is the free $\mathbb{F}_q$ -algebra on X and $F_i R$ is the positive, separated filtration on R such that $F_i R$ is the two sided ideal generated by

$$\{x_{k_1} x_{k_2} \dots x_{k_j} : \sum_{l=1}^j v(k_l) \geq i \text{ and } k_1, \dots, k_j \geq 1\}$$

for each $i \geq 1$. By Proposition 4.1.6, we have a topological isomorphism $\widehat{R} \simeq \mathbb{F}_q\langle\langle X \rangle\rangle$. We denote by $\mathcal{Q}$ the universal division R -ring of fractions of R .

Proposition 4.2.1. *Let $f: X \rightarrow \mathcal{Q}[t]$ be the map $f(x_i) = x_i t^{v(i)}$. Then, f extends to a strict filtered ring homomorphism $\psi: \widehat{R} \rightarrow \mathcal{Q}[[t]]$ under the t -adic filtration of $\mathcal{Q}[[t]]$.*

Proof. Endow $\mathcal{Q}[t]$ with the t -adic filtration, so that $\mathcal{Q}[[t]]$ is its completion. To obtain the existence of ψ it suffices to show that the induced map $\psi_0: R \rightarrow \mathcal{Q}[t]$ is strict filtered by Lemma 1.3.4 and Proposition 4.1.6. This follows immediately from

$$\psi_0(x_{k_1} \dots x_{k_j}) = x_{k_1} \dots x_{k_j} t^{v(k_1) + \dots + v(k_j)} \in F_{v(k_1) + \dots + v(k_j)} \mathcal{Q}[t]$$

and the fact that any element in $\psi_0(R) \subset \mathbb{F}_q\langle X \rangle[t]$ can be uniquely expressed as a finite sum of monomials in X times a power of t . $\square$

We let $\mathcal{D} = \mathcal{Q}((t))$ be the division ring of formal Laurent series over $\mathcal{Q}$, endowed with a filtration $F_i \mathcal{D} = t^i \mathcal{Q}[[t]]$ for every $i \in \mathbb{Z}$. The inclusion map $\iota: \mathcal{Q}[[t]] \rightarrow \mathcal{D}$ is then a strict filtered injective map. We define $\varphi: \widehat{R} \rightarrow \mathcal{D}$ to be the composition of ψ with ι . It is clear that φ is a strict filtered ring homomorphism.³² The main result we want to prove in this section is:³³

Theorem 4.2.2. *We have $\varphi^\# \text{rk}_{\mathcal{D}} = \text{irk}$. In particular, the envelope φ is universal.*

³¹Jaikin-Zapirain, “An Explicit Construction of the Universal Division Ring of Fractions of $\mathbb{F}_q\langle\langle X \rangle\rangle$ ”.

³²Since ι is injective, we have

$$\varphi(\widehat{R}) \cap F_i \mathcal{D} = \iota(\psi(\widehat{R})) \cap \iota(F_i \mathcal{Q}[[t]]) = \iota(\psi(\widehat{R}) \cap F_i \mathcal{Q}[[t]]) = \varphi(F_i \widehat{R}).$$

³³Jaikin-Zapirain, “An Explicit Construction of the Universal Division Ring of Fractions of $\mathbb{F}_q\langle\langle X \rangle\rangle$ ”, Thm. 1.3.

We will do so by using Jaikin's criterion of Theorem 4.1.14 together with the methods developed in Section 1.4 to see that we satisfy the necessary homological hypothesis. We start by studying the associated graded rings:

Proposition 4.2.3. *We have $\text{Gr } \mathcal{D} = \mathbb{Q}[\bar{t}, \bar{t}^{-1}]$ as graded rings, which is a principal ideal domain, where $\bar{t}$ is the principal symbol of t . If $\bar{X} = \{\bar{x}_1, \bar{x}_2, \dots\}$ denotes the set of principal symbols of each x_i in $\text{Gr } \hat{R}$, then we have $\text{Gr } \hat{R} \simeq \mathbb{F}_q\langle\bar{X}\rangle$ as graded rings where the homogeneous elements of degree i is given by the linear $\mathbb{F}_q$ -span of the monomials $\bar{x}_{k_1} \cdots \bar{x}_{k_j}$ where $v(k_1) + \dots + v(k_j) = i$. The graded map $\text{Gr } \varphi: \mathbb{F}_q\langle\bar{X}\rangle \rightarrow \text{Gr } \mathcal{D}$ is given by $\bar{x}_i \mapsto x_i \bar{t}^{v(i)}$, and the composition with the embedding of $\mathbb{Q}[\bar{t}, \bar{t}^{-1}]$ in its Ore division ring of fractions is universal.*

Proof. The equality $\text{Gr } \mathcal{D} = \mathbb{Q}[\bar{t}, \bar{t}^{-1}]$ is clear since multiplication by t is an isomorphism between $F_i \mathcal{D} / F_{i+1} \mathcal{D} \simeq \mathbb{Q}$ and $F_{i+1} \mathcal{D} / F_{i+2} \mathcal{D} \simeq \mathbb{Q}$ for all $i \in \mathbb{Z}$. For $\text{Gr } \hat{R}$, observe that it is isomorphic as a graded ring to $\text{Gr } R$ by Proposition 1.4.2. There is a graded surjective homomorphism $\mathbb{F}_q\langle\bar{X}\rangle \rightarrow \text{Gr } R$ by the universal property of the latter under the grading of the statement. If it were not injective, for some i there would be a non-zero homogeneous element

$$\bar{x} = \sum_{\substack{n \leq i \\ v(k_1) + \dots + v(k_n) = i}} a_{k_1, \dots, k_n} \bar{x}_{k_1} \cdots \bar{x}_{k_n}$$

in $\mathbb{F}_q\langle\bar{X}\rangle$ for which

$$x = \sum_{\substack{n \leq i \\ v(k_1) + \dots + v(k_n) = i}} a_{k_1, \dots, k_n} x_{k_1} \cdots x_{k_n} \in F_{i+1} R.$$

Since any element of R has a unique expression as a non-commutative monomial in X , this is a contradiction. From this, we get the desired graded isomorphism. To see that the given composition is universal, let S be the Ore set of non-zero elements of $\mathbb{Q}[\bar{t}, \bar{t}^{-1}]$. It suffices to see that there exists a specialization from $S^{-1}\mathbb{Q}[\bar{t}, \bar{t}^{-1}]$ to $\mathbb{Q}$ as $\text{Gr } R$ -division rings. Inside $\mathbb{Q}[\bar{t}]$, the set of all polynomials that are not multiples of the central polynomial $t - 1$ form a two-sided Ore set, and its Ore localization is the desired local subring of $S^{-1}\mathbb{Q}[\bar{t}, \bar{t}^{-1}]$. $\square$

By construction, $\hat{R}$ is a separated, complete and bounded ring. Hence, any finitely generated profinite $\hat{R}$ -module M admits a complete, separated, good filtration, which *a fortiori* must be bounded. In this case, Proposition 1.4.15 tells us that there exists semicompact weighted sets X_i , $i \geq 0$, and a strict exact sequence

$$\dots \rightarrow \widehat{R(X_2)} \rightarrow \widehat{R(X_1)} \rightarrow \widehat{R(X_0)} \rightarrow M \rightarrow 0.$$

We shall improve this by making it finite:³⁴

³⁴Jaikin-Zapirain, "An Explicit Construction of the Universal Division Ring of Fractions of $E\langle\langle x_1, \dots, x_d \rangle\rangle$ ", Prop. 4.11.

Proposition 4.2.4. *Let V be a semicompact weighted set. Then, any closed filtered submodule of $\widehat{R(V)}$ is isomorphic as a filtered module to $\widehat{R(U)}$ for some semicompact weighted set U .*

Proof. Let N be a closed filtered submodule of $\widehat{R(V)}$, so that N is also a separated, complete and bounded R -module. From Proposition 1.4.8, the induced map $\text{Gr } N \rightarrow (\text{Gr } \widehat{R})[\overline{V}]$ is injective. Since $\text{Gr } \widehat{R} \simeq \mathbb{F}_q\langle \overline{X} \rangle$ by Proposition 4.2.3 and this is a fir, we know that $\text{Gr } N$ must be free as an abstract $\text{Gr } \widehat{R}$ -module. By Lemma 1.4.14, it has a basis $\overline{U}$ of homogeneous elements. As the ring $\widehat{R}$ is bounded and V is semicompact, each homogeneous component of $(\text{Gr } \widehat{R})[\overline{V}]$ is finite dimensional over $\mathbb{F}_q$, and hence U contains only finitely many elements in each given degree. This means that we can lift $\overline{U}$ to a semicompact weighted subset U of N . By Proposition 1.4.8 once again, the map $\widehat{R(U)} \rightarrow N$ must be an isomorphism. $\square$

Corollary 4.2.5. *Let M be a complete, separated and bounded $\widehat{R}$ -module. Then, there exists semicompact weighted sets Y_0, Y_1 and a strict short exact sequence*

$$0 \rightarrow \widehat{R(Y_1)} \rightarrow \widehat{R(Y_0)} \rightarrow M \rightarrow 0.$$

The last missing ingredient for the proof of Theorem 4.2.2 is the following lemma.³⁵

Lemma 4.2.6. *Let M be a finitely generated right or left $\widehat{R}$ -submodule of $\mathcal{D}$. Then, for any exact sequence*

$$0 \rightarrow I \rightarrow \widehat{R}^d \rightarrow M \rightarrow 0$$

of R -modules, I is free of finite rank.

Proof. Again, we prove the claim for right modules, with the left proof being symmetric. Since the inclusion $\widehat{R} \rightarrow \mathcal{D}$ is strict, it is continuous with respect to the topology induced by the filtrations. As the ring $\widehat{R}$ is profinite in said topology and $\mathcal{D}$ is Hausdorff, any finitely generated $\widehat{R}$ -submodule M of $\mathcal{D}$ is closed in $\mathcal{D}$, with a profinite topology induced from the filtration. In particular: (1) M is complete, bounded and separated as a filtered $\widehat{R}$, where the filtration is induced from $\mathcal{D}$, (2) I is a closed and hence profinite $\widehat{R}$ -submodule of $\widehat{R}^d$.

Let

$$0 \rightarrow \widehat{R(Y_1)} \xrightarrow{\alpha} \widehat{R(Y_0)} \xrightarrow{\beta} M \rightarrow 0$$

be the strict short exact sequence from Corollary 4.2.5 and consider now the filtered sequence

$$0 \rightarrow \widehat{R(Y_1)} \hat{\otimes}_{\widehat{R}} \mathbb{Q}[[t]] \xrightarrow{\alpha \hat{\otimes} \text{id}} \widehat{R(Y_0)} \hat{\otimes}_{\widehat{R}} \mathbb{Q}[[t]] \xrightarrow{\beta \hat{\otimes} \text{id}} M \hat{\otimes}_{\widehat{R}} \mathbb{Q}[[t]] \rightarrow 0 \quad (4.1)$$

³⁵Ibid., Cor. 4.15.

of right, complete, separated $\mathcal{Q}[[t]]$ -modules. Observe that $\widehat{R(Y_k)} \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]] \simeq \widehat{\mathcal{Q}[[t]](Y_k)}$ as a filtered right $\mathcal{Q}[[t]]$ -modules. Upon passing to the graded ring, we get a commutative diagram

$$\begin{array}{ccccc} (\mathrm{Gr} \widehat{R})[\overline{Y_1}] \otimes_{\mathrm{Gr} \widehat{R}} \mathcal{Q}[t] & \xrightarrow{\mathrm{Gr}(\alpha) \otimes \mathrm{id}} & (\mathrm{Gr} \widehat{R})[\overline{Y_0}] \otimes_{\mathrm{Gr} \widehat{R}} \mathcal{Q}[t] & \xrightarrow{\mathrm{Gr}(\beta) \otimes \mathrm{id}} & \mathrm{Gr} M \otimes_{\mathrm{Gr} \widehat{R}} \mathcal{Q}[t] \\ \downarrow \gamma & & \downarrow \gamma' & & \downarrow \gamma'' \\ \mathcal{Q}[t][\overline{Y_1}] & \xrightarrow{\mathrm{Gr}(\alpha \hat{\otimes} \mathrm{id})} & \mathcal{Q}[t][\overline{Y_0}] & \xrightarrow{\mathrm{Gr}(\beta \hat{\otimes} \mathrm{id})} & \mathrm{Gr}(M \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]]) \end{array}$$

of right graded $\mathcal{Q}[t]$ -modules. It is clear that γ and γ' are isomorphisms, and that γ'' is surjective. On the top row, $\mathrm{Gr}(\beta) \otimes \mathrm{id}$ is surjective and $\ker(\mathrm{Gr}(\beta) \otimes \mathrm{id}) = \mathrm{im}(\mathrm{Gr}(\alpha) \otimes \mathrm{id})$ by the fact that the tensor product is right exact. The injectivity of $\mathrm{Gr}(\alpha) \otimes \mathrm{id}$ follows from the fact that

$$\mathrm{Tor}_1^{\mathrm{Gr} \widehat{R}}(M, \mathcal{Q}[t]) = 0$$

by part (1) Proposition 4.1.11. Hence, a diagram chase shows that $\mathrm{Gr}(\alpha \hat{\otimes} \mathrm{id})$ is injective and $\mathrm{Gr}(\beta \hat{\otimes} \mathrm{id})$ is surjective. By parts (2) and (3) of Corollary 1.4.9, $\alpha \hat{\otimes} \mathrm{id}$ is a strict injection and $\beta \hat{\otimes} \mathrm{id}$ is a strict surjection.

To see that the sequence (4.1) is exact at $\widehat{R(Y_0)} \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]]$, that is, $\ker \beta = \mathrm{im}(\alpha)$, observe that it is the inverse limit of the exact sequence of surjective inverse systems

$$\begin{aligned} \widehat{R(Y_1)} / \alpha^{-1}(F_i \widehat{R(Y_0)}) \otimes_{\widehat{R}} \mathcal{Q}[[t]] / (t^i) &\rightarrow \widehat{R(Y_0)} / F_i \widehat{R(Y_0)} \otimes_{\widehat{R}} \mathcal{Q}[[t]] / (t^i) \\ &\rightarrow M / \beta(F_i \widehat{R(Y_0)}) \otimes_{\widehat{R}} \mathcal{Q}[[t]] / (t^i), \end{aligned}$$

This implies for every $x \in \ker(\beta \hat{\otimes} \mathrm{id})$ the existence of a sequence y_i in $\widehat{R(Y_1)} \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]]$ such that $\lim \alpha(y_i) = x$. Since the filtrations on $\widehat{R(Y_k)} \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]]$ are bounded, completed and separated, topologically α is a homeomorphism into its image by Lemma 1.1.3. Hence, $y = \lim y_i$ exists and we have $\alpha(y) = x$.

Therefore, the sequence (4.1) is strict exact. Since M is finitely generated, we have a surjection

$$\mathcal{Q}[[t]]^d \rightarrow M \otimes_{\widehat{R}} \mathcal{Q}[[t]] \rightarrow 0$$

and therefore the latter is complete and finitely generated (although possibly not separated). Hence, the map

$$M \otimes_{\widehat{R}} \mathcal{Q}[[t]] \rightarrow M \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]]$$

is surjective, showing that the latter is also finitely generated as a $\mathcal{Q}[[t]]$ -module. But $\mathcal{Q}[[t]]$ is a principal ideal domain³⁶, an in particular Noetherian. This implies that the Tor group $\mathrm{Tor}_1^{\mathcal{Q}[[t]]}(M \hat{\otimes}_{\widehat{R}} \mathcal{Q}[[t]], \mathcal{Q})$ is finite dimensional over $\mathcal{Q}$.

³⁶Any power series is t^k times an invertible power series.

The finitely generated $\mathbb{Q}[[t]]$ -module $\mathbb{Q}$ is automatically of type FP_∞ . Since the sequence (4.1) is strict exact, by Theorem 1.4.17 we know that $\mathrm{Tor}_1^{\mathbb{Q}[[t]]}(M \hat{\otimes}_{\hat{R}} \mathbb{Q}[[t]], \mathbb{Q})$ can be obtained by applying $\hat{\otimes}_{\mathbb{Q}[[t]]} \mathbb{Q}$ to the sequence (4.1). Since we have a natural isomorphism of functors

$$-\hat{\otimes}_{\hat{R}} \mathbb{Q}[[t]] \hat{\otimes}_{\mathbb{Q}[[t]]} \mathbb{Q} \simeq -\hat{\otimes}_{\hat{R}} \mathbb{Q},$$

we can compute these Tor groups as the homology of the complex

$$0 \rightarrow \widehat{\mathbb{Q}(Y_1)} \xrightarrow{\alpha_{\mathbb{Q}}} \widehat{\mathbb{Q}(Y_0)} \rightarrow 0.$$

This means precisely that we have an exact sequence

$$0 \rightarrow \mathrm{Tor}_1^{\mathbb{Q}[[t]]}(M \hat{\otimes}_{\hat{R}} \mathbb{Q}[[t]], \mathbb{Q}) \rightarrow \widehat{\mathbb{Q}(Y_1)} \xrightarrow{\alpha_{\mathbb{Q}}} \widehat{\mathbb{Q}(Y_0)} \rightarrow M \hat{\otimes}_{\hat{R}} \mathbb{Q}[[t]] \hat{\otimes}_{\mathbb{Q}[[t]]} \mathbb{Q} \rightarrow 0.$$

On the other hand, the sequence (4.1) is a free resolution of M in the category of profinite $\hat{R}$ -modules. Hence, taking the completed tensor product with $\mathbb{F}_q$ over $\hat{R}$ we get a long exact sequence:

$$0 \longrightarrow \mathfrak{Tor}_1^{\hat{R}}(M, \mathbb{F}_q) \longrightarrow \widehat{\mathbb{F}_q(Y_1)} \xrightarrow{\alpha_{\mathbb{F}_q}} \widehat{\mathbb{F}_q(Y_0)} \longrightarrow M \hat{\otimes}_{\hat{R}} \mathbb{F}_q \longrightarrow 0$$

Observe that, as an $\mathbb{F}_q$ -vector space, we have a retraction $\pi: \mathbb{Q} \rightarrow \mathbb{F}_q$ of the inclusion $\iota: \mathbb{F}_q \rightarrow \mathbb{Q}$. Moreover, for any non-zero element $x \in \mathbb{Q}$, this retraction can be arranged so that $\pi(x) \neq 0$. As $\alpha_{\mathbb{Q}}$ is the map induced from $\alpha_{\mathbb{F}_q}$ through the complete tensor product $\widehat{\mathbb{F}_q(Y_1)} \hat{\otimes}_{\mathbb{F}_q} \mathbb{Q} \xrightarrow{\alpha \hat{\otimes} \mathrm{id}} \widehat{\mathbb{F}_q(Y_0)} \hat{\otimes}_{\mathbb{F}_q} \mathbb{Q}$, we get the commutative diagram

$$\begin{array}{ccc} \widehat{\mathbb{Q}(Y_1)} & \xrightarrow{\alpha_{\mathbb{Q}}} & \widehat{\mathbb{Q}(Y_0)} \\ \iota(Y_1) \uparrow & & \uparrow \iota(Y_0) \\ \widehat{\mathbb{F}_q(Y_1)} & \xrightarrow{\alpha_{\mathbb{F}_q}} & \widehat{\mathbb{F}_q(Y_0)} \\ \pi(Y_1) \downarrow & & \downarrow \pi(Y_0) \end{array}$$

where $\iota(Y_i)$ and $\pi(Y_i)$ are induced from ι and π by the universal property of the completion and weighted sum constructions, and moreover $\pi(Y_i) \circ \iota(Y_i) = \mathrm{id}$.

We now claim that this implies

$$\dim_{\mathbb{F}_q} \mathrm{Tor}_1^{\hat{R}}(M, \mathbb{F}_q) = \dim_{\mathbb{F}_q} \ker(\alpha_{\mathbb{F}_q}) \leq \dim_{\mathbb{Q}} \ker(\alpha_{\mathbb{Q}}) = \dim_{\mathbb{Q}} \mathrm{Tor}_1^{\mathbb{Q}[[t]]}(M \hat{\otimes}_{\hat{R}} \mathbb{Q}[[t]], \mathbb{Q}) < \infty.$$

Indeed, if $v_1, \dots, v_k$ are $\mathbb{F}_q$ -linearly independent elements in $\ker(\alpha_{\mathbb{F}_q})$, then the elements

$$\widehat{\iota(Y_1)}(v_1), \dots, \widehat{\iota(Y_1)}(v_k)$$

must be $\mathbb{Q}$ -linearly independent elements in $\ker(\alpha_{\mathbb{Q}})$. Otherwise, there are elements $q_1, \dots, q_k \in \mathbb{Q}$ not all zero such that

$$q_1 \widehat{\iota(Y_1)}(v_1) + \dots + q_k \widehat{\iota(Y_1)}(v_k) = 0,$$

and choosing the retraction $\pi: \mathbb{Q} \rightarrow \mathbb{F}_q$ such that $\pi(q_1), \dots, \pi(q_k)$ are not all zero we get a linear dependence

$$\pi(q_1)v_1 + \dots + \pi(q_k)v_k = 0$$

in $\ker(\alpha_{\mathbb{F}_q})$.

However, we also have the long exact sequence

$$0 \rightarrow \mathfrak{Tor}_1^{\widehat{R}}(M, \mathbb{F}_q) \rightarrow I \hat{\otimes}_{\widehat{R}} \mathbb{F}_q \rightarrow \mathbb{F}_q^d \rightarrow M \hat{\otimes}_{\widehat{R}} \mathbb{F}_q \rightarrow 0,$$

which implies that I is a finitely generated $\widehat{R}$ -module by Nakayama's lemma. Since $\widehat{R}$ has global dimension one in the category of profinite modules, I must be free of finite rank. $\square$

We are now ready to prove:

Proof of Theorem:4.2.2. We want to apply Theorem 4.1.14 to the embedding $\varphi: \widehat{R} \rightarrow \mathcal{D}$ of Proposition 4.2.1. The second part of the criteria is satisfied by Lemma 4.2.6, so it remains to prove that $\mathfrak{Tor}_1^{\widehat{R}}(\mathcal{D}, \mathcal{D})$ vanishes. Since the Tor functor commutes with direct limits, it suffices to prove that if M and N are finitely generated right and left $\widehat{R}$ -submodules of $\mathcal{D}$ respectively, then

$$\mathfrak{Tor}_1^{\widehat{R}}(M, N) = 0.$$

Observe that M and N are both of type FP_2 by Lemma 4.2.6, and under the induced filtration from $\mathcal{D}$ are complete, separated and bounded $\widehat{R}$ -modules.³⁷ Since $\text{Gr } M$ and $\text{Gr } N$ are submodules of $\text{Gr } \mathcal{D}$, by part (1) of Proposition 4.1.11 we must have

$$\mathfrak{Tor}_1^{\text{Gr } \widehat{R}}(\text{Gr } M, \text{Gr } N) = 0.$$

Hence, the desired Tor vanishes by Theorem 1.4.18. $\square$

³⁷They are separated because the filtration on $\mathcal{D}$ is separated, and since they are profinite this filtration must be complete as they are closed in $\mathcal{D}$ and bounded as they are compact.

4.3 Pro- p Sylvester domains under commensurability

The inner rank function does not behave functorially under the change of rings. If $\varphi: R \rightarrow S$ is a ring homomorphism, the best one can say is $\varphi^\# \text{irk}_S \leq \text{irk}_R$, even if φ is injective, epic or both.³⁸ One can also directly see that the Sylvester condition, in general, does not pass to subrings or overrings.

Example 4.3.1. The ring $\mathbb{Q}[x, y]$ is a Sylvester domain, but its subring $\mathbb{Z}[x, y]$ is not, since it has a non-vanishing Tor-group in dimension three:³⁹

$$\text{Tor}_3^{\mathbb{Z}[x, y]}(\mathbb{F}_2, \mathbb{F}_2) \simeq \mathbb{F}_2 \neq 0.$$

In the same spirit, $\mathbb{Z}[x]$ is a Sylvester domain, but its overring $\mathbb{Z}[x, y]$ is not.

For a more striking example, consider the subring

$$\mathbb{Z}[2i] = \{a + 2bi \in \mathbb{C} : a, b \in \mathbb{Z}\}$$

of the complex numbers. It is a finitely generated free $\mathbb{Z}$ -module, and has no zero divisors. However, it is of infinite weak global dimension.⁴⁰ Hence, it is not a Sylvester domain. On the other hand, the overring $\mathbb{Z}[i]$ is a Sylvester domain: it is a projective-free Noetherian domain of weak global dimension 1, since it is a principal ideal domain. This shows that the Sylvester condition in general does not go up or down. ■

Example 4.3.2. If G is a finitely generated pro- p group, U is an open subgroup of G , $\mathbb{F}_q$ is a finite field of characteristic p and $\mathbb{F}_q[[U]]$ is a Sylvester domain, then the inclusion of completed group algebras $\varphi: \mathbb{F}_q[[U]] \rightarrow \mathbb{F}_q[[G]]$ preserves the inner rank. It suffices to prove this for $|G:U| = p$, in which case U is normal in G . Identifying $\mathbb{F}_q[[G]]$ with the free right $\mathbb{F}_q[[U]]$ -module of rank p , we have a map $\psi: \mathbb{F}_q[[G]] \rightarrow \text{Mat}_{p \times p}(\mathbb{F}_q[[U]])$ given by left multiplication. If A is a full matrix over $\mathbb{F}_q[[U]]$, then $\psi(\varphi(A))$ is the diagonal sum of the conjugates A^g as g runs through a set of representatives for the cosets G/U . Since each A^g must also be full and the set of full matrices is closed under diagonal sums⁴¹ on $\mathbb{F}_q[[U]]$, $\psi(\varphi(A))$ must be a full matrix.

³⁸For an injective example, consider $\mathcal{M}$ the $\mathbb{F}_q$ -vector space on an infinite countable basis and the embedding $\varphi: \mathbb{F}_q \rightarrow S = \text{End}_{\mathbb{F}_q}(\mathcal{M})$ and observe that $\varphi^\# \text{irk}(\text{id}_n) = 1$ for all identity matrices, since $S^n \simeq S$ as S -modules for all $n > 0$. For an epic example, one could take the matrix $\text{id}_n \oplus p \text{id}_m$ and $\varphi: \mathbb{Z} \rightarrow \mathbb{F}_p$ the reduction mod p . For an injective epic example, recall that there are Ore domains that do not admit universal division rings, see Passman, “Universal fields of fractions for polycyclic group algebras”

³⁹One can compute this Tor group using the Koszul resolution

$$0 \rightarrow \mathbb{Z}[x, y] \xrightarrow{\alpha} \mathbb{Z}[x, y]^3 \xrightarrow{\beta} \mathbb{Z}[x, y]^3 \xrightarrow{\gamma} \mathbb{Z}[x, y] \rightarrow \mathbb{F}_2 \rightarrow 0$$

where $\gamma(a, b, c) = 2a + xb + yc$, $\beta(a, b, c) = (-xa - yb, 2a - yc, 2b + xc)$ and $\alpha(a) = (ya, -xa, 2a)$.

⁴⁰Observe that over Noetherian rings, weak global dimension and global dimension coincide. If $S = \mathbb{Z}[2i]$ had finite global dimension, then so would any localization of S at a maximal ideal by the flatness of localization. By the Auslander-Buchsbaum-Serre Theorem, Noetherian local rings with finite global dimension must be regular. However, the localization of S at $\mathfrak{m} = (2, 2i)$ is not regular, since the Krull dimension of $S_{\mathfrak{m}}$ is one, being an integral extension of $\mathbb{Z}_{(p)}$, its residue field is $\mathbb{F}_2$ but $\mathfrak{m}$ does not localize to a principal ideal.

⁴¹This follows immediately from the axioms of its complement being a prime matrix ideal.

Hence, $\varphi(A)$ is also a full matrix. Essentially the same argument shows that if $\mathbb{F}_q \leq \mathbb{F}_{q'}$ is a finite field extension and $\mathbb{F}_q[[G]]$ is a Sylvester domain, then the inclusion $\mathbb{F}_q[[G]] \rightarrow \mathbb{F}_{q'}[[G]]$ preserves the inner rank. ■

Although said examples illustrate that one should be careful when trying to pass the Sylvester condition to subrings or overrings, no such examples of bad behaviour are known for group algebras, specially completed group algebras. In the general case, the following questions remain largely open:

Question 4.3.3. Let $\mathbb{F}_q$ be a finite field of characteristic p , G a pro- p group and H a closed subgroup of G . Does the inclusion map $\mathbb{F}_q[[H]] \rightarrow \mathbb{F}_q[[G]]$ preserve the inner rank?

Question 4.3.4. Let $\mathbb{F}$ be a field, G be a torsion-free group and H a subgroup of finite index in G . If $\mathbb{F}[G]$ is a Sylvester domain, must $\mathbb{F}[H]$ be a Sylvester domain? If $\mathbb{F}[H]$ is a Sylvester domain, does the same hold for $\mathbb{F}[G]$?

One surprising result is that the latter question has a positive answer for *completed group algebras*.⁴² Proving this is the main objective of this section. We start by recalling how the group ring of a group extension can be expressed as a *crossed product* of rings.

Definition 4.3.5. Let R be a ring and G be a group. We say that S is a *crossed product* of R and G if there exists a decomposition $S = \bigoplus_{g \in G} S_g$ of abelian groups such that for every $g, h \in G$ we have $S_g S_h \subseteq S_{gh}$, $S_1 = R$ and for every $g \in G$ there exists a unit $u_g \in S_g$. We will write $S \simeq R * G$.

It is clear that the multiplication on $S = R * G$ is uniquely determined by the rules

$$u_{g_1} u_{g_2} = u_{g_1 g_2} \tau(g_1, g_2), \quad r u_{g_1} = u_{g_1} \sigma(g_1)(r),$$

where $\tau: G \times G \rightarrow R^\times$ and $\sigma: G \rightarrow \text{Aut}(R)$ are functions satisfying the identities

$$\begin{aligned} \tau(g_1 g_2, g_3) \sigma(g_3)(\tau(g_1, g_2)) &= \tau(g_1, g_2 g_3) \tau(g_2, g_3), \\ \sigma(g_2)(\sigma(g_1)(r)) &= \tau(g_1, g_2)^{-1} \sigma(g_1 g_2)(r) \tau(g_1, g_2), \end{aligned} \quad (4.2)$$

for every $g_1, g_2, g_3 \in G$ and $r \in R$: this follows directly from the associativity of the product in S .⁴³ Since the u_g are invertible in S , $\{u_g: g \in G\}$ is a free basis of S as a left and as a right R -module.

We can see any group algebra $R[G]$ as a crossed product $R * G$ where τ is the constant map to 1 and σ is the constant map to the identity. More generally, if N is a normal subgroup of G , we always have

$$R * G \simeq (R * N) * (G/N).$$

⁴²Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Thm. 3.5 and Cor. 3.6.

⁴³Donald S Passman. *Infinite crossed products*. Dover Publications, July 2013, Lem. 1.1.

Let R be a profinite ring, G be a profinite group, U a normal open subgroup of G and T a transversal of U in G . Since⁴⁴ $R[[G]] = \bigoplus_{t \in T} R[[U]]t$, we obtain that $R[[G]] \simeq R[[U]] * G/U$. For every $g \in G$, let $\bar{g}$ be its representative in T . For any $t \in T$, we put $u_{tU} = t$. In this case, the maps τ and σ can be explicitly described as follows: $\sigma(g_1 U)$ is conjugation by $\bar{g}_1$ and

$$\tau(g_1 U, g_2 U) = (\overline{g_1 g_2})^{-1} \cdot \bar{g}_1 \cdot \bar{g}_2.$$

Recall that if $R[[G]]$ has no zero divisors, then R and G must both be pro- p . If moreover G is finitely generated and it is a Sylvester domain, by Corollary 4.1.13 either G is free pro- p , a property which is a commensurability invariant for torsion-free pro- p groups, or R is a finite field $\mathbb{F}_q$ of characteristic p . Hence, it suffices to study how the Sylvester property behaves under torsion-free finite group extensions for completed group algebras over $\mathbb{F}_q$.⁴⁵

Proposition 4.3.6. *Let G be a pro- p group, U a normal open subgroup of G . Suppose that $\mathbb{F}_q[[U]]$ has a universal division ring of fractions $\mathbb{Q}$. Then:*

- (a) *The conjugation action of G on $\mathbb{F}_q[[U]]$ extends to a homomorphism $G \rightarrow \text{Aut}(\mathbb{Q})$.*
- (b) *For a fixed transversal T of G/U , the maps from the crossed product decomposition $\mathbb{F}_q[[G]] \simeq \mathbb{F}_q[[U]] * G/U$ can be extended to maps $\sigma: G/U \rightarrow \text{Aut}(\mathbb{Q})$ and $\tau: G/U \times G/U \rightarrow \mathbb{Q}^\times$ satisfying the identities (4.2). In particular, the crossed product $\mathbb{Q} * G/U$ exists and $\mathbb{F}_q[[U]]$ embeds into it.*
- (c) *If E is the ring of right $\mathbb{Q}$ -endomorphisms of $\mathbb{Q} * G/U$, then left multiplication induces an embedding of rings $\mathbb{Q} * G/U \rightarrow E$ such that it makes E into a free $\mathbb{Q} * G/U$ -module on both sides.*
- (d) *As a right $\mathbb{F}_q[[G]]$ -module, the crossed product $\mathbb{Q} * G/U$ is isomorphic to the induced module $\mathbb{Q} \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]]$. Analogously, $\mathbb{Q} * G/U$ is isomorphic to $\mathbb{F}_q[[G]] \otimes_{\mathbb{F}_q[[U]]} \mathbb{Q}$ as a left $\mathbb{F}_q[[G]]$ -module.*

Proof. For (a), fix $g \in G$ and let $\alpha: \mathbb{F}_q[[U]] \rightarrow \mathbb{F}_q[[U]]$ by the automorphism induced by conjugation by g . If we denote by $\iota: \mathbb{F}_q[[U]] \rightarrow \mathbb{Q}$ the universal embedding, then $\iota \circ \alpha$ must also be a universal embedding. By the universal property of $\mathbb{Q}$, ι and $\iota \circ \alpha$ are $\mathbb{F}_q[[U]]$ -isomorphic as epic division $\mathbb{F}_q[[U]]$ -rings. This means precisely the existence and uniqueness of some $\bar{\alpha}: \mathbb{Q} \rightarrow \mathbb{Q}$ such that $\bar{\alpha} \circ \iota = \iota \circ \alpha$, that is, extending α . By uniqueness, the map $g \mapsto \bar{\alpha}$ must be a group homomorphism.

(b) Since $U \leq \mathbb{Q}^\times$, one can take the same τ as in the crossed product decomposition $\mathbb{F}_q[[G]] \simeq \mathbb{F}_q[[U]] * G/U$. It is then only a matter of checking whether or not the map $G/U \rightarrow \text{Aut}(\mathbb{Q})$ induced through σ by T also satisfies the second identity in (4.2). However,

⁴⁴This follows from the finite case by taking the inverse limit, but the hypothesis that U is open is essential for the crossed product to be compact.

⁴⁵Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Prop. 3.4.

for each fixed pair $g_1, g_2 \in G$, both the left and the right hand side of that identity define ring automorphisms of $\mathcal{Q}$ which coincide on $\mathbb{F}_q[[U]]$. Since the inclusion of $\mathbb{F}_q[[U]]$ into $\mathcal{Q}$ is an epimorphism of rings, both sides must indeed be equal as automorphisms of $\mathcal{Q}$.

(c) Let T be a transversal of U in G . Then $T = \{g_1, \dots, g_n\}$ is a $\mathcal{Q}$ -basis of $\mathcal{Q} * G/U$ as a right $\mathcal{Q}$ -module. For every $i = 1, \dots, n$, we define $\gamma_i \in E$ such that $\gamma_i(g_j) = \delta_{ij}g_j$. We identify the elements of the ring $\mathcal{Q} * G/U$ with its images in E under the left multiplication map.

We have that $E = \bigoplus_{i=1}^n (\mathcal{Q} * G/U) \circ \gamma_i$. To see it, it is enough to show that if $\alpha_1, \dots, \alpha_n \in \mathcal{Q} * G/U$ and $\alpha = \sum_{i=1}^n \alpha_i \circ \gamma_i = 0$, then $\alpha_1 = \dots = \alpha_n = 0$. This follows from the equality $\alpha(g_i) = \alpha_i g_i$. We also have that $E = \bigoplus_{i=1}^n \gamma_i \circ (\mathcal{Q} * G/U)$, for which it is enough to show that if $\alpha_1, \dots, \alpha_n \in \mathcal{Q} * G/U$ and $\alpha = \sum_{i=1}^n \gamma_i \circ \alpha_i = 0$ then $\alpha_1 = \dots = \alpha_n = 0$. Write $\alpha_i = \sum_{j=1}^n g_j \alpha_{ij}$ with $\alpha_{ij} \in \mathcal{Q}$. Then:

$$\alpha(g_k) = \sum_{g_i^{-1}g_j g_k \in U} g_i \tau(g_j, g_k) \sigma^{-1}(g_k)(\alpha_{ij}).$$

From this, it is clear that if all $\alpha(g_k) = 0$ then for every i, j we have $\alpha_{ij} = 0$.

(d) Just observe that the map $\mathcal{Q} \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]] \rightarrow \mathcal{Q} * G/U$ sending $a \otimes b$ to ab is an isomorphism of right $\mathbb{F}_q[[G]]$ -modules, whose inverse is given by sending $q_1 g_1 + \dots + q_n g_n$ to $q_1 \otimes g_1 + \dots + q_n \otimes g_n$. The same argument works for the left module structure. $\square$

We are now ready to state and prove the main theorem of the section.

Theorem 4.3.7. *Let G be a torsion-free finitely generated pro- p group. Then, the following are equivalent:*

- (A) $\mathbb{F}_q[[G]]$ is a Sylvester domain for any finite field $\mathbb{F}_q$ of characteristic p .
- (B) $\mathbb{F}_q[[U]]$ is a Sylvester domain for every open subgroup U of G and any finite field $\mathbb{F}_q$ of characteristic p .
- (C) $\mathbb{F}_q[[U]]$ is a Sylvester domain for some open subgroup U of G and some finite field $\mathbb{F}_q$ of characteristic p .

Moreover, in case one of the above holds and U is an open normal subgroup of G with a universal embedding $\mathbb{F}_p[[U]] \rightarrow \mathcal{Q}$, then the crossed product $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is a division ring and the induced map $\mathbb{F}_q[[G]] \rightarrow (\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is also a universal embedding.

Proof. (A) $\implies$ (B) Suppose first that $\mathbb{F}_q[[G]]$ is a Sylvester domain with universal division ring of fractions $\mathcal{Q}$. We want to show that the induced embedding $\mathbb{F}_q[[U]] \rightarrow \mathbb{F}_q[[G]] \rightarrow \mathcal{Q}$ satisfies the hypothesis of Theorem 4.1.14.

First, we claim that the right $\mathbb{F}_q[[G]]$ -module

$$\mathcal{D} = \mathcal{Q} \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]] \simeq \mathcal{Q} \otimes_{\mathbb{F}_q} \mathbb{F}_q[U \backslash G]$$

admits a filtration by $\mathbb{F}_q[[G]]$ -submodules

$$\mathcal{D} = \mathcal{D}_0 > \mathcal{D}_1 > \dots > \mathcal{D}_k \simeq \mathcal{Q}$$

whose successive quotients are isomorphic to $\mathcal{Q}$. Indeed, $\mathbb{F}_q[U \setminus G]$ is a finite $\mathbb{F}_q[[G]]$ -module with a composition series whose successive quotients are isomorphic to $\mathbb{F}_q$. Tensoring over $\mathbb{F}_q$ with $\mathcal{Q}$, we obtain the desired filtration of $\mathcal{D}$. Since $\mathrm{Tor}_1^{\mathbb{F}_q[[G]]}(\mathcal{Q}, \mathcal{Q}) = 0$ by part (1) of Proposition 4.1.11, an easy induction on the length of the filtration shows that

$$\mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(\mathcal{Q}, \mathcal{Q}) = \mathrm{Tor}_1^{\mathbb{F}_q[[G]]}(\mathcal{D}, \mathcal{Q}) = 0.$$

Now take a short exact sequence

$$0 \rightarrow I \rightarrow \mathbb{F}_q[[U]]^n \rightarrow M \rightarrow 0$$

of right $\mathbb{F}_q[[U]]$ -modules where M is a finitely generated submodule of $\mathcal{Q}$. Since $\mathbb{F}_q[[G]] \simeq \mathbb{F}_q[[U]]^{[G:U]}$ is flat as a left $\mathbb{F}_q[[U]]$ -module, tensoring with $\mathbb{F}_q[[G]]$ we get a short exact sequence

$$0 \rightarrow I \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]] \rightarrow \mathbb{F}_q[[G]]^n \rightarrow M \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]] \rightarrow 0.$$

As $M \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]]$ is a submodule of $\mathcal{D}$ and $\mathbb{F}_q[[G]]$ has weak global dimension at most 2, we have that

$$\mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(I, \mathbb{F}_q) = \mathrm{Tor}_2^{\mathbb{F}_q[[U]]}(M, \mathbb{F}_q) \simeq \mathrm{Tor}_2^{\mathbb{F}_q[[G]]}(M \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]], \mathbb{F}_q) = 0$$

once again by induction on the filtration of $\mathcal{D}$ and now part (2) of Proposition 4.1.11. Since G and U are both finitely presented by Corollary 4.1.13, we get by Theorem 1.4.17 that

$$\mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(I, \mathbb{F}_q) \simeq \mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(I, \mathbb{F}_q) = 0,$$

that is, I is a free profinite $\mathbb{F}_q[[U]]$ -module.

By Proposition 4.1.12, the $\mathbb{F}_q[[G]]$ -module $\mathcal{Q}$ is *pseudo-coherent*: any finitely generated $\mathbb{F}_q[[G]]$ -submodule of $\mathcal{Q}$ is finitely presented. It is known that this property is preserved under module extensions,⁴⁶ so $\mathcal{D}$ is also pseudo-coherent. Hence, the module $I \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]]$ is finitely generated over $\mathbb{F}_q[[G]]$. Therefore,

$$I \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q[[G]] \otimes_{\mathbb{F}_q[[G]]} \mathbb{F}_q \simeq I \otimes_{\mathbb{F}_q[[U]]} \mathbb{F}_q$$

⁴⁶Another reformulation of the pseudo-coherent property on an R -module M is that any map $\varphi: R^n \rightarrow M$ has a finitely generated kernel. If $N \leq M$ and both N and M/N are pseudo-coherent with $\pi: M \rightarrow M/N$ the projection, then $\ker(\pi \circ \varphi) = L \leq R^n$ is finitely generated, and the restriction of φ to L has its image inside N . Since N is also pseudo-coherent, the kernel of this restriction is also finitely generated, and coincides with $\ker(\varphi)$.

is finite and by Nakayama's lemma I is finitely generated over $\mathbb{F}_q[[U]]$. The same argument works for a finitely generated left $\mathbb{F}_q[[U]]$ -submodule of $\mathcal{Q}$. By Theorem 4.1.14, $\mathbb{F}_q[[U]]$ is a Sylvester domain and the rational closure $\mathcal{Q}_0$ inside $\mathcal{Q}$ is $\mathbb{F}_q[[U]]$ -isomorphic to its universal division ring of fractions.

(B) $\implies$ (C) is immediate.

(C) $\implies$ (A) First, we claim that if $\mathbb{F}_q[[U]]$ is a Sylvester domain with universal division ring of fractions $\mathbb{F}_q[[U]] \rightarrow \mathcal{Q}$, then the composition

$$\mathbb{F}_p[[U]] \rightarrow \mathbb{F}_q[[U]] \rightarrow \mathcal{Q}$$

satisfies the hypothesis of Theorem 4.1.14. Indeed, we have an isomorphism

$$\mathcal{Q} \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_q[[U]] \simeq \mathcal{Q} \otimes_{\mathbb{F}_p} \mathbb{F}_q \simeq \mathcal{Q}^{\log_p q}$$

of right $\mathbb{F}_q[[U]]$ -modules, and analogously for $\mathbb{F}_q[[U]] \otimes_{\mathbb{F}_p[[U]]} \mathcal{Q}$ as a left module. Hence:

$$\mathrm{Tor}_1^{\mathbb{F}_p[[U]]}(\mathcal{Q}, \mathcal{Q}) \simeq \mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(\mathcal{Q} \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_q[[U]], \mathcal{Q}) \simeq \mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(\mathcal{Q}, \mathcal{Q})^{\log_p q} = 0$$

by part (I) of Proposition 4.1.11.

If M is a finitely generated right $\mathbb{F}_p[[U]]$ -submodule of $\mathcal{Q}$ and

$$0 \rightarrow I \rightarrow \mathbb{F}_p[[U]]^n \rightarrow M \rightarrow 0$$

is a short exact sequence, then

$$I \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_q[[U]] \simeq I \otimes_{\mathbb{F}_p} \mathbb{F}_q$$

is a free $\mathbb{F}_q[[U]]$ -module of finite rank by Proposition 4.1.12. However:

$$\mathrm{Tor}_1^{\mathbb{F}_p[[U]]}(I, \mathbb{F}_p)^{\log_p q} \simeq \mathrm{Tor}_1^{\mathbb{F}_p[[U]]}(I, \mathbb{F}_q) \simeq \mathrm{Tor}_1^{\mathbb{F}_q[[U]]}(I \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_q[[U]], \mathbb{F}_q) = 0,$$

so I must also be free of finite rank over $\mathbb{F}_p[[U]]$. The same argument works for left submodules, and the claim now follows from Theorem 4.1.14.

We now claim that one can also go up: if this implies $\mathbb{F}_p[[G]]$ is a Sylvester domain, then so is $\mathbb{F}_q[[G]]$. Let $\mathcal{Q}$ be the universal division ring of fractions of $\mathbb{F}_p[[G]]$ and consider the induced map $\mathbb{F}_q[[G]] \rightarrow \mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q} \rightarrow E$, where the middle one has a ring structure induced by that of $\mathcal{Q}$, E is the ring of right $\mathcal{Q}$ -endomorphisms of $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$ and the embedding $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q} \rightarrow E$ is given by left multiplication. The ring E is von Neumann regular as it is isomorphic to a matrix ring over a division ring.⁴⁷ Observe that $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$ is a finitely generated $\mathcal{Q}$ -module on both sides, and $E \simeq (\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q})^{\log_p q}$ as a $\mathcal{Q}$ -bimodule.

⁴⁷For any $n \times n$ matrix A over $\mathcal{Q}$, there are invertible matrices B and C over $\mathcal{Q}$ for which $A = B(\mathrm{id}_r \oplus 0_{n-r})C$ where $r = \mathrm{rk} A$. If we set $D = C^{-1}B^{-1}$, then $ADA = A$.

Since $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q} \simeq \mathbb{F}_q[[G]] \otimes_{\mathbb{F}_p[[G]]} \mathbb{Q}$ as left $\mathbb{F}_q[[G]]$ -modules and $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q} \simeq \mathbb{Q}^{\log_p q}$ as right $\mathbb{F}_p[[G]]$ -modules, we have

$$\mathrm{Tor}_1^{\mathbb{F}_q[[G]]}(E, E) \simeq \mathrm{Tor}_1^{\mathbb{F}_q[[G]]}(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}, \mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q})^{(\log_p q)^2} \simeq \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{Q}, \mathbb{Q})^{(\log_p q)^3} = 0$$

by part (i) of Proposition 4.1.11. If M is a finitely generated right $\mathbb{F}_q[[G]]$ -submodule of E and

$$0 \rightarrow I \rightarrow \mathbb{F}_q[[G]]^n \rightarrow M \rightarrow 0$$

is a short exact sequence, then I is a free $\mathbb{F}_p[[G]]$ -module of finite rank by Proposition 4.1.12. However:

$$\mathrm{Tor}_1^{\mathbb{F}_q[[G]]}(I, \mathbb{F}_q) \simeq \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(I, \mathbb{F}_p) = 0,$$

so I must also be free of finite rank over $\mathbb{F}_q[[G]]$. The same argument works for left submodules, and the claim now follows from Theorem 4.1.14.

Now, the argument is reduced to proving that if $\mathbb{F}_p[[U]]$ is a Sylvester domain then so is $\mathbb{F}_p[[G]]$. Suppose that $\mathbb{F}_p[[U]]$ is a Sylvester domain with universal division ring of fractions $\mathbb{Q}$ for some open subgroup U of G . It suffices to prove the theorem for the case $|G:U| = p$, so we may assume that U is a normal subgroup of G . Then, by Proposition 4.3.6, the embedding $\mathbb{F}_p[[U]] \rightarrow \mathbb{Q}$ induces an embedding $\mathbb{F}_p[[G]] \rightarrow \mathbb{Q} * G/U \rightarrow E$ where E is the ring of right $\mathbb{Q}$ -endomorphisms of $\mathbb{Q} * G/U$, and we will check the conditions of Theorem 4.1.14 for this embedding. Once again, the ring E is von Neumann regular as it is isomorphic to a matrix ring over a division ring, and since $\mathbb{Q} * G/U$ is isomorphic to the induced module $\mathbb{Q} \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]]$ as right $\mathbb{F}_p[[G]]$ -modules, we have

$$\begin{aligned} \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(E, E) &\simeq \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}((\mathbb{Q} * G/U)^p, (\mathbb{Q} * G/U)^p) \\ &\simeq \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{Q} * G/U, \mathbb{Q} * G/U)^{p^2} \\ &\simeq \mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{Q} \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]], \mathbb{Q} * G/U)^{p^2} \\ &\simeq \mathrm{Tor}_1^{\mathbb{F}_p[[U]]}(\mathbb{Q}, \mathbb{Q})^{p^3} = 0 \end{aligned}$$

by part (i) of Proposition 4.1.11.

Take any finitely generated right $\mathbb{F}_p[[G]]$ -submodule M of $E \simeq (\mathbb{Q} * G/U)^p$ and a short exact sequence

$$0 \rightarrow I \rightarrow \mathbb{F}_p[[G]]^n \rightarrow M \rightarrow 0.$$

As an $\mathbb{F}_p[[U]]$ -module, I must be free of finite rank by Proposition 4.1.12. Hence, I is finitely generated over $\mathbb{F}_p[[G]]$. Taking another short exact sequence

$$0 \rightarrow J \rightarrow \mathbb{F}_p[[G]]^k \rightarrow I \rightarrow 0$$

and repeating this argument, one obtains that J must also be finitely generated, so that I is finitely presented and inductively both I and M are of type FP_∞ . In particular, Theorem 1.4.17 and the long exact sequence shows that

$$\mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(I, \mathbb{F}_p) \simeq \mathfrak{Tor}_1^{\mathbb{F}_p[[G]]}(I, \mathbb{F}_p) \simeq \mathfrak{Tor}_2^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p) \simeq \mathrm{Tor}_2^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p).$$

By Corollary 4.1.13, $\mathrm{cd}(U) \leq 2$, and since G is torsion-free, a theorem of Serre⁴⁸ says that $\mathrm{cd}(G)$ is also at most 2. Thus, since $\mathrm{Tor}_3^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p) \simeq \mathfrak{Tor}_3^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p) \simeq H_3(G, M)$ vanishes, there is an exact sequence of Tor groups:

$$0 \rightarrow \mathrm{Tor}_2^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p) \rightarrow \mathrm{Tor}_2^{\mathbb{F}_p[[G]]}(\mathcal{Q} * G/U, \mathbb{F}_p)^p \simeq \mathrm{Tor}_2^{\mathbb{F}_p[[U]]}(\mathcal{Q}, \mathbb{F}_p)^p.$$

By part (2) of Proposition 4.1.11, the $\mathbb{F}_p[[U]]$ -module $\mathcal{Q}$ has weak global dimension at most 1, so that all the Tor groups above vanish. Since $\mathfrak{Tor}_1^{\mathbb{F}_p[[G]]}(I, \mathbb{F}_p)$ vanishes, once more we conclude that I must be free of finite rank. The same argument works for left submodules, and so we can apply Theorem 4.1.14.

Finally, observe that $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is then a domain, since it embeds into a division ring. Given that it is also a finite-dimensional $\mathbb{Q}$ -vector space, it must be itself a division ring, so we have shown that $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}) * G/U$ is the universal division ring of fractions of $\mathbb{F}_q[[G]]$. $\square$

Recall the statement of Atiyah's conjecture and its pro- p analogue (Conjectures 3.1.14 and 3.1.15): if G is a countable abstract group (resp. pro- p group) such that $\mathrm{lcm}(G) = n$ is finite, then rk_G takes values in $\frac{1}{n}\mathbb{Z}$. In particular, for G torsion-free, this rank function should have integer values. Since these rank function behaves functorially under subgroups, it is clear that if a torsion-free group G satisfies Atiyah's conjecture, then so does any closed subgroup H of G . The questions of lifting this property to overgroups or passing to subgroups when G has torsion is much more subtle.

Both for abstract and pro- p groups, it is not known universally whether the validity of Atiyah's conjecture passes to finite extensions, even if we assume such extensions are torsion-free.⁴⁹ In the case of Sylvester domains $\mathbb{F}_q[[G]]$ for which rk_G is universal (see Question 3.2.4), we are able to give a positive answer:

Corollary 4.3.8. *Let G be a torsion-free finitely generated pro- p group. Then, the following are equivalent:*

$$(A') \quad \mathrm{rk}_G = \mathrm{irk}_{\mathbb{F}_p[[G]]}.$$

$$(B') \quad \mathrm{rk}_U = \mathrm{irk}_{\mathbb{F}_p[[U]]} \text{ for every open subgroup } U \text{ of } G.$$

⁴⁸Serre, *Galois Cohomology*, Prop. I.3.14'.

⁴⁹Peter Linnell and Thomas Schick. "Finite Group Extensions and the Atiyah Conjecture". In: *Journal of the American Mathematical Society* 20.4 (2007), pp. 1003–1051.

(C') $\text{rk}_U = \text{irk}_{\mathbb{F}_q[[U]]}$ for every open subgroup U of G and any finite field $\mathbb{F}_q$ of characteristic p .

(D') $\text{rk}_U = \text{irk}_{\mathbb{F}_q[[U]]}$ for some open subgroup U of G and some finite field $\mathbb{F}_q$ of characteristic p .

Proof. By Theorem 4.3.7, any one of the four conditions imply that the completed group algebras $\mathbb{F}_p[[G]]$, $\mathbb{F}_p[[U]]$, $\mathbb{F}_q[[G]]$ and $\mathbb{F}_q[[U]]$ are Sylvester domains for every open subgroup U of G and any finite field $\mathbb{F}_q$ of characteristic p . We will prove the equivalences assuming this.

(A') $\implies$ (B') Let $\mathcal{Q}$ be the universal division ring of fractions of $\mathbb{F}_p[[G]]$, A be an $n \times m$ matrix over $\mathbb{F}_p[[U]]$ (which we also see as a matrix over $\mathbb{F}_p[[G]]$) and define the right $\mathbb{F}_p[[U]]$ -module $M = \mathbb{F}_p[[U]]^n / A \mathbb{F}_p[[U]]^m$. Note that $M \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]]$ is isomorphic to $\mathbb{F}_p[[G]]^n / A \mathbb{F}_p[[G]]^m$ as right $\mathbb{F}_p[[G]]$ -modules. Hence:

$$\begin{aligned} \dim_U M &= \frac{\dim_U M \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]]}{|G:U|} \\ &= \dim_G M \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]] \\ &= \dim_{\mathcal{Q}} M \otimes_{\mathbb{F}_p[[U]]} \mathbb{F}_p[[G]] \otimes_{\mathbb{F}_p[[G]]} \mathcal{Q}, \text{ by hypothesis} \\ &= \dim_{\mathcal{Q}} M \otimes_{\mathbb{F}_p[[U]]} \mathcal{Q} \end{aligned}$$

so that the embedding $\mathbb{F}_p[[U]] \rightarrow \mathcal{Q}$ induces the rank function $\dim_U$. By the computations of Example 4.3.2, this embedding must also induce the inner rank function, so $\text{irk}_{\mathbb{F}_p[[U]]} = \text{rk}_U$.

(B') $\implies$ (C') If $\mathbb{F}_p[[U]] \rightarrow \mathcal{Q}$ is a universal embedding such that

$$\text{rk}_U = \text{irk}_{\mathbb{F}_p[[U]]} = \text{rk}_{\mathcal{Q}}$$

as rank functions on $\mathbb{F}_p[[U]]$, then for any finite field $\mathbb{F}_q$ of characteristic p we've shown that $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$ is also a division ring and the embedding $\mathbb{F}_q[[U]] \rightarrow \mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$ is universal and induces the inner rank $\text{irk}_{\mathbb{F}_q[[U]]}$. The left action of $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$ on itself induces a map $\psi: \mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q} \rightarrow \text{End}_{\mathcal{Q}}(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q})$, and by unicity of the rank functions on matrix rings (Example 3.1.5) we have that

$$\text{rk}_{\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}}(A) = \frac{\text{rk}_{\mathcal{Q}}(\psi(A))}{\log_p q}$$

for any matrix A over $\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathcal{Q}$.

Let $f = \log_p q = \dim_{\mathbb{F}_p} \mathbb{F}_q$. Observe that ψ induces a map

$$\psi: \text{Mat}_{n \times m}(\mathbb{F}_q[[U]]) \rightarrow \text{Mat}_{nf \times mf}(\mathbb{F}_p[[U]]).$$

We claim that

$$\text{rk}_U(\psi(A)) = f \cdot \text{rk}_U(A).$$

Indeed, $\psi(A)$ is the matrix representing the left multiplication by A map between the right $\mathbb{F}_p[[U]]$ -modules $\mathbb{F}_q[[U]]^m$ and $\mathbb{F}_q[[U]]^n$. In particular, for any open normal subgroup V of U , the image of $\psi(A)$ modulo V represents the map between the right $\mathbb{F}_p[U/V]$ -modules $\mathbb{F}_q[U/V]^m$ and $\mathbb{F}_q[U/V]^n$ given by left multiplication by $A \pmod{V}$. Hence:

$$\begin{aligned} \mathrm{rk}_U(\psi(A)) &= \sup_{V \trianglelefteq_o U} \frac{\mathrm{rk}_{\mathbb{F}_p} \psi(A) \pmod{V}}{|U: V|} \\ &= \sup_{V \trianglelefteq_o U} \frac{\dim_{\mathbb{F}_p} A \mathbb{F}_q[U/V]^m}{|U: V|} \\ &= f \cdot \sup_{V \trianglelefteq_o U} \frac{\dim_{\mathbb{F}_q} A \mathbb{F}_q[U/V]^m}{|U: V|} \\ &= f \cdot \mathrm{rk}_U(A). \end{aligned}$$

Combining this equality with the one for the previous paragraph, we get

$$\mathrm{rk}_{\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}}(A) = \frac{\mathrm{rk}_{\mathbb{Q}}(\psi(A))}{f} = \frac{\mathrm{rk}_U(\psi(A))}{f} = \mathrm{rk}_U(A).$$

(C') $\implies$ (D') is immediate.

(D') $\implies$ (A') We have shown in Example 4.3.2 that in this case we also have $\mathrm{rk}_U = \mathrm{irk}_{\mathbb{F}_p[[U]]}$. If $\mathbb{Q}$ is the universal division ring of $\mathbb{F}_p[[U]]$, then the embedding of $\mathbb{F}_p[[G]]$ on $\mathbb{Q} * G/U$ is universal. Hence, for any $n \times m$ matrix A over $\mathbb{F}_p[[G]]$, which we see also as an $n \times m$ matrix over $\mathbb{Q} * G/U$ and as an $n|G: U| \times m|G: U|$ matrix over $\mathbb{F}_p[[U]]$:

$$\mathrm{irk}_{\mathbb{F}_p[[G]]} A = \mathrm{rk}_{\mathbb{Q} * G/U} A = \frac{\mathrm{rk}_{\mathbb{Q}} A}{|G: U|} = \frac{\mathrm{irk}_{\mathbb{F}_p[[U]]} A}{|G: U|} = \frac{\mathrm{rk}_U A}{|G: U|} = \mathrm{rk}_G A. \quad \square$$

4.4 Mild flag pro- p groups

By Proposition 4.1.13 and the results of Section 4.3, in order build new examples of completed finitely generated group algebras which are Sylvester domains, it suffices to look for $\mathbb{F}_p[[U]]$ where U is an open subgroup of some torsion-free finitely generated pro- p group G of cohomological dimension at most 2.

A natural class of such groups comes from procyclic extensions of free pro- p groups: a pro- p group G is said to be *free-by- $\mathbb{Z}_p$* if it contains a normal closed subgroup $N \trianglelefteq G$ that is free pro- p and $G/N \simeq \mathbb{Z}_p$. In this case, we have a semidirect product decomposition $G \simeq N \rtimes \mathbb{Z}_p$, and $\mathrm{cd}(G) \leq 2$.⁵⁰ We do not assume N is finitely generated.

⁵⁰Serre, *Galois Cohomology*, Prop. I.3.15.

Example 4.4.1. Non-trivial free pro- p groups are free-by- $\mathbb{Z}_p$ since any closed subgroup is again free pro- p . Another class of free-by- $\mathbb{Z}_p$ groups are Demushkin groups: finitely generated one-relator pro- p groups G for which the cup product

$$H^1(G, \mathbb{F}_p) \otimes H^1(G, \mathbb{F}_p) \xrightarrow{\cup} H^2(G, \mathbb{F}_p)$$

is a non-degenerate bilinear form. It is known⁵¹ that for odd primes p the group G admits a presentation on $n \in 2\mathbb{Z}$ generators given by

$$G \simeq \langle x_1, \dots, x_n \mid x_1^q [x_1, x_2] \cdots [x_{n-1}, x_n] = 1 \rangle,$$

where q is either 0 or a power of p . This class includes the pro- p completions of fundamental groups of closed orientable surfaces, as well as non-orientable ones when $p = 2$. From their presentation, it is immediate that they surject onto $\mathbb{Z}_p$. Since any closed subgroup of infinite index in a Demushkin group is free pro- p , the kernel N is a free pro- p group.⁵² Unless $G \simeq \mathbb{Z}_p \rtimes \mathbb{Z}_p$, Demushkin groups do not decompose as a non-trivial direct product, so by the virtual local retraction property⁵³ N is seldom finitely generated. ■

Example 4.4.2. The class of free-by- $\mathbb{Z}_p$ pro- p groups also appear naturally in the study of absolute Galois groups. If K is a global field, that is, either a number field or a global function field, then for every prime p distinct from $\text{char}(K)$ any p -Sylow subgroup of the absolute Galois group $\text{Gal}(\bar{K}/K)$ has a free-by- $\mathbb{Z}_p$ subgroup of index at most 2. If K^p denotes a subfield of $\bar{K}$ fixed by some p -Sylow subgroup and μ_{p^∞} denotes the group generated by all p^i -th roots of unity for all i , we have a short exact sequence

$$0 \rightarrow \text{Gal}(K^p/K(\mu_{p^\infty})) \rightarrow \text{Gal}(K^p/K) \rightarrow \text{Gal}(K(\mu_{p^\infty})/K) \rightarrow 0$$

called the *cyclotomic decomposition* of $\text{Gal}(K^p/K)$. One immediately checks that the Galois group $\text{Gal}(K(\mu_{p^\infty})/K)$ is an infinite pro- p subgroup of $\text{Aut}(\mu_{p^\infty}) \simeq \mathbb{Z}_p^\times$, so it contains a subgroup isomorphic to $\mathbb{Z}_p$ with index at most 2. It is a theorem of Lior Bary-Soroker, Moshe Jarden and Danny Neftin⁵⁴ that $\text{Gal}(K^p/K(\mu_{p^\infty}))$ is free pro- p of countable rank. Hence, G is virtually free-by- $\mathbb{Z}_p$. While $\text{Gal}(K^p/K)$ is not finitely generated itself, all its finitely generated subgroups are again virtually free-by- $\mathbb{Z}_p$, so our results apply locally to $\text{Gal}(K^p/K)$.

⁵¹John Labute. “Classification of Demushkin Groups”. In: *Canadian Journal of Mathematics* 19 (1967), pp. 106–132. ISSN: 1496-4279. DOI: 10.4153/cjm-1967-007-8.

⁵²Serre, *Galois Cohomology*, Exer. I.4.5.5.

⁵³Mark Shusterman and Pavel Zalesskii. “Virtual retraction and Howson’s theorem in pro- p groups”. In: *Transactions of the American Mathematical Society* 373.3 (Dec. 2019), pp. 1501–1527. ISSN: 1088-6850. DOI: 10.1090/tran/7784, Cor. 1.2.

⁵⁴Lior Bary-Soroker, Moshe Jarden, and Danny Neftin. “The Sylow subgroups of the absolute Galois group $\text{Gal}(\mathbb{Q})$ ”. In: *Advances in Mathematics* 284 (Oct. 2015), pp. 186–212. ISSN: 0001-8708. DOI: 10.1016/j.aim.2015.05.017, Obs. 3.1.

We recall a previous theorem of Ido Efrat⁵⁵ that characterizes the finitely generated subgroups of $\text{Gal}(K^p/K)$ as the free pro- p products of $\mathbb{Z}/2\mathbb{Z}$, finitely generated free pro- p groups and groups of the form $\mathbb{Z}_p \rtimes \mathbb{Z}_p$. It also follows from the standard results of profinite Bass-Serre theory that such groups are virtually free-by- $\mathbb{Z}_p$.⁵⁶ ■

The class of torsion-free virtually free-by- $\mathbb{Z}_p$ groups is the pro- p analogue of the well studied class of abstract virtually free-by-cyclic groups.⁵⁷ For abstract complex group algebras $\mathbb{C}[\Gamma]$ of free-by- $\mathbb{Z}$ groups, the question of whether they are Sylvester domains has previously been studied by Fabian Henneke and Diego López-Álvarez. They showed that this is the case if and only if the ring $\mathbb{Z}[\Gamma]$ admits stably free cancellation: any finitely generated module that is stably free⁵⁸ must be already free.⁵⁹ Both situations are possible: $\mathbb{C}[\mathbb{Z} \times \mathbb{Z}]$ admits stably free cancellation while $\mathbb{C}[\pi_1(S)]$ where S is the Klein bottle does not.⁶⁰ It is not known even if $\mathbb{Z}[\pi_1(S)]$ for surfaces S of higher genus are Sylvester domains or not. It was previously shown by A. Jaikin-Zapirain that the von Neumann rank function rk_Γ is universal for all locally indicable⁶¹ groups, so the equality $\text{rk}_\Gamma = \text{irk}_{\mathbb{C}[\Gamma]}$ holds automatically whenever $\mathbb{C}[\Gamma]$ is a Sylvester domain.⁶²

The goal of this chapter is to prove that completed group of algebras of finitely generated torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups G over finite fields $\mathbb{F}_q$ are Sylvester domains, and moreover rk_G is universal in $\mathbb{F}_q[[G]]$. We will break this goal into three smaller objectives, that will each be developed in its own section. The first objective is to define and describe a special type of free-by- $\mathbb{Z}_p$ pro- p group called *mild flag*, and the computation of its graded ring is the goal of this section. Afterwards, we will construct a series of embeddings into a division ring with a valuation in Section 4.5. We will prove those embeddings are universal in Section 4.6.

Example 4.4.3. First, observe that any non-trivial closed subgroup of a free-by- $\mathbb{Z}_p$ pro- p group

⁵⁵Ido Efrat. “Finitely Generated Pro- p Absolute Galois Groups over Global Fields”. In: *Journal of Number Theory* 77.1 (July 1999), pp. 83–96. ISSN: 0022-314X. DOI: 10.1006/jnth.1999.2379, Main Thm.

⁵⁶For each factor $\mathbb{Z}_p \rtimes \mathbb{Z}_p$, choose a surjection onto $\mathbb{Z}_p$, and glue all of those together to obtain a non-trivial homomorphism from their pro- p product to $\mathbb{Z}_p \times \mathbb{Z}/2\mathbb{Z}$ that is injective on each free $\mathbb{Z}/2\mathbb{Z}$ factor. The kernel of this map acts on the standard Bass-Serre pro- p tree associated to the free pro- p product with free pro- p vertex stabilizers, so it must be free pro- p itself. See Luis Ribes. *Profinite Graphs and Groups*. Springer International Publishing, 2017. ISBN: 9783319611990. DOI: 10.1007/978-3-319-61199-0, Thm. 9.6.2

⁵⁷Dawid Kielak and Marco Linton. “Virtually Free-by-Cyclic Groups”. In: *Geometric and Functional Analysis* 34.5 (July 2024), pp. 1580–1608. ISSN: 1420-8970. DOI: 10.1007/s00039-024-00687-6, Sec. 1.

⁵⁸A finitely generated module M over a ring R is stably free if $M \oplus R^k$ is finitely generated free for some k .

⁵⁹Fabian Henneke and Diego López-Álvarez. “Pseudo-Sylvester domains and skew Laurent polynomials over firs”. In: *Journal of Algebra and Its Applications* 21.08 (May 2021). ISSN: 1793-6829. DOI: 10.1142/S0219498822501687, Thm. B.

⁶⁰Fabian M. Henneke. “Embeddings of group rings and L^2 -invariants”. en. PhD thesis. Bonn, Germany: Friedrich-Wilhelms-Universität Bonn, 2020, Sec. 5.3.2.

⁶¹A group Γ is locally indicable if every non-trivial finitely generated subgroup $\Delta \leq \Gamma$ surjects onto $\mathbb{Z}$. Every free-by- $\mathbb{Z}$ group is locally indicable.

⁶²Andrei Jaikin-Zapirain. “The universality of Hughes-free division rings”. In: *Selecta Mathematica* 27.4 (July 2021). ISSN: 1420-9020. DOI: 10.1007/s00029-021-00691-w, Thm. 1.1.

G is again free-by- $\mathbb{Z}_p$. Indeed, such subgroup $H \leq G$ is either contained in the free kernel N of a surjection $\varphi: G \rightarrow \mathbb{Z}_p$, hence H is free itself, or we have a decomposition $H \simeq (H \cap N) \rtimes \varphi(H)$ and $\varphi(H) \simeq \mathbb{Z}_p$. While the class of free pro- p groups is closed under finite torsion-free extensions, this is no longer true for the class of free-by- $\mathbb{Z}_p$ pro- p groups.⁶³

Indeed, let $G = A \amalg_C B$ be the free pro- p product of two free pro- p groups A and B of rank 2 with bases $\{a_1, a_2\}$ and $\{b_1, b_2\}$ respectively amalgamated along the procyclic subgroup $C \simeq \langle [a_1, a_2]^p \rangle \simeq \langle [b_1, b_2]^p \rangle$. This free pro- p product with amalgamation is proper⁶⁴, that is, both A and B embed naturally into G as closed subgroups, and G is torsion-free.⁶⁵ If Γ is the discrete group with the same presentation as G , so that $G \simeq \hat{\Gamma}_p$, we have that Γ is residually p -finite⁶⁶ and therefore embeds into G . As $[a_1, a_2]$ and $[b_1, b_2]$ are two distinct elements of $[\Gamma, \Gamma] \leq [G, G]$ with the same p -th power, the group $[G, G]$ cannot be free pro- p . Hence, G itself cannot be free-by- $\mathbb{Z}_p$.

Let us prove now that G is virtually free-by- $\mathbb{Z}_p$. Consider F another free pro- p subgroup of rank two with basis $\{x_1, x_2\}$, and let $\varphi: G \rightarrow F$ be the continuous surjection defined by $\varphi(a_i) = \varphi(b_i) = x_i$. If we take $V = \Phi(F)$ the Frattini subgroup of F , we have that $[x_1, x_2]$ maps onto a non-zero element of $M = V/[V, V] \simeq \mathbb{Z}_p^n$. Observe that $[x_1, x_2][V, V]$ has finitely many F -conjugates which are linearly independent in M , and so one can find a homomorphism $\psi_0: M \rightarrow \mathbb{Z}_p$ such that $\psi_0(x^{-1}[x_1, x_2]x) = 1$ for all $x \in F$. We let $\psi: V \rightarrow \mathbb{Z}_p$ be the composition of the projection $V \rightarrow M$ with ψ_0 , $U = \varphi^{-1}(V)$ and $\theta: U \rightarrow \mathbb{Z}_p$ be the composition $\theta = \psi \circ \varphi$. Observe that U is a finite index subgroup of G .

Let $K = \ker \theta \trianglelefteq U$. We have that G acts⁶⁷ on the pro- p tree Y with vertex set $V(Y) = G/A \cup G/B$, edge set $E(Y) = G/C$ and incidence relations

$$d_0(gC) = gA, \quad d_1(gC) = gB$$

for every edge gC . In particular, G/Y consists of an edge between two vertices, the vertex stabilizers are conjugate in G to either A or B and the edge stabilizers are conjugate in G to C . Let $Z = K/Y$ and $\mathcal{Z}$ be the associated graph of pro- p groups on Z given by the K -action on Y . The edge stabilizers of the K -action on Y are trivial by construction, and the vertex stabilizers are all free pro- p . Hence, the pro- p fundamental group $\pi_1(Z, \mathcal{Z})$ is free pro- p .

By construction, U is a normal subgroup of G containing $[a_1, a_2]$ and $[b_1, b_2]$ and $G/U \simeq F/\Phi(F)$. Hence, $UA = UB = G$. Therefore,

$$G/A = UA/A \simeq U/(U \cap A)$$

⁶³Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Sec. 2.2.

⁶⁴Ribes and Zalesskii, *Profinite Groups*, Exer. 9.2.6(a).

⁶⁵Ibid., Thm. 9.4.2(b).

⁶⁶Goansu Kim and James McCarron. “On Amalgamated Free Products of Residually p -Finite Groups”. In: *Journal of Algebra* 162.1 (Dec. 1993), pp. 1–11. ISSN: 0021-8693. DOI: 10.1006/jabr.1993.1237, Cor. 3.5.

⁶⁷Ribes, *Profinite Graphs and Groups*, Sec. 4.

and similarly for G/B . Since K is normal in U with quotient $\mathbb{Z}_p$, we get homeomorphisms

$$V(Z) \simeq K \backslash U / (U \cap A) \cup K \backslash U / (U \cap B) \simeq \mathbb{Z}_p / \theta(U \cap A) \cup \mathbb{Z}_p / \theta(U \cap B),$$

so that $V(Z)$ consists of two points. Moreover, C is contained in U , and since it generates the image of θ we have $U \simeq K \rtimes C$. Also note that by the definition of θ and K we have $[G, U] \leq K$. Combining this information with the homeomorphism $K \backslash G \simeq K \backslash U \times U \backslash G$, we get that the induced right action of C in $K \backslash U \times U \backslash G$ is given by

$$(Kc, Ug)c' = (Kcc', Ug)$$

for $c, c' \in C$ and $g \in G$. Therefore

$$E(Z) = K \backslash G / C \simeq K \backslash U / C \times U \backslash G \simeq U \backslash G$$

is also finite, and so the map $\pi_1(Z, \mathcal{X}) \rightarrow K$ is an isomorphism.⁶⁸ This proves that U is free-by- $\mathbb{Z}_p$, and the class of torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p groups is strictly bigger than the class of free-by- $\mathbb{Z}_p$ pro- p groups. ■

By Corollary 4.3.8, to attain the goal of this chapter, it suffices to find an open subgroup U of a finitely generated torsion-free free-by- $\mathbb{Z}_p$ group G for which $\mathbb{F}_p[[U]]$ is a Sylvester domain and satisfies $\text{rk}_U = \text{irk}_{\mathbb{F}_p[[U]]}$. Once U is identified within G , we shall use a description of the group algebra $\mathbb{F}_p[[U]]$ as a skew power series ring and the results of Section 4.2 to build an explicit embedding of $\mathbb{F}_p[[U]]$ into a division ring satisfying the hypotheses of Theorem 4.1.14. We start with the power series ring description:⁶⁹

Proposition 4.4.4. *Let G be a pro- p group containing a normal closed subgroup $N \trianglelefteq G$ such that $G/N \simeq \mathbb{Z}_p$ is infinite procyclic. If $g \in G$ is an element that generates G/N , $\sigma: \mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p[[N]]$ is induced by right conjugation by g and $\delta = \sigma - \text{id}$, then σ^{-1} is also continuous, δ and $\delta' = \sigma^{-1} - \text{id}$ are both topologically nilpotent and the map that sends s to $g - 1$ is a continuous isomorphism*

$$\mathbb{F}_p[[N]][[s; \sigma, \delta]] \rightarrow \mathbb{F}_p[[G]]$$

of profinite rings.

Proof. From the homeomorphism $G \simeq G/N \times N$ given by the existence of a continuous section⁷⁰ of the projection $G \rightarrow G/N$, we see that $\mathbb{F}_p[[G]]$ is the free profinite right $\mathbb{F}_p[[N]]$ -module on the profinite space G/N . Through a base change, we see that the set of powers of $s = g - 1$ also form a topological basis for $\mathbb{F}_p[[G]]$ over $\mathbb{F}_p[[N]]$, from which we already get an isomorphism $\mathbb{F}_p[[N]][[s; \sigma, \delta]] \rightarrow \mathbb{F}_p[[G]]$ of right topological $\mathbb{F}_p[[N]]$ -modules, where

⁶⁸Ribes, *Profinite Groups and Groups*, Thm. 6.6.1.

⁶⁹Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Prop. 4.1.

⁷⁰Ribes and Zalesskii, *Profinite Groups*, Prop. 2.2.2.

$\mathbb{F}_p[[N]][[s; \sigma, \delta]]$ is regarded simply as the $\mathbb{F}_p[[N]]$ -module of all right power series on s over $\mathbb{F}_p[[N]]$. Alternatively, since the powers of s converge to zero in $\mathbb{F}_p[[G]]$, there is a well defined map $\mathbb{F}_p[[N]][[s; \sigma, \delta]] \rightarrow \mathbb{F}_p[[G]]$ of right $\mathbb{F}_p[[N]]$ -modules, and the decomposition $\mathbb{F}_p[[G]] \simeq \mathbb{F}_p[[N]][[G/N]]$ as right $\mathbb{F}_p[[N]]$ gives the desired inverse map $\mathbb{F}_p[[G]] \rightarrow \mathbb{F}_p[[N]][[s; \sigma, \delta]]$ by sending $g^i N$ to $(s+1)^i$ for all $i \in \mathbb{Z}$ and extending to all of G/N by continuity.

Hence, it suffices to prove that the formula

$$\left(\sum_{i \geq 0} s^i a_i \right) \left(\sum_{j \geq 0} s^j b_j \right) = \sum_{m \geq 0} s^m \left(\sum_{n=0}^m \sum_{k \geq n} \binom{k}{n} \delta^{k-n} (\sigma^n(a_{m-n})) b_k \right)$$

always converges and thus defines a continuous multiplication on $\mathbb{F}_p[[N]][[s; \sigma, \delta]]$, since the relation

$$as = a(g-1) = (g-1)(g^{-1}ag) + (g^{-1}ag - a) = s\sigma(a) + \delta(a)$$

already holds in $\mathbb{F}_p[[G]]$. We already know that this is the case if δ is topologically nilpotent, that is, if $\delta^k(x) \rightarrow 0$ as $k \rightarrow \infty$ for every $x \in \mathbb{F}_p[[N]]$. Let I_N be the augmentation ideal of $\mathbb{F}_p[[N]]$. Since N is pro- p , I_N is topologically nilpotent as an ideal. In particular, we have $\bigcap_{k \geq 0} I_N^k = \{0\}$.

Now let $x \in \mathbb{F}_p[[N]]$ be arbitrary and $\epsilon: \mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p$ be the augmentation map. We have

$$\epsilon(\delta(x)) = \epsilon(g^{-1}xg - x) = \epsilon(g^{-1})\epsilon(x)\epsilon(g) - \epsilon(x) = \epsilon(x) - \epsilon(x) = 0,$$

that is, $\delta(\mathbb{F}_p[[N]]) \leq I_N$. Moreover, for any $h \in N$ we have

$$\begin{aligned} \delta(h-1) &= (g^{-1}hg - 1) - (h-1) = g^{-1}hg - h \\ &= g^{-1}(hg - gh) = g^{-1}((h-1)(g-1) - (g-1)(h-1)). \end{aligned}$$

Since I_N^2 is the closure of the abstract two-sided ideal generated by $(h_1-1)(h_2-1)$ for all $h_1, h_2 \in N$, we get that $\delta(I_N) \leq I_N^2$. Since δ is a σ -derivation and $\sigma(I_N^n) \leq I_N^n$ for all n , we get

$$\delta(I_N^k) \subseteq \delta(I_N^{k-1})I_N + I_N^{k-1}\delta(I_N) \subseteq I_N^{k+1}$$

for all $k \geq 0$. Hence, δ is topologically nilpotent. Finally, observe that σ^{-1} is given by right conjugation with g^{-1} , which also generates G/N . Hence, σ^{-1} is also continuous, $\delta' = \sigma^{-1} - \text{id}$ is again topologically nilpotent and we conclude that the right and left skew power series rings constructed are isomorphic. $\square$

We now introduce the following definition in order to identify our special subgroup U . To shorten the notation, we stack commutators on the right, that is

$$[x_1, x_2, \dots, x_n] = [[[x_1, x_2], x_3], \dots, x_n].$$

Definition 4.4.5. A *flag presentation* of a pro- p group G is a finite presentation given by the quotient of the free pro- p group F on a set of generators $\{x_1, \dots, x_n, g\}$ by l relations of the form

$$[x_i, \underbrace{g, \dots, g}_{a_i \text{ times}}] = h_i \text{ for } h_i \in \Phi(\widetilde{N}) \text{ and } 1 \leq i \leq n, \quad (4.3)$$

where the a_i are positive integers and $\widetilde{N}$ is the closed subgroup of F generated by $\{x_1, \dots, x_n\}$. If all the a_i are equal to 1, we shall say that the flag presentation is *mild*. A pro- p group with a mild flag presentation will be called a *mild flag pro- p group*.

With this definition in hand, we are ready to single out our subgroup U :⁷¹

Proposition 4.4.6. *Every finitely generated free-by- $\mathbb{Z}_p$ pro- p group G has a flag presentation. Moreover, G has an open normal subgroup U , pre-image of an open subgroup of $\mathbb{Z}_p$, such that U has a mild flag presentation.*

Proof. Let $G \simeq N \rtimes G/N$ be a free-by- $\mathbb{Z}_p$ pro- p group with $G/N \simeq \overline{\langle gN \rangle} \simeq \mathbb{Z}_p$ and N free pro- p . Let $A = N/\Phi(N)$ and consider it as a $\mathbb{F}_p[[G/N]]$ -module through the action of conjugation by g . The semi-direct product decomposition gives us

$$G/\Phi(G) \simeq (A \hat{\otimes}_{\mathbb{F}_p[[G/N]]} \mathbb{F}_p) \oplus \mathbb{F}_p, \quad (4.4)$$

and Nakayama's lemma implies that G is finitely generated as a pro- p group if and only if A is finitely generated as an $\mathbb{F}_p[[G/N]]$ -module. Since $\mathbb{F}_p[[G/N]] \simeq \mathbb{Z}_p[[t]]$ is a principal ideal domain, we can then decompose A as a direct sum

$$A \simeq \mathbb{F}_p[[G/N]]^k \oplus \bigoplus_{i=1}^l \mathbb{F}_p[[G/N]]/(d_i)$$

with $0 \neq d_i$ contained in the augmentation ideal $I_{G/N}$ of $\mathbb{F}_p[[G/N]]$ for all i . Its Pontryagin dual $\text{Hom}(A, \mathbb{Q}/\mathbb{Z})$ must then decompose as

$$Q^k \oplus \bigoplus_{i=1}^l \mathbb{F}_p[[G/N]]/(d_i)$$

where Q is some injective discrete $\mathbb{F}_p[[G/N]]$ -module.

Since $0 \neq d_i \in I_{G/N}$, we obtain that $H^1(G/N, \mathbb{F}_p[[G/N]]/(d_i)) \simeq \mathbb{F}_p$ for all i . By the Lyndon-Hochschild-Serre spectral sequence, we have:

$$\begin{aligned} H^2(G, \mathbb{F}_p) &\simeq H^1(G/N, H^1(N, \mathbb{F}_p)) \\ &\simeq H^1(G/N, \text{Hom}(A, \mathbb{Q}/\mathbb{Z})) \\ &\simeq \bigoplus_{i=1}^l H^1(G/N, \mathbb{F}_p[[G/N]]/(d_i)) \simeq \mathbb{F}_p^l. \end{aligned} \quad (4.5)$$

⁷¹Jaikin-Zapirain and Souza, "Sylvester domains and pro- p groups", Lem. 4.6.

Choose representatives $x_1, \dots, x_k, z_1, \dots, z_l$ in N for the $\mathbb{F}_p[[G/N]]$ -cyclic generators of each factor in A . Then, $\{x_1, \dots, x_k, z_1, \dots, z_l, g\}$ is a minimal generating set for G by the isomorphism 4.4. If A is a free $\mathbb{F}_p[[G/N]]$ -module, then the isomorphism 4.5 shows that $H^2(G, \mathbb{F}_p) = 0$ and therefore G is a free pro- p group with the mild flag presentation

$$G \simeq \langle x_1, \dots, x_k, z_1, \dots, z_l, g \mid \emptyset \rangle.$$

Hence, we can assume that A is not a free $\mathbb{F}_p[[G/N]]$ -module, and in particular that $H^2(G, \mathbb{F}_p)$ is non-zero.

Since the G/N -action on the finite module $\mathbb{F}_p[[G/N]]/(d_i)$ is unipotent, there exists some positive a_i such that $[z_i, \underbrace{g, \dots, g}_{a_i \text{ times}}]$ is a fixed point, or equivalently, there exists $b_i \in \Phi(N)$ such that

$$[z_i, \underbrace{g, \dots, g}_{a_i+1 \text{ times}}] = b_i. \quad (4.6)$$

Then, by the isomorphism 4.5, we find that the identities 4.6 for $1 \leq i \leq l$, form a generating set of relations for G , yielding the desired flag presentation.

By taking in G the intersection U of the stabilizers of the G -action in each finite module $\mathbb{F}_p[[G/N]]/(d_i)$, we can assume that each non-free direct factor of A is isomorphic to $\mathbb{F}_p$, that is,

$$A \simeq \mathbb{F}_p[[U/N]]^{k|G:U|} \oplus \bigoplus_{i=1}^{l'} \mathbb{F}_p$$

as $\mathbb{F}_p[[U/N]]$ -modules. By repeating the steps above, we obtain a mild flag presentation for U . $\square$

The following pro- p analogue of a classical theorem of Mark Feighn and Michael Handel is an immediate consequence of the previous proposition.

Corollary 4.4.7. *Free-by- $\mathbb{Z}_p$ pro- p groups are coherent, that is, any finitely generated subgroup is finitely presented.*

Proof. Any closed subgroup of a free-by- $\mathbb{Z}_p$ pro- p group is again free-by- $\mathbb{Z}_p$, so the result follows from Proposition 4.4.6. $\square$

Remark 4.4.8. The adjective “flag” in flag presentation is inspired by the theory of Lie algebras. Indeed, the fact that we obtain this a_i -fold simple commutator as a piece of the relation isolating the generator g is directly connected to the unipotent action of g on A : it stabilizes a flag $A = A_0 > A_1 > \dots > A_j = 0$ of subspaces such that $\dim_{\mathbb{F}_p} A_i / A_{i+1} = 1$. It is reminiscent of Lie’s theorem that over algebraically closed fields of characteristic zero any finite dimensional representation of a solvable Lie algebra stabilizes a flag—in other words, the image of the representation is unipotent.

Also note that flag presentations characterize finitely generated free-by- $\mathbb{Z}_p$ pro- p groups.⁷² Let

$$G \simeq \langle x_1, \dots, x_n, g \mid r_1, \dots, r_l \rangle$$

be a flag presentation where

$$r_i = [x_1, \underbrace{g, \dots, g}_{a_i \text{ times}}] b_i^{-1}$$

with each $b_i \in \Phi(\widetilde{N})$, and denote by N the normal subgroup of G generated by the $x_1, \dots, x_n$. Taking the procyclic subgroup generated by g as the continuous section $\mathbb{Z}_p \simeq G/N \rightarrow G$, the Reidemeister-Schreier⁷³ rewriting process shows that N has a presentation with generating set converging to 1 given by all the elements

$$x_{i,k} = [x_i, \underbrace{g, \dots, g}_{k \text{ times}}]$$

and relations

$$x_{i,a_i+k} = \left(\prod_{j=0}^{k-1} g^{j-k+1} x_{i,a_i+k-1-j} g^{k-1-j} \right) \cdot g^{-k} b_i g^k$$

for $1 \leq i \leq l$ and $k \geq 0$. These relations allows us to eliminate the generators x_{i,a_i+k} for $1 \leq i \leq l$ and $k \geq 0$, showing that N is free pro- p . Alternatively, the LHS spectral sequence gives us

$$\begin{aligned} l &\geq \dim_{\mathbb{F}_p} H^2(G, \mathbb{F}_p) \\ &= \dim_{\mathbb{F}_p} H^2(N, \mathbb{F}_p)^{G/N} \oplus H^1(G/N, H^1(N, \mathbb{F}_p)) \\ &\geq l + \dim_{\mathbb{F}_p} H^2(N, \mathbb{F}_p)^{G/N}, \end{aligned}$$

where the last inequality is obtained by a computation of $H^1(G/N, H^1(N, \mathbb{F}_p))$ as in the proof of Proposition 4.4.6. Hence, $H^2(N, \mathbb{F}_p)^{G/N} = 0$, and since the cohomology groups are discrete torsion G/N -modules, one gets $H^2(N, \mathbb{F}_p) = 0$ as desired. ■

The reason we shall restrict ourselves to mild flag pro- p groups is that such groups are *mild* in the sense of John Labute.⁷⁴ Recall the Magnus isomorphism of Theorem 4.1.7: if F is the free pro- p group on a set X converging to 1, then $\mathbb{F}_p[[F]] \simeq \mathbb{F}_p\langle\langle X \rangle\rangle$. Moreover, if X is finite, under the filtration induced by powers of the augmentation ideal, we have an isomorphism $\text{Gr } \mathbb{F}_p[[F]] \simeq \mathbb{F}_p\langle X \rangle$.⁷⁵

⁷²Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Rmk. 4.7.

⁷³Ribes and Zalesskii, *Profinite Groups*, Thm. 3.6.2.

⁷⁴Patrick Forré. “Strongly free sequences and pro- p -groups of cohomological dimension 2”. In: *Journal für die reine und angewandte Mathematik (Crelles Journal)* 2011.658 (Jan. 2011). ISSN: 1435-5345. DOI: 10.1515/crelle.2011.067, Def. 1.1.

⁷⁵See Section 4.2

Definition 4.4.9. Let $\mathbb{F}_p\langle a_1, \dots, a_n \rangle$ be the free $\mathbb{F}_p$ -algebra on n generators with augmentation ideal $I = (a_1, \dots, a_n)$. Choose l elements $\rho_1, \dots, \rho_l \in I$ and let $J = (\rho_1, \dots, \rho_l)$ be the two-sided ideal they generate. We say that the sequence of elements $\rho_1, \dots, \rho_l$ is *strongly free* if J/JI is a free $\mathbb{F}_p\langle a_1, \dots, a_n \rangle/J$ -module with basis $\{\rho_i + JI\}$.

Let F be a free pro- p group on n generators $g_1, \dots, g_n$ and identify the graded ring $\text{Gr } \mathbb{F}_p[[F]]$ with the free $\mathbb{F}_p$ -algebra on n generators $a_i = \overline{g_i - 1}$. A minimal pro- p presentation

$$\langle g_1, \dots, g_n \mid r_1, \dots, r_n \rangle$$

is called a *strongly free* presentation if the homogeneous components $\rho_i = \overline{r_i - 1}$ form a strongly free sequence in $\text{Gr } \mathbb{F}_p[[F]]$. A pro- p group is *mild* if it has a strongly free presentation.

The relevance of mild presentations of a pro- p group G is that they allow us to explicitly describe the graded ring $\text{Gr } \mathbb{F}_p[[G]]$ with respect to the filtration by powers of the augmentation ideal.⁷⁶

Proposition 4.4.10. *If*

$$G \simeq \langle g_1, \dots, g_n \mid r_1, \dots, r_n \rangle$$

is a strongly free presentation, then $\text{Gr } \mathbb{F}_p[[G]] \simeq \text{Gr } \mathbb{F}_p[[F]]/J$ where F is the free pro- p group on $g_1, \dots, g_n$, both $\mathbb{F}_p[[G]]$ and $\mathbb{F}_p[[F]]$ are filtered by powers of the augmentation ideal and J is the two-sided ideal generated by the homogeneous components $\rho_i = \overline{r_i - 1}$.

We turn now to the problem of showing that mild flag of G presentations are strongly free, and to characterize the graded ring $\mathbb{F}_p[[G]]$. Until the end of this section, we assume that

$$G = \langle x_1, \dots, x_n, g \mid r_1, \dots, r_n \rangle$$

is a mild flag presentation for G , and we let F be the free pro- p group on the set $\{x_1, \dots, x_n, g\}$. Denote by I_G the augmentation ideal of $\mathbb{F}_p[[G]]$ and by

$$D_i(G) = \{g \in G : g - 1 \in I_G^i\}$$

the i -th dimension subgroup mod p . Each $D_i(G)$ is a normal open subgroup of G and their intersection is the trivial subgroup.

We recall that

$$\mathfrak{Q} = \bigoplus_{i \geq 1} D_i(F)/D_{i+1}(F)$$

has the structure of a free restricted Lie algebra over $\mathbb{F}_p$ in the sense of Nathan Jacobson.⁷⁷ A free generating set of this Lie algebra is the image $\xi_1, \dots, \xi_n, \gamma$ of $x_1, \dots, x_n, g$, and we can identify $\text{Gr } \mathbb{F}_p[[F]]$ with the universal restricted enveloping algebra of $\mathfrak{Q}$.^{78,79} If the elements

⁷⁶Ján Mináč et al. “Mild pro- p groups and the Koszulity conjectures”. In: *Expositiones Mathematicae* 40.3 (Sept. 2022), pp. 432–455. ISSN: 0723-0869. DOI: 10.1016/j.exmath.2022.03.004, Prop. 4.5.

⁷⁷Nathan Jacobson. *Lie Algebras*. en. Dover Books on Mathematics. Mineola, NY: Dover Publications, Dec. 1979, Sec. V.7.

⁷⁸Michel Lazard. “Sur les groupes nilpotents et les anneaux de Lie”. In: *Annales scientifiques de l’École normale supérieure* 71.2 (1954), pp. 101–190. ISSN: 1873-2151. DOI: 10.24033/asens.1021, Thm. 6.5.

⁷⁹Lazard, “Groupes analytiques p -adiques”, Thm. A.3.5.

$\rho_1, \dots, \rho_l$ associated to a presentation of G lie in $\mathfrak{Q}$ and $\mathfrak{r}$ is the restricted Lie ideal of $\mathfrak{Q}$ that they generate, then for J the ideal of $\text{Gr } \mathbb{F}_p[[F]]$ generated by the ρ_i the $\text{Gr } \mathbb{F}_p[[F]]/J$ -modules $J/J \text{Gr}(I_F)$ and $M = \mathfrak{r}/[\mathfrak{r}, \mathfrak{r}] + \mathfrak{r}^{[p]}$ are isomorphic. We also recall Lazard's elimination theorem for free (unrestricted) Lie algebras.⁸⁰

Theorem 4.4.11. *Let R be a non-zero commutative ring, X a set and S a subset of X . If $\mathfrak{Q}(X)$ is the free Lie R -algebra on X , then $\mathfrak{Q}(X)$ is isomorphic as a Lie algebra to $\mathfrak{Q}(S) \oplus \mathfrak{h}$ where $\mathfrak{h}$ is the ideal of $\mathfrak{Q}(X)$ generated by $X \setminus S$ and is isomorphic to the free Lie algebra on the set*

$$\{[x, s_1, \dots, s_k] : x \in X \setminus S, s_1, \dots, s_k \in S\}.$$

Note that since $\mathfrak{Q}$ is a free restricted Lie algebra it is also a free unrestricted $\mathbb{F}_p$ -Lie algebra on the p -th powers $\xi_i^{p^j}$ and γ^{p^j} of the restricted basis $\xi_1, \dots, x_n, \gamma$. The following is an immediate consequence of Lazard's elimination theorem.⁸¹

Theorem 4.4.12. *If $\mathfrak{h}$ is the restricted ideal generated by the ξ_i , then*

$$\mathfrak{Q} \simeq \langle \gamma^{p^j} : j \geq 0 \rangle \oplus \mathfrak{h}$$

and $\mathfrak{h}$ itself is a free restricted Lie algebra with free generating set

$$\left\{ [\xi_i, \underbrace{\gamma, \dots, \gamma}_{k \text{ times}}] : 1 \leq i \leq n, k \geq 0 \right\}.$$

Our main ingredient to proving that mild flag presentations are strongly free is the following restricted analogue of a criterion by Labute.^{82,83}

Proposition 4.4.13. *If $\rho_1, \dots, \rho_l$ are homogeneous elements of $\mathfrak{h}$ with respect to the canonical grading of $\mathfrak{Q}$ which are linearly independent over $\mathbb{F}_p[\gamma]$ modulo $[\mathfrak{h}, \mathfrak{h}] + \mathfrak{h}^{[p]}$, then they are strongly free. Moreover, the restricted Lie algebra $\mathfrak{h}/(\rho_1, \dots, \rho_l)$ is free, so that*

$$\mathfrak{Q}/(\rho_1, \dots, \rho_l) \simeq \mathfrak{h}/(\rho_1, \dots, \rho_l) \rtimes \langle \gamma^{p^i} : i \geq 0 \rangle$$

is a free-by-(free of rank one) restricted Lie algebra.

⁸⁰Bourbaki, *Lie Groups and Lie Algebras*, Prop. 10 in Sec. II.2.9.

⁸¹Ralph Stöhr. "Restricted Lazard elimination and modular Lie powers". In: *Journal of the Australian Mathematical Society* 71.2 (Oct. 2001), pp. 259–278. ISSN: 1446-8107. DOI: 10.1017/s1446788700002901, Sec. 1.2.

⁸²Jaikin-Zapirain and Souza, "Sylvester domains and pro- p groups", Prop. 4.10.

⁸³John Labute. "Mild pro- p -groups and Galois groups of p -extensions of $\mathbb{Q}$ ". In: *Journal für die reine und angewandte Mathematik (Crelles Journal)* 2006.596 (Jan. 2006). ISSN: 1435-5345. DOI: 10.1515/crelle.2006.058, Thm. 3.3.

Proof. If $\mathfrak{r}$ and J denote the restricted ideal of $\mathfrak{L}$ and the graded two-sided ideal of $\text{Gr } \mathbb{F}_p \llbracket F \rrbracket$ generated by the ρ_i , we must show that

$$\mathcal{M} \simeq \mathfrak{r} / [\mathfrak{r}, \mathfrak{r}] + \mathfrak{r}^{[p]}$$

is a free $\text{Gr } \mathbb{F}_p \llbracket F \rrbracket / J$ -module with basis given by the image of the ρ_i . Observe that $\mathfrak{r}$ is generated, as a restricted ideal of $\mathfrak{h}$, by the elements

$$\rho_{i,k} = [\rho_i, \underbrace{\gamma, \dots, \gamma}_{k \text{ times}}]$$

with $k \geq 0$ and $1 \leq i \leq l$. Hence, it suffices to show that these elements are a part of a free restricted generating set of $\mathfrak{h}$, for which it also suffices to show that the $\rho_1, \dots, \rho_l$ are $\mathbb{F}_p[\gamma]$ -independent modulo $[\mathfrak{h}, \mathfrak{h}] + \mathfrak{h}^{[p]}$. Since this holds by hypothesis, we are done. Moreover, as the $\rho_{i,k}$ form part of a free restricted basis of $\mathfrak{h}$, the quotient $\mathfrak{h} / (\rho_1, \dots, \rho_l)$ is again a free restricted Lie algebra. $\square$

Corollary 4.4.14. *The n elements $\rho_i = [\xi_i, \gamma]$ are strongly free, as so is, modulo $[\mathfrak{h}, \mathfrak{h}] + \mathfrak{h}^{[p]}$, any $\mathbb{F}_p$ -linearly independent subset of the $\mathbb{F}_p$ -vector space that they span.*

Proof. The first part follows from Theorem 4.4.12 and Proposition 4.4.13. For the second part, note that an $\mathbb{F}_p$ -linearly independent subset of their span will remain $\mathbb{F}_p[\gamma]$ -linearly independent modulo $[\mathfrak{h}, \mathfrak{h}] + \mathfrak{h}^{[p]}$. $\square$

We are now ready to prove the main result of this section.⁸⁴

Proposition 4.4.15. *Every mild flag presentation of a free-by- $\mathbb{Z}_p$ pro- p group is strongly free, and $\text{Gr}(\mathbb{F}_p \llbracket G \rrbracket)$ is the skew polynomial ring $(\text{Gr } \mathbb{F}_p \llbracket N \rrbracket)[\overline{g-1}, \text{id}, \overline{\delta}]$ generated by $\overline{g-1}$ over a free algebra $\text{Gr}(\mathbb{F}_p \llbracket N \rrbracket)$, where $\mathbb{F}_p \llbracket N \rrbracket$ is given the induced filtration from the filtration on $\mathbb{F}_p \llbracket G \rrbracket$, σ is right conjugation by g and $\delta = \sigma - \text{id}$. In particular, it is a domain.*

Proof. The l relations in (4.3) of a mild flag presentation lie in the $\mathbb{F}_p$ -span of $[\xi_i, \gamma]$ modulo $[\mathfrak{h}, \mathfrak{h}] + \mathfrak{h}^{[p]}$ in the language of Corollary 4.4.14. This can be seen by observing that (1) the ρ_i have degree two and (2) writing them in the basis of $D_2(F)/D_3(F)$ given by the commutators $[x_i, x_j]$, $[x_i, g]$ and possibly the squares x_i^2 and g^2 if $p = 2$, the image of $\Phi(\widetilde{N})$ in this quotient does not include expressions involving $[x_i, g]$ and g^2 . The last part follows from the universal property of restricted enveloping algebras and the identification

$$\text{Gr } \mathbb{F}_p \llbracket G \rrbracket \simeq \text{Gr } \mathbb{F}_p \llbracket F \rrbracket / J \simeq \mathcal{U}(\mathfrak{L} / \mathfrak{r})$$

of Proposition 4.4.10, while observing that the embedding $\mathbb{F}_p \llbracket N \rrbracket \rightarrow \mathbb{F}_p \llbracket G \rrbracket$ is strict by definition and hence identifies $\text{Gr } \mathbb{F}_p \llbracket N \rrbracket$ with the subalgebra $\mathcal{U}(\mathfrak{h} / \mathfrak{r})$ of $\text{Gr } \mathbb{F}_p \llbracket G \rrbracket$ generated by the ξ_i . One can also apply Proposition 1.4.5 to see that $\text{Gr } \mathbb{F}_p \llbracket G \rrbracket$ is a skew polynomial ring over $\text{Gr } \mathbb{F}_p \llbracket N \rrbracket$, since conjugation is a strict automorphism under the filtration by powers of the augmentation ideal of $\mathbb{F}_p \llbracket G \rrbracket$. $\square$

⁸⁴Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Cor. 4.12.

Now, let $G = N \rtimes \mathbb{Z}_p$ be a mild flag pro- p group, I_G be the augmentation ideal of $\mathbb{F}_p[[G]]$ and define the function $w: \mathbb{F}_p[[G]] \rightarrow \{p^i: i \geq 0\} \cup \{0\}$ by

$$w(x) = p^{-\max\{i \geq 0: x \in I_G^i\}}.$$

Since $\text{Gr } \mathbb{F}_p[[G]]$ has no zero divisors, this function w is a valuation on $\mathbb{F}_p[[G]]$ in the sense of Mikhail Ershov and Jaikin-Zapirain,⁸⁵ that is, it satisfies:

- (i) $w(x) = 0$ if and only if $x = 0$;
- (ii) $w(1) = 1$;
- (iii) $w(x + y) \leq \max\{w(x), w(y)\}$, with equality if $w(x) \neq w(y)$;
- (iv) $w(xy) = w(x)w(y)$.

In particular, $\mathbb{F}_p[[G]]$ does not have zero divisors.⁸⁶ The proof of Proposition 4.4.4 also showed that $w(\partial(x)) < w(x)$ for any non-zero $x \in \mathbb{F}_p[[G]]$.

By the inclusion $\mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p[[G]]$, w restricts to a valuation on $\mathbb{F}_p[[N]]$ as well, and the induced filtration on $\mathbb{F}_p[[N]]$ is described by

$$F_i \mathbb{F}_p[[N]] = \{x \in \mathbb{F}_p[[N]]: w(x) \leq p^{-i}\}.$$

If we set

$$x_{i,j} = [x_i, \underbrace{g, \dots, g}_{j \text{ times}}] \text{ and } X_{i,j} = x_{i,j} - 1,$$

then the group algebra $\mathbb{F}_p[[N]]$ is isomorphic to the free pro- p algebra $\mathbb{F}_p\langle\langle X \rangle\rangle$ where

$$X = \{X_{i,j}: 1 \leq i \leq n, j \geq 0 \text{ if } i > 1 \text{ and } j = 0 \text{ otherwise}\}$$

by Theorem 4.1.7. The following definition is from Ershov and Jaikin-Zapirain.

Definition 4.4.16. Let X be a countable set, $\text{Mon}(X)$ be the set of all non-commutative monomials in X and $w: \mathbb{F}_p\langle\langle X \rangle\rangle \rightarrow \{p^i: i \in \mathbb{Z}\} \cup 0$ be a valuation. Then, we say that w is a *weight function* on $\mathbb{F}_p\langle\langle X \rangle\rangle$ if the equality

$$w\left(\sum_{m \in \text{Mon}(X)} \lambda_m X\right) = \max\{w(X): \lambda_m \neq 0\} \quad (4.7)$$

holds for any coefficients $\lambda_m \in \mathbb{F}_p$.⁸⁷

⁸⁵Mikhail Ershov and Andrei Jaikin-Zapirain. “Groups of Positive Weighted Deficiency and Their Applications”. en. In: *Journal für die reine und angewandte Mathematik (Crelles Journal)* 2013.677 (Apr. 2013), pp. 71–134. DOI: 10.1515/crelle.2012.013, Sec. 2.2.

⁸⁶See also Corollary 1.4.4

⁸⁷Observe that, since the set X is topologically nilpotent, all such sums converge in $\mathbb{F}_p\langle\langle X \rangle\rangle$.

One of their results is a characterization of weight functions among valuations on free pro- p algebras:⁸⁸

Proposition 4.4.17. *A valuation w on $\mathbb{F}_p\langle\langle X \rangle\rangle$ is a weight function if and only if $\text{Gr } \mathbb{F}_p\langle\langle X \rangle\rangle$ is a free $\mathbb{F}_p$ -algebra on the set $\overline{X}$, where the filtration on $\mathbb{F}_p\langle\langle X \rangle\rangle$ is induced by w :*

$$F_i \mathbb{F}_p\langle\langle X \rangle\rangle = \{x \in \mathbb{F}_p\langle\langle X \rangle\rangle : w(x) \leq p^{-i}\}.$$

Corollary 4.4.18. *Let w be the valuation constructed from the powers of the augmentation ideal in $\mathbb{F}_p[[G]]$, where $G \simeq N \rtimes \mathbb{Z}_p$ is a mild flag pro- p group. Then, the restriction of w to $\mathbb{F}_p[[N]]$ is a weight function.*

By Proposition 4.1.6, this free pro- p algebra itself can be seen as a completion of the free $\mathbb{F}_p$ -algebra $\mathbb{F}_p\langle X \rangle$ under the filtration where we take the function $v(X_{i,j}) = -\log_p(w(X_{i,j}))$.⁸⁹ It then follows from Corollary 4.4.18 that the filtration induced from this completion and the filtration induced from w agree.

4.5 Embeddings of the group algebra

For this section, we fix a mild flag pro- p group $G \simeq N \rtimes \mathbb{Z}_p$ with a mild flag presentation

$$G \simeq \langle x_1, \dots, x_n, g \mid r_1, \dots, r_l \rangle,$$

and let $w: \mathbb{F}_p[[G]] \rightarrow \{p^i : i \in \mathbb{Z}\} \cup \{0\}$ be the valuation defined by powers of the augmentation ideal I_G .⁹⁰ This restricts to a valuation $w: \mathbb{F}_p[[N]] \rightarrow \mathbb{Z}_p$ on the free pro- p algebra $\mathbb{F}_p[[N]] \simeq \mathbb{F}_p\langle\langle X \rangle\rangle$ on the variables $X = \{X_{i,j} : 1 \leq i \leq n, j \geq 0 \text{ if } i > l \text{ or } j = 0 \text{ otherwise}\}$ defined by

$$x_{i,j} = [x_i, \underbrace{g, \dots, g}_{j \text{ times}}], \quad X_{i,j} = x_{i,j} - 1.$$

By Corollary 4.4.18, w is a weight function on $\mathbb{F}_p[[N]]$. If we denote $g - 1 \in \mathbb{F}_p[[G]]$ by s , the right conjugation with g by $\sigma: \mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p[[N]]$, $\delta = \sigma - \text{id}$ and by $\bar{s}$ the principal symbol of s in $\text{Gr } \mathbb{F}_p[[G]]$, we have

$$\mathbb{F}_p[[G]] \simeq \mathbb{F}_p[[N]][[s; \sigma, \delta]], \quad \text{Gr } \mathbb{F}_p[[G]] \simeq (\text{Gr } \mathbb{F}_p[[N]])[\bar{s}, \text{id}, \bar{\delta}]$$

by Propositions 4.4.4 and 4.4.15. We can also identify $\mathbb{F}_p[[N]]$ with the completion of the free $\mathbb{F}_p$ -algebra $\mathbb{F}_p\langle X \rangle$ under the filtration induced by $v(X_{i,j}) = -\log_p(w(X_{i,j}))$, and this filtration

⁸⁸ Ershov and Jaikin-Zapirain, “Groups of Positive Weighted Deficiency and Their Applications”, Prop. 3.2.

⁸⁹ This function has finite fibers since $v(X_{i,j}) = 1 + j$.

⁹⁰ As we’ve observe in the end of the last section, it is indeed a valuation. This follows from Proposition 4.4.15.

agree with the one induced by w since it is a weight function. We also have that $w(\delta(x)) < w(x)$ for any $x \in \mathbb{F}_p[[N]]$.

Our goal in this section is to use the results of Section 4.2 to embed $\mathbb{F}_p[[G]]$ in a division ring to which the valuation w extends. Let $\mathcal{Q}$ be the universal division $\mathbb{F}_p\langle X \rangle$ -ring. We know from Proposition 4.2.1 that there is a strict embedding $\varphi: \mathbb{F}_p[[N]] \rightarrow \mathcal{Q}[[t]]$ given by

$$\varphi(X_{i,j}) = X_{i,j} t^{v(X_{i,j})},$$

whose image is contained in $\mathbb{F}_p\langle X \rangle[[t]]$. If we define $w: \mathcal{Q}[[t]] \rightarrow \{p^i: i \in \mathbb{Z}\} \cup \{0\}$ by

$$w(x) = p^{-\max\{i: x \in t^i \mathcal{Q}[[t]]\}},$$

since φ is strict we see that w extends the valuation on $\mathbb{F}_p[[N]]$.

Proposition 4.5.1. *Both σ and δ extend to $\mathcal{Q}[[t]]$ in such a way that σ is an automorphism, $\delta = \sigma - \text{id}$, $\sigma(t) = t$, $\delta(t) = 0$,*

$$\sigma(X_{i,j}) = \frac{\varphi(\sigma(X_{i,j}))}{t^{v(X_{i,j})}}, \quad \delta(X_{i,j}) = \frac{\varphi(\delta(X_{i,j}))}{t^{v(X_{i,j})}}.$$

Moreover, $\delta(\mathcal{Q}[[t]]) \subseteq t\mathcal{Q}[[t]]$ and therefore it is also topologically nilpotent on $\mathcal{Q}[[t]]$. In particular, φ extends to an embedding $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathcal{Q}[[s, \sigma, \delta]]$.

Proof. Once we set $\sigma(t) = t$, by the universal property of $\mathbb{F}_p\langle X \rangle$ we can define σ arbitrarily on the generators $X_{i,j}$ to obtain a homomorphism $\sigma: \mathbb{F}_p\langle X \rangle[[t]] \rightarrow \mathbb{F}_p\langle X \rangle[[t]]$. For this map to extend σ as defined in $\mathbb{F}_p[[N]]$, the formula must be as given. Hence, the derivation $\delta = \sigma - \text{id}$ also extends, and since $\delta(t) = 0$ one sees that

$$\delta(\varphi(X_{i,j})) = \delta(X_{i,j} t^{v(X_{i,j})}) = \delta(X_{i,j}) t^{v(X_{i,j})}.$$

As $w(\delta(X_{i,j})) < w(X_{i,j})$ and φ is strict, we get that $\delta(\mathbb{F}_p\langle X \rangle[[t]]) \subseteq t\mathbb{F}_p\langle X \rangle[[t]]$, and thus the successive compositions $\delta^i(x)$ converge to 0 for any $x \in \mathbb{F}_p\langle X \rangle$. Hence, the ring of skew power series $\mathbb{F}_p\langle X \rangle[[t]][[s, \sigma, \delta]]$ exists and we get an embedding

$$\varphi: \mathbb{F}_p[[G]] \simeq \mathbb{F}_p[[N]][[s, \sigma, \delta]] \rightarrow \mathbb{F}_p\langle X \rangle[[t]][[s, \sigma, \delta]]$$

that maps s to s extending $\varphi: \mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p\langle X \rangle[[t]]$.

We now want to extend σ to $\mathcal{Q}[[t]]$. Let us consider σ as a map $\mathbb{F}_p\langle X \rangle \rightarrow \mathcal{Q}[[t]]$ by taking the composition

$$\mathbb{F}_p\langle X \rangle \xrightarrow{\varphi} \mathbb{F}_p\langle X \rangle[[t]] \xrightarrow{\sigma} \mathbb{F}_p\langle X \rangle[[t]] \rightarrow \mathcal{Q}[[t]].$$

If we show that the image of every full matrix through σ is invertible, by the universal property of $\mathcal{Q}$ we get an extension of σ to a map $\sigma: \mathcal{Q} \rightarrow \mathcal{Q}[[t]]$. Moreover, this suffices to define σ and hence $\delta = \sigma - \text{id}$ on $\mathcal{Q}[[t]]$, since we have $\sigma(t) = t$.

Since $\delta(\mathbb{F}_p\langle X \rangle[[t]]) \leq t\mathbb{F}_p\langle X \rangle[[t]]$, for any matrix A over $\mathbb{F}_p\langle X \rangle$ there exists a matrix B of the same size such that $\sigma(A) = A + Bt$. However, a matrix over $\mathbb{Q}[[t]]$ is invertible if and only if its projection over $\mathbb{Q}$ is invertible, which is the case for $\sigma(A)$ if A is full. Therefore, we can extend σ and $\delta = \sigma - \text{id}$ to $\mathbb{Q}[[t]]$.

To show that $\delta(\mathbb{Q}[[t]]) \leq t\mathbb{Q}[[t]]$ and thus it is still topologically nilpotent, we use that $\mathbb{Q}$ is the division closure of $\mathbb{F}_p\langle X \rangle$ and hence any element of $\mathbb{Q}$ can be recursively obtained from the elements of $\mathbb{F}_p\langle X \rangle$ by taking inverses of all non-zero elements and closing under the ring operations. This implies that it suffices to prove that $\delta(a^{-1}) \in t\mathbb{Q}[[t]]$ whenever $\delta(a) \in t\mathbb{Q}[[t]]$ for any non-zero $a \in \mathbb{Q}$. We write $\delta(a) = tb$ for some $b \in \mathbb{Q}[[t]]$ and observe that

$$\delta(a^{-1}) = -\sigma(a)^{-1}\delta(a)a^{-1} = -\sigma(a)ba^{-1}t.$$

Hence, δ is topologically nilpotent on $\mathbb{Q}[[t]]$ and we get the desired extension

$$\varphi: \mathbb{F}_p[[G]] \rightarrow \mathbb{Q}[[t]]. \quad \square$$

In order to shorten the notation, we will denote the filtered ring $\mathbb{Q}[[t]][[s; \sigma, \delta]]$ simply by $\mathbb{D}$. We now turn to the problem of extending the valuation w from $\mathbb{F}_p[[G]]$ to one on $\mathbb{D}$, for which we need to compute its associated graded ring with respect to the linear, positive, complete and separated filtration⁹¹

$$F_i\mathbb{D} = \sum_{j+k=i} s^k \mathbb{D} t^j = \left\{ \sum_{j \geq 0} s^j t^{\min\{0, i-j\}} a_j : a_j \in \mathbb{Q}[[t]] \right\}.$$

Proposition 4.5.2. *If $\bar{t} = t + F_2\mathbb{D}$ and $\bar{s} = s + F_2\mathbb{D}$, then the graded ring $\text{Gr } \mathbb{D}$ is isomorphic to the skew polynomial ring $\mathbb{Q}[\bar{t}][\bar{s}; \text{id}, \bar{\delta}]$. In particular, the induced map $\text{Gr } \varphi: \text{Gr } \mathbb{F}_p[[G]] \rightarrow \text{Gr } \mathbb{D}$ is injective and $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathbb{D}$ is strict.*

Proof. The first part is the contents of Proposition 1.4.5, observing that $\sigma: \mathbb{Q}[[t]] \rightarrow \mathbb{Q}[[t]]$ is strict since $\sigma(t) = t$ and $\sigma(a) = a + bt$ for some $b \in \mathbb{Q}[[t]]$ by Proposition 4.5.1, the induced filtration on $\mathbb{Q}[[t]]$ by the inclusion into $\mathbb{D}$ is the t -filtration and under this filtration we have $\text{Gr } \mathbb{Q}[[t]] \simeq \mathbb{Q}[\bar{t}]$. Since $\text{Gr } \mathbb{F}_p[[N]] \rightarrow \mathbb{Q}[\bar{t}]$ is injective and $\text{Gr } \mathbb{F}_p[[G]] \simeq (\text{Gr } \mathbb{F}_p[[N]])[\bar{s}; \text{id}, \bar{\delta}]$, we get the injectivity of $\text{Gr } \varphi$. That φ is a strict injection now follows from Corollary 1.4.9. $\square$

Corollary 4.5.3. *The function $w: \mathbb{D} \rightarrow \{p^i : i \in \mathbb{Z}\} \cup \{0\}$ given by*

$$w(x) = p^{-\max\{i : x \in F_i\mathbb{D}\}}$$

is a valuation on $\mathbb{D}$ that extends the valuation w on $\mathbb{F}_p[[G]]$. In particular, $\mathbb{D}$ is a Noetherian domain and has a Ore division ring of fractions $\mathcal{D}$, to which the valuation w extends uniquely.

⁹¹See the discussion at the end of Section 1.3.

Proof. The fact that w satisfies the axioms of a valuation is precisely the fact that $\text{Gr } \mathbb{D} \simeq \mathcal{Q}[\bar{t}][\bar{s}; \text{id}, \bar{\delta}]$ has no zero divisors. Since φ is strict, it preserves the filtration, and hence the valuation w . Moreover, since $\text{Gr } \mathbb{D}$ is a Noetherian domain, then so is $\mathbb{D}$ by Corollary 1.4.4 and Proposition 1.4.12. By Proposition 2.1.3, the set of non-zero elements in $\mathbb{D}$ satisfies both the left and right Ore condition, so $\mathcal{D}$ exists.

Any element of $\mathcal{D}$ can be written as $b^{-1}a$ for some $a, b \in \mathbb{D}$ with b non-zero. We define $w(b^{-1}a) = w(a)/w(b)$. In $\mathcal{D}$, we have $b^{-1}a = d^{-1}c$ if and only if there exists $e, f \in \mathbb{D}$ with $f \neq 0$ such that $fd = eb$ and $fc = ea$. Hence, assuming that $a \neq 0$, we have $c, e \neq 0$ and therefore

$$\begin{aligned} \frac{w(a)}{w(b)} &= \frac{w(a)w(e)}{w(b)w(e)} \\ &= \frac{w(ea)}{w(eb)} \\ &= \frac{w(fc)}{w(fd)} \\ &= \frac{w(f)w(c)}{w(f)w(d)} \\ &= \frac{w(c)}{w(d)}, \end{aligned}$$

so that w is well defined. Given $b^{-1}a$ and $d^{-1}c$ in $\mathcal{D}$, we find $f^{-1}e$ and $h^{-1}g$ in $\mathcal{D}$ for which $ed = fa$ and $hb = gd$. The sum and product of $b^{-1}a$ and $d^{-1}c$ are defined as

$$b^{-1}a \cdot d^{-1}c = (fb)^{-1}(ec), \quad b^{-1}a + d^{-1}c = (hb)^{-1}(ha + gc).$$

We then compute:

$$\begin{aligned} w(b^{-1}ad^{-1}c) &= \frac{w(ec)}{w(fb)} \\ &= \frac{w(e)w(c)w(d)}{w(f)w(b)w(d)} \\ &= \frac{w(ed)w(c)}{w(f)w(b)w(d)} \\ &= \frac{w(fa)w(c)}{w(f)w(b)w(d)} \\ &= \frac{w(a)w(c)}{w(b)w(d)} = w(b^{-1}a) \cdot w(d^{-1}c), \end{aligned}$$

$$\begin{aligned}
w(b^{-1}a + d^{-1}c) &= \frac{w(ha + gc)}{w(hb)} \\
&\leq \frac{\max\{w(ha), w(gc)\}}{w(hb)} \\
&= \max\left\{\frac{w(ha)}{w(hb)}, \frac{w(gc)}{w(gd)}\right\} \\
&= \max\{w(b^{-1}a), w(d^{-1}c)\},
\end{aligned}$$

as desired. It is clear that w is uniquely determined by this rule. $\square$

4.6 Homological properties of the embedding

We keep the notation of Section 4.5:

$$G \simeq \langle x_1, \dots, x_n, g \mid r_1, \dots, r_l \rangle$$

is a mild flag presentation, N is the closed normal subgroup of G generated by the x_i , $g \in G$ is an element that generates $G/N \simeq \mathbb{Z}_p$ and $s = g - 1 \in \mathbb{F}_p[[G]]$. If $\sigma: \mathbb{F}_p[[N]] \rightarrow \mathbb{F}_p[[N]]$ is right conjugation by g , we have

$$\mathbb{F}_p[[G]] \simeq \mathbb{F}_p[[N]][[s; \sigma, \delta]]$$

where $\delta = \sigma - \text{id}$. If I_G denotes the augmentation ideal of $\mathbb{F}_p[[G]]$, the function

$$w(x) = p^{-\max\{i: x \in I_G^i\}}$$

is a valuation on $\mathbb{F}_p[[G]]$. Denoting by X the set of the $X_{i,j}$ defined by

$$x_{i,j} = [x_i, \underbrace{g, \dots, g}_{j \text{ times}}], X_{i,j} = x_{i,j} - 1,$$

with $j = 0$ for $i \leq l$ and $j \geq 0$ otherwise, we can identify $\mathbb{F}_p[[N]]$ with the free pro- p algebra $\mathbb{F}_p\langle X \rangle$. Under the induced I_G -adic filtration, we have $\text{Gr } \mathbb{F}_p[[N]] \simeq \mathbb{F}_p\langle X \rangle$ and

$$\text{Gr } \mathbb{F}_p[[G]] \simeq (\text{Gr } \mathbb{F}_p[[N]])[\bar{s}; \text{id}, \bar{\delta}].$$

Let $\mathcal{Q}$ be the universal division ring of fractions of $\mathbb{F}_p\langle X \rangle$. We have constructed in Section 4.5 a series of strict embeddings

$$\mathbb{F}_p[[G]] \xrightarrow{\varphi} \mathcal{Q}[[t]][[s; \sigma, \delta]] = \mathbb{D} \rightarrow \mathcal{D}$$

and extensions of the valuation w all the way to the division ring $\mathcal{D}$. Moreover, we know that

$$\text{Gr } \varphi: \mathbb{F}_p\langle X \rangle[\bar{s}; \text{id}, \bar{\delta}] = \text{Gr } \mathbb{F}_p[[G]] \rightarrow \text{Gr } \mathbb{D} = \mathcal{Q}[\bar{t}][\bar{s}; \text{id}, \bar{\delta}]$$

is injective.

To prove that $\mathbb{F}_p[[G]]$ is a Sylvester domain, we will apply Theorem 4.1.14 to the embedding $\mathbb{F}_p[[G]] \rightarrow \mathcal{D}$ using the results of Sections 1.4 and 4.2 to see that it satisfies the necessary homological conditions. We begin by analyzing the homological properties of $\text{Gr } \mathcal{D}$ as a module over $\text{Gr } \mathbb{F}_p[[G]]$.

Lemma 4.6.1. *As a left (resp. right) $\text{Gr } \mathbb{F}_p[[G]]$ -module, we have $\text{Gr } \mathcal{D} \simeq \text{Gr } \mathbb{F}_p[[G]] \otimes_{\mathbb{F}_p\langle X \rangle} \mathbb{Q}[\bar{t}]$ (resp. $\mathbb{Q}[\bar{t}] \otimes_{\mathbb{F}_p\langle X \rangle} \text{Gr } \mathbb{F}_p[[G]]$).*

Proof. The multiplication maps

$$\text{Gr } \mathbb{F}_p[[G]] \times \mathbb{Q}[\bar{t}], \mathbb{Q}[\bar{t}] \times \text{Gr } \mathbb{F}_p[[G]] \rightarrow \text{Gr } \mathcal{D}$$

are bilinear and $\mathbb{F}_p\langle X \rangle$ -balanced, and hence define the said homomorphisms. Observe that this homomorphisms are bimodule homomorphisms, commuting with the $\mathbb{F}_p[[G]]$ action on one side and the $\mathbb{Q}[\bar{t}]$ action on the other. Moreover, both are free $\mathbb{Q}[\bar{t}]$ -modules, and the multiplication map sends the bases $\{\bar{s}^i \otimes 1 : i \geq 0\}$ and $\{1 \otimes \bar{s}^i : i \geq 0\}$ of the formers to the basis $\{\bar{s}^i\}$ of the latter. Hence, they are isomorphisms. $\square$

Proposition 4.6.2. *Let M (resp. M') be a left (resp. right) $(\text{Gr } \mathbb{F}_p[[G]])$ -module and N (resp. N') be a left (resp. right) $(\text{Gr } \mathbb{F}_p[[G]])$ -submodule of $(\text{Gr } \mathcal{D})^k$ for some $k \geq 1$. Then, the following holds:*

- (1) $\text{Gr } \mathbb{F}_p[[G]]$ has global dimension at most two;
- (2) $\text{Gr } \mathcal{D}$ has projective dimension at most one as a left and right $\text{Gr } \mathbb{F}_p[[G]]$ -module;
- (3) N and N' have projective dimension at most one;
- (4) $\text{Tor}_2^{\text{Gr } \mathbb{F}_p[[G]]}(\text{Gr } \mathcal{D}, M) = \text{Tor}_2^{\text{Gr } \mathbb{F}_p[[G]]}(M', \text{Gr } \mathcal{D}) = 0$;
- (5) $\text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(\text{Gr } \mathcal{D}, N'') = 0$ for every left $(\text{Gr } \mathcal{D})$ -module N'' ;
- (6) $\text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(\text{Gr } \mathcal{D}, N) = 0$;
- (7) $\text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(N', \text{Gr } \mathcal{D}) = 0$;
- (8) $\text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(N', N) = 0$;
- (9) $\text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(\text{Gr } \mathcal{D}, \text{Gr } \mathcal{D}) = 0$.

Proof. (1) The global dimension of a skew polynomial ring $\text{Gr } \mathbb{F}_p[[G]] \simeq (\text{Gr } \mathbb{F}_p[[N]])[\bar{s}; \text{id}, \bar{\delta}]$ increases by at most one,⁹² so

$$\text{gl. dim}(\text{Gr } \mathbb{F}_p[[G]]) \leq \text{gl. dim}(\text{Gr } \mathbb{F}_p[[N]]) + 1 = 2$$

since $\text{Gr } \mathbb{F}_p[[N]] \simeq \mathbb{F}_p\langle X \rangle$ is a fir.

(2) We prove the claim for the structure as a left module, the proof for the right side being analogous. Since $\mathbb{F}_p\langle X \rangle$ has global dimension at most 1 and is projective-free, there are free $\mathbb{F}_p\langle X \rangle$ -modules F_0 and F_1 such that

$$0 \rightarrow F_1 \rightarrow F_0 \rightarrow \mathcal{Q}[\bar{t}] \rightarrow 0$$

is a resolution of the left $\mathbb{F}_p\langle X \rangle$ -module $\mathcal{Q}[\bar{t}]$. By Lemma 4.6.1, tensoring with $\text{Gr } \mathbb{F}_p[[G]]$ over $\mathbb{F}_p\langle X \rangle$ we get the long exact sequence

$$0 \rightarrow \text{Tor}_1^{\mathbb{F}_p\langle X \rangle}(\text{Gr } \mathbb{F}_p[[G]], \mathcal{Q}[\bar{t}]) \rightarrow F_1' \rightarrow F_0' \rightarrow \text{Gr } \mathbb{D} \rightarrow 0$$

of left $\text{Gr } \mathbb{F}_p[[G]]$ -modules where F_1' and F_0' are free. As

$$\text{Gr } \mathbb{F}_p[[G]] \simeq (\text{Gr } \mathbb{F}_p[[N]])[\bar{s}; \text{id}, \bar{\delta}] \simeq \mathbb{F}_p\langle X \rangle[\bar{s}; \text{id}, \bar{\delta}]$$

is free as an $\mathbb{F}_p\langle X \rangle$ -module, the Tor group vanishes. This gives a free resolution of $\text{Gr } \mathbb{D}$ with length one.

(3) Again, we prove the claim for N , and the same argument works for N' . The short exact sequence

$$0 \rightarrow N \rightarrow (\text{Gr } \mathbb{D})^k \rightarrow Q \rightarrow 0$$

induces a long exact sequence of Ext groups

$$\dots \rightarrow \text{Ext}_{\text{Gr } \mathbb{F}_p[[G]]}^2((\text{Gr } \mathbb{D})^k, M) \rightarrow \text{Ext}_{\text{Gr } \mathbb{F}_p[[G]]}^2(N, M) \rightarrow \text{Ext}_{\text{Gr } \mathbb{F}_p[[G]]}^3(Q, M) \rightarrow \dots$$

The first Ext group vanishes by (2), and the third one by (1). Hence, $\text{Ext}_{\text{Gr } \mathbb{F}_p[[G]]}^2(N, M) = 0$ for any left $\text{Gr } \mathbb{F}_p[[G]]$ -module M , showing that the projective dimension of N is at most one.

(4) follows immediately from (2).

(5) We have isomorphisms:

$$\begin{aligned} \text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(\text{Gr } \mathbb{D}, N'') &= \text{Tor}_1^{\text{Gr } \mathbb{F}_p[[G]]}(\mathcal{Q}[\bar{t}] \otimes_{\mathbb{F}_p\langle X \rangle} \text{Gr } \mathbb{F}_p[[G]], N'') \\ &= \text{Tor}_1^{\mathbb{F}_p\langle X \rangle}(\mathcal{Q}[\bar{t}], N''). \end{aligned}$$

⁹²James C. Robson and John. C. McConnell. *Noncommutative Noetherian rings*. 2nd ed. Graduate studies in mathematics 30. American Mathematical Society, 2001. ISBN: 978-0-8218-2169-5, Thm. 7.5.3.

It was shown by Aidan Schofield⁹³ that since $\mathbb{F}_p\langle X \rangle \rightarrow \mathcal{Q}[\bar{t}]$ is an universal epimorphism, then $\mathrm{Tor}_1^{\mathbb{F}_p\langle X \rangle}(A, B) \simeq \mathrm{Tor}_1^{\mathcal{Q}[\bar{t}]}(A, B)$ for any $\mathcal{Q}[\bar{t}]$ -modules A and B . Hence:

$$\mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, N'') \simeq \mathrm{Tor}_1^{\mathcal{Q}[\bar{t}]}(\mathcal{Q}[\bar{t}], N'') = 0.$$

(6) The short exact sequence

$$0 \rightarrow N \rightarrow (\mathrm{Gr}\mathbb{D})^k \rightarrow Q \rightarrow 0$$

induces the long exact sequence

$$\dots \rightarrow \mathrm{Tor}_2^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, Q) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, N) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, \mathrm{Gr}\mathbb{D})^k \rightarrow \dots.$$

The first Tor group vanishes by (4), and the last one by (5). Hence, the middle one also vanishes.

(7) By the same argument, a short exact sequence

$$0 \rightarrow N' \rightarrow (\mathrm{Gr}\mathbb{D})^k \rightarrow Q' \rightarrow 0$$

induces the long exact sequence

$$\dots \rightarrow \mathrm{Tor}_2^{\mathrm{Gr}\mathbb{F}_p[[G]]}(Q', \mathrm{Gr}\mathbb{D}) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(N', \mathrm{Gr}\mathbb{D}) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, \mathrm{Gr}\mathbb{D})^k \rightarrow \dots.$$

The first Tor group vanishes by (4) and the last one by (5).

(8) As in the previous two cases, a short exact sequence

$$0 \rightarrow N' \rightarrow (\mathrm{Gr}\mathbb{D})^k \rightarrow Q' \rightarrow 0$$

induces the long exact sequence

$$\dots \rightarrow \mathrm{Tor}_2^{\mathrm{Gr}\mathbb{F}_p[[G]]}(Q', N) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(N', N) \rightarrow \mathrm{Tor}_1^{\mathrm{Gr}\mathbb{F}_p[[G]]}(\mathrm{Gr}\mathbb{D}, N)^k \rightarrow \dots$$

The first Tor group vanishes by (3) and the last Tor group vanishes by (6).

(9) is a particular case of (5). □

The following is an analogue of Corollary 4.2.5 for $\mathbb{F}_p[[G]]$:

Corollary 4.6.3. *Let M be a complete, separated and bounded filtered $\mathbb{F}_p[[G]]$ -module. Then, there are semicompact sets V_0, V_1 and V_2 and a strict exact sequence of filtered $\mathbb{F}_p[[G]]$ -modules*

$$0 \rightarrow \widehat{\mathbb{F}_p[[G]](V_2)} \rightarrow \widehat{\mathbb{F}_p[[G]](V_1)} \rightarrow \widehat{\mathbb{F}_p[[G]](V_0)} \rightarrow M \rightarrow 0.$$

⁹³Schofield, *Representations of Rings over Skew Fields*, Thm. 4.8.

Proof. By Proposition 1.4.15, we know that there exists semicompact sets $U_0, U_1, \dots$ and a strict exact sequence

$$\dots \rightarrow \widehat{\mathbb{F}_p[[G]](U_2)} \rightarrow \widehat{\mathbb{F}_p[[G]](U_1)} \xrightarrow{\alpha} \widehat{\mathbb{F}_p[[G]](U_0)} \rightarrow M \rightarrow 0.$$

We take $V_0 = U_0$ and $V_1 = U_1$, and we define $K = \ker \alpha$ with the induced filtration. Upon passing to the graded ring, we get an exact sequence

$$0 \rightarrow \text{Gr } K \rightarrow (\text{Gr } \mathbb{F}_p[[G]])[\overline{V_1}] \rightarrow (\text{Gr } \mathbb{F}_p[[G]])[\overline{V_0}] \rightarrow \text{Gr } M \rightarrow 0.$$

As V_1 is semicompact, $(\text{Gr } \mathbb{F}_p[[G]])[\overline{V_1}]$ is bounded from below and thus so is $\text{Gr } K$. Since $\text{Gr } \mathbb{F}_p[[G]]$ has projective dimension at most two by Proposition 4.6.2, by Schanuel's lemma we know that $\text{Gr } K$ must be projective as an abstract $\text{Gr } \mathbb{F}_p[[G]]$ -module. By Lemma 1.4.14, $\text{Gr } K$ must be graded-free: there exists a homogeneous set $\overline{V_2} \subseteq \text{Gr } K$ and an isomorphism $(\text{Gr } \mathbb{F}_p[[G]])[\overline{V_2}] \simeq \text{Gr } K$ extending the inclusion $\overline{V_2} \rightarrow \text{Gr } K$. With the degree induced from $\text{Gr } K$, the set $\overline{V_2}$ has naturally the structure of a semicompact set as K is bounded. We lift this to a semicompact subset $V_2 \leq K$. Given that K is bounded, complete and separated, we get a filtered homomorphism $\psi: \widehat{\mathbb{F}_p[[G]](V_2)} \rightarrow K$ extending the inclusion $V_2 \rightarrow K$. By construction, $\text{Gr } \psi$ is bijective, and so ψ is a strict bijection by Corollary 1.4.9. $\square$

The main homological property that we shall need of the embedding $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathbb{D}$ is the following:

Proposition 4.6.4. *Let M be a finitely generated left (resp. right) $\mathbb{F}_p[[G]]$ -submodule of $\mathbb{D}^k$ for some k . Then, $\text{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M)$ (resp. $\text{Tor}_1^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p)$) is finite and $\text{Tor}_2^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M)$ (resp. $\text{Tor}_2^{\mathbb{F}_p[[G]]}(M, \mathbb{F}_p)$) vanishes.*

Proof. We will prove the proposition for left modules only, since the proof for right modules follows the same arguments. We will also follow closely the argument used in Lemma 4.2.6. Under the induced filtration from $\mathbb{D}^k$, M becomes a bounded, complete, separated filtered $\mathbb{F}_p[[G]]$ -module, and the topology induced by the filtration is profinite.⁹⁴ Let us take

$$0 \rightarrow \widehat{\mathbb{F}_p[[G]](V_2)} \xrightarrow{d_2} \widehat{\mathbb{F}_p[[G]](V_1)} \xrightarrow{d_1} \widehat{\mathbb{F}_p[[G]](V_0)} \xrightarrow{d_0} M \rightarrow 0 \quad (\dagger)$$

⁹⁴Since the inclusion $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathbb{D}$ is strict, it is a homeomorphism of $\mathbb{F}_p[[G]]$ onto its image. Given that the filtration on $\mathbb{D}^k$ is separated, the image of any compact set under a continuous function is compact. This applies to any surjection $\mathbb{F}_p[[G]]^d \rightarrow M \rightarrow \mathbb{D}^k$, so M is compact with the topology coming from the induced filtration of $\mathbb{D}^k$. The filtration is automatically separated and complete. To see that it is bounded, any quotient by an open submodule must be a compact and discrete $\hat{R}$ -module, and thus finite.

the strict exact sequence of Corollary 4.6.3, and note that this is a free resolution of M in the category of profinite $\mathbb{F}_p[[G]]$ -modules. By passing to the graded ring, we get the exact sequence

$$\begin{aligned} 0 \rightarrow (\mathrm{Gr} \mathbb{F}_p[[G]])[\overline{V}_2] &\xrightarrow{\mathrm{Gr} d_2} (\mathrm{Gr} \mathbb{F}_p[[G]])[\overline{V}_1] \\ &\xrightarrow{\mathrm{Gr} d_1} (\mathrm{Gr} \mathbb{F}_p[[G]])[\overline{V}_0] \xrightarrow{\mathrm{Gr} d_0} \mathrm{Gr} M \rightarrow 0. \quad (\mathrm{Gr} \dagger) \end{aligned}$$

Since $\mathrm{Tor}_i^{\mathrm{Gr} \mathbb{F}_p[[G]]}(\mathrm{Gr} \mathbb{D}, M) = 0$ for all $i > 0$ by Proposition 4.6.2, we get exactness of

$$\begin{aligned} 0 \rightarrow (\mathrm{Gr} \mathbb{D})[\overline{V}_2] &\xrightarrow{\mathrm{id} \otimes (\mathrm{Gr} d_2)} (\mathrm{Gr} \mathbb{D})[\overline{V}_1] \\ &\xrightarrow{\mathrm{id} \otimes (\mathrm{Gr} d_1)} (\mathrm{Gr} \mathbb{D})[\overline{V}_0] \xrightarrow{\mathrm{id} \otimes (\mathrm{Gr} d_0)} \mathrm{Gr} \mathbb{D} \otimes_{\mathrm{Gr} \mathbb{F}_p[[G]]} \mathrm{Gr} M \rightarrow 0 \\ &\quad (\mathrm{Gr} \mathbb{D} \otimes_{\mathrm{Gr} \mathbb{F}_p[[G]]} \mathrm{Gr} \dagger) \end{aligned}$$

We now claim that the sequence

$$0 \rightarrow \widehat{\mathbb{D}(V_2)} \xrightarrow{\mathrm{id} \hat{\otimes} d_2} \widehat{\mathbb{D}(V_1)} \xrightarrow{\mathrm{id} \hat{\otimes} d_1} \widehat{\mathbb{D}(V_0)} \xrightarrow{\mathrm{id} \hat{\otimes} d_0} \mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} M \rightarrow 0 \quad (\mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} \dagger)$$

is again strict exact. Since we have isomorphisms

$$\mathrm{Gr}(\widehat{\mathbb{D}(V_i)}) \simeq (\mathrm{Gr} \mathbb{D})[\overline{V}_i] = \mathrm{Gr} \mathbb{D} \otimes_{\mathrm{Gr} \mathbb{F}_p[[G]]} (\mathrm{Gr} \mathbb{F}_p[[G]])[\overline{V}_i]$$

for all $i \in \{0, 1, 2\}$, the exactness of $\mathrm{Gr} \mathbb{D} \otimes_{\mathrm{Gr} \mathbb{F}_p[[G]]} \mathrm{Gr} \dagger$ shows that $\mathrm{id} \hat{\otimes} d_2$ is strict injective, $\mathrm{id} \hat{\otimes} d_1$ is strict and $\ker(\mathrm{id} \hat{\otimes} d_1) = \mathrm{im}(\mathrm{id} \hat{\otimes} d_2)$ by Proposition 1.4.8. We also have the commutative diagram

$$\begin{array}{ccc} (\mathrm{Gr} \mathbb{D})[\overline{V}_0] & \xrightarrow{\mathrm{Gr}(\mathrm{id} \hat{\otimes} d_0)} & \mathrm{Gr}(\mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} M) \longrightarrow 0 \\ \uparrow \simeq & & \uparrow \gamma \\ (\mathrm{Gr} \mathbb{D})[\overline{V}_0] & \xrightarrow{\mathrm{id} \otimes \mathrm{Gr} d_0} & \mathrm{Gr} \mathbb{D} \otimes_{\mathrm{Gr} \mathbb{F}_p[[G]]} \mathrm{Gr} M \longrightarrow 0 \end{array}$$

and given that γ is surjective it follows that $\mathrm{Gr}(\mathrm{id} \hat{\otimes} d_0)$ is also surjective. Hence, $\mathrm{id} \hat{\otimes} d_0$ is a strict surjection by Corollary 1.4.9.

It remains to show that $\mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} \dagger$ is exact at $\widehat{\mathbb{D}(V_0)}$. Let $x \in \ker(\mathrm{id} \hat{\otimes} d_0)$. For each $i \geq 0$, we have the following short exact sequence:

$$\widehat{\mathbb{D}(V_1)} / (\mathrm{id} \hat{\otimes} d_1)^{-1}(F_i \widehat{\mathbb{D}(V_0)}) \rightarrow \widehat{\mathbb{D}(V_0)} / F_i \widehat{\mathbb{D}(V_0)} \rightarrow (\mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} M) / (\mathrm{id} \hat{\otimes} d_0)(F_i \widehat{\mathbb{D}(V_0)}).$$

Hence, there exists a sequence $y_i \in \widehat{\mathbb{D}(V_1)} / \ker(\mathrm{id} \hat{\otimes} d_1)$ such that $\lim_{i \rightarrow \infty} (\mathrm{id} \hat{\otimes} d_1)(y_i) = x$. Given that $\mathrm{id} \hat{\otimes} d_1$ is strict, the topology on $\widehat{\mathbb{D}(V_1)} / \ker(\mathrm{id} \hat{\otimes} d_1)$ is complete and also is

induced from the topology on $\widehat{\mathbb{D}(V_0)}$, so the y_i converge in $\widehat{\mathbb{D}(V_1)} / \ker(\text{id} \hat{\otimes} d_1)$ to some y such that $(\text{id} \hat{\otimes} d_1)(y) = x$. Lifting this y to some element in $\widehat{\mathbb{D}(V_1)}$ shows that $\ker(\text{id} \hat{\otimes} d_0) = \text{im}(\text{id} \hat{\otimes} d_1)$ as desired.

Note that $\mathbb{D} \simeq \mathbb{Q}[[t]][[s; \sigma, \delta]]$ surjects onto $\mathbb{Q}$: the map that $\mathbb{D} \rightarrow \mathbb{Q}$ that kills t and s is a well defined ring map since $\delta(a) = \sigma(a) - a \in t\mathbb{D}$ for all $a \in \mathbb{D}$. Moreover, the composition $\mathbb{F}_p[[G]] \rightarrow \mathbb{D} \rightarrow \mathbb{Q}$ factors through $\mathbb{F}_p$ by construction, so that we can indentify the right $\mathbb{D}$ -module $\mathbb{Q}$ with the induced module $\mathbb{F}_p \otimes_{\mathbb{F}_p[[G]]} \mathbb{D}$.⁹⁵ Since $\mathbb{D}$ is Noetherian, $\mathbb{Q}$ is of type FP_∞ . By Theorem 1.4.17, we can compute $\text{Tor}_i^{\mathbb{D}}(\mathbb{Q}, \mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} M)$ as the homology of the sequence

$$0 \rightarrow \widehat{\mathbb{Q}(V_2)} \rightarrow \widehat{\mathbb{Q}(V_1)} \rightarrow \widehat{\mathbb{Q}(V_0)} \rightarrow 0. \quad (\mathbb{Q} \hat{\otimes}_{\mathbb{D}} \mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} \dagger)$$

The fact that $\mathbb{D}$ is Noetherian implies that these Tor groups are all finite.

On the other hand, as G is mild, the right $\mathbb{F}_p[[G]]$ -module $\mathbb{F}_p$ has type FP_∞ .⁹⁶ Applying Theorem 1.4.17 again we deduce that $\text{Tor}_i^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M)$ can be computed as the homology of the sequence

$$0 \rightarrow \widehat{\mathbb{F}_p(V_2)} \rightarrow \widehat{\mathbb{F}_p(V_1)} \rightarrow \widehat{\mathbb{F}_p(V_0)} \rightarrow 0. \quad (\mathbb{F}_p \hat{\otimes}_{\mathbb{F}_p[[G]]} \dagger)$$

As in the proof of Lemma 4.2.6, the inclusion map $\iota: \mathbb{F}_p \rightarrow \mathbb{Q}$ has a retraction $\pi: \mathbb{Q} \rightarrow \mathbb{F}_p$ as $\mathbb{F}_p$ -vector spaces, and given any non-zero $x \in \mathbb{Q}$ this retraction can be chosen such that $\pi(x) \neq 0$. We have a commutative diagram of $\mathbb{F}_p$ -vector spaces

$$\begin{array}{ccccc} \widehat{\mathbb{Q}(V_2)} & \xrightarrow{d_2^{\mathbb{Q}}} & \widehat{\mathbb{Q}(V_1)} & \xrightarrow{d_1^{\mathbb{Q}}} & \widehat{\mathbb{Q}(V_0)} \\ \uparrow \scriptstyle \iota(V_2) & \searrow \scriptstyle \pi(V_2) & \uparrow \scriptstyle \iota(V_1) & \searrow \scriptstyle \pi(V_1) & \uparrow \scriptstyle \iota(V_0) \\ \widehat{\mathbb{F}_p(V_2)} & \xrightarrow{d_2^{\mathbb{F}_p}} & \widehat{\mathbb{F}_p(V_1)} & \xrightarrow{d_1^{\mathbb{F}_p}} & \widehat{\mathbb{F}_p(V_0)} \end{array}$$

where $\widehat{\iota(V_i)}$ and $\widehat{\pi(V_i)}$ are induced from ι and π respectively, and moreover $\widehat{\pi(V_i)} \circ \widehat{\iota(V_i)} = \text{id}$.

We claim that this implies

$$\dim_{\mathbb{F}_p} \text{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M) \leq \dim_{\mathbb{Q}} \text{Tor}_1^{\mathbb{D}}(\mathbb{F}_p, \mathbb{D} \hat{\otimes}_{\mathbb{F}_p[[G]]} M) < \infty.$$

Indeed, if $v_1, \dots, v_k$ are elements of $\ker(d_1^{\mathbb{F}_p})$ and $\widehat{\iota(V_1)}(v_1), \dots, \widehat{\iota(V_1)}(v_k)$ are $\mathbb{Q}$ -linearly dependent modulo the image of $d_2^{\mathbb{Q}}$, then there exists $q_1, \dots, q_k \in \mathbb{Q}$ and some $w \in \widehat{\mathbb{Q}(V_2)}$ such that

$$q_1 \widehat{\iota(V_1)}(v_1) + \dots + q_k \widehat{\iota(V_1)}(v_k) = d_2^{\mathbb{Q}}(w).$$

⁹⁵The two-sided ideal generated by t and s in $\mathbb{D}$ coincides with the left ideal generated by t and s by the relation $s\mathbb{D} \subseteq \mathbb{D}s + \mathbb{D}t$, so it is automatically closed.

⁹⁶Labute, “Mild pro-p-groups and Galois groups of p-extensions of $\mathbb{Q}$ ”, Thm. 5.1.

Choosing the retraction $\pi: \mathcal{Q} \rightarrow \mathbb{F}_p$ such that $\pi(q_1), \dots, \pi(q_k)$ are not all zero we get

$$\pi(q_1)v_1 + \dots + \pi(q_k)v_k = \widehat{\pi(V_1)}(d_2^{\mathcal{Q}}(w)) = d_2^{\mathbb{F}_p}(\widehat{\pi(V_2)}(w)) \in \text{im}(d_2^{\mathbb{F}_p}),$$

as desired. For the vanishing of $\text{Tor}_2^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M)$, we apply Theorem 1.4.18 observing that

$$\text{Tor}_2^{\text{Gr } \mathbb{F}_p[[G]]}(\mathbb{F}_p, \text{Gr } M) = 0$$

by part (3) of Proposition 4.6.2. □

Corollary 4.6.5. *Let M be a finitely generated left or right $\mathbb{F}_p[[G]]$ -submodule of $\mathcal{D}$. Then, for any exact sequence*

$$0 \rightarrow I \rightarrow \mathbb{F}_p[[G]]^d \rightarrow M \rightarrow 0,$$

the $\mathbb{F}_p[[G]]$ -module I is free of finite rank.

Proof. Assume that M is a left module, and the same argument works from the right side. By multiplying with a common denominator of a $\mathbb{F}_p[[G]]$ -generating set of M , we may assume that M is contained in $\mathbb{D}$. As we've seen in the proof of Proposition 4.6.4, M becomes a profinite module under the topology induced from $\mathbb{D}$. Since $\mathbb{F}_p$ is of type FP_∞ as a $\mathbb{F}_p[[G]]$ -module, by Theorem 1.4.17 and Proposition 4.6.4 we have isomorphisms

$$0 = \text{Tor}_2^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M) \simeq \mathfrak{Tor}_2^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M) \simeq \mathfrak{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, I).$$

Hence, I is free. The long exact sequence

$$0 \rightarrow \text{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M) \simeq \mathfrak{Tor}_1^{\mathbb{F}_p[[G]]}(\mathbb{F}_p, M) \rightarrow \mathbb{F}_p \hat{\otimes}_{\mathbb{F}_p[[G]]} I \rightarrow \mathbb{F}_p^d \rightarrow \mathbb{F}_p \hat{\otimes}_{\mathbb{F}_p[[G]]} M \rightarrow 0$$

and Nakayama's lemma shows that I is also finitely generated, and hence free of finite rank. □

We are now ready to prove the main and final theorem of the present chapter.

Theorem 4.6.6. *Let G be a finitely generated torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p group. Then $\mathbb{F}_q[[G]]$ is a Sylvester domain and $\text{rk}_G = \text{irk}_{\mathbb{F}_q[[G]]}$.*

Proof. By Theorem 4.3.7, Corollary 4.3.8 and Proposition 4.4.6 we may assume that G is a mild flag pro- p group and $q = p$. We will check the embedding $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathcal{D}$ constructed in Section 4.5 satisfies the hypotheses of Theorem 4.1.14. The second condition is given by Corollary 4.6.5, so it remains to show that

$$\text{Tor}_1^{\mathbb{F}_p[[G]]}(\mathcal{D}, \mathcal{D}) = 0.$$

Since $\mathcal{D}$ is flat as a $\mathbb{D}$ -module by Proposition 2.1.5, if $P_* \rightarrow \mathcal{D}$ is a projective resolution of $\mathcal{D}$ over $\mathbb{F}_p[[G]]$, we have isomorphisms:

$$\begin{aligned} \mathcal{D} \otimes_{\mathbb{D}} \mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathbb{D}, \mathcal{D}) &= \mathcal{D} \otimes_{\mathbb{D}} H_i(\mathbb{D} \otimes_{\mathbb{F}_p[[G]]} P_*) \\ &\simeq H_i(\mathcal{D} \otimes_{\mathbb{D}} \mathbb{D} \otimes_{\mathbb{F}_p[[G]]} P_*) \\ &\simeq H_i(\mathcal{D} \otimes_{\mathbb{F}_p[[G]]} P_*) \\ &= \mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathcal{D}, \mathcal{D}). \end{aligned}$$

Hence, if $\mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathbb{D}, \mathcal{D})$ vanishes, then so does $\mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathcal{D}, \mathcal{D})$. The same reasoning, now applied to the right factor, gives us that the vanishing of $\mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathbb{D}, \mathbb{D}) = 0$ implies

$$\mathrm{Tor}_i^{\mathbb{F}_p[[G]]}(\mathcal{D}, \mathcal{D}) = 0.$$

Since Tor commutes with direct limits, it suffices to show that for any finitely generated right and left $\mathbb{F}_p[[G]]$ -modules M and N of $\mathbb{D}$, one has

$$\mathrm{Tor}_1^{\mathbb{F}_p[[G]]}(M, N) = 0.$$

As we have argued in the proof of Proposition 4.6.4, under the induced filtration from $\mathbb{D}$, M and N become complete, bounded, separated filtered $\mathbb{F}_p[[G]]$ -modules. From Corollary 4.6.5, we know that both M and N are of type FP_{∞} : a presentation

$$0 \rightarrow I \rightarrow \mathbb{F}_p[[G]]^d \rightarrow M \rightarrow 0$$

implies that I is free of finite rank, so one can iterate this argument with a presentation of I *ad infinitum*, and the same for N . The desired vanishing of Tor now follows from Theorem 1.4.18 and Proposition 4.6.2.

Finally, we know that irk is the universal Sylvester matrix rank function on $\mathbb{F}_p[[G]]$, so $\mathrm{rk}_G \leq \mathrm{irk}_G$. The other inequality $\mathrm{irk}_G = \varphi^{\#} \mathrm{rk}_{\mathcal{D}} \leq \mathrm{rk}_G$ follows from Theorem 3.2.5. $\square$

Chapter 5

p -adic Lie groups

(...) using algebraic methods such as
Harmonic Analysis (...)

From a paper J.D. was reading.

Having grown up physically and mathematically in Brasília, Brazil, the author would say that his algebraic upbringing had a different bias from that of his current European peers. As an undergraduate, he first learned about finite groups, then infinite profinite groups, and only later the classical combinatorial theory of infinite abstract groups *à la* group presentations. It was the same with homological algebra: first it was the cohomology of finite groups through cochains, then the continuous cohomology of profinite groups, and only after that he was acquainted with the classical methods of algebraic topology and cellular/simplicial/singular complexes. No different with Bass-Serre theory, where I picked up Luis Ribes' book on profinite trees¹ before I ever caught a sniff of Jean-Pierre Serre's original *œuvre*.² The reader may take a guess at how I came to learn about Lie groups and Lie algebras.

The quick and dirty way to define a finite dimensional p -adic Lie group is as a topological group containing an open subgroup topologically isomorphic to a closed subgroup of $GL_n(\mathbb{Z}_p)$ for some n . The study of p -adic Lie groups is older than it may first seem to the unacquainted. In the first half of the 20th century, Claude Chevalley was committed to writing a series of books on Lie groups and algebraic groups. He soon realized that compact Real Lie groups admitted a natural structure of an algebraic group, that is, one defined by the vanishing of polynomial equations in $GL_n(\mathbb{R})$. Because of this, he decided to scrap the project halfway through and start from zero using the language of algebraic geometry to study linear algebraic groups in more generality.³ Perhaps the first definition of a p -adic Lie group appeared in the

¹Ribes, *Profinite Groups and Groups*.

²Jean-Pierre Serre. *Arbres, amalgames, SL_2* . Astérisque 46. Société mathématique de France, 1977.

³Jean Dieudonné and Jacques Tits. "Claude Chevalley (1909-1984)". In: *Bulletin of the American Mathemat-*

work of his PhD student Robert Hooke, published later in 1942.⁴

Throughout the 1950's and 1960's, as the theory of classical modular forms on the complex upper half-plane was morphing into its more abstract theory of automorphic forms on reductive groups,⁵ it became clear through the adelic picture that p -adic points of said groups would play a central role. Hence, the study of complex representations of p -adic Lie groups arose naturally from number theorists' need of doing "harmonic analysis"⁶ over adelic points of reductive groups. We shall have more to say about automorphic forms in Chapter 6.

The study of p -adic representations of p -adic Lie groups, that is, continuous ring homomorphisms $\mathbb{Z}_p[[G]] \rightarrow \text{End}(V)$ for some $\mathbb{Z}_p$ -module V , also has its origins in arithmetic—although of a different sort. Recall that the ring of integers $\mathcal{O}_F$ often fails to have unique factorization, and this failure is captured in the ideal class group $\text{Cl}(F)$. In the mid 50's, Kenkichi Iwasawa as interested in the following question: given a tower $F = F_0 \subset F_1 \subset F_2 \subset \dots$ of field extensions, how does the size of the class group $\text{Cl}(F_i)$ grow?

In 1956, he gave a talk titled "*A theorem on Abelian groups and its application in algebraic number theory*" on the Summer Meeting of the American Mathematical Society in Seattle. In this talk, he announced the following remarkable theorem:⁷

Theorem 5.0.1. *Let F be a number field, and $F = F_0 \subset F_1 \subset F_2 \subset \dots$ be a tower of finite extensions of F such that for $L = \bigcup_{i \geq 0} F_i$ we have $\text{Gal}(L/F) \simeq \mathbb{Z}_p$. For each n , let L_n be the subfield of L fixed by $p^n \mathbb{Z}_p$ and p^{e_n} be the highest power of p dividing the order of the class group $\text{Cl}(L_n)$. Then, there exists integers c , l and m such that*

$$e_n = ln + mp^n + c$$

*for sufficiently large n .*⁸

ical Society 17.1 (1987), pp. 1–7. ISSN: 1088-9485. DOI: 10.1090/S0273-0979-1987-15509-1.

⁴Robert Hooke. "Linear p -Adic Groups and Their Lie Algebras". In: *The Annals of Mathematics* 43.4 (Oct. 1942), p. 641. ISSN: 0003-486X. DOI: 10.2307/1968957.

⁵A pivotal point was J. Tate's thesis from 1950 that reframes Hecke's work in an adelic point of view, which can be seen as a theory of automorphic forms over GL_1 . The first modern definition of automorphic forms appeared in a letter from R. Langlands to Weil dated from 1967, and Langlands monograph from the same year marks the birth date of what's now called the "Langlands program". For an account of the history of automorphic forms and its connection with Langlands, see James G. Arthur. "The work of Robert Langlands". In: *The Abel Prize 2018-2022*. Springer International Publishing, 2024, pp. 31–230. ISBN: 9783031339738. DOI: 10.1007/978-3-031-33973-8_2

⁶The meaning of this sentence comes from the fact that the description of the irreducible constituents of $\ell^2(G)$ where G is some compact abelian Lie group can be more easily described through the isomorphism $\ell^2(G) \simeq \ell^2(\widehat{G})$ induced by the Fourier transform, where $\widehat{G}$ is the now discrete Pontryagin dual of G . What was needed to better understand the theory of automorphic forms is to obtain a description of the space of ℓ^2 -integrable complex unitary representations of adelic points of reductive groups, and the lack of "good notion" of Pontryagin duality in the non-commutative setting is a major obstruction.

⁷Kenkichi Iwasawa. "On Γ -Extensions of Algebraic Number Fields". In: *Bulletin of the American Mathematical Society* 65.4 (1959), pp. 183–26, Thm. II.

⁸There is an interpretation of the constants l and m in terms of arithmetic information from the field extension L/F , but we refer the reader to Iwasawa's paper for this data.

The main ingredient in Iwasawa's proof is purely algebraic. Taking $G = \mathbb{Z}_p$, although $\mathbb{Z}_p[[G]]$ is not a principal ideal domain, it behaves very similarly to one. For every finitely generated $\mathbb{Z}_p[[G]]$ -module M , there are unique elements $f_1, \dots, f_n \in \mathbb{Z}_p[[G]]$ and a $\mathbb{Z}_p[[G]]$ -module homomorphism $\varphi: M \rightarrow \bigoplus \mathbb{Z}_p[[G]]/(f_i)$ such that $\ker \varphi$ and $\operatorname{coker} \varphi$ are both finite.⁹ Thus, up to a finite constant error, it suffices to check Theorem 5.0.1 on modules of the form $\mathbb{Z}_p[[G]]/(f_i)$. Such modules are very easy to describe since $\mathbb{Z}_p[[G]]$ is a unique factorization domain.¹⁰ Iwasawa's theorem was perhaps the first investigation into the module category of completed group algebras of profinite groups, and it is for this reason that completed group algebras $\mathbb{Z}_p[[G]]$ and $\mathbb{F}_p[[G]]$ of pro- p groups are sometimes referred to as *Iwasawa algebras* in the literature.

Our main goal in this chapter is to study the asymptotic behaviour of rank functions on $\mathbb{Z}_p[[G]]$ induced from rational representations of G when G is the congruence subgroup of a simply connected split semisimple p -adic Lie group. This chapter is organized as follows. In Section 5.1, we collect mostly without proofs the main facts about semisimple Lie groups, Lie algebras and Chevalley groups that will be needed throughout this chapter. The reader that is familiar with either the split semisimple theory of algebraic groups over a field or the theory of Chevalley groups can safely skip this section; its main purpose is to set up the nomenclature and definitions for the objects studied in the remaining sections.

In Section 5.2 we specialize to the study of $\mathbb{Z}_p$ -points $G(\mathbb{Z}_p)$ of a universal split semisimple Chevalley group G . We prove that if p is odd, then its first congruence subgroup $G_1 = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$ is a uniform pro- p group, and compare the four Lie algebras naturally associated with G_1 . We finish with a proof of a theorem of Michael Harris showing that the Ore rank function on $\mathbb{Z}_p[[G_1]]$ can be approximated by the normalized rank functions on certain finite quotients of G_1 .

Section 5.3 is dedicated to the relation between the Iwasawa algebra $\mathbb{Z}_p[[G_1]]$ and a Banach completion of the universal enveloping $\mathbb{Q}_p$ -algebra $U(\mathfrak{g}_{\mathbb{Z}_p})$ of the Lie algebra of G_1 . This relation can be traced back to Michel Lazard's work on the saturation of the group algebra $\mathbb{Z}_p[G_1]$, but we follow the modern exposition of Konstantin Ardakov and Simon Wadsley. The main technique is the microlocalization of Section 2.2. We finish this chapter in Section 5.4, where we show that for $G = \prod_{i=1}^k \mathrm{SL}_2$ the Ore rank function on $\mathbb{Z}_p[[G_1]]$ can also be approximated by the rank functions coming from the rational representations of G .

5.1 Universal split semisimple Chevalley groups

The groups studied in this chapter are “ p -adic points” of products of the classical complex matrix groups $\mathrm{SL}_n(\mathbb{C})$, $\mathrm{SO}_n(\mathbb{C})$ and $\mathrm{Sp}_n(\mathbb{C})$ and their exceptional cousins G_2 , F_4 , E_6 , E_7 and

⁹Such map is often called a pseudo-isomorphism. Hence, up to pseudo-isomorphism, $\mathbb{Z}_p[[G]]$ behaves like a PID.

¹⁰The Weierstrass preparation theorem describes any f_i as a unique product of a power of p , some invertible power series and a distinguished polynomial.

E_8 . The goal in this section is to make sense of this notion of p -adic points, as well as to provide a unified language to work with all these groups and their representations simultaneously and rigorously. This is achieved by exchanging the classical groups with a so called *group functor* given by Chevalley's construction. Hence, in this first section, we will lay out this language and the main results of this theory, mostly without proofs, for the convenience of the reader. Our main references are Jean-Pierre Serre's book¹¹ and Nicolas Bourbaki's Elements¹² for semisimple Lie algebras, Dong Hoon Lee's book¹³ for complex Lie groups and Robert Steinberg's lecture notes for Chevalley groups.¹⁴

We start with a complex finite-dimensional Lie algebra $\mathfrak{g}$. It is said to be *simple* if it is not abelian¹⁵ and its only ideals are 0 and $\mathfrak{g}$. The left adjoint action

$$\mathrm{ad}(x)(y) = [x, y]$$

gives us a Lie algebra representation $\mathrm{ad}: \mathfrak{g} \rightarrow \mathrm{gl}(\mathfrak{g})$ of $\mathfrak{g}$ acting on itself. This endows $\mathfrak{g}$ with a symmetric bilinear form $(\cdot, \cdot): \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ given by

$$(x, y) = \mathrm{tr}(\mathrm{ad}(x) \circ \mathrm{ad}(y))$$

called the *Killing form*¹⁶ of $\mathfrak{g}$. The Lie algebra $\mathfrak{g}$ is *semisimple* if its Killing form is non-degenerate. Every simple Lie algebra is semisimple,¹⁷ and a Lie algebra is semisimple if and only if it is isomorphic to the direct sum of its minimal ideals, all of which are simple Lie algebras.¹⁸

A *Cartan subalgebra* $\mathfrak{h} \leq \mathfrak{g}$ of a semisimple Lie algebra is a maximal abelian subalgebra of $\mathfrak{g}$ for which $\mathrm{ad}(x)$ is diagonalizable for every $x \in \mathfrak{h}$.¹⁹ A standard fact²⁰ of Lie theory is that

¹¹Jean-Pierre Serre. *Complex Semisimple Lie Algebras*. en. Springer Monographs in Mathematics. Berlin Heidelberg: Springer-Verlag, 2001. ISBN: 978-3-540-67827-4. DOI: 10.1007/978-3-642-56884-8.

¹²Bourbaki, *Lie Groups and Lie Algebras*; Nicolas Bourbaki. *Lie groups and Lie algebras. Chapters 7-9*. 1st ed. Elements of mathematics. Springer, 2004. ISBN: 978-3-540-43405-4.

¹³Dong Hoon Lee. *The Structure of Complex Lie Groups*. Chapman and Hall/CRC, Aug. 2001. ISBN: 9780429128622. DOI: 10.1201/9781420035452.

¹⁴Robert Steinberg. *Lectures on Chevalley Groups*. Vol. 66. University Lecture Series. American Mathematical Society, Dec. 2016. DOI: 10.1090/ulect/066.

¹⁵Meaning $[\mathfrak{g}, \mathfrak{g}] = 0$

¹⁶Named after Wilhelm Killing, nothing sinister.

¹⁷By the relation

$$(x, [y, z]) = ([x, y], z),$$

the set of all $x \in \mathfrak{g}$ for which $(x, -)$ is identically zero is an ideal of $\mathfrak{g}$. If $\mathfrak{g}$ is simple, either this ideal is zero, so $\mathfrak{g}$ is semisimple, or the Killing form is identically zero everywhere. By a theorem of Cartan, the latter only happens if $\mathfrak{g}$ is *solvable*, which is not the case. See Serre, *Complex Semisimple Lie Algebras*, Sec. I.8

¹⁸The complement of any ideal $\mathfrak{h}$ with respect to the Killing form is again a Lie ideal. That the restriction of the Killing form to $\mathfrak{h}$ is again non-degenerate is more subtle but still true, it follows from another characterization of semisimplicity as “does not contain non-trivial solvable ideals” — see *ibid.*, Sec. II.1 and II.2

¹⁹A subalgebra $\mathfrak{h} \leq \mathfrak{g}$ is called *toral*, or a *torus*, if $\mathrm{ad}(x)$ is diagonalizable for every $x \in \mathfrak{h}$. This implies that $\mathfrak{h}$ is abelian. A Cartan subalgebra is then a maximal torus.

²⁰Serre, *Complex Semisimple Lie Algebras*, Sec. III.4.

any two Cartan subalgebras of $\mathfrak{g}$ are conjugate, that is, if $\mathfrak{h}, \mathfrak{h}' \leq \mathfrak{g}$ are Cartan subalgebras then there exists some Lie algebra automorphism²¹ $\alpha: \mathfrak{g} \rightarrow \mathfrak{g}$ such that $\alpha(\mathfrak{h}) = \mathfrak{h}'$.

We henceforth fix $\mathfrak{h}$ a Cartan subalgebra of $\mathfrak{g}$ and observe that any properties of $\mathfrak{h}$ and of $\mathfrak{g}$ as an $\mathfrak{h}$ -Lie module are isomorphism invariants of $\mathfrak{g}$. We call $n = \dim_{\mathbb{C}} \mathfrak{h}$ the *rank* of $\mathfrak{g}$. Since $\text{ad}(\mathfrak{h}) \leq \mathfrak{gl}(\mathfrak{g})$ is abelian and all of its elements are diagonalizable, they can be simultaneously diagonalized. Denoting by $\mathfrak{h}^*$ the complex dual of $\mathfrak{h}$ as a vector space, this means precisely that we have an $\mathfrak{h}$ -module decomposition

$$\mathfrak{g} \simeq \bigoplus_{\lambda \in \mathfrak{h}^*} \mathfrak{g}_{\lambda}$$

where each $\mathfrak{g}_{\lambda}$ consists exactly of the elements $x \in \mathfrak{g}$ for which

$$\text{ad}(\gamma)(x) = [\gamma, x] = \lambda(\gamma)x$$

for all $\gamma \in \mathfrak{h}$. In particular, $\mathfrak{h} \leq \mathfrak{g}_0$.

The elements of the set

$$R = \{\lambda \in \mathfrak{h}^* : \lambda \neq 0, \mathfrak{g}_{\lambda} \neq 0\}$$

are called the *roots* of $\mathfrak{h}$, and the non-zero elements x in each $\mathfrak{g}_{\lambda}$ are called *root vectors* with root λ . Since $\mathfrak{g}$ is finite-dimensional, R is a finite set. Moreover, for each non-zero λ we have $\dim_{\mathbb{C}} \mathfrak{g}_{\lambda} \leq 1$, that is, a root vector for a given root is unique up to a scalar factor, and shall be denoted by e_{λ} .²² Following Serre,²³ for a complex vector space V we define a *symmetry* of V with respect to an element $\lambda \in V$ to be the linear automorphism $s_{\lambda}: V \rightarrow V$ with the following properties:

1. $s_{\lambda}(\lambda) = -\lambda$;
2. The subspace of V fixed by s_{λ} is a hyperplane, that is, it has codimension one.

Definition 5.1.1. Let V be a finite-dimensional complex vector space and R a finite subset of non-zero vectors of V . We say that R is a *root system* in V if it satisfies:

- (RS1) R spans V as a complex vector space;
- (RS2) For any $\lambda \in R$, there exists a symmetry $s_{\lambda}: V \rightarrow V$ with respect to λ such that $s_{\lambda}(R) = R$.
- (RS3) For any $\lambda, \mu \in R$ the symmetry s_{λ} satisfies

$$s_{\lambda}(\mu) - \mu \in \mathbb{Z}\lambda.$$

²¹In fact, α can be taken to be inner: of the form $\exp(\text{ad}(x))$ for some $x \in \mathfrak{g}$ with $\text{ad}(x)$ nilpotent.

²²Serre, *Complex Semisimple Lie Algebras*, Thm. VI.1.2.

²³Ibid., Sec. V.1.

(RS4) For any $\lambda \in R$, the only scalar multiples of λ in R are λ and $-\lambda$.

In any root system, the symmetry s_λ from (RS2) is necessarily unique.²⁴ We call $\dim V$ the rank of the root system. The set R is an example²⁵ of a root system in $\mathfrak{h}^*$, called the *root system* of $\mathfrak{g}$, and by definition its rank coincides with the rank of $\mathfrak{g}$, that is, the dimension of $\mathfrak{h}$. We say that two root systems R_1 and R_2 in V_1 and V_2 respectively are isomorphic if there exists an isomorphism $\alpha: V_1 \rightarrow V_2$ of complex vector spaces such that $\alpha(R_1) = R_2$ and $\alpha \circ s_\lambda \circ \alpha^{-1} = s_{\alpha(\lambda)}$ for all $\lambda \in R_1$. The main result of the classification of semisimple complex Lie algebras is the following:²⁶

Theorem 5.1.2. *Two semisimple complex Lie algebras are isomorphic if and only if their root systems are isomorphic.*

Therefore, to classify the semisimple complex Lie algebras $\mathfrak{g}$, it suffices to classify the root systems in complex finite dimensional vector spaces. Before exploring two examples, let us do one further reduction. If R_1 and R_2 are two root systems in V_1 and V_2 respectively, then $R_1 \cup R_2$ is a root system in $V_1 \oplus V_2$ called the sum of R_1 and R_2 and denoted by $R_1 \times R_2$. The root system of a semisimple Lie algebra is the sum of the root systems of its simple summands. A root system is irreducible if it cannot be expressed as a sum of two non-empty subsets, and any root system can be uniquely expressed as a sum of irreducible root systems. As we shall soon see, every irreducible root system is the root system of some simple Lie algebra, so that reduces the classification of semisimple Lie algebras to the classification of irreducible root systems corresponding to its simple summands.

Example 5.1.3. The simplest simple Lie algebra is $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$, the Lie algebra of all traceless 2×2 matrices with the Lie bracket

$$[A, B] = AB - BA.$$

A basis of this Lie algebra consists of the matrices

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ and } F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

It has complex dimension 3, and in this basis its adjoint representation is the Lie algebra homomorphism $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}_3(\mathbb{C})$ given by

$$\text{ad} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} 2a & -2b & 0 \\ -c & 0 & b \\ 0 & -2c & -2a \end{pmatrix}.$$

²⁴If s_λ and t_λ are two symmetries, then $u = s_\lambda t_\lambda$ is a finite order automorphism of V by (RS2) all of whose eigenvalues are one by (RS1). Therefore, it is the identity.

²⁵Serre, *Complex Semisimple Lie Algebras*, Sec. VI.1.

²⁶Ibid., Sec. VI.5.

It is a simple exercise to verify that $\mathfrak{g}$ is a simple Lie algebra, and in particular semisimple. Another way of checking that it is semisimple is to check that the matrix representing the Killing form in the ordered basis (H, E, F) is given by the invertible matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

A Cartan subalgebra of $\mathfrak{g}$ is $\mathfrak{h} = \mathbb{C}H$. Hence, $\mathfrak{g}$ has rank one. With respect to this Cartan subalgebra, the elements E and F are already eigenvectors:

$$[H, E] = 2E, [H, F] = -2F.$$

If ω is the element of $\mathfrak{h}^*$ that takes the value 1 on H , one sees that the roots of $\mathfrak{g}$ are $\alpha = 2\omega$ and $-\alpha$, with the symmetry with respect to either of them being multiplication by -1 . We denote this root system by A_1 , which is clearly irreducible. ■

Example 5.1.4. We now move onto $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, the Lie algebra of all traceless 3×3 complex matrices. A basis of this Lie algebra is given by the matrices

$$\begin{aligned} H_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & H_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \\ E_{12} &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & E_{23} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, & E_{13} &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ F_{21} &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & F_{32} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & F_{31} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \end{aligned}$$

It is a simple 8-dimensional Lie algebra, and thus also semisimple.²⁷

A Cartan subalgebra of $\mathfrak{g}$ is $\mathfrak{h} = \mathbb{C}H_1 \oplus \mathbb{C}H_2$. Hence, $\mathfrak{g}$ has rank two. With respect to this Cartan subalgebra, the elements E_{ij} and F_{ij} are also eigenvectors for the adjoint operator:

$$\begin{aligned} [H_1, E_{12}] &= 2E_{12}, [H_1, E_{23}] = -E_{23}, [H_1, E_{13}] = E_{13}, \\ [H_2, E_{12}] &= -E_{12}, [H_2, E_{23}] = 2E_{23}, [H_2, E_{13}] = E_{13}, \\ [H_1, F_{21}] &= -2F_{21}, [H_1, F_{32}] = F_{32}, [H_1, F_{31}] = -F_{31}, \\ [H_2, F_{21}] &= F_{21}, [H_2, F_{32}] = -2F_{32}, [H_2, F_{31}] = -F_{31}. \end{aligned}$$

²⁷We leave the computation of the Killing form as an exercise in patience to the reader.

Let ω_1, ω_2 be the dual basis of H_1, H_2 in $\mathfrak{h}^*$, that is, $\omega_i(H_j) = \delta_{ij}$. Then, the six roots of $\mathfrak{g}$ are

$$\pm\alpha_1 = \pm(2\omega_1 - \omega_2), \pm\alpha_2 = \pm(2\omega_2 - \omega_1) \text{ and } \pm(\alpha_1 + \alpha_2) = \pm(\omega_1 + \omega_2).$$

We call this root system A_2 . Its symmetries interchange the roots on opposite sides of the perpendicular dashed lines depicted in Figure 5.1. From this image, it is clear that this root system is irreducible. ■

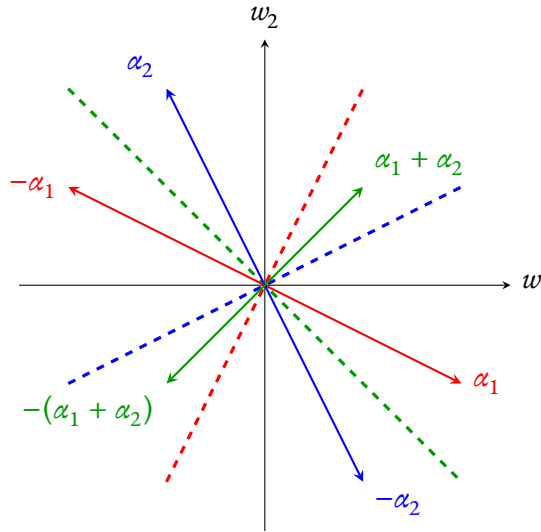


Figure 5.1: The root system A_2 associated to $\mathfrak{sl}_3(\mathbb{C})$.

There are four infinite families of irreducible root systems: A_n, B_n, C_n and D_n , all described in the Table 5.1.²⁸ They give rise to the four infinite families of isomorphism classes of simple Lie algebras, also called the classical algebras. Those are the algebra $\mathfrak{sl}_n(\mathbb{C})$ of traceless $n \times n$ complex matrices, the algebra $\mathfrak{so}_n(\mathbb{C})$ of $n \times n$ complex matrices that are skew-symmetric²⁹ with respect to a non-degenerate bilinear symmetric form of maximal index on $\mathbb{C}^n$ and the algebra $\mathfrak{sp}_n(\mathbb{C})$ of $n \times n$ complex matrices that are skew-symmetric with respect to a non-degenerate alternating bilinear form on $\mathbb{C}^n$. Apart from those, there are five exceptional root systems G_2, F_4, E_6, E_7 and E_8 . Their Lie algebras lack a description as simple as the classical ones, and since we will not need them for our latter applications, we refer the reader to the work of Jacques Tits for a uniform approach to their construction.³⁰

²⁸Bourbaki, *Lie groups and Lie algebras. Chapters 7-9*, Sec. VIII.13.

²⁹Meaning $B(Ax, y) + B(x, Ay) = 0$, or equivalently $A^* = -A$ where A^* is the adjoint matrix of A with respect to the non-degenerate symmetric or antisymmetric bilinear form B .

³⁰Jacques Tits. "Algebres alternatives, algebres de Jordan et algebres de Lie exceptionnelles". In: *Indagationes Mathematicae (Proceedings)* 69 (1966), pp. 223-237. ISSN: 1385-7258. DOI: 10.1016/S1385-7258(66)50028-2.

Root system	Simple Lie algebra $\mathfrak{g}$	Description	$\dim_{\mathbb{C}} \mathfrak{g}$
$A_n (n > 0)$	$\mathfrak{sl}_{n+1}(\mathbb{C})$	Traceless $(n+1) \times (n+1)$ complex matrices.	$n(n+2)$
$B_n (n > 1)$	$\mathfrak{so}_{2n+1}(\mathbb{C})$	$(2n+1) \times (2n+1)$ complex matrices A satisfying $B(Ax, y) + B(x, Ay) = 0$, where $B(x, y) = \sum_{i=1}^{n+1} x_i y_i - \sum_{j=n+2}^{2n+1} x_j y_j$.	$n(2n+1)$
$C_n (n > 1)$	$\mathfrak{sp}_{2n}(\mathbb{C})$	$2n \times 2n$ complex matrices A satisfying $B(Ax, y) + B(x, Ay) = 0$, where $B(x, y) = \sum_{j=1}^n (x_j y_{n+j} - x_{n+j} y_j)$.	$n(2n+1)$
$D_n (n > 2)$	$\mathfrak{so}_{2n}(\mathbb{C})$	$2n \times 2n$ complex matrices A satisfying $B(Ax, y) + B(x, Ay) = 0$, where $B(x, y) = \sum_{i=1}^n x_i y_i - \sum_{j=n+1}^{2n} x_j y_j$.	$n(2n-1)$

Table 5.1: The classical (non-exceptional) irreducible root systems and their associated simple Lie algebras. The subscript indicates their rank.

By a complex Lie group we understand a complex analytic manifold G with a group structure whose operations are complex analytic. The tangent space $T_1 G = \mathfrak{g}$ at the identity admits a natural complex Lie algebra structure by means of the second derivative of the commutator map.³¹ We say that two connected complex Lie groups G_1 and G_2 are *locally isomorphic* if they have isomorphic Lie algebras. If G is a connected complex Lie group, then its universal cover $\tilde{G}$ also admits a complex Lie group structure such that the covering map $\tilde{G} \rightarrow G$ is a surjective Lie group homomorphism inducing the identity on their Lie algebras.³² In particular, G and $\tilde{G}$ are locally isomorphic. A fundamental result of the Lie group-Lie algebra correspondence is the following.³³

³¹Consider the commutator map $\mu: G \times G \rightarrow G$ given by $\mu(x, y) = xyx^{-1}y^{-1}$ and denote by TG the tangent bundle of G . Its first derivative $D\mu$ can be seen as an analytic map from $G \times G$ to the space $\text{Hom}(TG \times TG, TG)$ of vector bundle maps. Its second derivative, evaluated at the identity $(1, 1)$, is then a function $D^2\mu(1, 1)$ from $\mathfrak{g} \times \mathfrak{g}$ to $\text{Hom}(\mathfrak{g} \times \mathfrak{g}, \mathfrak{g})$. We define the Lie bracket of $\mathfrak{g}$ as $[x, y] = D^2\mu(1, 1)(x, 0)(0, y)$. The antisymmetry follows from the fact that $\mu(x, y) = \mu(y, x)^{-1}$ and the chain rule, and the Jacobi identity follows from taking the third derivative of the Hall-Witt commutator identity for groups and evaluating at the three triples of tangent vectors $(x, 0, 0)$, $(0, y, 0)$ and $(0, 0, z)$.

³²Lee, *The Structure of Complex Lie Groups*, Cor. 1.14.

³³Bourbaki, *Lie Groups and Lie Algebras*, Thm. III.6.1.

Theorem 5.1.5. *If G and H are two complex connected Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ respectively and G is simply-connected, then for any Lie algebra homomorphism $\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$ there exists a unique Lie group homomorphism $\Phi: G \rightarrow H$ such that φ is the derivative of Φ at the identity.*

From this theorem, it follows that if G_1 and G_2 are two connected, simply connected and locally isomorphic complex Lie groups, then they are in fact isomorphic as Lie groups. Hence, for each finite dimensional complex Lie algebra $\mathfrak{g}$, there exists a *unique* simply connected, connected complex Lie group G having $\mathfrak{g}$ as its Lie algebra.

On the other extreme, if G is connected and simply-connected, *any* other complex connected Lie group G' locally isomorphic to G is obtained as a quotient of G by a discrete central³⁴ subgroup Γ . If G is semisimple³⁵, then its center $Z(G)$ is finite.³⁶ Hence, for every semisimple complex Lie algebra $\mathfrak{g}$, there exists a *unique* centerless and connected complex Lie group $G' = G/Z(G)$ having $\mathfrak{g}$ as its Lie algebra, called the Lie group of *adjoint type*³⁷ with Lie algebra $\mathfrak{g}$. Any other connected Lie group G'' locally isomorphic to G is finitely covered by G and finitely covers G' , and moreover is uniquely determined by the embedding of its fundamental group as a central subgroup of G . The simply connected and adjoint type classical groups are summarized in the Table 5.2.

We will be interested in the Lie group representations of a semisimple complex Lie group G , that is, Lie group homomorphisms $G \rightarrow GL_n(\mathbb{C})$ for some n . Since any such group is covered by the unique simply connected one, it suffices to study the representations of the simply connected ones. By Theorem 5.1.5, these are in bijection with Lie algebra representations $\mathfrak{g} \rightarrow gl_n(\mathbb{C})$. The first essential result we will need about the representations of $\mathfrak{g}$ is their complete reducibility.³⁸

Theorem 5.1.6. *Any representation $\mathfrak{g} \rightarrow gl(V)$ of a complex semisimple Lie algebra is completely reducible, that is, it decomposes uniquely as a direct sum $V = V_1 \oplus \dots \oplus V_n$ of irreducible subrepresentations up to ordering of the factors.*

If $\varphi: \mathfrak{g} \rightarrow gl(V)$ is a finite-dimensional representation of a semisimple complex Lie algebra $\mathfrak{g}$ with a Cartan subalgebra $\mathfrak{h}$ in a finite dimensional complex vector space V , then $\varphi(x)$ is also diagonalizable for every $x \in \mathfrak{h}$.³⁹ What this means is that we have

$$V \simeq \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda$$

³⁴A priori, this subgroup is just normal. However, for each fixed $x \in \Gamma$, the conjugation map $G \rightarrow \Gamma$ is continuous. Since G is connected and Γ is discrete, it must be constant.

³⁵That is, its Lie algebra is semisimple

³⁶Lee, *The Structure of Complex Lie Groups*, Cor. 4.17.

³⁷It coincides precisely with the image of the adjoint map $\text{Ad}: G \rightarrow \text{Aut}(\mathfrak{g})$.

³⁸Serre, *Complex Semisimple Lie Algebras*, Thm. II.6.8.

³⁹Ibid., Prop. VII.3.

Simple Lie algebra	Simply connected group	Adjoint type group	Center/Fundamental group
$\mathfrak{sl}_n(\mathbb{C})$ ($n > 1$)	$SL_n(\mathbb{C})$	$PSL_n(\mathbb{C})$	$\{e^{2\pi i k/n} \text{Id} : 0 \leq k \leq n-1\} \simeq \mathbb{Z}/n\mathbb{Z}$
$\mathfrak{so}_{2n+1}(\mathbb{C})$ ($n > 1$)	$Spin_{2n+1}(\mathbb{C})$	$SO_{2n+1}(\mathbb{C})$	$\simeq \mathbb{Z}/2\mathbb{Z},$
$\mathfrak{so}_{2n}(\mathbb{C})$ ($n > 2$)	$Spin_{2n}(\mathbb{C})$	$PSO_{2n}(\mathbb{C})$	$\begin{cases} \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, & n \text{ even}, \\ \mathbb{Z}/4\mathbb{Z}, & n \text{ odd}, \end{cases}$
$\mathfrak{sp}_{2n}(\mathbb{C})$ ($n > 1$)	$Sp_{2n}(\mathbb{C})$	$PSp_{2n}(\mathbb{C})$	$\{\pm \text{Id}\} \simeq \mathbb{Z}/2\mathbb{Z},$

Table 5.2: The simply connected and adjoint type classical groups, together with the centers of the simply connected forms (equivalently, the fundamental groups of the adjoint forms). Observe that if $\mathfrak{g} = \bigoplus_{i=1}^n \mathfrak{g}_i$ is a sum of simple Lie algebras, then both the simply connected and the adjoint type (i.e. centerless) groups of $\mathfrak{g}$ are the products of their respective groups for each $\mathfrak{g}_i$ by uniqueness. Moreover, for the classical special orthogonal group $SO_k(\mathbb{C})$, it is never simply connected, and it is centerless precisely when k is odd. Its universal cover, the spin groups $Spin_k(\mathbb{C})$, can be explicitly described as certain subgroups of unit groups in complex Clifford algebras.

as $\mathfrak{h}$ -modules, where V_λ denotes the space of elements $v \in V$ for which

$$\varphi(x)v = \lambda(x)v$$

for all $x \in \mathfrak{h}$. The elements of

$$\{\lambda \in \mathfrak{h}^* : V_\lambda \neq 0\}$$

are called the *weights* of the representation V , and the elements of each V_λ are called *weight vectors*. Since V is finite-dimensional, there are only finitely many weights. Observe that if $v \in V$ is a weight vector with weight λ and $x \in \mathfrak{g}$ is a root vector with root μ , then $\varphi(x)v$ is again a weight vector, now with weight $\lambda + \mu$.

To describe the weights of an irreducible representation of $\mathfrak{g}$, we shall need some extra concepts. If R denotes the roots of $\mathfrak{g}$ in $\mathfrak{h}^*$, we shall decompose R as a disjoint union $R = R^+ \cup R^-$ where the set R^+ has the following property: if α and β are in R^+ and their sum $\alpha + \beta$ is a root, then $\alpha + \beta \in R^+$. The elements of R^+ are called the *positive roots* and the elements of R^- are called the *negative roots*. Such a decomposition always exists.⁴⁰ A non-zero weight vector v in a representation $\varphi: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is said to be a *primitive element* if $\varphi(e_\lambda)v = 0$ for all root vectors e_λ associated to positive roots $\lambda \in R^+$.

Proposition 5.1.7. *Let $\varphi: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a non-trivial finite dimensional irreducible representation of $\mathfrak{g}$. Then V contains a unique primitive element $v \in V$ up to scalar multiplication, and its weight uniquely determines V up to isomorphism.⁴¹ Conversely, any representation generated by a primitive element is irreducible, regardless of it being finite dimensional or not.⁴²*

The weight $\lambda \in \mathfrak{h}^*$ of the primitive element in V is called the *highest weight* of V . Since V is uniquely determined by λ , we shall denote by V_λ the unique irreducible representation of $\mathfrak{g}$ with highest weight λ . We summarize some properties of the highest weights in the following proposition.⁴³

Proposition 5.1.8. *Let $P(\mathfrak{g}) \subset \mathfrak{h}^*$ be the set of highest weights of the irreducible finite-dimensional representations of $\mathfrak{g}$, and $n = \dim \mathfrak{h}$ be the rank of $\mathfrak{g}$.*

1. *There are unique elements $\omega_1, \dots, \omega_n \in P(\mathfrak{g})$ such that $P(\mathfrak{g})$ consists precisely of the integral linear combinations*

$$k_1\omega_1 + \dots + k_n\omega_n$$

with $k_i \geq 0$ not all zero. Hence, we can also specify any element in $P(\mathfrak{g})$ by n non-negative integral coordinates $k_1, \dots, k_n$.

⁴⁰It can be constructed inductively since R is finite. Start with $R^+ = \{\alpha\}$ and $R^- = \{-\alpha\}$ for some root α , and for some $\beta \neq \pm\alpha$ add to R^+ all the positive linear combinations of α and β that lie in R , and to R^- all their opposites. This process eventually exhausts R .

⁴¹Serre, *Complex Semisimple Lie Algebras*, Thm. VII.3.1.

⁴²Ibid., Prop. VII.2.2.

⁴³Ibid., Sec. VII.4.

2. For each positive root $\alpha \in R^+$, let h_α be the unique element of $\mathbb{C}[e_\alpha, e_{-\alpha}]$ for which $\alpha(h_\alpha) = 2$. Then, for any non-zero $\lambda \in \mathfrak{h}$, we have $\lambda \in P(\mathfrak{g})$ if and only if $\lambda(h_\alpha)$ is a non-negative integer for all $\alpha \in R^+$.

3. If λ is a weight in some finite dimensional representation of $\mathfrak{g}$, then λ is a unique linear integral combination of $\omega_1, \dots, \omega_n$.

The weights in $\mathfrak{h}^*$ satisfying the second condition of Proposition 5.1.8 are called the *dominant integral weights* of $\mathfrak{g}$, and the unique elements $\omega_1, \dots, \omega_n$ that generate $P(\mathfrak{g})$ are called its *fundamental weights*. We refer the reader to Serre's book for the explicit construction of each V_λ for a given dominant integral weight in the general case.⁴⁴

Example 5.1.9. Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with the Cartan subalgebra $\mathfrak{h} = \mathbb{C} \cdot H$, $\omega \in \mathfrak{h}^*$ be the linear functional that takes the value one in H and $\alpha = 2\omega$, so that the roots of $\mathfrak{g}$ are $R = \{\pm\alpha\}$. If we take α as the positive root, since

$$[e_\alpha, e_{-\alpha}] = [E, F] = H$$

and $\alpha(H) = 2$, we see that $h_\alpha = H$. Hence, by Proposition 5.1.8, the dominant integral weights of $\mathfrak{g}$ are precisely the positive multiples $k\omega$ of ω . We shall construct explicitly the representation V_k of $\mathfrak{sl}_2(\mathbb{C})$ given by the dominant integral element $k\omega$ for each $k \geq 1$.

Let $V_k = \text{Sym}^k(\mathbb{C}^2)$ be the set of all complex homogeneous polynomials $f \in \mathbb{C}[x, y]$ in two variables of degree k . For a matrix $A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ in $\mathfrak{g}$, we define

$$(Af)(x, y) = (ax + cy) \frac{\partial f}{\partial x}(x, y) + (bx - ay) \frac{\partial f}{\partial y}(x, y).$$

One then checks⁴⁵ that this map defines a Lie algebra homomorphism $\mathfrak{g} \rightarrow \mathfrak{gl}(V_k)$, and thus a representation of $\mathfrak{g}$ with dimension $\dim V_k = k + 1$. We now claim that $f(x, y) = x^k$ is a primitive element of V_k .

Indeed, it is a weight vector with weight $k\omega$:

$$(Hf)(x, y) = x \cdot kx^{k-1} = (k\omega)(H)f(x, y).$$

Moreover, for the root vector E associated with the positive root α , we have

$$(Ef)(x, y) = x \cdot \frac{\partial f}{\partial y}(x, y) = 0.$$

Finally, observe that by applying F multiple times to f we obtain the polynomials $kx^{k-1}y, k(k-1)x^{k-2}y, \dots$ all the way down to y^k , that is, V_k is generated by $f(x, y)$ as a $\mathfrak{g}$ -module. Hence, by Proposition 5.1.7, V_k must be the unique irreducible representation of $\mathfrak{g}$ with highest weight $k\omega$. ■

⁴⁴Ibid., Thm. VII.3.2.

⁴⁵Identifying $\mathfrak{g}$ with the tangent space of $\text{SL}_2(\mathbb{C})$ at the identity, this map is the derivative of the $\text{SL}_2(\mathbb{C})$ action on V_k by pre-composing f with a linear change of variables.

Example 5.1.10. We now consider the case of $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ with the Cartan subalgebra $\mathfrak{h} = \mathbb{C}H_1 \oplus \mathbb{C}H_2$. Let ω_1 and ω_2 be the dual basis of H_1 and H_2 in $\mathfrak{h}^*$, that is, $\omega_i(H_j) = \delta_{ij}$, and choose as our positive roots the set

$$R^+ = \{\alpha_1 = 2\omega_1 - \omega_2, \alpha_2 = 2\omega_2 - \omega_1, \alpha_1 + \alpha_2 = \omega_1 + \omega_2\}.$$

We have

$$\begin{aligned} [e_{\alpha_1}, e_{-\alpha_1}] &= [E_{12}, F_{21}] = H_1 = h_{\alpha_1}, \\ [e_{\alpha_2}, e_{-\alpha_2}] &= [E_{23}, F_{32}] = H_2 = h_{\alpha_2}, \\ [e_{\alpha_1+\alpha_2}, e_{-\alpha_1-\alpha_2}] &= [E_{13}, F_{31}] = H_1 + H_2 = h_{\alpha_1+\alpha_2}. \end{aligned}$$

Once again, by Proposition 5.1.8 we see that the dominant integral weights of $\mathfrak{g}$ are the linear combinations $\lambda = k_1\omega_1 + k_2\omega_2$ with $k_1, k_2 \geq 0$ not both zero. The dimension of the irreducible representation V_λ of highest weight λ is given by the Weyl dimension formula:⁴⁶

$$\dim V_\lambda = \frac{(k_1 + 1)(k_2 + 1)(k_1 + k_2 + 2)}{2}.$$

Although there is not a nice description of V_λ for general k_1 and k_2 as we do have for $\mathfrak{sl}_2(\mathbb{C})$, we can certainly describe some particular cases.

Fix some $k \geq 1$ and let V_k be the set of all complex polynomials $f(x, y, z) \in \mathbb{C}[x, y, z]$ in three variables that are homogeneous of degree k . For a matrix A with columns a_1, a_2 and a_3 in $\mathfrak{g}$, we define

$$(Af)(x, y, z) = a_1^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial x}(x, y, z) + a_2^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial y}(x, y, z) + a_3^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial z}(x, y, z).$$

As it was the case for $\mathfrak{sl}_2(\mathbb{C})$, this defines a Lie algebra representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V_k)$ now of dimension $\binom{k+2}{2}$, and we claim that this is irreducible.

Consider the vector $f(x, y, z) = x^k$. Since all the positive root vectors E_{12}, E_{23} and E_{13} are strictly upper triangular, we have

$$(E_{12}f)(x, y, z) = (E_{23}f)(x, y, z) = (E_{13}f)(x, y, z) = 0.$$

Moreover, one computes

$$(H_1f)(x, y, z) = kf(x, y, z), (H_2f)(x, y, z) = 0$$

to see that f is a weight vector with weight $k\omega_1$. Since

$$x^{k-i-j}y^iz^j = \frac{1}{k(k-1)\dots(k-i-j+1)}(F_{31}^jF_{21}^if)(x, y, z),$$

⁴⁶Bourbaki, *Lie groups and Lie algebras. Chapters 7-9*, Thm. VIII.2.2.

we see that V_k is generated as a $\mathfrak{g}$ -module by $f(x, y, z)$. Hence, Proposition 5.1.7 tells us that V_k is the unique irreducible $\mathfrak{g}$ -module with highest weight $k\omega_1$. This induces the representation $\mathrm{SL}_3(\mathbb{C}) \rightarrow \mathrm{GL}(V_k)$ given by precomposing $f(x, y, z)$ with the linear change of variables determined by a matrix $A \in \mathrm{SL}_3(\mathbb{C})$. One can also build fairly explicitly the irreducible representation of $\mathfrak{g}$ with highest weight $k\omega_2$ by considering the complex dual V_k^* of V_k as a $\mathfrak{g}$ -module, or alternatively by precomposing the representation $\mathfrak{g} \rightarrow \mathrm{gl}(V_k)$ with the automorphism $A \mapsto -A^T$ of $\mathfrak{g}$. The induced representation of $\mathrm{SL}_3(\mathbb{C})$ is then the dual of previous induced representation, and can also be obtained by first composing the representation with the automorphism $A \mapsto (A^T)^{-1}$ of the complex Lie group $\mathrm{SL}_3(\mathbb{C})$. ■

Now that we have the necessary language, we can say how Chevalley groups enter the picture. The astute reader will have noticed that the simple complex Lie groups in the Table 5.2 are all *affine linear algebraic groups*: that is, they can be defined as subgroups of $\mathrm{GL}_n(\mathbb{C})$ for some n that satisfy a given set of polynomial equations in their entries.⁴⁷ Moreover, the representations of $\mathrm{SL}_2(\mathbb{C})$ and $\mathrm{SL}_3(\mathbb{C})$ constructed in Examples 5.1.9 and 5.1.10 are all *rational*, that is, as functions between linear algebraic groups their coefficients are expressible as polynomial equations in the matrix entries. This is a general phenomenon.⁴⁸

Theorem 5.1.11. *Let G be a complex semisimple Lie group and $\varphi: G \rightarrow \mathrm{GL}(V)$ a finite dimensional Lie group representation of G . Then:*

1. *The group G is a complex linear algebraic group, that is, it embeds as a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$ for some n under the Zariski topology.*
2. *The representation φ is rational: as a function from the affine variety G to the affine variety $\mathrm{GL}(V)$, it is given by polynomial equations.*

Moreover, this algebraic structure on G is unique.

The main insight of Claude Chevalley was that the equations defining G and its representations can be chosen to have *integer coefficients*. In a geometric language, it means that the algebraic varieties and morphisms in question are all *defined over $\mathbb{Z}$* . This can then be used to define their analogues in any ring R by considering the varieties those same equations cut out in the affine R -space. We will now proceed to formalize this concept.

Definition 5.1.12. *A group functor G is a functor from the category of commutative, associative unital rings to the category of groups. A morphism of group functors $\Phi: G \rightarrow H$ is a*

⁴⁷For example, $\mathrm{SL}_2(\mathbb{C})$ consists of the matrices A in $\mathrm{GL}_2(\mathbb{C})$ satisfying $\det A = 1$, and $\mathrm{PSL}_2(\mathbb{C})$ embeds under the adjoint representation as a subgroup of $\mathrm{GL}_3(\mathbb{C})$. In fact, since the adjoint representation preserves the Killing form, it embeds as a connected subgroup conjugate to the special orthogonal group $\mathrm{SO}_3(\mathbb{C})$, which is clearly algebraic. Those two coincide since they have isomorphic Lie algebras and are both centerless. This generalizes to all adjoint groups once one sees that the subgroup of $\mathrm{GL}(\mathfrak{g})$ that preserves the bracket is precisely the image of the adjoint representation, see Lee, *The Structure of Complex Lie Groups*, Sec. 1.4

⁴⁸Ibid., Thm. 5.11.

natural transformation, that is, for every ring R an assignment of a group homomorphism $\Phi(R): G(R) \rightarrow H(R)$ such that, for every ring homomorphism $\varphi: R \rightarrow S$ the diagram

$$\begin{array}{ccc} G(R) & \xrightarrow{\Phi(R)} & H(R) \\ G(\varphi) \downarrow & & \downarrow H(\varphi) \\ G(S) & \xrightarrow{\Phi(S)} & H(S) \end{array}$$

commutes.

A natural example of a group functor is $G = GL_n$, the functor that takes for any ring its group of $n \times n$ invertible matrices. We shall call a *representation* of a group functor any morphism $\Phi: G \rightarrow GL_n$ for some n . A *linear group functor* is a subfunctor G of GL_n , meaning that (1) for any ring R , $G(R)$ is a subgroup of $GL_n(R)$ and (2) for any ring homomorphism $R \rightarrow S$ the group homomorphism $G(R) \rightarrow G(S)$ is simply the restriction of the group homomorphism $GL_n(R) \rightarrow GL_n(S)$. A *rank n torus functor* is the group functor $\underbrace{GL_1 \times \dots \times GL_1}_{n \text{ times}}$, and a torus

in a linear group functor is a subfunctor⁴⁹ $T \leq G$ isomorphic to a torus functor. We denote the additive group functor $R \mapsto (R, +)$ by G_a .

Lemma 5.1.13. *Let T be a functor isomorphic to a rank n torus functor. Then, after identifying T with $\underbrace{GL_1 \times \dots \times GL_1}_{n \text{ times}}$, any morphism of functors $\Phi: T \rightarrow GL_1$ is a character, that is, there exists integers $a_1, \dots, a_n$ such that for every ring R we have*

$$\Phi(R)(t_1, \dots, t_n) = t_1^{a_1} \dots t_n^{a_n}.$$

Conversely, every n -tuple of integers define a unique morphism of functors $\Phi: T \rightarrow GL_1$.

Moreover, any morphism of functors $\Psi: GL_1 \rightarrow T$ is a cocharacter, that is, there exists integers $a_1, \dots, a_n$ such that for every ring R we have $\Psi(R)(t) = (t^{a_1}, \dots, t^{a_n})$, and any n -tuple of integers define a unique cocharacter $\Psi: GL_1 \rightarrow T$.

Proof. Let S be ring $\mathbb{Z}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$ of integral Laurent polynomials in n variables. Then, by definition, $\Phi(S)(x_1, \dots, x_n) = u$ is some unit of S . The only units of S are the elements of the form $\pm x_1^{a_1} \dots x_n^{a_n}$ for some choice of n integers $a_1, \dots, a_n$. On one hand, $\Phi(S)$ is a group homomorphism, which forces the sign to be positive. On the other hand, if R is any other ring and $t_1, \dots, t_n$ are n units in R , then there exists a ring homomorphism $\varphi: S \rightarrow R$ mapping each x_i to t_i . We then have:

$$\begin{aligned} \Phi(R)(t_1, \dots, t_n) &= (\Phi(R) \circ T(\varphi))(x_1, \dots, x_n) \\ &= (GL_1(\varphi) \circ \Phi(S))(x_1, \dots, x_n) \\ &= GL_1(\varphi)(x_1^{a_1} \dots x_n^{a_n}) \\ &= t_1^{a_1} \dots t_n^{a_n}. \end{aligned}$$

⁴⁹Again, meaning that $T(R) \rightarrow G(R)$ is always injective and $T(R) \rightarrow T(S)$ is always the restriction of $G(R) \rightarrow G(S)$.

That the formula in the statement defines a morphism of functors is immediate.

The same proof holds for cocharacters, instead now one considers the ring $S = \mathbb{Z}[x^{\pm 1}]$. We must have $\Psi(S)(x) = (\pm x^{a_1}, \dots, \pm x^{a_n})$ for some choice of signs and exponents. The fact that $\Psi(S)$ is a group homomorphism forces the sign to be positive, and for any ring R and unit $t \in \mathrm{GL}_1(R)$ there exists a ring homomorphism $S \rightarrow R$ mapping x to t . $\square$

Hence, by Lemma 5.1.13, the groups $X^*(T) = \mathrm{Hom}(T, \mathrm{GL}_1)$ and $X_*(T) = \mathrm{Hom}(\mathrm{GL}_1, T)$ are *non-canonically* isomorphic to $\mathbb{Z}^n$ for any rank n torus functor T . We do, however, have a canonical isomorphism $\mathrm{Hom}(\mathrm{GL}_1, \mathrm{GL}_1) \simeq \mathbb{Z}$ given by choosing the identity morphism as the generator. Hence, by Lemma 5.1.13, the composition of morphisms gives us a pairing

$$\langle \cdot, \cdot \rangle : X^*(T) \times X_*(T) \rightarrow \mathrm{Hom}(\mathrm{GL}_1, \mathrm{GL}_1) \simeq \mathbb{Z}$$

which is perfect, that is, a bilinear map of abelian groups that induces isomorphisms between their $\mathbb{Z}$ -duals.

Definition 5.1.14. A *reduced root datum* is a quadruple $(X, \Phi, Y, \Phi^\vee)$ where X and Y are free abelian groups of finite rank with a perfect pairing

$$\langle \cdot, \cdot \rangle : X \times Y \rightarrow \mathbb{Z},$$

$\Phi \subset X$ and $\Phi^\vee \subset Y$ are finite subsets with a bijection $(-)^\vee : \Phi \rightarrow \Phi^\vee$ satisfying:

(RD1) For all $\alpha \in \Phi$, $\langle \alpha, \alpha^\vee \rangle = 2$.

(RD2) For all $\alpha, \beta \in \Phi$, we have

$$\beta - \langle \beta, \alpha^\vee \rangle \alpha \in \Phi, \beta^\vee - \langle \alpha, \beta^\vee \rangle \alpha^\vee \in \Phi^\vee.$$

(RD3) For all $\alpha \in \Phi$, $2\alpha \notin \Phi$.

If R is a root system, we say that $(X, \Phi, Y, \Phi^\vee)$ is a root datum of type R if Φ is a root system in $X \otimes_{\mathbb{Z}} \mathbb{C}$ isomorphic to R .⁵⁰

We finally have all the ingredients to define the Chevalley groups.

Definition 5.1.15. Let R be a root system of rank n . A *split semisimple Chevalley group of type R* consists of a linear group functor G , a torus $T \leq G$ of rank n , a reduced root datum $(X^*(T), \Phi, X_*(T), \Phi^\vee)$ of type R (so we identify R with Φ) and for each root $\alpha \in R$ a subfunctor $x_\alpha \leq G$ isomorphic to G_α called the root subgroup for α such that:

(CG1) For every ring S , the group $G(S)$ is generated by the root subgroups $\{x_\alpha(S) : \alpha \in R\}$ and the torus $T(S)$.

⁵⁰In particular, we can identify the elements of Φ with R .

(CG2) For each root $\alpha \in R$, each ring S and each $t \in T(S)$, the conjugation by t on the root subgroup for α is controlled by the root datum:

$$tx_\alpha(u)t^{-1} = x_\alpha(\alpha(S)(t) \cdot u)$$

for all $u \in G_\alpha(S) \simeq (S, +)$.

(CG3) For all roots $\alpha, \beta \in R$ such that $\alpha + \beta \neq 0$, there are integer constants $c_{\alpha, \beta, i, j}$ independent of the ring S such that

$$[x_\alpha(u), x_\beta(v)] = \prod_{i, j \geq 0} x_{i\alpha + j\beta}(c_{\alpha, \beta, i, j} u^i v^j)$$

holds for all rings S and all elements $u, v \in G_\alpha(S)$, where the product is computed in some fixed order of the factors.

The Chevalley group G is said to be *universal* if it satisfies:

(CG4) The abelian group $X_*(T)$ is generated by $\Phi^\vee$.

We can now state the main theorem of this section.

Theorem 5.1.16. *Let G^{an} be the connected and simply connected complex semisimple Lie group with associated root system R . Then, there exists a universal split semisimple Chevalley group G of type R and an isomorphism of complex Lie groups $G^{\text{an}} \simeq G(\mathbb{C})$, and moreover G is unique up to a unique isomorphism commuting with the torus and the root subgroups. If moreover $\varphi: G^{\text{an}} \rightarrow \text{GL}_n(\mathbb{C})$ is a finite dimensional Lie group representation of G^{an} , then there exists a unique representation $\Phi: G \rightarrow \text{GL}_n$ such that $\varphi = \Phi(\mathbb{C})$.*

Sketch of the proof. The existence of a G as in the statement is standard material on Chevalley groups.⁵¹ We will describe the main steps, which start by fixing $\mathfrak{g}$ the complex semisimple Lie algebra with root system R and $\mathfrak{h}$ a Cartan subalgebra of dimension n , identifying R with the roots of $\mathfrak{g}$ in $\mathfrak{h}^*$. We fix a decomposition $R = R^+ \cup R^-$ into positive and negative roots. Recall that the root spaces $\mathfrak{g}_\alpha$ are one dimensional, for each positive root α there is exactly one element $h_\alpha \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ for which $\alpha(h_\alpha) = 2$. The next step is to prove the existence⁵² of a Chevalley basis for $\mathfrak{g}$, that is, a choice of generator e_α in each root space $\mathfrak{g}_\alpha$ and a basis $h_{\alpha_1}, \dots, h_{\alpha_n}$ of $\mathfrak{h}$ satisfying:

(CB1) h_α is an integral linear combination of the h_{α_i} for all $\alpha \in R^+$.

(CB2) $[h_{\alpha_i}, e_\alpha] \in \mathbb{Z} e_\alpha$ for all $1 \leq i \leq n$ and all $\alpha \in R$;

⁵¹Michel Demazure. *Structure des Schémas en Groupes Réductifs*. Springer Berlin Heidelberg, 1970. ISBN: 9783540363392. DOI: 10.1007/bfb0059027, Sec. XXV.1.

⁵²Steinberg, *Lectures on Chevalley Groups*, Thm. 1.1.

(CB3) $[e_\alpha, e_{-\alpha}] = h_\alpha$ for all $\alpha \in R^+$

(CB4) For all α and β in R :

$$[e_\alpha, e_\beta] \begin{cases} = 0, & \text{if } \alpha + \beta \notin R, \\ \in \mathbb{Z} e_{\alpha+\beta} & \text{otherwise.} \end{cases}$$

Suppose we have now fixed a Chevalley basis in $\mathfrak{g}$. The next step is to form a subring $U(\mathfrak{g})_{\mathbb{Z}}$ of the universal enveloping algebra $U(\mathfrak{g})$ called Kostant's $\mathbb{Z}$ -form. By definition, $U(\mathfrak{g})_{\mathbb{Z}}$ will be the subring of $U(\mathfrak{g})$ generated by all the elements $\frac{e_\alpha^m}{m!}$ with $\alpha \in R$ and $m \geq 1$. Recall from the Poincaré-Birkhoff-Witt Theorem (Theorem I.4.6) that a basis for $U(\mathfrak{g})$ over $\mathbb{C}$ is given by all the products

$$X_1 \cdots X_n, n \geq 0$$

of ordered symbols $X_1 \leq X_2 \leq \cdots \leq X_n$ in an ordered basis of $\mathfrak{g}$. A fundamental result⁵³ is that $U(\mathfrak{g})_{\mathbb{Z}}$ is a free abelian group with basis given by all the products of elements

$$\frac{H_i(H_i - 1) \cdots (H_i - m + 1)}{m!} \quad \text{and} \quad \frac{X_\alpha^m}{m!}$$

for some choices of m and some total ordering on our Chevalley basis.

We now let $\omega_1, \dots, \omega_n$ be the fundamental weights of $P(\mathfrak{g})$, and we denote by $V = V_1 \oplus \cdots \oplus V_n$ the direct sum of corresponding irreducible finite dimensional representations of $\mathfrak{g}$. We have a natural structure of a $U(\mathfrak{g})$ -module on V , and it can be shown⁵⁴ that V contains a lattice $V_{\mathbb{Z}}$ that is invariant under the action of $U(\mathfrak{g})_{\mathbb{Z}}$ and still decomposes as the direct sum $V_{\mathbb{Z}} = \bigoplus_{\lambda \in \mathfrak{h}^*} (V_{\mathbb{Z}} \cap V_\lambda)$ of its weight components.

We are now ready to define the group functor G . We fix a ring S and we let $V_S = V_{\mathbb{Z}} \otimes_{\mathbb{Z}} S$. For each $\alpha \in R$, we have that e_α is a nilpotent operator in $\text{End}_R(V_S)$. Since $V_{\mathbb{Z}}$ is invariant under the $U(\mathfrak{g})_{\mathbb{Z}}$ -action, the element $\frac{e_\alpha^m}{m!}$ is a well defined element of $\text{End}_R(V_S)$ for each $m \geq 1$. Hence, for each $t \in S$, we can define the automorphism

$$x_\alpha(t) = \exp(te_\alpha) = \sum_{m=0}^{\infty} \frac{t^m e_\alpha^m}{m!}$$

inside $\text{GL}(V_S) = \text{Aut}_S(V_S)$. Our group $G(S)$ is defined as the subgroup of $\text{GL}(V_S)$ generated by all the $x_\alpha(t)$ for all $\alpha \in R$ and all $t \in S$. It is not obvious that neither $G(S)$ is functorial in S nor that it is a universal split semisimple Chevalley group.

We start by addressing the latter issue, although it might not make much sense before establishing functoriality. It is clear that the x_α will be the root subgroups, and it can be shown that the subgroup generated by all the elements

$$h_\alpha(t) = x_\alpha(t)x_{-\alpha}(-t^{-1})x_\alpha(t)x_\alpha(1)^{-1}x_{-\alpha}(-1)^{-1}x_\alpha(1)^{-1}$$

⁵³Ibid., Thm. 2.2.

⁵⁴Ibid., Cor. 2.1.

for $\alpha \in R^+$ and $t \in S^\times$ is a rank n torus,⁵⁵ that is, isomorphic to $\underbrace{S^\times \times \dots \times S^\times}_{n \text{ times}}$. The root datum is given as follows: for any two roots $\alpha, \beta \in R$, it can be shown⁵⁶ that

$$h_\alpha(t)x_\beta(u)h_\alpha(t)^{-1} = x_\beta(t^{c(\alpha,\beta)}u)$$

for some integer $c(\alpha, \beta)$ that only depends on α and β , not on S . We define the element $\beta \in X^*(T)$ by the condition that

$$\beta(S)(h_\alpha(t)) = t^{c(\alpha,\beta)}$$

for any root α , any ring S and any $t \in S^\times$. Moreover, for each $\alpha \in R$, there is a unique map⁵⁷ $\iota_\alpha: \text{SL}_2(S) \rightarrow G(S)$ extending

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \mapsto x_\alpha(t), \quad \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} \mapsto x_{-\alpha}(t).$$

We define the element $\alpha^\vee \in X_*(T)$ by the condition that

$$\alpha^\vee(S)(t) = \iota_\alpha \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}$$

for any ring S and any $t \in S^\times$.

From there, one can verify properties (CG1) to (CG3) for any given ring S using the properties of the exponential map and the Chevalley basis.⁵⁸ Now, the fact that it is functorial in S is established by giving a presentation of $G(S)$ in terms of the generators $x_\alpha(t)$ and $h_\alpha(t)$ with relations that are naturally functorial in S .⁵⁹ The universality property (CG4) stems from our choice of V , using the fact that cocharacters associated to simple roots⁶⁰ are precisely dual to the fundamental weights under the duality

$$X^*(T) \times X_*(T) \rightarrow \mathbb{Z}$$

given by composition and the identification $\text{Hom}(\text{GL}_1, \text{GL}_1) \simeq \mathbb{Z}$ as groups of natural transformations. The uniqueness of G also follows from the fact that the defining relations on each ring are consequences of the axioms (CG1) through (CG4).⁶¹

For the existence of representations, we take the Lie algebra map $D\varphi: \mathfrak{g} \rightarrow \mathfrak{gl}_n(\mathbb{C})$ induced by said representation. As in the previous discussion, there exists a lattice $M_{\mathbb{Z}} \simeq \mathbb{Z}^n \leq \mathbb{C}^n$ that

⁵⁵Steinberg, *Lectures on Chevalley Groups*, Lem. 2.19 and Lem. 3.28.

⁵⁶Ibid., Lem. 3.20.

⁵⁷Ibid., Cor. 3.6.

⁵⁸Ibid., Lem. 2.15 and Lem. 2.20.

⁵⁹Ibid., Cor. 3.3 and Lem. 3.28.

⁶⁰Positive roots that cannot be written as a sum of two other positive roots.

⁶¹Steinberg, *Lectures on Chevalley Groups*, Cor. 3.3 and and Cor. to Lemma 3.28.

is invariant under the $U(\mathfrak{g})_{\mathbb{Z}}$ action and still decomposes as a direct sum of weight spaces, and for any ring S we can form $M_S = M_{\mathbb{Z}} \otimes_{\mathbb{Z}} S$. The map $\Phi(S): G(S) \rightarrow GL_n(S)$ is now defined by mapping each $x_{\alpha}(t)$ in $G(S)$ to the automorphism

$$\exp(tD\varphi(e_{\alpha})) = \sum_{m=0}^{\infty} \frac{t^m D\varphi(e_{\alpha}^m)}{m!} \in \text{End}_S(M_S).$$

The fact that this extends to a group homomorphism follows from the description of $G(S)$ in terms of generators and relations, plus the fact that the image in $\text{End}_S(M_S)$ satisfies the same relations as they are consequences of root and weight dynamics in the Lie algebra and the representation. Moreover, this definition of $\Phi(S)$ is forced. Let $S_0 = \mathbb{Z}[t]$ and consider the universal element $x_{\alpha}(t) \in G(S_0)$. By construction, $\Phi(S_0)(x_{\alpha}(t))$ is a matrix in $GL_n(S_0)$ whose entries are polynomials in t with coefficients in $\mathbb{Z}$. For any complex number $z \in \mathbb{C}$ there is a ring map $\theta_z: S_0 \rightarrow \mathbb{C}$ sending $t \mapsto z$, and naturality implies

$$\Phi(\mathbb{C})(x_{\alpha}(z)) = GL_n(\theta_z)(\Phi(S_0)(x_{\alpha}(t))) = \exp(zD\varphi(e_{\alpha})).$$

Thus each matrix entry of $\Phi(S_0)(x_{\alpha}(t))$ is the unique polynomial in $\mathbb{Z}[t]$ whose evaluation at every $z \in \mathbb{C}$ coincides with the corresponding entry of $\exp(zD\varphi(e_{\alpha}))$. This determines $\Phi(S_0)(x_{\alpha}(t))$ uniquely, and hence, by naturality, determines $\Phi(S)(x_{\alpha}(s))$ for every ring S and every $s \in S$, since any such s lies in the image of a ring homomorphism $S_0 \rightarrow S$ sending $t \mapsto s$. As the elements $x_{\alpha}(s)$ generate $G(S)$, the whole family $\Phi(S)$ is uniquely determined. Finally, that $G(\mathbb{C}) \simeq G^{\text{an}}$ and $\Phi(\mathbb{C}) = \varphi$ are standard facts from the Lie group – Lie algebra correspondence. $\square$

The universal split semisimple Chevalley group of type R is just a particular case – in fact, the existence and uniqueness of a split semisimple Chevalley group functor G can be shown for any root datum $(X^*(T), \Phi, X_*(T), \Phi^{\vee})$ of type R . Since we shall only need the universal ones, we will not push this topic further.

Example 5.1.17. The universal split semisimple Chevalley group for the root systems A_n are simply the special linear group functors

$$SL_{n+1}(S) = \{A \in GL_{n+1}(S) : \det(A) = 1\}.$$

Analogously, the universal split semisimple Chevalley group for the root systems C_n are the symplectic linear group functors

$$Sp_{2n}(S) = \{A \in GL_{2n}(S) : B(Ax, Ay) = B(x, y)\}$$

where B is the standard non-degenerate alternating bilinear form in S^{2n} . More surprising is the existence of Spin group functors for the root systems B_n and D_n , as well as for the exceptional groups. More importantly, finite-dimensional complex representations of semisimple complex Lie algebras $\mathfrak{g}$ all have functorial models over all rings S , although we might lack an “efficient” description of them in the complex case. $\blacksquare$

The last fact that we will need about split semisimple Chevalley groups G is the following deep result of Chevalley.⁶²

Proposition 5.1.18. *For each split semisimple Chevalley group G , the functor G is representable, that is, there exists a Hopf $\mathbb{Z}$ -algebra $\mathcal{O}(G)$ and an isomorphism of functors*

$$G(-) \simeq \operatorname{Hom}_{\operatorname{Rings}}(\mathcal{O}(G), -).$$

In particular, if $R = \varprojlim R_i$, then $G(R) = \varprojlim G(R_i)$.

In the language of algebraic geometry, the above proposition says that G is an affine group scheme over $\mathbb{Z}$, and $\mathcal{O}(G)$ is the Hopf $\mathbb{Z}$ -algebra that defines it. It is not hard to show that $G(-)$ is representable in the category of fields of any given characteristic⁶³, but representability over arbitrary rings is subtler. The only feature of Proposition 5.1.18 that we will use is its last conclusion: that G commutes with inverse limits. The skeptical reader can also check this property directly for many models of G , for instance SL_n .

5.2 p -adic points in Chevalley groups

We shall keep some of the notation from Section 5.1: we will fix a root system R and let G denote the universal split semisimple Chevalley group functor of type R . Our goal in this section is to record some fundamental group theoretic properties of $G(\mathbb{Z}_p)$, where $\mathbb{Z}_p$ is the ring of p -adic integers, that shall be needed for the rest of this chapter. We start by recalling the definition of a compact p -adic Lie group. Throughout this section, p will always be **an odd prime**.⁶⁴

Definition 5.2.1. Let G be a profinite group. We say that G is a *compact p -adic Lie group* or a *p -adic analytic profinite group* if it satisfies any of the following equivalent properties:⁶⁵

- (a) It is a compact analytic $\mathbb{Q}_p$ -manifold endowed with a group structure for which multiplication and inversion are analytic functions.
- (b) G is topologically isomorphic to a closed subgroup of $GL_n(\mathbb{Z}_p)$ for some $n \geq 1$.
- (c) G contains an open pro- p subgroup U that is a *uniform* pro- p group, that is, U is finitely generated, $[G, G] \leq G^p$ and all successive quotients of the lower p -central series of U have the same order.

⁶²Claude Chevalley. “Certains schémas de groupes semi-simples”. fre. In: *Séminaire Bourbaki* 6 (1961), pp. 219–234. URL: <http://eudml.org/doc/109612>.

⁶³Steinberg, *Lectures on Chevalley Groups*, Chap. 5.

⁶⁴This is for simplicity of exposition: dealing with 2-adic Lie groups involves handling several exceptions and cases in the statements of the results, although almost all of them have analogues for $p = 2$. For our applications, we can safely avoid the even primes.

⁶⁵Dixon et al., *Analytic Pro- p Groups*, Interlude A.

(d) There exists a positive integer d such that all closed subgroups $H \leq G$ can be topologically generated by d elements.

Associated to a compact p -adic Lie group G , one can associate several Lie algebras. The first one is a finite-dimensional $\mathbb{Q}_p$ -Lie algebra $\text{Lie}(G)$ defined through the $\mathbb{Q}_p$ -manifold structure of (a).⁶⁶ Indeed, there exists some open subgroup U in G such that U is homeomorphic to $(p\mathbb{Z}_p)^n$ for some n and the multiplication rule $U \times U \rightarrow U$ is given by an everywhere convergent power series $F(X, Y)$, called a formal group law over $\mathbb{Z}_p$. Such a power series is of the form

$$F(X, Y) = X + Y + B(X, Y) + \text{terms of degree} \geq 3,$$

where $B(X, Y)$ is bilinear. Identifying $\text{Lie}(G) = \mathbb{Q}_p^n$ with the “tangent $\mathbb{Q}_p$ -space at 0” of the “clopen ball of radius 1” given by $U \leq \mathbb{Q}_p^n$, we define a Lie bracket on $\text{Lie}(G)$ by means of

$$[X, Y] = B(X, Y) - B(Y, X).$$

The second one is a $\mathbb{Z}_p$ -algebra $\mathcal{L}(U)$ defined on an uniform open subgroup U of G , called the *Lazard-Lie algebra* of U , that is defined as follows: as a set it is equal to U , and an addition operation on U is defined by

$$x + y = \lim_{n \rightarrow \infty} (x^{p^n} y^{p^n})^{p^{-n}}.$$

This operation makes U topologically isomorphic to $\mathbb{Z}_p^n$ as an additive group for some $n \geq 0$.⁶⁷ This already makes U naturally into a $\mathbb{Z}_p$ -module, and the Lie bracket is defined analogously⁶⁸ through

$$[x, y] = \lim_{n \rightarrow \infty} [x^{p^n}, y^{p^n}]^{p^{-n}}.$$

Although the isomorphism class of $\mathcal{L}(U)$ may depend on the choice of U , these Lie algebras are related to $\text{Lie}(G)$ by the formula⁶⁹

$$\text{Lie}(G) = \mathcal{L}(U) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p.$$

The $\mathbb{Z}_p$ -Lie algebra $\mathcal{L}(U)$ is an example of a *powerful* Lie algebra, that is, it is finitely generated and free as a $\mathbb{Z}_p$ -module and moreover satisfies⁷⁰

$$[\mathcal{L}(U), \mathcal{L}(U)] \leq p\mathcal{L}(U).$$

⁶⁶Jean-Pierre Serre. *Lie Algebras and Lie Groups: 1964 Lectures given at Harvard University*. 2nd ed. Lecture Notes in Mathematics 1500. Springer-Verlag Berlin Heidelberg, 1992. ISBN: 978-0-387-55008-4, Sec. V.1.

⁶⁷Dixon et al., *Analytic Pro- p Groups*, Thm. 4.17.

⁶⁸Ibid., Thm. 4.30.

⁶⁹Ibid., Exer. 9.13.

⁷⁰Ibid., Exer. 4.2.(ii).

A third Lie algebra we can associate to G comes from *associative Banach envelopes*. Let A is a Banach $\mathbb{Q}_p$ -algebra, that is, a complete normed $\mathbb{Q}_p$ -algebra, and assume that topologically G embeds as a closed subgroup of the unit group $A^\times$ such that the topology on G is induced by the norm. Let G_1 be the subgroup of G given by all the elements $g \in G$ for which $g - 1$ has norm strictly less than one. Then, the series

$$\log(g) = \log((g - 1) + 1) = \sum_{i=1}^{\infty} (-1)^{i+1} \frac{(g - 1)^i}{i}$$

converges in A for every $g \in G_1$. If we moreover assume that G_1 is *uniform*, with the commutator induced by the commutator $[X, Y] = XY - YX$ in A we have that

$$\log(G_1) = \{\log(g) : g \in G_1\}$$

is a $\mathbb{Z}_p$ -Lie algebra.⁷¹ Under these hypothesis, we call A an associative Banach envelope of G . Moreover, identifying $\mathcal{L}(G_1)$ with an open subset of G , the logarithm map $\log : \mathcal{L}(G_1) \rightarrow \log(G_1)$ is an isomorphism of $\mathbb{Z}_p$ -Lie algebras.⁷²

The main source of examples of compact p -adic Lie groups in our work will come from p -adic points of the Chevalley groups G constructed in Section 5.1.

Lemma 5.2.2. *The group of $\mathbb{Z}_p$ points $G(\mathbb{Z}_p)$ of a universal split semisimple Chevalley group G is a closed subgroup of $\mathrm{GL}_n(\mathbb{Z}_p)$ in its p -adic topology. In particular, it is a compact p -adic Lie group.*

Proof. We have $\mathrm{GL}_n(\mathbb{Z}_p) = \varprojlim \mathrm{GL}_n(\mathbb{Z} / p^i \mathbb{Z})$, and a subgroup $H \leq \mathrm{GL}_n(\mathbb{Z}_p)$ is closed if and only if $H = \varprojlim H_i$ where H_i is the image of H in $\mathrm{GL}_n(\mathbb{Z} / p^i \mathbb{Z})$. By the definition of a linear group functor, the image H_i of $H = G(\mathbb{Z}_p)$ in $\mathrm{GL}_n(\mathbb{Z} / p^i \mathbb{Z})$ is contained in $G(\mathbb{Z} / p^i \mathbb{Z})$, and we have

$$\begin{aligned} G(\mathbb{Z}_p) &= \varprojlim G(\mathbb{Z} / p^i \mathbb{Z}), \text{ by Proposition 5.1.18,} \\ &= \varprojlim H_i \end{aligned}$$

where in the last equality we use the property that if $G = \varprojlim G_i$ with possibly non-surjective projection maps $\varphi_i : G \rightarrow G_i$, then $G = \varprojlim \varphi_i(G)$. $\square$

Recall that any split semisimple Chevalley group G comes equipped with a unique semisimple complex Lie algebra $\mathfrak{g}$, and this complex Lie algebra contains a lattice $\mathfrak{g}_{\mathbb{Z}}$ that is itself a $\mathbb{Z}$ -Lie algebra. We can compare all the Lie algebras involved in case of $\mathbb{Z}_p$ -points:

Proposition 5.2.3. *Let p be an odd prime, G be a universal split semisimple Chevalley group given as a subfunctor of GL_n , $\mathfrak{g}_{\mathbb{Z}}$ the lattice given by a Chevalley basis in its complex Lie algebra $\mathfrak{g}$, $G = G(\mathbb{Z}_p)$ the compact p -adic Lie group of $\mathbb{Z}_p$ -points. Then:*

⁷¹Dixon et al., *Analytic Pro-P Groups*, Lem. 7.12.

⁷²Ibid., Cor. 7.14.

(a) The embedding $G \rightarrow \text{Mat}_n(\mathbb{Q}_p)$ is an associative Banach envelope under the supremum norm on $\text{Mat}_n(\mathbb{Q}_p) \simeq \mathbb{Q}_p^{n^2}$.

(b) We have $G_1 = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$.

(c) As $\mathbb{Z}_p$ -Lie subalgebras of $\text{Mat}_n(\mathbb{Q}_p)$, we have

$$\mathcal{L}(G_1) \simeq \log(G_1) = p(\mathfrak{g}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}_p).$$

Proof. We already saw in Lemma 5.2.2 that $G(\mathbb{Z}_p)$ is a closed subgroup of $\text{GL}_n(\mathbb{Z}_p)$, and hence a closed subgroup of $\text{GL}_n(\mathbb{Q}_p)$. Under the norm topology

$$\|(a_{ij})\| = \max_{i,j} \|a_{ij}\|,$$

a set of fundamental neighborhoods of the identity in $\text{GL}_n(\mathbb{Z}_p)$ consists of the subgroups

$$1 + p^i \text{Mat}_n(\mathbb{Z}_p),$$

which induces the profinite topology on $\text{GL}_n(\mathbb{Z}_p)$ and thus also on $G(\mathbb{Z}_p)$. We clearly have

$$G_1 = (1 + p \text{Mat}_n(\mathbb{Z}_p)) \cap G(\mathbb{Z}_p) = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{F}_p))$$

since G is a subfunctor of GL_n , showing (b).

For (a), it remains to check that G_1 is uniform. We sketch the argument of Jishnu Ray.⁷³, which follows from an exercise in DDMS.⁷⁴ We fix a basis $\alpha_1, \dots, \alpha_m$ of positive roots for the root system R of G and we define the height function $h: R \rightarrow \mathbb{Z}$ given by

$$h(\lambda_1 \alpha_1 + \dots + \lambda_m \alpha_m) = |\lambda_1 + \dots + \lambda_m|.$$

The critical input^{75,76} from the theory of universal Chevalley groups is that any element g in G_1 can be uniquely written as a product

$$g = \prod_{\alpha \in R^+} x_{\alpha}(t_{\alpha}) \times \prod_{i=1}^m h_{\alpha_i}(1 + s_i) \times \prod_{\alpha \in R^-} x_{\alpha}(t_{\alpha})$$

⁷³Jishnu Ray. “Presentation of the Iwasawa algebra of the first congruence kernel of a semi-simple, simply connected Chevalley group over $\mathbb{Z}_p$ ”. In: *Journal of Algebra* 511 (Oct. 2018), pp. 405–419. ISSN: 0021-8693. DOI: 10.1016/j.jalgebra.2018.05.039, postnote.

⁷⁴Dixon et al., *Analytic Pro-P Groups*, Exer. 13.11.

⁷⁵Ray, “Presentation of the Iwasawa algebra of the first congruence kernel of a semi-simple, simply connected Chevalley group over $\mathbb{Z}_p$ ”, Lem. 2.1.

⁷⁶See also Eiichi Abe. “Chevalley groups over local rings”. In: *Tohoku Mathematical Journal* 21.3 (Jan. 1969). ISSN: 0040-8735. DOI: 10.2748/tmj/1178242958, Cor. 3.3

where t_α and $s_1, \dots, s_m$ are elements in $p\mathbb{Z}_p$ and the product is taken on some ascending fixed order respecting the height function $h(\alpha)$. The conjugation relations⁷⁷ among the elements $x_\alpha(t_\alpha)$ and $h_{\alpha_i}(1 + s_i)$ show that G_1 is a standard group in the sense of Serre.⁷⁸ Hence, it is uniform of dimension $m + |R|$.⁷⁹

By the previous remarks, we have that $\mathcal{L}(G_1)$ can be identified with the $\mathbb{Z}_p$ -Lie subalgebra $\log(G_1)$ of $\text{Mat}_n(\mathbb{Q}_p)$ under the logarithm map. Recall that a $\mathbb{Z}$ -basis of $\mathfrak{g}$ is given by elements e_α for $\alpha \in R$ and $h_{\alpha_i} = [e_{\alpha_i}, e_{-\alpha_i}]$ of a Chevalley basis of $\mathfrak{g}$. A $\mathbb{Z}_p$ -basis of $\log(G_1)$ is given by elements

$$X_\alpha = \log(x_\alpha(p)) = \log(\exp(pe_\alpha)) = pe_\alpha$$

and

$$H_i = \log(h_{\alpha_i}(1 + p))$$

Recall that we are identifying $\text{GL}_n(\mathbb{Z}_p)$ with the automorphism group of a representation $V_{\mathbb{Z}_p}$ of $\mathfrak{g}_{\mathbb{Z}_p}$ which decomposes as a sum of weight spaces for the Cartan subalgebra $\mathfrak{h}_{\mathbb{Z}_p} = \langle h_{\alpha_1}, \dots, h_{\alpha_m} \rangle_{\mathbb{Z}_p}$. Hence, each h_{α_i} is a diagonal matrix over $\mathbb{Z}_p$ with entries $\lambda(\alpha_i)$ for some ordering of the weight spaces. Moreover, for any weight vector $v_\lambda \in V_{\mathbb{Z}_p}$ we have⁸⁰

$$h_{\alpha_i}(t) \cdot v_\lambda = t^{\lambda(\alpha_i)} v_\lambda$$

for any $t \in \mathbb{Z}_p^\times$. Hence, in that same ordering of the weight spaces, $h_{\alpha_i}(1 + p)$ is a diagonal matrix with entries $(1 + p)^{\lambda(\alpha_i)}$. Therefore:

$$H_i = \log(h_{\alpha_i}(1 + p)) = h_{\alpha_i} \log(1 + p) = ph_{\alpha_i} + p^2(\text{something in } \mathbb{Z}_p h_{\alpha_i}).$$

This shows that $\log(G_1)$ and $p(\mathfrak{g}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}_p)$ generate the same $\mathbb{Z}_p$ -submodule of $\text{End}_{\mathbb{Z}_p}(V_{\mathbb{Z}_p})$. Since both are Lie subalgebras of this matrix algebra, part (c) follows. $\square$

In this chapter, we shall study rank functions on the Iwasawa algebra $\mathbb{Z}_p[[G_1]]$. Some natural rank functions that can be defined on this ring are:

- As we have seen in Example 1.4.13, the algebra $\mathbb{Z}_p[[G_1]]$ is a Noetherian domain. Hence, by Proposition 2.1.3, the set of non-zero elements in $\mathbb{Z}_p[[G_1]]$ is a two-sided Ore set, and $\mathbb{Z}_p[[G_1]]$ embeds in an Ore division ring of fractions $\mathcal{Q}$. This embedding induces by pullback a Sylvester matrix rank function $\text{rk}_{\mathcal{Q}}$ on $\mathbb{Z}_p[[G_1]]$. By the computations carried out in the Examples 3.1.5 and 3.1.10, this rank function is uniquely characterized by $\text{rk}_{\mathcal{Q}}(x) = 1$ for any non-zero element $x \in \mathbb{Z}_p[[G_1]]$.

⁷⁷Steinberg, *Lectures on Chevalley Groups*, Lem. 20.

⁷⁸Dixon et al., *Analytic Pro-P Groups*, Def. 8.22.

⁷⁹Ibid., Thm. 8.31.

⁸⁰Steinberg, *Lectures on Chevalley Groups*, Lem. 19.

- If U is a normal open subgroup of G_1 , we can embed the group ring $\mathbb{Z}_p[G_1/U]$ into the finite-dimensional $\mathbb{Q}_p$ -algebra $\mathbb{Q}_p[G_1/U]$. The pullback of the normalized dimension of G_1/U under the composition

$$\mathbb{Z}_p[[G_1]] \rightarrow \mathbb{Z}_p[G_1/U] \rightarrow \mathbb{Q}_p[G_1/U]$$

is a Sylvester matrix rank function $\text{rk}_{G_1/U}$ on $\mathbb{Z}_p[[G_1]]$.

- Let $\varphi: G(\mathbb{C}) \rightarrow \text{GL}_m(\mathbb{C})$ be a Lie group representation and $\Phi: G \rightarrow \text{GL}_m$ the unique representation of group functors associated to it from Theorem 5.1.16. If for any ring S we associate the space $V(S) \simeq S^m$ endowed with its natural left $\text{GL}_m(S)$ -action, we obtain a functor from the category of rings to the category of abelian groups such that the natural diagram

$$\begin{array}{ccc} G(S) \times V(S) & \xrightarrow{(g, v) \mapsto \Phi(S)(g)v} & V(S) \\ \downarrow G(\psi) \times V(\psi) & & \downarrow V(\psi) \\ G(S') \times V(S') & \xrightarrow{(g', v') \mapsto \Phi(S')(g')v'} & V(S') \end{array}$$

commutes for any ring map $\psi: S \rightarrow S'$. Hence, G_1 acts continuously on $V(\mathbb{Z}_p)$ and therefore also on $V(\mathbb{Q}_p)$. As we've seen in Example 3.1.8, the composition

$$\mathbb{Z}_p[[G_1]] \rightarrow \text{End}_{\mathbb{Z}_p}(V(\mathbb{Z}_p)) \rightarrow \text{End}_{\mathbb{Q}_p}(V(\mathbb{Q}_p))$$

induces a rank function $\text{rk}_{V(\mathbb{Q}_p)}$ on $\mathbb{Z}_p[[G_1]]$ by taking the pullback of the normalized $\mathbb{Q}_p$ -rank on $\text{End}_{\mathbb{Q}_p}(V(\mathbb{Q}_p)) \simeq \text{Mat}_m(\mathbb{Q}_p)$. In particular, for every irreducible finite dimensional representation $V_\lambda(\mathbb{C})$ of $G(\mathbb{C})$ with highest weight $\lambda \in P(\mathfrak{g})$, we obtain a rank function rk_λ on $\mathbb{Z}_p[[G_1]]$ coming from the composition of the ring maps

$$\mathbb{Z}_p[[G_1]] \rightarrow \text{End}_{\mathbb{Z}_p}(V_\lambda(\mathbb{Z}_p)) \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)).$$

Our goal is to compare these natural rank functions on $\mathbb{Z}_p[[G_1]]$. To do so, we begin with a proposition concerning the Ore rank function on Ore domains that is inspired by Weibo Fu's work.^{81,82}

⁸¹Weibo Fu. "Sharp bounds for multiplicities of Bianchi modular forms". In: *Annals of Mathematics* 200.1 (July 2024). ISSN: 0003-486X. DOI: 10.4007/annals.2024.200.1.3, Sec. 6.

⁸²Sánchez and Souza, *Asymptotics of rational representations for algebraic groups*, Lem. 5.7.

Proposition 5.2.4. *Let S be a Ore domain with an Ore division ring of fractions $\mathbb{Q}$ inducing the rank function $\text{rk}_{\mathbb{Q}} \in \mathbb{P}(S)$. For any matrix A over S there are non-zero elements $x_1, \dots, x_r \in A$ such that for any Sylvester matrix rank function $\text{rk} \in \mathbb{P}(S)$ we have*

$$|\text{rk}(A) - \text{rk}_{\mathbb{Q}}(A)| \leq r - \sum_{i=1}^r \text{rk}(x_i).$$

Proof. Let $\dim_{\mathbb{Q}}$ and $\dim$ be the Sylvester module rank functions associated to $\text{rk}_{\mathbb{Q}}$ and rk respectively. For any $A \in \text{Mat}_{m \times n}(S)$ we have

$$|\text{rk}(A) - \text{rk}_{\mathbb{Q}}(A)| = |\dim(S^m / AS^n) - \dim_{\mathbb{Q}}(S^m / AS^n)|.$$

Hence, let M be the right S -module S^m / AS^n . Let $m_1, \dots, m_k$ be elements in M that form a $\mathbb{Q}$ -basis of the right $\mathbb{Q}$ -module $M \otimes_S \mathbb{Q}$. Since $\mathbb{Q}$ is flat over S by Proposition 2.1.5, tensoring the short exact sequence

$$0 \rightarrow \sum_{i=1}^k m_i S \rightarrow M \rightarrow Q \rightarrow 0$$

with $\mathbb{Q}$ over S shows that

$$(Q \otimes_S \mathbb{Q}) = 0.$$

Hence, Q is a torsion S -module.

Moreover, the submodule of M generated by the m_i must be free on the basis $\{m_1, \dots, m_k\}$, since any relation of S -linear dependence between them would imply a relation of $\mathbb{Q}$ -linear dependence in $M \otimes_S \mathbb{Q}$. Iterating this process using the finite presentation $S^n \rightarrow S^m \rightarrow M$ as a quotient of a free right S -module, we obtain three short exact sequences

$$0 \rightarrow S^k \rightarrow M \rightarrow Q \rightarrow 0, \quad (5.1)$$

$$0 \rightarrow N \rightarrow S^m \rightarrow M \rightarrow 0, \quad (5.2)$$

$$0 \rightarrow S^j \rightarrow N \rightarrow P \rightarrow 0, \quad (5.3)$$

where N is finitely generated and Q and P are torsion R -modules. Observe that we have $m = k + j$ since $\dim_{\mathbb{Q}}$ is additive under short exact sequences.

Since both Q and P are finitely generated, we can find filtrations

$$Q = Q_0 > Q_1 > \dots > Q_e = 0,$$

$$P = P_0 > P_1 > \dots > P_f = 0$$

such that all the quotients Q_{i-1}/Q_i and P_{i-1}/P_i are cyclic and torsion. If $\gamma_1, \dots, \gamma_e$ and $z_1, \dots, z_f$ are non-zero elements of S that annihilate the quotients Q_{i-1}/Q_i and P_{i-1}/P_i respectively, by a repeated application of (SDim₃) we obtain the bounds

$$\dim(Q) \leq \sum_{i=1}^e \dim(S/\gamma_i S) \text{ and } \dim(P) \leq \sum_{i=1}^f \dim(S/z_i S)$$

for any Sylvester module rank function $\dim$.

Let $r = e + f$ and $(x_1, \dots, x_r) = (\gamma_1, \dots, \gamma_e, z_1, \dots, z_f)$. By (SDim₃) and the short exact sequences 5.1 and 5.3, for any Sylvester module rank function $\dim$ we have:

$$\dim M \leq k + \dim Q \leq \dim_{\mathcal{Q}} M + \sum_{i=1}^e \dim(S/\gamma_i S), \quad (5.4)$$

$$\dim N \leq j + \dim P \leq \dim_{\mathcal{Q}} N + \sum_{i=1}^f \dim(S/z_i S). \quad (5.5)$$

Now, by (SDim₃) and the short exact sequence 5.2 we obtain

$$\dim_{\mathcal{Q}} M + \dim_{\mathcal{Q}} N = m = \dim S^m \leq \dim N + \dim M,$$

which we reorganize as

$$\dim_{\mathcal{Q}} N - \dim N \leq \dim M - \dim_{\mathcal{Q}} M. \quad (5.6)$$

Now, by the inequality 5.4 we already get the upper bound

$$\dim M - \dim_{\mathcal{Q}} M \leq \sum_{i=1}^e \dim(S/\gamma_i S) \leq \sum_{i=1}^r \dim(S/x_i S).$$

From Inequality 5.5 we get the lower bound

$$-\sum_{i=1}^r \dim(S/x_i S) \leq -\sum_{i=1}^f \dim(S/z_i S) \leq \dim_{\mathcal{Q}} N - \dim N.$$

Combining both of this bounds with the inequality 5.6 we obtain

$$-\sum_{i=1}^r \dim(S/x_i S) \leq \dim M - \dim_{\mathcal{Q}} M \leq \sum_{i=1}^r \dim(S/x_i S).$$

The claim now follows from the identity

$$\dim(S/x_i S) = 1 - \text{rk}(x_i)$$

for all $1 \leq i \leq r$. □

Corollary 5.2.5. *For any sequence rk_i of Sylvester matrix rank functions on an Ore domain S with Ore division ring of fractions $\mathcal{Q}$, we have $\lim_{i \rightarrow \infty} \text{rk}_i = \text{rk}_{\mathcal{Q}}$ as rank functions on S if and only if*

$$\lim_{i \rightarrow \infty} \text{rk}_i(x) = 1$$

for all non-zero $x \in S$.

From Example 3.1.10, it is clear that $\text{rk}_{\mathcal{Q}}$ is the only rank function on $\mathbb{P}(S)$ for which $\text{rk}_{\mathcal{Q}}(x) = 1$ for all non-zero $x \in S$. Hence, the non-trivial contribution of Corollary 5.2.5 is that if

$$\lim_{i \rightarrow \infty} \text{rk}_i(x) = 1$$

for all non-zero $x \in S$, then the sequence $\text{rk}_i(A)$ converges for *any* S -matrix A .

The study of the rank functions rk_{λ} for $\lambda \in P(\mathfrak{g})$ is the theme of the following sections. We finish this section by comparing the rank functions $\text{rk}_{G_1/U}$ and $\text{rk}_{\mathcal{Q}}$. We recall the definition of the big- O notation: for two real valued functions $f, g: X \subseteq \mathbb{Z}_{\geq 0}^m \rightarrow \mathbb{R}$, we say that $f = O(g)$ if there exists some positive constant M such that

$$f(x) \leq M g(x)$$

for all x sufficiently large in the domain. The following was first proven by Michael Harris for arbitrary torsion-free subgroups of $\ker(\text{GL}_m(\mathbb{Z}_p) \rightarrow \text{GL}_m(\mathbb{F}_p))$.⁸³

Theorem 5.2.6. *Let $G_n = \ker(\text{G}(\mathbb{Z}_p) \rightarrow \text{G}(\mathbb{Z}/p^n\mathbb{Z}))$, so that each G_n is an open normal subgroup of G_1 and $G_1 = \varprojlim G_1/G_n$. Then, for any matrix A over $\mathbb{Z}_p[[G_1]]$, we have*

$$|\text{rk}_{G_1/G_n}(A) - \text{rk}_{\mathcal{Q}} A| = O\left(\frac{1}{p^{n-1}}\right).$$

In particular, $\lim_{n \rightarrow \infty} \text{rk}_{G_1/G_n} = \text{rk}_{\mathcal{Q}}$.

Proof. By Proposition 5.2.4, it suffices to show that

$$1 - \text{rk}_{G_1/G_n}(x) = O\left(\frac{1}{p^{n-1}}\right)$$

for any non-zero x in $\mathbb{Z}_p[[G_1]]$. Let $M = \mathbb{Z}_p[[G_1]]/x\mathbb{Z}_p[[G_1]]$. Since

$$1 - \text{rk}_{G_1/G_n}(x) = \dim_{G_1/G_n}(M) = \frac{\dim_{\mathbb{Q}_p} M \otimes_{\mathbb{Z}_p[[G_1]]} \mathbb{Q}_p[G_1/G_n]}{|G_1 : G_n|}$$

and

$$M \otimes_{\mathbb{Z}_p[[G_1]]} \mathbb{Q}_p[G_1/G_n] = (M \otimes_{\mathbb{Z}_p[[G_1]]} \mathbb{Z}_p[G_1/G_n]) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p,$$

we get the bound

$$\dim_{G_1/G_n}(M) \leq \frac{\dim_{\mathbb{F}_p} M \otimes_{\mathbb{Z}_p[[G_1]]} \mathbb{F}_p[G_1/G_n]}{|G_1 : G_n|}.$$

⁸³Harris, “Correction to p-adic Representations Arising from Descent on Abelian Varieties”, Thm. 1.10.

On one hand, we have $G_n = \exp(p^{n-1} \log(G_1))$, and so the G_n coincide with the $(n-1)$ -th term of the lower p -central series of G_1 . In particular, if $d = \dim_{\mathbb{Z}_p} \log(G_1)$ denotes the dimension of G_1 as a p -adic manifold, then $|G_1 : G_n| = p^{d(n-1)}$. On the other hand, we have

$$M \otimes_{\mathbb{Z}_p[[G_1]]} \mathbb{F}_p[G_1/G_n] \simeq \mathbb{F}_p[G_1/G_n]/x \mathbb{F}_p[G_1/G_n] \simeq \mathbb{F}_p[[G_1]]/x \mathbb{F}_p[[G_1]] \otimes_{\mathbb{F}_p[[G_1]]} \mathbb{F}_p[G_1/G_n],$$

so we are reduced to showing that

$$\dim_{\mathbb{F}_p} N \otimes_{\mathbb{F}_p[[G_1]]} \mathbb{F}_p[G_1/G_n] = O(p^{(d-1)(n-1)})$$

for any torsion cyclic $\mathbb{F}_p[[G_1]]$ -module N .

Recall that if we filter $\mathbb{F}_p[[G_1]]$ by powers of its augmentation ideal, the graded ring $\text{Gr } \mathbb{F}_p[[G_1]]$ is isomorphic to the polynomial algebra $\mathbb{F}_p[\overline{x}_1, \dots, \overline{x}_d]$ where $x_i + 1 = g_i$ is a minimal generating set for G_1 , which we take to be the exponentials $\exp(y)$ as y runs through a $\mathbb{Z}_p$ -basis of $\log(G_1)$.⁸⁴ We denote by J the graded ideal of $\text{Gr } \mathbb{F}_p[[G_1]]$ generated by the $\overline{x}_i$, so that

$$\dim_{\mathbb{F}_p} N/F_i N = \dim_{\mathbb{F}_p} \text{Gr } N / \text{Gr } N J^i.$$

Since N is a torsion cyclic $\mathbb{F}_p[[G_1]]$ -module, if we endow it with the quotient filtration coming from $\mathbb{F}_p[[G_1]]$ we obtain that $\text{Gr } N$ is a torsion $\text{Gr } \mathbb{F}_p[[G_1]]$ -module. As $\text{Gr } N$ is torsion, it has Krull dimension at most $d-1$, and therefore the function

$$f(i) = \dim_{\mathbb{F}_p} N/F_i N$$

is a polynomial of degree at most $d-1$ for large enough i .⁸⁵

If I_n denotes the kernel of the map $\mathbb{F}_p[[G_1]] \rightarrow \mathbb{F}_p[G_1/G_n]$, then I_n contains all the elements $g_i^{p^{n-1}} - 1 = x_i^{p^{n-1}}$. Hence, the graded ideal $\text{Gr } I_n$ contains all the polynomials in $\text{Gr } \mathbb{F}_p[[G_1]]$ divisible by some $\overline{x}_i^{p^{n-1}}$ under the induced filtration, and thus $J^{dp^{n-1}} \leq \text{Gr } I_n$. This gives us a surjection

$$\text{Gr } N / \text{Gr } N J^{dp^{n-1}} \rightarrow \text{Gr } N / \text{Gr } N I_n$$

and therefore the bounds

$$\begin{aligned} \dim_{\mathbb{F}_p} N \otimes_{\mathbb{F}_p[[G_1]]} \mathbb{F}_p[G_1/G_n] &= \dim_{\mathbb{F}_p} N / N I_n \\ &= \dim_{\mathbb{F}_p} \text{Gr}(N / N I_n) \\ &= \dim_{\mathbb{F}_p} \text{Gr } N / \text{Gr } N \text{Gr } I_n \\ &\leq \dim_{\mathbb{F}_p} \text{Gr } N / \text{Gr } N J^{dp^{n-1}} \\ &= f(dp^{n-1}). \end{aligned}$$

⁸⁴Dixon et al., *Analytic Pro-P Groups*, Thm. 7.24.

⁸⁵Winfried Bruns and Jürgen Herzog. *Cohen-Macaulay Rings*. Cambridge University Press, June 1998. ISBN: 9780511608681. DOI: 10.1017/cbo9780511608681, Thm. 4.1.3.

Therefore, for large enough n , we have

$$\dim_{\mathbb{F}_p} N \otimes_{\mathbb{F}_p[[G_1]]} \mathbb{F}_p[G_1/G_n] = O\left((dp^{n-1})^{d-1}\right) = O\left(p^{(n-1)(d-1)}\right)$$

as desired. $\square$

5.3 Microlocalization

We shall keep the notation of the previous section: p is an *odd prime*, G is a universal split semisimple Chevalley group and $G_n = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{Z}/p^n\mathbb{Z}))$ is the n -th congruence kernel of its p -adic points. If $\mathfrak{g}$ is the complex semisimple Lie algebra of $G(\mathbb{C})$, we denote by $P(\mathfrak{g})$ the set of highest weights of the irreducible representations of $\mathfrak{g}$. If $\phi_\lambda: G(\mathbb{C}) \rightarrow GL(V_\lambda(\mathbb{C}))$ is the Lie group representation associated to $\lambda \in P(\mathfrak{g})$, we denote by $\Phi_\lambda: G \rightarrow GL(V_\lambda)$ the representation defined by it through Theorem 5.1.16. The Iwasawa algebra $\mathbb{Z}_p[[G_1]]$ is a Noetherian domain with an Ore field of fractions $\mathbb{Q}$ and rank functions $\text{rk}_{\mathbb{Q}}$, rk_{G_1/G_n} and rk_λ coming from the ring maps from $\mathbb{Z}_p[[G]]$ to $\mathbb{Q}$, $\mathbb{Q}_p[G_1/G_n]$ and $\text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ respectively.

We have already related the rank functions $\text{rk}_{\mathbb{Q}}$ and rk_{G_1/G_n} through Harris' Theorem 5.2.6, so now we turn our eye to comparing the rank functions rk_λ and $\text{rk}_{\mathbb{Q}}$. To do so, we shall introduce an overring to $\mathbb{Z}_p[[G_1]]$ that is a natural envelope for the rank functions rk_λ and has a more explicit presentation in terms of generators and relations.

Consider the $\mathbb{Z}_p$ -Lie algebra $\mathfrak{g}_{\mathbb{Z}_p} = \mathfrak{g}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}_p$ where $\mathfrak{g}_{\mathbb{Z}}$ is a Chevalley basis in $\mathfrak{g}$, and let $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ be its universal enveloping algebra over $\mathbb{Z}_p$. By the Poincaré-Birkhoff-Witt theorem (Theorem 1.4.6), if we fix a basis $\gamma_1, \dots, \gamma_d$ of $\log(G_1)$, the monomials $\gamma_1^{a_1} \dots \gamma_d^{a_d}$ form a basis for the $\mathbb{Q}_p$ -algebra $U(\mathfrak{g}_{\mathbb{Z}_p}) = U(\mathfrak{g}_{\mathbb{Z}_p})_0 \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. We define a norm on this $\mathbb{Q}_p$ vector space by declaring⁸⁶

$$\left\| \sum_{a_1, \dots, a_d} \alpha_{a_1, \dots, a_d} \gamma_1^{a_1} \dots \gamma_d^{a_d} \right\| = \max\{|\alpha_{a_1, \dots, a_d}| \}.$$

Moreover, $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ coincides with the clopen ball of radius one around 0 in the induced metric.

Definition 5.3.1. The *completed enveloping algebra* of $\mathfrak{g}_{\mathbb{Z}_p}$ is the completion $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ of $U(\mathfrak{g}_{\mathbb{Z}_p})$ with respect to the norm defined above.

The elements of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ admit a very explicit description.

⁸⁶We can of course identify $U(\mathfrak{g}_{\mathbb{Z}_p})$ with the universal enveloping $\mathbb{Q}_p$ -algebra of the $\mathbb{Q}_p$ -Lie algebra $\mathfrak{g}_{\mathbb{Q}_p} = \mathfrak{g}_{\mathbb{Z}_p} \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. However, our choice of norm comes from a choice of $\mathfrak{g}_{\mathbb{Z}_p}$ as a lattice inside $\mathfrak{g}_{\mathbb{Q}_p}$, and this will be needed later.

Lemma 5.3.2. *Any element of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ can be uniquely represented as a power series*

$$\sum_{a_1, \dots, a_d} \alpha_{a_1, \dots, a_d} y_1^{a_1} \dots y_d^{a_d}$$

with $\mathbb{Q}_p$ -coefficients such that for each $\epsilon > 0$ we have $|\alpha_{a_1, \dots, a_d}| < \epsilon$ for all but finitely many $a_1, \dots, a_d$.

Proof. Let us consider first the completion $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ of $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ under the induced norm. This norm induces on $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ a linear topology in which a fundamental system of neighborhoods of zero are the two-sided ideals $p^i U(\mathfrak{g}_{\mathbb{Z}_p})_0$. Hence, in the completion $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$, a series as in the statement converges if and only if $|\alpha_{a_1, \dots, a_d}| \rightarrow 0$ along the filter of cofinite subsets of $\mathbb{Z}_{\geq 0}^d$. For uniqueness of expression, observe that by its universal property $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0 / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ is isomorphic to the universal enveloping algebra of the $\mathbb{Z} / p^i \mathbb{Z}$ -Lie algebra $\mathfrak{g}_{\mathbb{Z}_p} / p^i \mathfrak{g}_{\mathbb{Z}_p}$. This Lie algebra is free as a $\mathbb{Z} / p^i \mathbb{Z}$ -module on the basis $y_i + p^i \mathfrak{g}_{\mathbb{Z}_p}$, so reduction modulo p^i determines all the coefficients $\alpha_{a_1, \dots, a_d}$ such that $|\alpha_{a_1, \dots, a_d}| > |p^i|$.

Now, observe that the tensor product $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ embeds in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$, and we claim that its image is the whole completion. To do so, it suffices to show that it is closed as the image contains $U(\mathfrak{g}_{\mathbb{Z}_p})$ and therefore is dense. This is equivalent to the image of $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ being complete under the induced norm, which amounts to saying that if f_i is a sequence in $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ for which $\|f_i\| \rightarrow 0$, then the series $\sum_{i \geq 0} f_i$ converges in $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$.

For each f_i , there exists a minimal $b_i \geq 0$ for which $p^{b_i} f_i \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$. Since $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ coincides with the ball of radius zero in $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$, the condition that $\|f_i\| \rightarrow 0$ implies that $b_i = 0$ for almost all i . Hence, for some $N_0 > 0$, we have that $f_i \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ for all $i > N_0$ and therefore the series

$$\sum_{i \geq 0} f_i = f_0 + \dots + f_{N_0} + \sum_{i > N_0} f_i$$

converges to a limit in $\mathbb{Q}_p \otimes_{\mathbb{Z}_p} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$. □

Recall that the logarithm map identifies $\log(G_1)$ with $p \mathfrak{g}_{\mathbb{Z}_p}$ as $\mathbb{Z}_p$ -Lie subalgebras of $\text{Mat}_n(\mathbb{Q}_p)$ by Proposition 5.2.3. The goal in this section is to relate the rings $\mathbb{Z}_p[[G_1]]$ and $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$, which we do in the following theorem.⁸⁷ While its roots can be traced to Michel

⁸⁷Sánchez and Souza, *Asymptotics of rational representations for algebraic groups*, Prop. 4.3.

Lazard⁸⁸, it first appeared in this form in a paper by Konstantin Ardakov and Simon Wadsley.⁸⁹

Theorem 5.3.3. *Let $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ be the completed enveloping algebra of $\mathfrak{g}_{\mathbb{Z}_p}$. Then:*

(a) *For any $y \in \log(G_1)$, the series $\exp(y)$ converges to a unit in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$.*

(b) *The composition*

$$G_1 \xrightarrow{\log} \log(G_1) \xrightarrow{\exp} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$$

is a continuous group embedding which extends to a strict filtered ring homomorphism $\mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ where $\mathbb{Z}_p[[G_1]]$ is filtered by the powers of its radical J and $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ is filtered by $F_i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} = p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$.

(c) *The set*

$$T = \bigcup_{a \geq 0} p^a + J^{a+1}$$

is a microlocal Ore set in $\mathbb{Z}_p[[G_1]]$ and the map $\mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ induces an isomorphism $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} \simeq T^{-1} \widehat{\mathbb{Z}_p[[G_1]]}$. In particular, $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ is a flat $\mathbb{Z}_p[[G_1]]$ -module on both sides.

Proof. Since $\log(G_1) = p \mathfrak{g}_{\mathbb{Z}_p}$, we have $y = pz$ for some $z \in \mathfrak{g}_{\mathbb{Z}_p}$. As

$$\left| \frac{p^n}{n!} \right| \rightarrow 0$$

inside $\mathbb{Q}_p$, we get that the series

$$\exp(y) = \exp(pz) = \sum_{n \geq 0} \frac{p^n}{n!} z^n$$

converges in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$. Moreover,

$$\exp(y) = 1 + p \left(\sum_{n \geq 1} \frac{p^{n-1}}{n!} z^n \right) = 1 + px$$

with $x \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ is a unit with inverse

$$\exp(y)^{-1} = \sum_{n \geq 0} (-1)^n p^n x^n \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0.$$

⁸⁸Lazard, “Groupes analytiques p-adiques”, Sec. IV.3.2.5.

⁸⁹Konstantin Ardakov and Simon Wadsley. “On irreducible representations of compact p-adic analytic groups”. In: *Annals of Mathematics* 178.2 (2013), pp. 453–557. ISSN: 0003486X, Thm. 10.4.

This shows (a).

For (b), observe that for any $x, y \in \log(G_1)$ with the exponentials computed as a map $\log(G_1) \rightarrow G_1$ we have that

$$x * y = \log(\exp(x) \cdot \exp(y))$$

is an element of $\log(G_1)$ by the Baker-Campbell-Hausdorff formula⁹⁰ for powerful Lie subalgebras of complete normed $\mathbb{Q}_p$ -algebras. In this case, the operation $*$ endows $\log(G_1)$ with a profinite group structure⁹¹ and the logarithm is a topological group isomorphism⁹² $\log: (G_1, \cdot) \rightarrow (\log(G_1), *)$. On the other direction, computing the exponentials as a map $\log(G_1) \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}^\times$, the same Baker-Campbell-Hausdorff formula also tells us that

$$x *' y = \log(\exp(x) \cdot \exp(y))$$

is an element of $\log(G_1)$. Moreover, both $x *' y$ and $x * y$ can be expressed as a power series in the commutators of x and y which only depends on the Lie algebra structure of $\log(G_1)$, not on where the exponential is being computed.

This shows that the composition $G_1 \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0^\times$ is a group homomorphism. For continuity, it suffices to check that the exponential map $\exp: p \mathfrak{g}_{\mathbb{Z}_p} \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0^\times$ is continuous at zero. This is immediate from the power series definition.

From the existence of this homomorphism, we obtain a ring homomorphism $\varphi: \mathbb{Z}_p[G_1] \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ from the usual $\mathbb{Z}_p$ -group ring of G_1 . Denote by J_0 the kernel of the map $\mathbb{Z}_p[G_1] \rightarrow \mathbb{F}_p$. Since G_1 is uniform and topologically generated by $g_i = \exp(p\gamma_i)$ for $\gamma_1, \dots, \gamma_d$ a $\mathbb{Z}_p$ -basis of $\mathfrak{g}_{\mathbb{Z}_p}$, the elements of $\mathbb{Z}_p[G_1]$ can be uniquely expressed as a convergent sum⁹³

$$x = \sum_{a_1, \dots, a_d \geq 0} \alpha_{a_1, \dots, a_d} (g_1 - 1)^{a_1} \dots (g_d - 1)^{a_d}$$

under the J_0 -adic topology of $\mathbb{Z}_p[G_1]$. Moreover, we have that $x \in J_0^i$ if and only if⁹⁴

$$\sup \left\{ |\alpha_{a_1, \dots, a_d}| p^{-(a_1 + \dots + a_d)} : a_1, \dots, a_d \geq 0 \right\} \leq p^{-i}.$$

Since $\varphi(g_i) = \exp(p\gamma_i)$, the homomorphism φ satisfies

$$\begin{aligned} \varphi((g_1 - 1)^{a_1} \dots (g_d - 1)^{a_d}) &= (\exp(p\gamma_1) - 1)^{a_1} \dots (\exp(p\gamma_d) - 1)^{a_d} \\ &\equiv p^{a_1 + \dots + a_d} \gamma_1 \dots \gamma_d \pmod{F_{a_1 + \dots + a_d + 1} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}} \end{aligned}$$

⁹⁰Dixon et al., *Analytic Pro-P Groups*, Thm. 6.28.

⁹¹Ibid., Thm. 9.8.

⁹²Ibid., Thm. 9.10.

⁹³Ibid., Thm. 7.4.

⁹⁴Ibid., Thm. 7.5.

This shows that φ is a strict ring homomorphism. Since J is the closure of J_0 in $\mathbb{Z}_p[[G_1]]$, we can identify the filtered ring $\mathbb{Z}_p[[G_1]]$ as the J_0 -adic completion⁹⁵ of $\mathbb{Z}_p[G_1]$. Hence φ extends uniquely to a strict filtered ring homomorphism $\varphi: \mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ by Lemma 1.3.4. This shows (b).

We now turn to (c). It was already shown in Proposition 2.2.8 that T is a microlocal Ore set in $\mathbb{Z}_p[[G_1]]$ associated to the Ore set $T' = \{\pi^i: i \geq 0\}$ in $\text{Gr } \mathbb{Z}_p[[G_1]]$. We must now show that the image of T in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ is invertible. Since φ is filtered, for any $x \in J^{i+1}$ we have that $\varphi(p^i + x) = p^i + p^{i+1}y$ where $y \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$. Hence, its inverse is given by

$$(p^i + p^{i+1}y)^{-1} = p^{-i} \left(\sum_{j \geq 0} (-1)^j p^j y^j \right).$$

By the universal property of the microlocalization, φ extends uniquely to a filtered ring homomorphism $\varphi: \widehat{T^{-1} \mathbb{Z}_p[[G_1]]} \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ such that

$$\text{Gr } \varphi: \text{Gr}(\widehat{T^{-1} \mathbb{Z}_p[[G_1]]}) \simeq (T')^{-1} \text{Gr } \mathbb{Z}_p[[G_1]] \rightarrow \text{Gr } \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$$

is the map induced by the Ore localization of $\text{Gr } \mathbb{Z}_p[[G_1]]$ at T' .

However, by Proposition 1.4.7 and Lemma 5.3.2, we can identify $\text{Gr } \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} \simeq \text{Gr } U(\mathfrak{g}_{\mathbb{Z}_p})$ with the universal enveloping algebra of the filtered $\mathbb{Q}_p$ -Lie algebra $\text{Gr}(\mathfrak{g}_{\mathbb{Z}_p} \otimes_{\mathbb{Z}_p} \mathbb{Q}_p)$ under the filtration by $p^i \mathfrak{g}_{\mathbb{Z}_p}$ over the ring $\text{Gr } \mathbb{Q}_p \simeq \mathbb{F}_p[\pi^{\pm 1}]$. It is then immediate from the identity $\text{Gr } \varphi(\overline{g_i - 1}) = \pi \overline{y_i}$ and the the unique expression of the elements of $\mathbb{Z}_p[G_1]$ as a convergent sum that $\text{Gr } \varphi$ is a bijection. By Corollary 1.4.9, φ is a strict bijection. $\square$

We are now endowed with the task of relating the rank functions rk_λ to the embedding $\mathbb{Z}_p[[G_1]] \rightarrow \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$. If we denote by $D\varphi_\lambda: \mathfrak{g}_{\mathbb{Z}_p} \rightarrow \mathfrak{gl}(V_\lambda(\mathbb{Q}_p))$ the Lie algebra representation defined by the highest weight λ , then since $V_\lambda(\mathbb{Q}_p)$ is complete under the p -adic norm the $U(\mathfrak{g}_{\mathbb{Z}_p})$ -action extends to a $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ -action. Then, the commutative diagram

$$\begin{array}{ccc} G_1 & \xrightarrow{\Phi_\lambda} & \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) \\ \log \downarrow & \nearrow D\varphi_\lambda & \uparrow \exp \\ & \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} & \\ p \mathfrak{g}_{\mathbb{Z}_p} & \xrightarrow{D\varphi_\lambda} & \mathfrak{gl}(V_\lambda(\mathbb{Q}_p)) \end{array}$$

⁹⁵Dixon et al., *Analytic Pro-P Groups*, Lem. 7.1.

from Theorem 5.3.3 shows that the $\mathbb{Z}_p[[G_1]]$ action on $V_\lambda(\mathbb{Q}_p)$ factors through the embedding into $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$. In particular, rk_λ is in the image of the change of rings map $\mathbb{P}(\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}) \rightarrow \mathbb{P}(\mathbb{Z}_p[[G_1]])$ for any $\lambda \in P(\mathfrak{g})$.

We expect that a similar approximation theorem to that of Harris' Theorem 5.2.6 also holds for the rank functions rk_λ .⁹⁶ More specifically, if $\omega_1, \dots, \omega_n$ are the fundamental weights of $\mathfrak{g}$ and $\lambda = \lambda_1\omega_1 + \dots + \lambda_n\omega_n$ is the unique expression of λ as an integral linear combination of the ω_i from Proposition 5.1.8, we define $\min \lambda = \min\{\lambda_1, \dots, \lambda_n\}$. We then propose the following conjecture:

Conjecture 5.3.4. *Let G be a universal split semisimple Chevalley group, p and odd prime and G_1 be the first congruence subgroup of $G(\mathbb{Z}_p)$. Then, for any non-zero x in $\mathbb{Z}_p[[G_1]]$ we have*

$$1 - \text{rk}_\lambda(x) = O\left(\frac{1}{\min \lambda}\right).$$

By Proposition 5.2.4, this conjecture would imply that for any matrix A over $\mathbb{Z}_p[[G_1]]$ we have that

$$|\text{rk}_Q(A) - \text{rk}_\lambda(A)| = O\left(\frac{1}{\min \lambda}\right)$$

where Q is the Ore division ring of fractions of $\mathbb{Z}_p[[G_1]]$. It also implies the following weaker conjecture:

Conjecture 5.3.5. *Let G be a universal split semisimple Chevalley group, p and odd prime and G_1 be the first congruence subgroup of $G(\mathbb{Z}_p)$. Then, we have*

$$\lim_{\min \lambda \rightarrow \infty} \text{rk}_\lambda = \text{rk}_Q$$

as rank functions on $\mathbb{Z}_p[[G_1]]$.

As of the time of writing, Conjectures 5.3.4 and 5.3.5 are only known to hold for $G = \text{SL}_2 \times \dots \times \text{SL}_2$. This was first shown by Fu in 2022⁹⁷, and the proof of the conjecture in this case is the topic of the next section. The validity of any of the two conjectures above has far reaching consequences to the cohomology of lattices in real or complex semisimple Lie groups, as will be explained in Chapter 6.

The relevance of the microlocalization comes from Fu's approach to the Conjecture 5.3.4. An essential ingredient is the following theorem of Ardakov and Wadsley:

Theorem 5.3.6. *Let G be a universal split semisimple Chevalley group, p and odd prime and G_1 be the first congruence subgroup of $G(\mathbb{Z}_p)$. Then, for almost all primes p (depending on G), the following hold:*

⁹⁶Sánchez and Souza, *Asymptotics of rational representations for algebraic groups*, Cor. 5.8.

⁹⁷Fu, "Sharp bounds for multiplicities of Bianchi modular forms", Thm. 3.2.

(a) The center of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ is the closure of the center of $U(\mathfrak{g}_{\mathbb{Z}_p})$.^{98,99}

(b) Let $\chi: Z(\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}) \rightarrow \mathbb{Q}_p$ be a ring homomorphism and I_χ be the ideal of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ generated by the central elements $\ker(\chi)$. Then, each I_χ intersects trivially the image of $\mathbb{Z}_p[[G_1]]$.¹⁰⁰ In other words, $\mathbb{Z}_p[[G_1]]$ embeds into the quotient ring $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}/I_\chi$.

5.4 Asymptotics for SL_2 and the δ method

We keep the notation of the previous section. Henceforth, we **assume that** $\mathbf{G} = \prod_{i=1}^k SL_2$, so that $\mathfrak{g}_{(-)} = \prod_{i=1}^k \mathfrak{sl}_2(-)$. Our goal in this final section is to show the following reformulation of a theorem due to Weibo Fu¹⁰¹:

Theorem 5.4.1. *For almost all primes p , Conjecture 5.3.4 holds for $\mathbf{G} = \prod_{i=1}^k SL_2$.*

Fu's proof works by explicitly computing bases for the quotients $U(\mathfrak{g}_{\mathbb{Z}_p})/(I_\chi \cap U(\mathfrak{g}_{\mathbb{Z}_p}))$, working through the natural ascending filtration of $U(\mathfrak{g}_{\mathbb{Z}_p})$ by total degree and comparing back with elements in the completed universal enveloping algebra $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$. We refer to this approach as the δ -method, and we will explain its main features in this section, as well as why it does not seem to extend beyond SL_2 . Recall our canonical Chevalley basis

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

of $\mathfrak{sl}_2$ from Example 5.1.3. We collect in the following proposition some essential properties of $U(\mathfrak{g}_{\mathbb{Z}_p})$ that we will need throughout this section.

Proposition 5.4.2. *Let $\mathfrak{g}_{\mathbb{Z}_p} = \prod_{i=1}^k \mathfrak{sl}_2(\mathbb{Z}_p)$, $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ be its universal enveloping algebra over $\mathbb{Z}_p$ and $U(\mathfrak{g}_{\mathbb{Z}_p}) = U(\mathfrak{g}_{\mathbb{Z}_p})_0 \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$. The following hold:*

(a) *We have ring isomorphisms*

$$\begin{aligned} U(\mathfrak{g}_{\mathbb{Z}_p})_0 &= \underbrace{U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 \otimes_{\mathbb{Z}_p} \cdots \otimes_{\mathbb{Z}_p} U(\mathfrak{sl}_2(\mathbb{Z}_p))_0}_{k \text{ times}}, \\ U(\mathfrak{g}_{\mathbb{Z}_p}) &= \underbrace{U(\mathfrak{sl}_2(\mathbb{Z}_p)) \otimes_{\mathbb{Q}_p} \cdots \otimes_{\mathbb{Q}_p} U(\mathfrak{sl}_2(\mathbb{Z}_p))}_{k \text{ times}}. \end{aligned}$$

⁹⁸Konstantin Ardakov. “Krull dimension of Iwasawa algebras and some related topics”. PhD thesis. University of Cambridge, 2004, Thm. 5.2.1.

⁹⁹Konstantin Ardakov and Simon Wadsley. “Verma modules for Iwasawa algebras are faithful”. In: *Münster Journal of Mathematics* 7 (2014), pp. 5–26. ISSN: 1867-5778. urn: urn:nbn:de:hbz:6-22359595785, Thm. C.

¹⁰⁰Ibid., Thm. A.

¹⁰¹Fu, “Sharp bounds for multiplicities of Bianchi modular forms”, Thm. 3.2.

(b) For any three tuples $a = (a_1, \dots, a_k)$, $b = (b_1, \dots, b_k)$ and $c = (c_1, \dots, c_k)$ in $\mathbb{Z}_{\geq 0}^k$, define the element

$$F^a H^b E^c = F_1^{a_1} \dots F_k^{a_k} H_1^{b_1} \dots H_k^{b_k} E_1^{c_1} \dots E_k^{c_k}$$

where each one of the F_i, H_i, E_i form a Chevalley basis of one $\mathfrak{sl}_2(\mathbb{Z}_p)$ factor of $\mathfrak{g}_{\mathbb{Z}_p}$. Then, the monomials $F^a H^b E^c$ form a basis of the free modules $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ and $U(\mathfrak{g}_{\mathbb{Z}_p})$.

(c) For each $1 \leq i \leq k$, let

$$\Delta_i = H_i^2 + 2H_i + 4F_i E_i.$$

Then, the center of $U(\mathfrak{g}_{\mathbb{Z}_p})$ is the polynomial algebra $\mathbb{Q}_p[\Delta_1, \dots, \Delta_k]$.

(d) Let $\omega_1, \dots, \omega_k$ be the fundamental weights of each $\mathfrak{sl}_2$ factor of $\mathfrak{g}$, and write any element λ of $P(\mathfrak{g})$ as

$$\lambda = \lambda_1 \omega_1 + \dots + \lambda_k \omega_k$$

where each λ_i is a non-negative integer. Then, if $\chi_\lambda: Z(U(\mathfrak{g}_{\mathbb{Z}_p})) \rightarrow \mathbb{Q}_p$ is the central character associated to the irreducible representation $V_\lambda(\mathbb{Q}_p)$, we have $\chi_\lambda(\Delta_i) = \lambda_i(\lambda_i + 2)$ for all $1 \leq i \leq k$.

(e) Let $s = (s_1, \dots, s_k)$ be any sequence of k elements in $\mathbb{Z}_p$, and let $I_{s,0}$ be the ideal of $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ generated by the elements $\Delta_i - s_i$ for all $1 \leq i \leq k$. Then, the quotient ring

$$U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} = U(\mathfrak{g}_{\mathbb{Z}_p})_0 / I_{s,0}$$

has no zero divisors. The same is true for

$$U(\mathfrak{g}_{\mathbb{Z}_p})_s = U(\mathfrak{g}_{\mathbb{Z}_p}) / I_s$$

where I_s is the ideal generated by the central elements $\Delta_i - s_i$ for all $1 \leq i \leq k$.

(f) For any $s \in \mathbb{Z}_p^k$, the monomials

$$\{F^a H^b E^c : a, c \in \mathbb{Z}_{\geq 0}^k, b \in \{0, 1\}^k\}$$

form a $\mathbb{Z}_p$ -basis of the free $\mathbb{Z}_p$ -module $U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}$ as well as a $\mathbb{Q}_p$ -basis of $U(\mathfrak{g}_{\mathbb{Z}_p})_s$.

(g) The rings $U(\mathfrak{g}_{\mathbb{Z}_p})_0 / pU(\mathfrak{g}_{\mathbb{Z}_p})_0$ and $U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} / pU(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}$ do not have zero divisors.

Proof. Point (a) is standard for universal enveloping algebras after we identify $U(\mathfrak{g}_{\mathbb{Z}_p})$ with the universal enveloping $\mathbb{Q}_p$ -algebra of $\mathfrak{g}_{\mathbb{Q}_p}$.¹⁰² Point (b) is a consequence of the Poincaré-Birkhoff-Witt Theorem 1.4.6. It suffices to prove (c) for $U(\mathfrak{g}_{\mathbb{Z}_p})$ since each Δ_i belongs to $U(\mathfrak{g}_{\mathbb{Z}_p})_0$. A straightforward computation shows that

$$[E, H^2 + 2H + 4FE] = [F, H^2 + 2H + 4FE] = [H, H^2 + 2H + 4FE] = 0.$$

¹⁰²Bourbaki, *Lie Groups and Lie Algebras*, Prop. I.2.2.

Hence, $H^2 + 2H + 4FE$ commutes with all generators of $U(\mathfrak{g}_{\mathbb{Z}_p})_0$, and hence it is central. By (a), to show that the center of $U(\mathfrak{g}_{\mathbb{Z}_p})$ is isomorphic to $\mathbb{Q}_p[\Delta_1, \dots, \Delta_k]$, it suffices to show that the center of $U(\mathfrak{sl}_2(\mathbb{Z}_p))$ is isomorphic to $\mathbb{Q}_p[H^2 + 2H + 4FE]$. The Harish-Chandra isomorphism¹⁰³ identifies the center of $U(\mathfrak{sl}_2(\mathbb{Z}_p))$ with the graded algebra $\mathbb{Q}_p[H]^W$ of polynomials in H invariant under the Weyl group $W = \langle H \mapsto -H \rangle$ of $\mathfrak{sl}_2(\mathbb{Z}_p)$. Since $\mathbb{Q}_p[H]^W = \mathbb{Q}_p[H^2]$, we see that $Z(U(\mathfrak{sl}_2(\mathbb{Z}_p)))$ is a polynomial algebra in one variable generated by any element of degree 2, which we can take to be $H^2 + 2H + 4FE$.

For (d), let $v \in V_\lambda(\mathbb{Q}_p)$ be a highest weight vector of weight λ . Then $E_i v = 0$ and $H_i v = \lambda_i v$, so

$$\Delta_i v = (H_i^2 + 2H_i + 4F_i E_i) v = (\lambda_i^2 + 2\lambda_i) v = \lambda_i(\lambda_i + 2) v.$$

Since Δ_i is central, it acts by a scalar on the irreducible module $V_\lambda(\mathbb{Q}_p)$. This scalar is precisely $\chi_\lambda(\Delta_i)$, so $\chi_\lambda(\Delta_i) = \lambda_i(\lambda_i + 2)$ for all $1 \leq i \leq k$. For (e), note that by (a) we have isomorphisms:

$$\begin{aligned} U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} &\simeq \bigotimes_{\mathbb{Z}_p}^k U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 / (\Delta_i - s_i) U(\mathfrak{sl}_2(\mathbb{Z}_p))_0, \\ U(\mathfrak{g}_{\mathbb{Z}_p})_s &\simeq \bigotimes_{\mathbb{Q}_p}^k U(\mathfrak{sl}_2(\mathbb{Z}_p)) / (\Delta_i - s_i) U(\mathfrak{sl}_2(\mathbb{Z}_p)). \end{aligned}$$

Hence, it also suffices to prove it for a single $\mathfrak{sl}_2(\mathbb{Z}_p)$ factor. Also, since

$$U(\mathfrak{g}_{\mathbb{Z}_p})_s \simeq U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} \otimes_{\mathbb{Z}_p} \mathbb{Q}_p,$$

it suffices to prove it for the $\mathbb{Z}_p$ -algebra.

Hence, we fix $s \in \mathbb{Z}_p$ and let I be the ideal of $U(\mathfrak{sl}_2(\mathbb{Z}_p))_0$ generated by $\Delta - s = H^2 + 2H + 4FE - s$. By the PBW Theorem 1.4.6, $\text{Gr } U(\mathfrak{sl}_2(\mathbb{Z}_p))_0$ is isomorphic as a graded ring to the polynomial algebra $\mathbb{Z}_p[\overline{F}, \overline{H}, \overline{E}]$. In particular, it has no zero divisors, and so

$$I \cap F_i U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 = (\Delta - s) \cdot F_{i-2} U(\mathfrak{sl}_2(\mathbb{Z}_p))_0.$$

Hence, under the induced filtration, $\text{Gr } I$ is a principal ideal generated by the element $\overline{\Delta - s} = \overline{H}^2 + 4\overline{F}\overline{E}$. This is an irreducible polynomial, and so the quotient

$$\mathbb{Z}_p[\overline{F}, \overline{H}, \overline{E}] / (\overline{H}^2 + 4\overline{F}\overline{E}) \simeq \text{Gr}(U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 / I)$$

has no zero divisors. Therefore, neither does $U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 / I$, showing (e).

For point (f), using the previous isomorphisms, we see it also suffices to show that the set

$$\{\overline{F}^a \overline{H}^b \overline{E}^c : a, c \in \mathbb{Z}_{\geq 0}, b \in \{0, 1\}\}$$

¹⁰³Bourbaki, *Lie groups and Lie algebras. Chapters 7-9*, Thm. VIII.5.2.

is a $\mathbb{Z}_p$ -basis of $\mathbb{Z}_p[\overline{F}, \overline{H}, \overline{E}]/(\overline{H}^2 + 4\overline{F}\overline{E})$. If we denote by S the polynomial ring $\mathbb{Z}_p[\overline{E}, \overline{F}]$ and we see $\overline{H}^2 + 4\overline{F}\overline{E}$ as a monic polynomial in $S[\overline{H}]$, the division algorithm shows that $S[\overline{H}]/(\overline{H}^2 + 4\overline{F}\overline{E})$ is a free S -module on the basis $\{1, \overline{H}\}$, which is precisely what we wanted to prove. By tensoring with $\mathbb{Q}_p$, we get the desired basis for $U(\mathfrak{g}_{\mathbb{Z}_p})_s$.

Finally, for (g) we can identify $U(\mathfrak{g}_{\mathbb{Z}_p})_0/pU(\mathfrak{g}_{\mathbb{Z}_p})_0$ with the universal enveloping $\mathbb{F}_p$ -algebra $U(\mathfrak{g}_{\mathbb{F}_p})$ of the $\mathbb{F}_p$ -Lie algebra $\mathfrak{g}_{\mathbb{F}_p} = \mathfrak{g}_{\mathbb{Z}_p}/p\mathfrak{g}_{\mathbb{Z}_p}$. Under the filtration by the total degree, its graded algebra $\text{Gr } U(\mathfrak{g}_{\mathbb{F}_p})$ is isomorphic to the symmetric $\mathbb{F}_p$ -algebra on $\mathfrak{g}_{\mathbb{F}_p}$, and thus it doesn't have zero divisors. Once more, the isomorphism

$$U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}/pU(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} \simeq \bigotimes_{\mathbb{Z}_p}^k U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 / \left((\Delta_i - s_i)U(\mathfrak{sl}_2(\mathbb{Z}_p))_0 + pU(\mathfrak{sl}_2(\mathbb{Z}_p))_0 \right)$$

shows that it suffices to prove the final part of (g) for a single $\mathfrak{sl}_2$ factor.

We again first reduce modulo p and identify $U(\mathfrak{sl}_2(\mathbb{Z}_p))_0/pU(\mathfrak{sl}_2(\mathbb{Z}_p))_0$ with $U(\mathfrak{sl}_2(\mathbb{F}_p))$. To show that the quotient by $H^2 + 2H + 4FE - s$ also doesn't have zero divisors, we pass onto the graded ring $\text{Gr } U(\mathfrak{sl}_2(\mathbb{F}_p)) \simeq \mathbb{F}_p[\overline{F}, \overline{H}, \overline{E}]$ to get that

$$\text{Gr}((H^2 + 2H + 4FE - s)U(\mathfrak{sl}_2(\mathbb{F}_p))) = (\overline{H}^2 + 4\overline{F}\overline{E})$$

is a prime ideal. Hence, neither

$$\text{Gr}(U(\mathfrak{sl}_2(\mathbb{F}_p)))/(H^2 + 2H + 4FE - s)U(\mathfrak{sl}_2(\mathbb{F}_p))) \simeq \mathbb{F}_p[\overline{F}, \overline{H}, \overline{E}]/(\overline{H}^2 + 4\overline{F}\overline{E})$$

nor $U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}$ have zero divisors. □

We now endow $U(\mathfrak{g}_{\mathbb{Z}_p})_0$ and $U(\mathfrak{g}_{\mathbb{Z}_p})$ with their ascending $\mathbb{Z}^k$ filtrations $F_i U(\mathfrak{g}_{\mathbb{Z}_p})_0$ and $F_i U(\mathfrak{g}_{\mathbb{Z}_p})$ with $i = (i_1, \dots, i_k)$ given by the total degree in each $\mathfrak{sl}_2$ factor, and each quotient ring $U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}$ and $U(\mathfrak{g}_{\mathbb{Z}_p})_s$ with their quotient filtrations.

Definition 5.4.3. Let $\lambda = \lambda_1\omega_1 + \dots + \lambda_k\omega_k$ be the highest weight of an irreducible representation V_λ and define $s = (\lambda_1(\lambda_1 + 2), \dots, \lambda_k(\lambda_k + 2))$. A δ -invariant $\delta(\lambda)$ for λ is a tuple of integers $n \in \mathbb{Z}_{\geq 0}^k$ for which the map

$$F_n U(\mathfrak{g}_{\mathbb{Z}_p})_s \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$$

induced by the action is an embedding.

Observe that the action map $U(\mathfrak{g}_{\mathbb{Z}_p}) \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ factors through $U(\mathfrak{g}_{\mathbb{Z}_p})_s$ by part (d) of Proposition 5.4.2. Any δ -invariant can be used to give lower bounds on the size of the image for the multiplication by $x \in U(\mathfrak{g}_{\mathbb{Z}_p})$ map on each V_λ .

Lemma 5.4.4. *Let x be an element of $F_C \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})$ for some degree $C = (C_1, \dots, C_k)$, and $\lambda = \lambda_1 \omega_1 + \dots + \lambda_k \omega_k$. If $x \notin I_s$ where $s = (\lambda_1(\lambda_1 + 2), \dots, \lambda_k(\lambda_k + 2))$, then for any δ -invariant $\delta(\lambda)$ we have*

$$\frac{\dim_{\mathbb{Q}_p} xV_\lambda(\mathbb{Q}_p)}{\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p)} \geq \frac{\dim_{\mathbb{Q}_p} F_{\delta(\lambda)-C} \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})_s}{(\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p))^2}.$$

Proof. If $x \notin I_s$, then it represents a non-zero element of $F_C \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$. As this $\mathbb{Q}_p$ -algebra has no zero divisors by part (e) of Proposition 5.4.2, left multiplication by x induces an injective map

$$F_{\delta(\lambda)-C} \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})_s \xrightarrow{x} F_{\delta(\lambda)} \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})_s,$$

which by the definition of the δ -invariant implies that

$$\dim_{\mathbb{Q}_p} x \operatorname{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) \geq \dim_{\mathbb{Q}_p} F_{\delta(\lambda)-C} \mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})_s.$$

Now, as left $\mathbf{U}(\mathfrak{g}_{\mathbb{Z}_p})$ -modules, we have

$$\operatorname{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) \simeq V_\lambda(\mathbb{Q}_p) \otimes_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p)^* \simeq \bigoplus_{\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p)} V_\lambda(\mathbb{Q}_p).$$

This shows the equality

$$\dim_{\mathbb{Q}_p} x \operatorname{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) = \dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p) \cdot \dim_{\mathbb{Q}_p} xV_\lambda(\mathbb{Q}_p).$$

Dividing both sides by $(\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p))^2$ and applying the first inequality yields the statement of the lemma. $\square$

In particular, if $\delta(\lambda)$ is large compared to C , the right-hand side is bounded away from zero, so x acts with large image on $V_\lambda(\mathbb{Q}_p)$. This is the basic mechanism by which a δ -invariant influences the rank functions attached to the family $(V_\lambda)_\lambda$.

Example 5.4.5. For $\mathfrak{g}(\mathbb{Z}_p) = \prod_{i=1}^k \mathfrak{sl}_2(\mathbb{Z}_p)$ and $\lambda = \lambda_1 \omega_1 + \dots + \lambda_k \omega_k$, we claim that¹⁰⁴

$$\delta(\lambda) = (\lambda_1, \dots, \lambda_k)$$

is a δ -invariant for λ . Since

$$V_\lambda(\mathbb{Q}_p) \simeq \bigotimes_{\mathbb{Q}_p}^k \operatorname{Sym}^{\lambda_i}(\mathbb{Q}_p),$$

we see that

$$\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p) = \prod_{i=1}^k (\lambda_i + 1),$$

¹⁰⁴Fu, “Sharp bounds for multiplicities of Bianchi modular forms”, Prop. 3.4.

and therefore

$$\dim_{\mathbb{Q}_p} \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) = (\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p))^2 = \prod_{i=1}^k (\lambda_i + 1)^2.$$

Moreover, each weight subspace of $V_\lambda(\mathbb{Q}_p)$ has dimension at most one.

Let $A = F_{(\lambda_1, \dots, \lambda_k)} \text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$. We have that A is the $\mathbb{Q}_p$ -span of the monomials

$$\{F^a H^b E^c : a, c \in \mathbb{Z}_{\geq 0}^k, b \in \{0, 1\}^k, a_i + b_i + c_i \leq \lambda_i\}$$

inside $\text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$ where $s = (\lambda_1(\lambda_1 + 2), \dots, \lambda_k(\lambda_k + 2))$. We argue that $A \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ is surjective. Since $V_\lambda(\mathbb{Q}_p)$ is an irreducible $\text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$ -module, the map $\text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ is already surjective,¹⁰⁵ so we are reduced to proving that the image of any element of $\text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$ in $\text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ coincides with the image of some element of A . By part (f) of Proposition 5.4.2, we only need to consider monomials $F^a H^b E^c$ with $b \in \{0, 1\}^k$.

Since p is odd, inside $\text{U}(\mathfrak{g}_{\mathbb{Z}_p})_s$ we have

$$\begin{aligned} F_i^n E_i^n &= F_i^{n-1} F_i E_i E_i^{n-1} \\ &= F_i^{n-1} \left(\frac{s_i}{4} - \frac{H_i^2}{4} - \frac{H_i}{2} \right) E_i^{n-1} \\ &= F_i^{n-1} E_i^{n-1} \left(\frac{s_i}{4} - \frac{H_i^2}{4} - \frac{H_i}{2} \right) - (n-1) (F_i^{n-1} E_i^{n-1} - F_i^{n-1} E_i^{n-1} H_i - (n-1) F_i^{n-1} E_i^{n-1}) \end{aligned}$$

for all $1 \leq i \leq k$. By induction on n , we obtain that $F_i^n E_i^n$ is contained in the $\mathbb{Z}_p$ -subalgebra generated by H_i for all n . Analogously, we have

$$F_i^n H_i E_i^n = F_i^n E_i^n H_i + 2n F_i^n E_i^n,$$

and so $F_i^n H_i E_i^n$ is also contained in said subalgebra for all $1 \leq i \leq k$.

We can always rewrite

$$X = F^a H^b E^c = F^{a'} F^x H^b E^x E^{c'}$$

with $a', c', x \in \mathbb{Z}_{\geq 0}^k$ satisfying $a'_i c'_i = 0$ and $a'_i + 2x_i + b_i + c'_i = a_i + b_i + c_i$. Define the following set of weights:

$$\mathcal{M} = \{\mu = \mu_1 \omega_1 + \dots + \mu_k \omega_k \in \mathfrak{h}^* : (V_\lambda(\mathbb{Q}_p))_\mu \neq 0, -\lambda_i + 2a'_i \leq \mu_i \leq \lambda_i - 2c'_i\},$$

¹⁰⁵The module $V_\lambda(\mathbb{Q}_p)$ remains irreducible after passing to the algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ since it is a highest weight module. The result now follows from Thm. 3.7.5 of T. Y. Lam, *A First Course in Noncommutative Rings*, 2nd ed. Graduate Texts in Mathematics 131. Springer-Verlag New York, 2001. ISBN: 978-0-387-95325-0

and the associated $\mathbb{Q}_p$ -vector subspace

$$W = \bigoplus_{\mu \in M} (V_\lambda(\mathbb{Q}_p))_\mu.$$

We have that $XV_\lambda(\mathbb{Q}_p) \leq W$: by our construction of M , X annihilates any weight vector outside of W .

We know that $F^x H^b E^x$ can be expressed as a polynomial in the H_i . Since each H_i is diagonalizable and has at most $\lambda_i + 1 - (a'_i + c'_i)$ eigenvalues in W , all of which are integral, its minimal polynomial is integral and has degree at most $\lambda_i + 1 - (a'_i + c'_i)$. Hence, we can replace $F^x H^b E^x$ with an polynomial expression Y in the $H_1, \dots, H_k$ with $\mathbb{Z}_p$ -coefficients such that the maximal degree in H_i is at most $\lambda_i - (a'_i + c'_i)$. This shows that as operators in $V_\lambda(\mathbb{Q}_p)$ we have that X is equivalent to $F^{a'} Y E^{c'} \in A$.

Once again, part (f) of Proposition 5.4.2 guarantees that those monomials are $\mathbb{Q}_p$ -linearly independent, and so

$$\dim_{\mathbb{Q}_p} A = \prod_{i=1}^k \left(\binom{\lambda_i + 2}{2} + \binom{\lambda_i + 1}{2} \right) = \prod_{i=1}^k (\lambda_i + 1)^2.$$

Since the map $A \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ is surjective, by comparing dimensions we conclude it is a bijection. ■

Remark 5.4.6. In the previous Example 5.4.5, the argument also shows that the image of the $\mathbb{Z}_p$ -span A_0 of

$$\{F^a H^b E^c : a, c \in \mathbb{Z}_{\geq 0}^k, b \in \{0, 1\}^k, a_i + b_i + c_i \leq \lambda_i\}$$

in $\text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ coincides with the full image of $U(\mathfrak{g}_{\mathbb{Z}_p})_0$. ■

Example 5.4.7. For $\mathfrak{sl}_3(\mathbb{Z}_p)$ with its two fundamental weights ω_1 and ω_2 from Example 5.1.10, if we take $\lambda = k\omega_1$, we now claim that $\partial(\lambda) \leq 2$ independent of k . To do so, it suffices to produce an element $X \in U(\mathfrak{sl}_3(\mathbb{Z}_p))$ of degree three that is not contained in the ideal generated by the kernel of $\chi_\lambda : Z(U(\mathfrak{sl}_3(\mathbb{Z}_p))) \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p))$ but annihilates $V_\lambda(\mathbb{Q}_p)$ for all such λ . Keeping the notation of the Example 5.1.10, the element is as follows:

$$X = -E_{12}^2 E_{23} + E_{12} E_{13} H_2 + E_{13}^2 F_{32}.$$

The representation $V_\lambda(\mathbb{Q}_p)$ is given by the action of $\mathfrak{sl}_3(\mathbb{Z}_p)$ on the space $V_k \leq \mathbb{Q}_p[x, y, z]$ of homogeneous polynomials of degree k . This action is described as follows: for any matrix A with columns a_1, a_2 and a_3 in $\mathfrak{sl}_3(\mathbb{Z}_p)$, we define

$$(Af)(x, y, z) = a_1^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial x}(x, y, z) + a_2^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial y}(x, y, z) + a_3^T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \frac{\partial f}{\partial z}(x, y, z).$$

Under this action, the generators appearing in X are mapped to the following differential operators:

$$\begin{aligned} E_{12} &\mapsto x \frac{\partial}{\partial y}, & E_{23} &\mapsto y \frac{\partial}{\partial z}, & E_{13} &\mapsto x \frac{\partial}{\partial z}, \\ H_2 &\mapsto y \frac{\partial}{\partial y} - z \frac{\partial}{\partial z}, & F_{32} &\mapsto z \frac{\partial}{\partial y}. \end{aligned}$$

If we apply it to a monomial $x^a y^b z^c$, we get:

$$\begin{aligned} -E_{12}^2 E_{23}(x^a y^b z^c) &= -b(b+1)c x^{a+2} y^{b-1} z^{c-1}, \\ E_{12} E_{13} H_2(x^a y^b z^c) &= bc(b-c) x^{a+2} y^{b-1} z^{c-1}, \\ E_{13}^2 F_{32}(x^a y^b z^c) &= b(c+1)c x^{a+2} y^{b-1} z^{c-1}. \end{aligned}$$

Summing it up:

$$X(x^a y^b z^c) = (-b(b+1)c + bc(b-c) + b(c+1)c) x^{a+2} y^{b-1} z^{c-1} = 0.$$

To show now that X is not in the ideal I_λ generated by $\ker(\chi_\lambda)$, we use another standard fact from the theory of Lie algebras: we can identify the graded ring $\text{Gr } \mathcal{U}(\mathfrak{sl}_3(\mathbb{Z}_p)) \simeq S(\mathfrak{sl}_3(\mathbb{Q}_p))$ with the ring of $\mathbb{Q}_p$ -polynomial functions on $\mathfrak{sl}_3(\mathbb{Q}_p)$ if we associate to each basis element of $\mathfrak{sl}_3(\mathbb{Q}_p)$ the corresponding linear functional on the dual basis. Under this identification,¹⁰⁶ any function in $\text{Gr } I_\lambda$ is identically zero on any nilpotent element of $\mathfrak{sl}_3(\mathbb{Q}_p)$. However, if we consider the matrix

$$Y = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = E_{12} + E_{23},$$

we have

$$\overline{X}(Y) = -1 \neq 0.$$

This shows that Lemma 5.4.4 does not give any information about the asymptotic behaviour of the rank functions rk_λ along this particular family of representations. However, this example also does not contradict the Conjecture 5.3.4, since $\min \lambda = 0$ also stays fixed. ■

With a good δ -invariant for $\prod_{i=1}^k \mathfrak{sl}_2$ in hand, we are able to prove the final theorem of this chapter.

Proof of Theorem 5.4.1. Let p be a prime such that Theorem 5.3.6 holds for $G = \prod_{i=1}^k \text{SL}_2$, and x be a non-zero element of $\mathbb{Z}_p[[G_1]]$ seen as an element of $\widehat{\mathcal{U}(\mathfrak{g}_{\mathbb{Z}_p})}$. We must show that

$$1 - \text{rk}_\lambda(x) = 1 - \frac{\dim_{\mathbb{Q}_p} xV_\lambda(\mathbb{Q}_p)}{\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p)} = O\left(\frac{1}{\min \lambda}\right).$$

¹⁰⁶ Bourbaki, *Lie groups and Lie algebras. Chapters 7-9*, Prop. VIII.8.4 and VIII.8.5.

After multiplying by a suitable power of p , we can assume that $x \in \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0 \setminus p\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$. By Lemma 5.3.2, we can write x uniquely as a convergent sum

$$x = \sum_{a,b,c \in \mathbb{Z}_{\geq 0}^k} \alpha_{a,b,c} F^a H^b E^c$$

with $\alpha_{a,b,c}$ in $\mathbb{Z}_p$ such that for every $\epsilon > 0$ we have $|\alpha_{a,b,c}| < \epsilon$ for almost all a, b, c .

Let $s = (s_1, \dots, s_k) \in \mathbb{Z}_p^k$ and consider by part (a) of Theorem 5.3.6 and part (c) of Proposition 5.4.2 the central character $\chi_s: \mathbb{Z}_p \llbracket \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} \rrbracket \rightarrow \mathbb{Q}_p$ defined by $\chi_s(\Delta_i) = s_i$, and let I_s be the ideal of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}$ generated by $\ker(\chi_s)$. By part (b) of Theorem 5.3.6, the element x is not in I_s for any $s \in \mathbb{Z}_p^k$. Hence, we have a well defined function $\iota: \mathbb{Z}_p^k \rightarrow \mathbb{Z}_{\geq 1}$ that associates to each $s \in \mathbb{Z}_p^k$ the smallest integer $\iota(s)$ such that the image of x in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_{s,0} / p^{\iota(s)} \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_s$ is non-zero.

Since

$$\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_{s,0} / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_s \simeq (\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}) / \left(\sum_{i=1}^k (\Delta_i - s_i) (\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})} / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}) \right)$$

holds for every i , we see that the function ι is *locally constant*: if $\iota(s) = i$, then $\iota(s') = i$ for any $s' \in \mathbb{Z}_p^k$ satisfying $s \equiv s' \pmod{p^i}$. In particular, ι is a continuous function from the compact space $\mathbb{Z}_p^k$ to the discrete space $\mathbb{Z}_{\geq 1}$, so the image is bounded. This implies that there exists an $i \geq 1$ such that the image of x is non-zero in $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_{s,0} / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_{s,0}$ for all $s \in \mathbb{Z}_p^k$.

Let $\delta(\lambda) = (\lambda_1, \dots, \lambda_k)$ and C be such that x in $F_C(\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0 / p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0)$. We now claim that the composition of the maps

$$F_{\delta(\lambda)-C} U(\mathfrak{g}_{\mathbb{Z}_p})_s \xrightarrow{x} x F_{\delta(\lambda)-C} U(\mathfrak{g}_{\mathbb{Z}_p})_s \rightarrow \text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Q}_p)) \quad (5.7)$$

is injective as in Lemma 5.4.4. Let E_λ denote the image of $\widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_0$ inside $\text{End}_{\mathbb{Q}_p}(V_\lambda(\mathbb{Z}_p))$ and for each s write

$$x \equiv p^{\kappa(s)} x_s \pmod{p^i \widehat{U(\mathfrak{g}_{\mathbb{Z}_p})}_{s,0}}$$

with $x_s \in F_C U(\mathfrak{g}_{\mathbb{Z}_p})_0 \setminus p U(\mathfrak{g}_{\mathbb{Z}_p})$, so that $0 \leq \kappa(s) < i$. On one hand, if the composition in 5.7 were not injective, we would be able to find some $y \in F_{\delta(\lambda)-C} U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} \setminus p U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0}$ such that xy gets mapped to the zero endomorphism in E_λ . On the other hand, by part (g) of Proposition 5.4.2 the injectivity of

$$F_{\delta(\lambda)-C} \left(U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} / p U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} \right) \xrightarrow{x_s} F_{\delta(\lambda)} \left(U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} / p U(\mathfrak{g}_{\mathbb{Z}_p})_{s,0} \right) \rightarrow E_\lambda / p E_\lambda$$

shows that the image of $xy = p^{x(s)}x_y y$ is non-zero in $E_\lambda/p^i E_\lambda$, which is a contradiction. Arguing as in Lemma 5.4.4, we now obtain that

$$\frac{\dim_{\mathbb{Q}_p} xV_\lambda(\mathbb{Q}_p)}{\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p)} \geq \frac{\dim_{\mathbb{Q}_p} F_{\delta(\lambda)-C}U(\mathfrak{g}_{\mathbb{Z}_p})_s}{(\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p))^2}. \quad (5.8)$$

As in Example 5.4.5, we have that $\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p) = \prod_{i=1}^k (\lambda_i + 1)$. Moreover, it follows from part (f) of Proposition 5.4.2 as in Example 5.4.5 that

$$\dim_{\mathbb{Q}_p} F_{\delta(\lambda)-C}U(\mathfrak{g}_{\mathbb{Z}_p})_s = \prod_{i=1}^k (\lambda_i - C_i + 1)^{2k}.$$

Hence:

$$\frac{\dim_{\mathbb{Q}_p} F_{\delta(\lambda)}U(\mathfrak{g}_{\mathbb{Z}_p})_s}{(\dim_{\mathbb{Q}_p} V_\lambda(\mathbb{Q}_p))^2} = \prod_{i=1}^k \left(\frac{\lambda_i - C_i + 1}{\lambda_i + 1} \right)^2$$

and therefore:

$$\begin{aligned} 1 - \text{rk}_\lambda(x) &\leq 1 - \prod_{i=1}^k \left(\frac{\lambda_i - C_i + 1}{\lambda_i + 1} \right)^2 \\ &= \frac{\prod_{i=1}^k (\lambda_i + 1)^2 - \prod_{i=1}^k (\lambda_i - C_i + 1)^2}{\prod_{i=1}^k (\lambda_i + 1)^2} \\ &\leq \frac{\prod_{i=1}^k (\lambda_i + 1)^2 - \prod_{i=1}^k (\lambda_i - C_i + 1)^2}{\lambda_1^2 \cdots \lambda_k^2} \end{aligned}$$

If we write the numerator as a polynomial in the λ_i

$$\prod_{i=1}^k (\lambda_i + 1)^2 - \prod_{i=1}^k (\lambda_i - C_i + 1)^2 = \sum_{d_1, \dots, d_k \in \{0,1,2\}} \beta_{d_1, \dots, d_k} \lambda_1^{d_1} \cdots \lambda_k^{d_k},$$

then $\beta_{2, \dots, 2} = 0$. Hence:

$$\begin{aligned} 1 - \text{rk}_\lambda(x) &\leq \frac{\prod_{i=1}^k (\lambda_i + 1)^2 - \prod_{i=1}^k (\lambda_i - C_i + 1)^2}{\lambda_1^2 \cdots \lambda_k^2} \\ &= \sum_{\substack{d_1, \dots, d_k \in \{0,1,2\}^k \\ (d_1, \dots, d_k) \neq (2, \dots, 2)}} \beta_{d_1, \dots, d_k} \frac{\lambda_1^{d_1} \cdots \lambda_k^{d_k}}{\lambda_1^2 \cdots \lambda_k^2} \\ &= O\left(\frac{1}{\min \lambda}\right) \end{aligned}$$

as desired. □

Corollary 5.4.8. *Let $G = \prod_{i=1}^k \mathrm{SL}_2$, p and odd prime and G_1 be the first principal congruence subgroup of $G(\mathbb{Z}_p)$. For almost all primes p , we have for all matrices A over $\mathbb{Z}_p[[G_1]]$ that*

$$|\mathrm{rk}_{\mathcal{Q}}(A) - \mathrm{rk}_{\lambda}(A)| = O\left(\frac{1}{\min \lambda}\right),$$

where $\mathcal{Q}$ is the Ore division ring of fractions of $\mathbb{Z}_p[[G_1]]$. In particular,

$$\lim_{\min \lambda \rightarrow \infty} \mathrm{rk}_{\lambda} = \mathrm{rk}_{\mathcal{Q}}.$$

Proof. This follows from Proposition 5.2.4 and Theorem 5.4.I. □

Chapter 6

Applications to abstract groups

That's a well known example of a group.

D.S. on $\mathbb{Z}$.

Without a doubt, one of the main goals of completing a PhD is to get a job. As a PhD candidate working on group theory in the early 21st century, this means that after spending many pages completing, localizing and filtering everything in sight, one should eventually come back to honest abstract groups and ask what all the effort was for.¹ This final chapter is where we try to fulfill that promise.

Many of the questions that motivated this thesis arise from attempts to understand asymptotic or analytic invariants of groups via finite quotients. On the one hand, ℓ^2 -Betti numbers and von Neumann dimensions, born in the work of Michael Atiyah, relate cohomological information of a group or manifold with invariants of an associated operator algebra. On the other hand, approximation theorems such as Lück's classical result² for residually finite groups seek to recover these analytic quantities as limits of normalized dimensions of homologies over finite index subgroups. A recurring theme is that behind every such approximation statement there lurks a suitable notion of "rank". The techniques developed in the previous chapters were deliberately designed with this circle of ideas in mind. Here we apply them in four different directions, all of which concern abstract groups sitting inside richer topological objects.

Taken together, these four sections illustrate the guiding philosophy of the thesis: once an appropriate rank function has been constructed on a completed group algebra, a wide range of questions about abstract groups can be attacked by comparing that rank with more classical cohomological invariants along finite quotients or finite-dimensional representations.

¹In other words, most committees don't care about $R[\widehat{G}]$ and would just be interested in what can it say about G .

²See Theorem 3.1.16

6.1 Lück approximation in positive characteristic

Let Γ be a group and let K be a field. For every matrix $A \in \text{Mat}_{m \times n}(K[\Gamma])$ and every normal subgroup N of Γ of finite index let us define

$$\phi_{\Gamma/N}^A: K[\Gamma/N]^n \rightarrow K[\Gamma/N]^m$$

as the map induced from left multiplication by A . This allows us to define a Sylvester matrix rank function $\text{rk}_{\Gamma/N}$ on $K[\Gamma]$ by means of

$$\text{rk}_{\Gamma/N}(A) = \frac{\dim_K \text{im}(\phi_{\Gamma/N}^A)}{|\Gamma: N|} = n - \frac{\dim_K \ker \phi_{\Gamma/N}^A}{|\Gamma: N|}.$$

In the language of Section 3.1, $\text{rk}_{\Gamma/N}$ is the pullback of the normalized rank function on $K[\Gamma/N]$ along the projection $K[\Gamma] \rightarrow K[\Gamma/N]$. The following conjecture has been widely studied throughout the last decades.³

Conjecture 6.1.1 (The Lück approximation conjecture). *Let Γ be a group, K be a field and $\Gamma > N_1 > N_2 > \dots$ be a descending chain of normal subgroups of Γ of finite index with trivial intersection. Then, for any matrix A over $K[\Gamma]$, the following holds:*

- (1) *The sequence $\text{rk}_{\Gamma/N_i}(A)$ converges.*
- (2) *The limit $\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/N_i}(A)$ does not depend on the chain $\Gamma > N_1 > N_2 > \dots$.*
- (3) *If moreover Γ is locally indicable⁴, then there exists an universal epic division $K[\Gamma]$ -ring $\mathcal{Q}$ and*

$$\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/N_i}(A) = \text{rk}_{\mathcal{Q}}(A).$$

If K is of characteristic 0, then parts (1) and (2) of Conjecture 6.1.1 are known to be true (see Theorem 3.1.16). For part (3) in characteristic 0, it is known⁵ that there exists an embedding $K[\Gamma] \rightarrow \mathcal{Q}$ into a division ring $\mathcal{Q}$ such that $\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/N_i}(A) = \text{rk}_{\mathcal{Q}}(A)$, but we still do not know whether $\mathcal{Q}$ is universal in general.⁶ If K has characteristic $p > 0$, then parts (1), (2) and (3) are only known when Γ is amenable.⁷ Using the results of Chapter 4, we can extend this to the class of abstract subgroups of free-by- $\mathbb{Z}_p$ pro- p groups when K is finite of characteristic p . We start with a lemma.

³Jaikin-Zapirain, “L2-Betti Numbers and Their Analogues in Positive Characteristic”.

⁴Meaning that any non-trivial finitely generated subgroup surjects onto $\mathbb{Z}$

⁵Andrei Jaikin-Zapirain and Diego López-Álvarez. “The strong Atiyah and Lück approximation conjectures for one-relator groups”. In: *Mathematische Annalen* 376.3–4 (Nov. 2019), pp. 1741–1793. ISSN: 1432-1807. DOI: 10.1007/s00208-019-01926-0.

⁶Jaikin-Zapirain, “The universality of Hughes-free division rings”.

⁷Jaikin-Zapirain, “L2-Betti Numbers and Their Analogues in Positive Characteristic”, Thm. 7.1.

Lemma 6.1.2. *If G is a free-by- $\mathbb{Z}_p$ pro- p group and Γ is a finitely generated abstract subgroup of G , then Γ is locally indicable.*

Proof. Let Γ' be a finitely generated subgroup of Γ and $H = \overline{\Gamma'}$ be the closure of Γ' in G . Then, as we have observed in Example 4.4.3, H is also free-by- $\mathbb{Z}_p$. In particular, there exists a surjection $\varphi: H \rightarrow \mathbb{Z}_p$. If $\varphi(\Gamma') = \{0\}$, then by the density of Γ' in H we would have that φ is the trivial map, contradicting surjectivity. Hence, $\varphi(\Gamma')$ is a non-trivial finitely generated abelian subgroup of $\mathbb{Z}_p$. Since $\mathbb{Z}_p$ is torsion-free, the same applies to $\varphi(\Gamma')$, which is readily seen to admit a surjection onto $\mathbb{Z}$. $\square$

An embedding $K[\Gamma] \rightarrow \mathbb{Q}$ from the group algebra of Γ into a division ring $\mathbb{Q}$ is said to be *Hughes-free* if the following condition holds:⁸ for any subgroup $\Gamma' \leq \Gamma$ and any normal subgroup $N \trianglelefteq \Gamma'$ with Γ'/N infinite cyclic generated by some element $t \in \Gamma' \setminus N$, the set of powers $\{t^i : i \in \mathbb{Z}\}$ is $\mathbb{Q}_N$ -linearly independent in $\mathbb{Q}$, where $\mathbb{Q}_N$ is the division closure of $K[N]$ in $\mathbb{Q}$. The following is a particular case of a theorem of Andrei Jaikin-Zapirain:⁹

Theorem 6.1.3. *Let Γ be a locally indicable abstract group and K field. Assume that Γ contains a normal subgroup N such that Γ/N is amenable and locally indicable and N is such that $K[N]$ admits a universal Hughes-free division $K[N]$ -ring of fractions. Then, $K[\Gamma]$ admits a universal Hughes-free division $K[\Gamma]$ -ring of fractions.*

With this, we can prove:

Proposition 6.1.4. *Let Γ be a finitely generated abstract subgroup of a free-by- $\mathbb{Z}_p$ pro- p group G and $\mathbb{F}_q$ some finite field of characteristic p . Then, $\mathbb{F}_q[\Gamma]$ admits a universal Hughes-free division $\mathbb{F}_q[\Gamma]$ -ring of fractions $\mathbb{Q}$.*

Proof. Write $G = N' \rtimes \mathbb{Z}_p$ where N' is free pro- p and let $N = \Gamma \cap N'$. Since Γ/N is a finitely generated torsion-free abelian group, it is locally indicable and amenable. Moreover, since N' is a free pro- p group, it is residually-(torsion-free nilpotent): first, observe that N' is the inverse limit of finitely generated free pro- p groups, so it suffices to prove this for N' finitely generated, in which case it is a theorem of Michel Lazard.¹⁰ Hence, N is a locally indicable residually-(torsion-free nilpotent) group, and therefore $\mathbb{F}_q[N]$ admits a Hughes-free division ring of fractions.¹¹ By Theorem 6.1.3, so does $\mathbb{F}_q[\Gamma]$. $\square$

We can now establish the validity of parts (1) and (3) of the Conjecture 6.1.1 for $K = \mathbb{F}_q$ and $\Gamma \leq G$ for a chain of subgroups coming from G .¹²

⁸Jaikin-Zapirain and López-Álvarez, “The strong Atiyah and Lück approximation conjectures for one-relator groups”, Sec. 3.3.

⁹Jaikin-Zapirain, “The universality of Hughes-free division rings”, Thm. 3.7.

¹⁰Lazard, “Sur les groupes nilpotents et les anneaux de Lie”, Sec. 4.

¹¹Jaikin-Zapirain, “The universality of Hughes-free division rings”, Thm. 1.1.

¹²Jaikin-Zapirain and Souza, “Sylvester domains and pro- p groups”, Thm. 8.1.

Theorem 6.1.5. *Let G be a free-by- $\mathbb{Z}_p$ pro- p group and Γ be a finitely generated abstract subgroup of G . Let $G > U_1 > U_2 > \dots$ be a chain of open normal subgroups of G with trivial intersection and define $\Gamma_i = \Gamma \cap U_i$. Then, for any matrix A over $\mathbb{F}_q[\Gamma]$ we have*

$$\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/\Gamma_i}(A) = \text{rk}_{\mathcal{Q}}(A),$$

where $\mathcal{Q}$ is the universal division ring of fractions of $\mathbb{F}_q[\Gamma]$.

Proof. Since the closure of Γ in G is a finitely generated free-by- $\mathbb{Z}_p$ pro- p group, we can assume that Γ is dense in G , so that G is topologically finitely generated. This in particular also implies that the inclusion map $\Gamma \rightarrow G$ induces isomorphisms $\Gamma/\Gamma_i \simeq G/U_i$ for every $i \geq 1$. By Theorem 4.6.6, $\mathbb{F}_q[[G]]$ admits a universal division ring of fractions $\mathcal{D}$ and we have $\text{rk}_G = \text{rk}_{\mathcal{D}}$ as rank functions on $\mathbb{F}_q[[G]]$. Since $\text{rk}_{G/U_i}(A) = \text{rk}_{\Gamma/\Gamma_i}(A)$ for any matrix A over $\mathbb{F}_q[\Gamma]$, we obtain part (i) of Conjecture 6.1.1: the limit $\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/\Gamma_i}(A)$ exists and is equal to $\text{rk}_{\mathcal{D}}(A)$.

Write $G \simeq N \rtimes \mathbb{Z}_p$ and let $H = N \cap \Gamma$. Denote by $\mathcal{D}_{\Gamma}$ and $\mathcal{D}_H$ the division closure of $\mathbb{F}_q[\Gamma]$ and $\mathbb{F}_q[H]$ in $\mathcal{D}$ respectively, by R the subring of $\mathcal{D}$ generated by $\mathcal{D}_H$ and Γ . It was shown by Jaikin-Zapirain that the epic embedding of $\mathbb{F}_q[H]$ into $\mathcal{D}_H$ is universal.¹³ We also claim that the induced surjective map $\mathcal{D}_H * (\Gamma/H) \rightarrow R$ is an isomorphism. If $\gamma_1, \dots, \gamma_n$ are elements of Γ that lie in different classes modulo H , then there exists an open normal subgroup U of G containing N such that $\gamma_1, \dots, \gamma_n$ lie in different classes modulo U . By part (b) of Proposition 4.3.6, the sum $\sum_{i=1}^n \mathcal{D}_H \gamma_i$ is direct. Hence, it sends linearly independent elements to linearly independent elements, establishing its injectivity.

These two facts together imply that the ring R is isomorphic to a crossed product $\mathcal{D}_H * (\Gamma/H)$. Since Γ/H is a finitely generated abelian group, R is a Noetherian ring. Thus, $\mathcal{D}_{\Gamma}$ is isomorphic to the Ore division ring of fractions of R . It only remains to show that the embedding $\mathbb{F}_q[\Gamma] \rightarrow \mathcal{D}_{\Gamma}$ is universal. If $\mathcal{Q}$ is the universal Hughes-free division $\mathbb{F}_q[\Gamma]$ -ring, then by the universality of $\mathcal{D}_H$ we know that there exists a unique local subring $L \leq \mathcal{D}_H$ containing $\mathbb{F}_q[H]$ for which the embedding $\mathbb{F}_q[H] \rightarrow \mathcal{Q}$ factors through

$$\mathbb{F}_q[H] \rightarrow L \rightarrow \mathcal{Q}$$

and the last map identifies the division closure $\mathcal{Q}_H$ of $\mathbb{F}_q[H]$ in $\mathcal{Q}$ with the residue division ring of L . Observe that by the uniqueness of L it must be invariant under the crossed product structure of R , that is, we can extend the map $L \rightarrow \mathcal{Q}$ to an epic map $\varphi: L * (\Gamma/H) \rightarrow \mathcal{Q}$. The Hughes-free condition on $\mathcal{Q}$ imply that for any tuple $\gamma_1, \dots, \gamma_n$ in Γ that lie in different classes modulo H we have $\varphi(r_1 \gamma_1 + \dots + r_n \gamma_n) = 0$ if and only if

$$\varphi(r_1) = \dots = \varphi(r_n) = 0$$

for any $r_1, \dots, r_n \in L$. This shows that the set of elements not in the kernel of φ is a two-sided Ore set in $L * (\Gamma/H)$, and the Ore localization of $L * (\Gamma/H)$ at this Ore set is a local subring of

¹³ Andrei Jaikin-Zapirain. “Free $\mathbb{Q}$ -groups are residually torsion-free nilpotent”. In: *Annales Scientifiques de l’École Normale Supérieure* (July 2024). ISSN: 1873-2151. DOI: 10.24033/asens.2587, Prop. 3.5.

$\mathcal{D}_\Gamma$ through which the embedding $\mathbb{F}_q[\Gamma] \rightarrow \mathcal{Q}$ factors. This is precisely an specialization of $\mathcal{D}_\Gamma$ onto $\mathcal{Q}$ as explained in Section 2.3. Since $\mathcal{Q}$ was universal, so is $\mathcal{D}_\Gamma$. In particular, $\mathcal{Q} \simeq \mathcal{D}_\Gamma$ as epic division $\mathbb{F}_q[\Gamma]$ -rings. $\square$

6.2 Subgroups of semisimple Lie groups

Discrete subgroups of complex semisimple Lie groups and their cohomology are ubiquitous in modern mathematics. In the context of geometry and topology, they appear as fundamental groups of locally symmetric Riemannian manifolds and their cohomology groups have a natural interpretation in terms of differential forms. In the context of number theory, arithmetic lattices encode critical information about number fields and their cohomology is related to the theory of automorphic forms. At this point, it goes without saying that there is an extensive and deep literature on this broad topic, and just to name a few major works we cite the classical books of Madabusi Santanam Raghunathan¹⁴ and Armand Borel and Nolan Wallach¹⁵.

Within those contexts, there is significance in the study of the asymptotic behaviour of the cohomology groups $H^i(\Gamma, V)$ when Γ is a discrete subgroup of a semisimple algebraic group G and V is a finite dimensional complex representation of $G(\mathbb{C})$. This asymptotic behaviour could be considered either with respect to going down a residual chain of finite index subgroups of Γ or with respect to taking larger and larger representations V of $G(\mathbb{C})$. The former is the realm of Lück approximation¹⁶, and we are concerned with the latter.

Therefore, given a subgroup Γ and a sequence of irreducible finite dimensional complex representations V_i of a semisimple complex Lie group $G(\mathbb{C})$ with $\lim_{i \rightarrow \infty} \dim_{\mathbb{C}} V_i = \infty$, we are interested in the following questions:

(1) Does the sequence

$$\frac{\dim_{\mathbb{C}} H^j(\Gamma, V_i)}{\dim_{\mathbb{C}} V_i} \quad (*)$$

converge, and if so, can we describe the limit?

(2) Does this limit depend on the choice of V_i ?

(3) Can we estimate the rate of convergence?

¹⁴M. S. Raghunathan. *Discrete Subgroups of Lie Groups*. Vol. 68. Ergebnisse der Mathematik und ihrer Grenzgebiete. 2. Folge. Berlin–Heidelberg: Springer-Verlag, 1972. ISBN: 978-3-642-86428-5. DOI: 10.1007/978-3-642-86426-1.

¹⁵Armand Borel and Nolan R. Wallach. *Continuous Cohomology, Discrete Subgroups, and Representations of Reductive Groups*. 2nd ed. Vol. 67. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 2000. ISBN: 978-0-8218-0851-1. DOI: 10.1090/surv/067.

¹⁶Theorem 3.1.16. See also Jan Boschheidgen and Andrei Jaikin-Zapirain. “Twisted L^2 -Betti numbers for sofic groups”. In: *Bulletin of the London Mathematical Society* 56.6 (Apr. 2024), pp. 2178–2187. ISSN: 1469-2120. DOI: 10.1112/blms.13050

In this present section we will conjecture an answer for all three questions when $G(\mathbb{C})$ is a complex semisimple Lie group and Γ is a subgroup of $G(\mathbb{C})$ of type FP_{j+1} (not necessarily discrete). Moreover, we will use the results of Chapter 5 to provide positive answers when all the irreducible factors in the root system of G are of type A_1 .

We expect the answer to questions (1) and (2) depends on two factors. The first factor is how the central elements $Z(G) \cap \Gamma$ act on each representation. More precisely, each irreducible representation V of $G(\mathbb{C})$ induces a central character $\chi: Z(G(\mathbb{C})) \rightarrow \mathbb{C}^\times$. If the sequence (*) converges for a sequence of irreducible representations V_i , we need them to induce the same central character when restricted to Γ . An explicit illustration of this fact is given at the end of Remark A.o.4. Given that the center $Z(G(\mathbb{C}))$ is finite¹⁷, this is not an obstruction if Γ is torsion-free.

The second factor regards the direction along which the weight lattice $P(\mathfrak{g})$, where $\mathfrak{g}$ is the complex Lie algebra of $G(\mathbb{C})$, the sequence of representations V_i is growing. We recall from Proposition 5.1.8 that to each irreducible representation V we can associate an element $\lambda \in P(\mathfrak{g})$ called the highest weight of V , and this element characterizes V up to isomorphism. If $\omega_1, \dots, \omega_n$ are the fundamental weights of $\mathfrak{g}$, then

$$\lambda = \lambda_1 \omega_1 + \dots + \lambda_n \omega_n$$

for unique non-negative integers λ_i and we define $\min \lambda = \min\{\lambda_1, \dots, \lambda_n\}$. To ensure convergence, if $\lambda(i)$ denotes the highest weight of V_i , we insist that all the weight parameters of V_i grow to infinity, that is, $\lim_{i \rightarrow \infty} \min \lambda(i) = \infty$.

To state our conjectures, we fix Γ a subgroup of $G(\mathbb{C})$, $\chi: Z(G(\mathbb{C})) \rightarrow \mathbb{C}^\times$ a central character of G and we denote by $P(\mathfrak{g})_\chi$ the set of weights λ that are highest weights of irreducible representations V onto which $Z(G(\mathbb{C}))$ acts through the central character χ . The twisted von Neumann rank function rk_Γ^χ is defined in Appendix A, where some of its properties are also explored, and $\text{rk}_V \in \mathbb{P}(\mathbb{C}[\Gamma])$ is the rank function induced by the representation V as in Example 3.1.8. We also fix a subring S_0 of $\mathbb{C}$.

Conjecture 6.2.1. *For any $\lambda \in P(\mathfrak{g})_\chi$ and any matrix A over $S_0[\Gamma]$, if V_λ is the corresponding irreducible representation of G we have that*

$$\left| \text{rk}_{V_\lambda}(A) - \text{rk}_\Gamma^\chi(A) \right| = O\left(\frac{1}{\min \lambda}\right).$$

Conjecture 6.2.2. *As rank functions on $S_0[\Gamma]$, we have*

$$\lim_{\min \lambda \rightarrow \infty} \text{rk}_{V_\lambda} = \text{rk}_\Gamma^\chi,$$

where the limit is taken over $P(\mathfrak{g})_\chi$ and rk_Γ^χ is the twisted von Neumann rank of Γ by the central character χ restricted to Γ .

¹⁷Lee, *The Structure of Complex Lie Groups*, Cor. 4.17.

Since any matrix involves only finitely many coefficients, it is clear that Γ satisfies either conjecture over $\mathbb{C}$ if and only if it satisfies the conjecture over all finitely generated subrings S_0 of $\mathbb{C}$. The same argument applies to the subgroups: a group Γ satisfies either conjecture over S_0 if and only if every finitely generated subgroup of Γ satisfies the conjecture over S_0 by Lemma A.o.3. When Γ satisfies additional finiteness properties and intersects $Z(G(\mathbb{C}))$ trivially, these conjectures can be used to deduce approximation results in group homology.¹⁸ We denote by $\beta_j^{(2)}(\Gamma)$ the j -th ℓ^2 -Betti number of Γ .¹⁹

Proposition 6.2.3. *Let Γ be a subgroup of $G(\mathbb{C})$ of type $FP_{j+1}(S_0)$ such that $\Gamma \cap Z(G(\mathbb{C})) = \{1\}$.*

1. *If Γ satisfies Conjecture 6.2.1 over S_0 , then for any $\lambda \in P(\mathfrak{g})$ and V_λ the corresponding irreducible representation of $G(\mathbb{C})$ we have that*

$$\left| \frac{\dim_{\mathbb{C}} H_k(\Gamma, V_\lambda)}{\dim_{\mathbb{C}} V_\lambda} - \beta_k^{(2)}(\Gamma) \right| = O\left(\frac{1}{\min \lambda}\right)$$

for any $0 \leq k \leq j$.

2. *If Γ satisfies Conjecture 6.2.2 over S_0 and V_i is a sequence of irreducible representations of $G(\mathbb{C})$ of highest weight $\lambda(i)$ such that $\lim_{i \rightarrow \infty} \min \lambda(i) = \infty$, then*

$$\lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{C}} H_k(\Gamma, V_i)}{\dim_{\mathbb{C}} V_i} = \beta_k^{(2)}(\Gamma)$$

for any $0 \leq k \leq j$.

3. *If Γ virtually satisfies Conjecture 6.2.1 over S_0 and $\beta_k^{(2)}(\Gamma) = 0$ for some $0 \leq k \leq j$, then for any $\lambda \in P(\mathfrak{g})$ and V_λ the corresponding irreducible representation of $G(\mathbb{C})$ we have that*

$$\frac{\dim_{\mathbb{C}} H_k(\Gamma, V_\lambda)}{\dim_{\mathbb{C}} V_\lambda} = O\left(\frac{1}{\min \lambda}\right).$$

Proof. Let

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow S_0 \rightarrow 0$$

be a free resolution of S_0 over $S_0[\Gamma]$ such that F_k is of finite rank for every $0 \leq k \leq j+1$. For each $0 \leq k \leq j$, there exists an $s \times r$ matrix A and a $t \times s$ matrix B over $S_0[\Gamma]$ such that, after choosing bases for F_{k-1} , F_k and F_{k+1} , the maps ψ and ϕ in

$$\cdots \rightarrow F_{k+1} \xrightarrow{\psi} F_k \xrightarrow{\phi} F_{k-1} \rightarrow \cdots$$

are given by left multiplication with A and B respectively.

¹⁸Sánchez and Souza, *Asymptotics of rational representations for algebraic groups*, Prop. 3.1, 3.2 and 3.3.

¹⁹Lück, *L2-invariants*, Def. 6.50.

After tensoring it with $\ell^2(\Gamma)$ over $S_0[\Gamma]$, we get a complex of the form

$$\dots \rightarrow \ell^2(\Gamma)^r \xrightarrow{\tilde{\psi}} \ell^2(\Gamma)^s \xrightarrow{\tilde{\phi}} \ell^2(\Gamma)^t \rightarrow \dots$$

The kernel of $\tilde{\phi}$ is a closed Γ -invariant subspace of $\ell^2(\Gamma)^s$ which contains the image of $\tilde{\psi}$. The k -th ℓ^2 -Betti number of Γ is then defined as

$$\beta_k^{(2)}(\Gamma) = \dim_{\Gamma} \ker(\tilde{\phi}) / \overline{\text{im}(\tilde{\psi})} = \dim_{\Gamma} \ker(\tilde{\phi}) - \dim_{\Gamma} \overline{\text{im}(\tilde{\psi})}$$

where the last equality comes from the fact that $\dim_{\Gamma}$ is exact under short exact sequences.²⁰

Using that $\dim_{\Gamma} \overline{\text{im}(\tilde{\psi})} = \text{rk}_{\Gamma} A$ and $\dim_{\Gamma} \ker(\tilde{\phi}) = s - \text{rk}_{\Gamma} B$, we obtain that

$$\beta_k^{(2)}(\Gamma) = s - \text{rk}_{\Gamma} A - \text{rk}_{\Gamma} B.$$

On the other hand, tensoring it with V_{λ} over $S_0[\Gamma]$ gives us the complex

$$\dots \rightarrow V_{\lambda}^r \xrightarrow{\hat{\psi}} V_{\lambda}^s \xrightarrow{\hat{\phi}} V_{\lambda}^t \rightarrow \dots$$

and $H_k(\Gamma, V_{\lambda}) = \text{Tor}_k^{S_0[\Gamma]}(S_0, V_{\lambda})$ is isomorphic to $\ker(\hat{\phi}) / \text{im}(\hat{\psi})$. From this, we get that

$$\dim_{\mathbb{C}} H_k(\Gamma, V_{\lambda}) = \dim_{\mathbb{C}} \ker(\hat{\phi}) / \text{im}(\hat{\psi}) = \dim_{\mathbb{C}} \ker(\hat{\phi}) - \dim_{\mathbb{C}} \text{im}(\hat{\psi}).$$

Since $\dim_{\mathbb{C}} \text{im}(\hat{\psi}) = \dim_{\mathbb{C}} V_{\lambda} \cdot \text{rk}_{V_{\lambda}} A$ and $\dim_{\mathbb{C}} \ker(\hat{\phi}) = \dim_{\mathbb{C}} V_{\lambda}(s - \text{rk}_{V_{\lambda}} B)$, we conclude that

$$\dim_{\mathbb{C}} H_k(\Gamma, V_{\lambda}) = \dim_{\mathbb{C}} V_{\lambda}(s - \text{rk}_{V_{\lambda}} A - \text{rk}_{V_{\lambda}} B).$$

The statements 1. and 2. now follow since $\text{rk}_{\Gamma} = \text{rk}_{\Gamma}^{\chi}$ for any central character χ by Remark A.o.4.

For the third statement, let Γ' be a finite index subgroup of Γ that satisfies Conjecture 6.2.1. If $\text{cores}: H_k(\Gamma', V_{\lambda}) \rightarrow H_k(\Gamma, V_{\lambda})$ and $\text{res}: H_k(\Gamma, V_{\lambda}) \rightarrow H_k(\Gamma', V_{\lambda})$ are the corestriction and transfer maps in homology respectively, then the composition $\text{cores} \circ \text{res}$ equals²¹ multiplication by the index $|\Gamma: \Gamma'|$. As $H_k(\Gamma, V_{\lambda})$ is a complex vector space, this implies that the composition $\text{cores} \circ \text{res}$ is an isomorphism and therefore res is injective. Consequently,

$$\dim_{\mathbb{C}} H_k(\Gamma, V_{\lambda}) \leq \dim_{\mathbb{C}} H_k(\Gamma', V_{\lambda}).$$

Since $\beta_k^{(2)}(\Gamma') = |\Gamma: \Gamma'| \beta_k^{(2)}(\Gamma) = 0$, the claim follows from statement 1. $\square$

We also remark that it suffices to prove either conjecture for subgroups of the unique simply connected and connected semisimple complex Lie group with Lie algebra $\mathfrak{g}$.

²⁰Lück, *L2-invariants*, Thm. 6.7(4b).

²¹Joseph J. Rotman. *An introduction to homological algebra*. 2nd. Universitext. Springer, 2008. ISBN: 978-0-387-24527-0, Thm. 9.94.

Proposition 6.2.4. *Let $G(\mathbb{C})$ be a connected semisimple complex Lie group and $\widetilde{G(\mathbb{C})}$ its universal cover. Fix a central character $\chi: Z(\widetilde{G(\mathbb{C})}) \rightarrow \mathbb{C}^\times$ and assume that it factors through $Z(G(\mathbb{C}))$, that is, $\pi_1(G(\mathbb{C})) \leq \ker(\chi)$. Choose a finitely generated subring S_0 of $\mathbb{C}$ and Γ' a subgroup of $G(\mathbb{C})$. If Γ denotes the pre-image of Γ' through the projection $\widetilde{G(\mathbb{C})} \rightarrow G(\mathbb{C})$ and either Conjecture 6.2.1 or 6.2.2 holds for Γ over S_0 , then the same conjecture holds for Γ' over S_0 .*

Proof. Let $\lambda \in P(\mathfrak{g})_\chi$ and $s: \Gamma' \rightarrow \Gamma$ be a set-theoretic section such that $s(1) = 1$. Then, the map $S_0[\Gamma] \rightarrow \text{End}_{\mathbb{C}}(V_\lambda)$ induced by the action factors through the quotient map $S_0[\Gamma] \rightarrow S_0[\Gamma']$ by the definition of $P(\mathfrak{g})_\chi$. This shows that for any matrix A over $S_0[\Gamma']$, if $s(A)$ denotes the matrix over $S_0[\Gamma]$ obtained by applying s to A entrywise, then

$$\text{rk}_{V_\lambda} s(A) = \text{rk}_{V_\lambda} A.$$

On the other hand, the equality

$$\text{rk}_\Gamma^\chi(s(A)) = \text{rk}_{\Gamma'}^\chi(A)$$

follows from Proposition A.o.12. This shows that

$$|\text{rk}_{V_\lambda}(A) - \text{rk}_{\Gamma'}^\chi(A)| = |\text{rk}_{V_\lambda}(s(A)) - \text{rk}_\Gamma^\chi(s(A))|$$

and the claim follows. $\square$

Our goal in this section is to use the results of Chapter 5 to show:²²

Theorem 6.2.5. *Let Γ be a finitely generated subgroup of $\prod_{i=1}^n \text{SL}_2(\mathbb{C})$. Then, for any central character χ and each finitely generated subring S_0 of $\mathbb{C}$, Conjecture 6.2.1 holds over S_0 for some finite index subgroup of Γ and Conjecture 6.2.2 holds over $\mathbb{C}$ for Γ .*

Recall that for any connected and simply connected complex semisimple Lie group $G(\mathbb{C})$, by Theorem 5.1.16 we can see $G(\mathbb{C})$ as the complex points of an universal split semisimple Chevalley group functor G . The link between Conjectures 6.2.1-6.2.2 and Chapter 5 is the following result.²³

Theorem 6.2.6. *Let G be a universal split semisimple Chevalley group and Γ be a finitely generated subgroup of $G(\mathbb{C})$. If Conjecture 5.3.5 holds for G and almost all primes p , then for all central characters χ , the group Γ satisfies Conjecture 6.2.2 over $\mathbb{C}$. If Conjecture 5.3.4 holds for G and almost all primes p , then for all central characters χ and each finitely generated subring S_0 of $\mathbb{C}$ the group Γ contains a finite index subgroup Γ' satisfying Conjecture 6.2.1 over S_0 .*

²²Sánchez and Souza, *Asymptotics of rational representations for algebraic groups*, Thm. 6.1.

²³Ibid., Thm. 5.5.

Proof. By Lemma A.o.3, the validity of both statements is inherited to subgroups, so we can assume that Γ is both finitely generated and Zariski-dense in $G(\mathbb{C})$. By looking at the entries in the generators of Γ , we can find a finitely generated subring S of $\mathbb{C}$ containing S_0 such that $\Gamma \leq G(S)$. In this case, Cassels' embedding theorem²⁴ tells us that S embeds as a subring of $\mathbb{Z}_p$ for infinitely many primes p . In particular, for some odd prime p we can assume that (1) either Conjecture 5.3.4 or 5.3.5 hold for p and (2) $\Gamma \leq G(S)$ where S embeds in $\mathbb{Z}_p$.

We now see Γ as a subgroup of $G(\mathbb{Z}_p)$. Let $G_i = \ker(G(\mathbb{Z}_p) \rightarrow G(\mathbb{Z}/p^i\mathbb{Z}))$ be the i -th congruence kernel of $G(\mathbb{Z}_p)$ and set $\Gamma' = \Gamma \cap G_1$, so that by Proposition 5.2.3 Γ' is a torsion-free finite index subgroup of Γ and $S[\Gamma']$ embeds in $\mathbb{Z}_p[[G_1]]$. Set $N_i = \Gamma' \cap G_i$, so that the N_i form a chain of normal subgroups of Γ' of finite index with trivial intersection. If $L \leq \mathbb{Q}_p$ denotes the field of fractions of S in $\mathbb{Q}_p$ and $\mathcal{Q}$ the Ore division ring of fractions of $\mathbb{Z}_p[[G_1]]$, for any $n \times m$ matrix A over $S[\Gamma']$ we have:

$$\begin{aligned}
\text{rk}_{\Gamma'} A &= \lim_{i \rightarrow \infty} \text{rk}_{\Gamma'/N_i} A, \text{ by Theorem 3.1.16,} \\
&= \lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{C}} A \mathbb{C}[\Gamma'/N_i]^m}{m|\Gamma' : N_i|} \\
&= \lim_{i \rightarrow \infty} \frac{\dim_L AL[\Gamma'/N_i]^m}{m|\Gamma' : N_i|} \\
&= \lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{Q}_p} A \mathbb{Q}_p[\Gamma'/N_i]^m}{m|\Gamma' : N_i|} \\
&= \lim_{i \rightarrow \infty} \frac{\dim_{\mathbb{Q}_p} A \mathbb{Q}_p[G_1/G_i]^m}{m|G_1 : G_i|}, \text{ since } - \otimes_{\mathbb{Q}_p[\Gamma'/N_i]} \mathbb{Q}_p[G_1/G_i] \text{ is exact,} \\
&= \text{rk}_{\mathcal{Q}} A \text{ by Theorem 5.2.6.}
\end{aligned}$$

By Theorem 5.1.16, we also know that there exists a unique morphism $G \rightarrow \text{GL}(W_\lambda)$ for which $V_\lambda = W_\lambda(\mathbb{C})$. Observe that $\text{rk}_{W_\lambda(\mathbb{Q}_p)} \in \mathbb{P}(S[\Gamma'])$ lies in the image of the change of rings

²⁴John William Scott Cassels. "An embedding theorem for fields". In: *Bulletin of the Australian Mathematical Society* 14.2 (Apr. 1976), pp. 193–198. ISSN: 1755-1633. DOI: 10.1017/S000497270002503x, Thm. I.

$S[\Gamma'] \rightarrow \mathbb{Z}_p[[G_1]]$. Moreover:

$$\begin{aligned}
\mathrm{rk}_{V_\lambda} A &= \frac{\dim_{\mathbb{C}} AV_\lambda^m}{m \dim_{\mathbb{C}} V_\lambda} \\
&= \frac{\dim_{\mathbb{C}} AW_\lambda(\mathbb{C})^m}{m \dim_{\mathbb{C}} W_\lambda(\mathbb{C})} \\
&= \frac{\dim_L AW_\lambda(L)^m}{m \dim_L W_\lambda(L)} \\
&= \frac{\dim_{\mathbb{Q}_p} AW_\lambda(\mathbb{Q}_p)^m}{m \dim_{\mathbb{Q}_p} W_\lambda(\mathbb{Q}_p)} \\
&= \mathrm{rk}_{W_\lambda(\mathbb{Q}_p)} A
\end{aligned}$$

Therefore, we can regard both $\mathrm{rk}_{\mathbb{Q}} A$ and $\mathrm{rk}_{W_\lambda(\mathbb{Q}_p)} A$ as ranks evaluated at the matrix A over the ring $\mathbb{Z}_p[[G_1]]$.

If Conjecture 5.3.4 holds for p , then

$$|\mathrm{rk}_{V_\lambda} A - \mathrm{rk}_{\Gamma'} A| = |\mathrm{rk}_{W_\lambda(\mathbb{Q}_p)} A - \mathrm{rk}_{\mathbb{Q}} A| = O\left(\frac{1}{\min \lambda}\right).$$

and so Γ' satisfies Conjecture 6.2.1 over S (and in particular over S_0) as $\mathrm{rk}_{\Gamma'} = \mathrm{rk}_{\Gamma}^\chi$ by Remark A.o.4. If Conjecture 5.3.5 holds for p , then

$$\lim_{\min \lambda \rightarrow \infty} |\mathrm{rk}_{V_\lambda} A - \mathrm{rk}_{\Gamma'} A| = \lim_{\min \lambda \rightarrow \infty} |\mathrm{rk}_{W_\lambda(\mathbb{Q}_p)} A - \mathrm{rk}_{\mathbb{Q}} A| = 0,$$

that is,

$$\lim_{\min \lambda \rightarrow \infty} \mathrm{rk}_{V_\lambda} = \mathrm{rk}_{\Gamma'} = \mathrm{rk}_{\Gamma}^\chi$$

as rank functions on $S[\Gamma']$. By Theorem A.o.11, this implies that

$$\lim_{\min \lambda \rightarrow \infty} \mathrm{rk}_{V_\lambda} = \mathrm{rk}_{\Gamma}^\chi$$

as rank functions on $S[\Gamma]$. Since S is an arbitrary finitely generated subring of $\mathbb{C}$, we get that Γ satisfies Conjecture 6.2.2 over $\mathbb{C}$. $\square$

Proof of Theorem 6.2.5. Follows from Theorem 5.4.1 and Theorem 6.2.6. $\square$

Corollary 6.2.7. *Let $G(\mathbb{C})$ be a connected semisimple complex Lie group such that its Lie algebra $\mathfrak{g}$ is isomorphic to $\prod_{i=1}^k \mathfrak{sl}_2(\mathbb{C})$ and Γ be a finitely generated subgroup of $G(\mathbb{C})$. Then, for any central character χ and each finitely generated subring S_0 of $\mathbb{C}$, Conjecture 6.2.1 holds over S_0 for some finite index subgroup of Γ and Conjecture 6.2.2 holds over $\mathbb{C}$ for Γ .*

Proof. Is an immediate consequence of Theorem 6.2.5 and Proposition 6.2.4 $\square$

6.3 Three-dimensional hyperbolic geometry

Let M be an orientable hyperbolic 3-manifold, i.e. a 3-manifold admitting a complete Riemannian metric of constant negative curvature. We assume that its fundamental group $\pi_1(M)$ is finitely generated. It is a theorem of Godfrey Peter Scott²⁵ that in this case $\pi_1(M)$ is isomorphic to the fundamental group of a compact 3-manifold, and in particular it is of type $FP_\infty(S)$ over any ring S .

Marden's tameness theorem, proven independently by Ian Agol²⁶ and Danny Calegari and David Gabai²⁷, says that M is homeomorphic to the interior of a compact 3-manifold $\overline{M}$. In particular, M can be decomposed into a compact core C that is homotopy equivalent to M and a union of finitely many ends E_i , each homeomorphic to a product $S_i \times (0, \infty)$ where S_i is a closed surface.²⁸ Moreover, if M is of finite volume, then all the S_i are tori, which follows from the Thick-Thin decomposition of M .²⁹ Observe that by the Poincaré duality for manifolds with boundary, if each S_i has genus g_i , then by “cutting” the cusps we see that the Euler characteristic of M is $E(M) = \sum (1 - g_i)$.

We can identify $\Gamma = \pi_1(M)$ with a torsion-free subgroup of the group $\text{Isom}^+(\mathbb{H}^3)$ of orientation preserving isometries of the hyperbolic 3-space $\mathbb{H}^3$. This group is isomorphic to $\text{PSL}_2(\mathbb{C})$ acting on $\mathbb{H}^3$ through Möbius transformations. Recall from Example 5.1.9 that the representations of $\text{PSL}_2(\mathbb{C})$ are parametrized by their highest weight $\lambda = k\omega_1$, where ω_1 is the fundamental weight of $\mathfrak{sl}_2(\mathbb{C})$ and k is an even non-negative integer. Moreover, this representation V_λ is isomorphic to the space $\text{Sym}^k(\mathbb{C}^2)$ of homogeneous polynomials of degree k in two variables over $\mathbb{C}$, onto which $\text{PSL}_2(\mathbb{C})$ acts by a linear change of variables. The homology of Γ with coefficients in V_λ has been computed by Pere Menal-Ferrer and Joan Porti.³⁰

Proposition 6.3.1. *Let M hyperbolic 3-manifold with finitely generated fundamental group*

²⁵Godfrey Peter Scott. “Finitely Generated 3-Manifold Groups are Finitely Presented”. In: *Journal of the London Mathematical Society* s2-6.3 (May 1973), pp. 437–440. ISSN: 0024-6107. DOI: 10.1112/jlms/s2-6.3.437.

²⁶Ian Agol. *Tameness of hyperbolic 3-manifolds*. 2004. arXiv: math/0405568 [math.GT].

²⁷Danny Calegari and David Gabai. “Shrinkwrapping and the taming of hyperbolic 3-manifolds”. In: *Journal of the American Mathematical Society* 19.2 (Dec. 2005), pp. 385–446. ISSN: 1088-6834. DOI: 10.1090/s0894-0347-05-00513-8.

²⁸Richard Canary. “Marden’s Tameness Conjecture: History and Applications”. In: *Geometry, Analysis and Topology of Discrete Groups*. Somerville, Mass: International Press of Boston Inc, Nov. 2008. ISBN: 978-1-57146-126-1, Sec. 5.

²⁹Bruno Martelli. *An Introduction to Geometric Topology*. Independently published, Aug. 2023. ISBN: 979-8-8587-9070-9, Cor. 4.2.18.

³⁰Pere Menal-Ferrer and Joan Porti. “Twisted cohomology for hyperbolic three manifolds”. In: *Osaka Journal of Mathematics* 49.3 (Sept. 2012), pp. 741–769. ISSN: 0030-6126, Cor. 3.6.

and whose ends contain exactly n toroidal cusps. For $k \in \mathbb{Z}_{\geq 0}$ even and $\lambda = k\omega_1$, we have that

$$\dim_{\mathbb{C}} H_i(\Gamma, V_\lambda) = \begin{cases} 0, & \text{if } i \in \{0, 3\}, \\ n, & \text{if } i = 2, \\ n - (k + 1)E(M), & \text{if } i = 1. \end{cases}$$

In particular, since any finitely generated subgroup of $\mathrm{PSL}_2(\mathbb{C})$ satisfies Conjecture 6.2.2 over $\mathbb{C}$ by Corollary 6.2.7, we can deduce the ℓ^2 -Betti numbers of M from this computation.

Corollary 6.3.2. *Let M be a hyperbolic 3-manifold with Euler characteristic $E(M)$ and such that its fundamental group $\pi_1(M)$ is finitely generated. Then:*

$$\beta_i^{(2)}(M) = \begin{cases} 0, & \text{if } i \neq 1, \\ -E(M), & \text{if } i = 1. \end{cases}$$

Proof. Since M is aspherical, we can identify³¹ the ℓ^2 -Betti numbers of M with the ℓ^2 -Betti numbers of its fundamental group $\Gamma = \pi_1(M)$. By Corollary 6.2.7 and part (b) of Proposition 6.2.3, for $\lambda = k\omega_1$ with k even we have:

$$\begin{aligned} \beta_i^{(2)}(\Gamma) &= \lim_{k \rightarrow \infty} \frac{\dim_{\mathbb{C}} H_i(\Gamma, V_\lambda)}{\dim_{\mathbb{C}} V_\lambda} \\ &= \begin{cases} \lim_{k \rightarrow \infty} 0, & \text{if } i \in \{0, 3\}, \\ \lim_{k \rightarrow \infty} \frac{n}{k+1}, & \text{if } i = 2, \\ \lim_{k \rightarrow \infty} \frac{n}{k+1} - E(M), & \text{if } i = 1. \end{cases} \end{aligned}$$

From this, the result follows since n is fixed. □

John Lott and Wolfgang Lück³² gave a general formula for the ℓ^2 -Betti numbers of every compact 3-manifold M . Using P. Scott's compact core theorem³³ to reduce the finitely generated case to the compact case, one recovers the formula in Corollary 6.3.2. By repeatedly applying the ℓ^2 -Mayer-Vietoris sequence, the prime decomposition theorem reduces the general case to that of an irreducible M with infinite $\pi_1(M)$ and the loop theorem reduces this even further to the case where the boundary of M is incompressible. Furthermore, by doubling M we may assume that it is closed. Then, the Geometrization theorem reduces the analysis even further to the cases where M is either Seifert fibered or hyperbolic.

Both cases are where the core of the calculation lies. That the ℓ^2 -Betti numbers of Seifert fibered 3-manifolds vanish can be seen, for instance, as a consequence of the existence of a

³¹Lück, *L2-invariants*, p. 6.50.

³²John Lott and Wolfgang Lück. " L^2 -topological invariants of 3-manifolds". In: *Inventiones Mathematicae* 120.1 (Dec. 1995), pp. 15–60. ISSN: 1432-1297. DOI: 10.1007/bf01241121.

³³Godfrey Peter Scott. "Compact Submanifolds of 3-Manifolds". In: *Journal of the London Mathematical Society* s2-7.2 (Nov. 1973), pp. 246–250. ISSN: 0024-6107. DOI: 10.1112/jlms/s2-7.2.246.

virtual S^1 -action³⁴ or of the virtual algebraic fibering of its fundamental group.³⁵ The vanishing of ℓ^2 -Betti numbers for closed hyperbolic 3-manifolds is implied by a theorem of Jozef Dodziuk, and thus the trickiest part of Lott and Lück's argument is the deduction of the finite volume hyperbolic case from the closed one (for more details, see Lück's book³⁶). This is precisely the case covered by Corollary 6.3.2.

6.4 Cusp forms for SL_2

The quantitative statement of Conjecture 6.2.1 has applications in the theory of automorphic forms for SL_2 . We will give here a brief account of the definitions of the theory of cusp forms necessary for stating our results, and we direct the reader to either the Corvalis volumes³⁷ or Haruzo Hida's papers³⁸ for more details. We mainly follow the exposition given by Hida.

Let F be a number field with $r_1 + 2r_2 = (F : \mathbb{Q})$ real and complex embeddings respectively, and let $I_{\mathbb{R}} = \{\tau_1, \dots, \tau_{r_1}\}$ and $I_{\mathbb{C}} = \{\sigma_1, \dots, \sigma_{r_2}\}$ be complete representative sets of real and complex embeddings corresponding to the archimedean places of K . We define $I_{\infty} = I_{\mathbb{R}} \cup I_{\mathbb{C}}$ and $I = I_{\infty} \cup \{\bar{\sigma}_1, \dots, \bar{\sigma}_{r_2}\}$, where $\bar{\sigma}_i$ denotes its complex conjugate. Consider the group

$$G = \mathrm{SL}_2(F \otimes_{\mathbb{Q}} \mathbb{R}) = \prod_{r_1} \mathrm{SL}_2(\mathbb{R}) \times \prod_{r_2} \mathrm{SL}_2(\mathbb{C}).$$

Letting $\mathcal{O}_F$ denote the ring of integers of F , we fix a congruence subgroup Γ of $\mathrm{SL}_2(\mathcal{O}_F)$ that will be the *level* of our automorphic forms. Then, we get an embedding $\Gamma \rightarrow G$ by taking the product of all the maps in I_{∞} .

By quotienting out the right action by the product $\prod_{r_1} \mathrm{SO}_2(\mathbb{R}) \times \prod_{r_2} \mathrm{SU}_2(\mathbb{C})$ of the standard maximal compact subgroups of each factor, we obtain the locally symmetric space

$$\mathbb{H} = \prod_{r_1} \mathbb{H}^2 \times \prod_{r_2} \mathbb{H}^3$$

associated to G on which Γ acts. The quotient $M = \Gamma \backslash \mathbb{H}$ is an $(2r_1 + 3r_2)$ -dimensional orbifold of finite volume, and it is a manifold if Γ is torsion-free. In this case, the Borel-Serre compactification attaches a boundary ∂M to the manifold $M = \Gamma \backslash \mathbb{H}$, and the resulting compact manifold has the same fundamental group Γ .

Let $\mathfrak{g} = \prod_{r_1+2r_2} \mathfrak{sl}_2(\mathbb{C})$ be the complexification of the real Lie algebra of G . As we have explained in Proposition 5.4.2 over $\mathbb{Q}_p$, the center of the complex universal enveloping algebra

³⁴Lück, *L2-invariants*, Thm. 1.40.

³⁵Ibid., Thm. 7.2(5).

³⁶Ibid., Sec. 4.2.

³⁷Armand Borel and William Casselman. *Automorphic forms, representations, and L-functions*. 1st. Proceedings of symposia in pure mathematics 33. American Mathematical Society, 1979. ISBN: 978-0-8218-1435-2.

³⁸Haruzo Hida. "p-ordinary cohomology groups for $\mathrm{SL}(2)$ over number fields". In: *Duke Mathematical Journal* 69.2 (Feb. 1993). ISSN: 0012-7094. DOI: 10.1215/S0012-7094-93-06914-1.

$U(\mathfrak{g})$ is a polynomial algebra in $r_1 + 2r_2$ generators called Casimir operators. For each embedding $\delta \in I$, we fix the Casimir operator Δ_δ given by the image under δ of $\frac{1}{4}(EF + FE + H^2/2)$, where

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form the standard basis of $\mathfrak{sl}_2(\mathbb{C})$.

Let $k = (k_\delta)_{\delta \in I_\infty}$ be an $(r_1 + r_2)$ tuple of non-negative integers indexed by the archimedean places of F . Let $V_k = \bigotimes_{\delta \in I_\infty} V_\delta$ be the complex vector space given by the tensor product over $\mathbb{C}$ of $V_\delta = \mathbb{C}$ for all real embeddings δ and $V_\delta = \text{Sym}^{2k_\delta-2}(\mathbb{C}^2)$ for all the complex embeddings δ . It naturally has the structure of a right module over $\prod_{r_1} \mathbb{C}^\times \times \prod_{r_2} \text{GL}_2(\mathbb{C})$.

Definition 6.4.1. Let J be a subset of $I_\mathbb{R}$. A *cuspidal form* of level Γ , type J and weight k is a smooth function $f: \Gamma \backslash G \rightarrow V_k$ that satisfies the following properties:

(M1) With the convention that $k_{\bar{\sigma}_i} = k_{\sigma_i}$, we have that f is an eigenfunction for the infinitesimal action of all the Casimir operator Δ_δ with eigenvalue

$$\Delta_\delta f = \left(\frac{(k_\delta - 2)^2}{2} + k_\delta - 2 \right) f.$$

(M2) If we write $h \in \prod_{r_1} \text{SO}_2(\mathbb{R}) \times \prod_{r_2} \text{SU}_2(\mathbb{C})$ as

$$h = \left(\left(\begin{pmatrix} \cos(2\pi\theta_\tau) & \sin(2\pi\theta_\tau) \\ -\sin(2\pi\theta_\tau) & \cos(2\pi\theta_\tau) \end{pmatrix} \right)_{\tau \in I_\mathbb{R}}, (h_\sigma)_{\sigma \in I_\mathbb{C}} \right),$$

then

$$f(xh) = f(x) \left((e^{k_\tau \theta_\tau})_{\tau \in J}, (e^{k_\tau \theta_\tau})_{\tau \notin J}, (h_\sigma)_{\sigma \in I_\mathbb{C}} \right).$$

(M3) For all $x \in G$, we have that

$$\int_{(\Gamma \cap U) \backslash U} f(ux) du = 0$$

where U is the product of all upper unitriangular factors of G and du is the Lebesgue measure on $U \simeq \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$.

Remark 6.4.2. The usual notion of a cuspidal form is that of a scalar valued function $f^0: \Gamma \backslash G \rightarrow \mathbb{C}$ that satisfies various finiteness and growth conditions.³⁹ The span of such functions is finite-dimensional under the action of $\prod_{r_1} \text{SO}_2(\mathbb{R}) \times \prod_{r_2} \text{SU}_2(\mathbb{C})$ on the right, and identifying this span with V_k gives us a vector valued cuspidal form according to our definition. Conversely, composing any vector valued cuspidal form with a linear functional gives us a scalar valued cuspidal form.⁴⁰ ■

³⁹Borel and Casselman, *Automorphic forms, representations, and L-functions*, p. 190.

⁴⁰Hida, “p-ordinary cohomology groups for $\text{SL}(2)$ over number fields”, Sec. 1.4.

We denote the space of all cusp forms of level Γ , type J and weight k by $S_{k,J}(\Gamma)$. If $J = I_{\mathbb{R}}$, we denote this space simply by $S_k(\Gamma)$. The current literature⁴¹ shows great interest in the asymptotic behaviour of the dimensions $\dim_{\mathbb{C}} S_{k,J}(\Gamma)$ either for a fixed weight k and a residual chain of levels Γ_i or for a fixed level Γ and an increasing sequence of weight k_i . We are concerned with the latter case of growing weights for a fixed level.

When F is totally real, the precise growth rate of $S_k(\Gamma)$ was obtained by Hideo Shimizu⁴² as:

$$\dim_{\mathbb{C}} S_k(\Gamma) \sim \prod_{\tau \in I} k_{\tau}.$$

When F is not totally real, the question of what is the precise growth rate of $S_k(\Gamma)$ is very much open. Currently, the best general upper bound is the one given by Weibo Fu.⁴³

Theorem 6.4.3. *If F is not totally real, then*

$$\dim_{\mathbb{C}} S_k(\Gamma) = O\left(\frac{\prod_{\tau \in I_{\mathbb{R}}} k_{\tau} \times \prod_{\sigma \in I_{\mathbb{C}}} k_{\sigma}^2}{\min\{k_{\delta} : \delta \in I_{\infty}\}}\right).$$

If F is imaginary quadratic, i.e. when Γ is a congruence subgroup of a Bianchi group, this result combined with the lower bound from Tobias Finis, Fritz Grunewald and Paulo Tirao⁴⁴ yields the precise growth rate.⁴⁵

Corollary 6.4.4. *If F is imaginary quadratic, then $k = (k)$ and*

$$\dim_{\mathbb{C}} S_k(\Gamma) \sim k.$$

Assume that Γ is torsion-free and that $k_{\delta} \geq 3$ for each $\delta \in I_{\infty}$, and let $\lambda_{\delta} = k_{\delta} - 2$ with the convention that $k_{\delta} = k_{\bar{\delta}}$. If we take $\lambda \in P(\mathfrak{g})$ the weight whose coordinates in the fundamental weights are precisely the λ_{δ} , then V_{λ} becomes a Γ -module through the embedding $\Gamma \rightarrow \prod_I \mathrm{SL}_2(\mathbb{C})$ given by the product of all the embeddings $\Gamma \rightarrow \mathrm{SL}_2(\mathbb{C})$ induced by each $\delta \in I$.

It is well known that the spaces $S_{k,J}(\Gamma)$ can be computed through the group cohomology of Γ , a result that goes by the names of the *Eichler-Shimura isomorphism*, the *Matsushima-Murakami formula*, and, in the present formulation, is due to Günter Harder. There is

⁴¹We point the reader to the following paper and the references therein. Tobias Finis, Fritz Grunewald, and Paulo Tirao. “The Cohomology of Lattices in $\mathrm{SL}(2, \mathbb{C})$ ”. In: *Experimental Mathematics* 19.1 (Jan. 2010), pp. 29–63. ISSN: 1944-950X. DOI: 10.1080/10586458.2010.10129067

⁴²Hideo Shimizu. “On Discontinuous Groups Operating on the Product of the Upper Half Planes”. In: *Annals of Mathematics* 77.1 (1963), pp. 33–71. ISSN: 0003486X, 19398980.

⁴³Fu, “Sharp bounds for multiplicities of Bianchi modular forms”, Thm. 1.2.

⁴⁴Finis, Grunewald, and Tirao, “The Cohomology of Lattices in $\mathrm{SL}(2, \mathbb{C})$ ”.

⁴⁵Fu, “Sharp bounds for multiplicities of Bianchi modular forms”, Cor. 1.3.

a natural procedure to attach to each cusp form $f \in S_{k,J}(\Gamma)$ a Γ -invariant V_λ -valued closed differential $(r_1 + 2r_2)$ -form $\omega(f)$ on $\mathbb{H}$, thus inducing a map

$$\omega \bigoplus_{J \subseteq I_{\mathbb{R}}} S_{k,J}(\Gamma) \rightarrow H^{r_1+2r_2}(\Gamma, V_\lambda).$$

We refer the reader to another paper of Hida for details.⁴⁶ The consequence of the Eichler-Shimura isomorphism that we will need is as follows:

Theorem 6.4.5. *The map ω is injective and the image is contained in the subspace $H_c^{r_1+2r_2}(\Gamma, V_\lambda)$ of compactly supported cohomology.*

Sketch of the proof. The Eichler-Shimura isomorphism states that ω induces an isomorphism from its domain onto the subspace $H_{sq}^{r_1+2r_2}(\Gamma, V_\lambda)$ of square-integrable forms.^{47,48} That this subspace is contained in the compactly supported cohomology $H_c^{r_1+2r_2}(\mathbb{C})$ is a standard fact about the cohomology of discrete subgroups of Lie groups.⁴⁹ $\square$

Fu's proof of Theorem 6.4.3 applies Theorem 5.4.1 to the completed homology of $\Gamma \backslash \mathbb{H}$ as constructed by Matthew Emerton and combines this with a spectral sequence in the profinite homology of G_1 to obtain the desired upper bound for $H^{r_1+2r_2}(\Gamma, V_\lambda)$. We propose a different proof of this result that avoids the use of completed cohomology entirely and depends solely on Theorems 5.4.1 and 6.4.5.

Proof of Theorem 6.4.3. The duality isomorphism⁵⁰ between the compactly supported cohomology groups $H_c^{r_1+2r_2}(\Gamma, V_\lambda)$ and the homology groups $H_{r_1+r_2}(\Gamma, V_\lambda)$ shows that, by the Eichler-Shimura map and Theorem 6.4.5, it suffices to prove Fu's upper bound on $H_{r_1+r_2}(\Gamma, V_\lambda)$. Due to the fact that Γ is the fundamental group of a compact manifold by the Borel-Serre construction, we conclude that Γ is of type $FP_\infty(\mathbb{Z})$.

By the proportionality principle of ℓ^2 -Betti numbers⁵¹, the $(r_1 + r_2)$ -th ℓ^2 -Betti numbers of any two lattices in G are proportional. Since G is a symmetric space of non-compact type and dimension $2r_1 + 3r_2$, it is a theorem of Borel⁵² that $b_{r_1+r_2}^{(2)}(\Gamma) = 0$. By Theorem 6.2.5, Γ contains

⁴⁶Haruzo Hida. "On the critical values of L-functions of $GL(2)$ and $GL(2) \times GL(2)$ ". In: *Duke Mathematical Journal* 74.2 (May 1994). ISSN: 0012-7094. DOI: 10.1215/S0012-7094-94-07417-6, Sec. 2.4.

⁴⁷Hida, "p-ordinary cohomology groups for $SL(2)$ over number fields", Thm. 1.1.

⁴⁸Hida, "On the critical values of L-functions of $GL(2)$ and $GL(2) \times GL(2)$ ", Cor. 2.2.

⁴⁹Armand Borel. "Stable Real Cohomology of Arithmetic Groups II". in: *Manifolds and Lie Groups: Papers in Honor of Yozô Matsushima*. Ed. by Jun-ichi Hano et al. Boston, MA: Birkhäuser, 1981, pp. 21–55. ISBN: 978-1-4612-5987-9. DOI: 10.1007/978-1-4612-5987-9_2, Cor. 5.5.

⁵⁰Allen Hatcher. *Algebraic Topology*. Inglès. Cambridge ; New York: Cambridge University Press, 2001. ISBN: 978-0-521-79540-1, Exer. 3.3.35.

⁵¹Lück, *L2-invariants*, Thm. 3.183 and Cor. 7.37.

⁵²Ibid., Thm. 5.12.

a finite index subgroup Γ' satisfying Conjecture 6.2.1 over $\mathbb{Z}$. By part (c) of Proposition 6.2.3, we deduce that

$$\dim_{\mathbb{C}} H_{r_1+r_2}(\Gamma, V_\lambda) = O\left(\frac{\dim_{\mathbb{C}} V_\lambda}{\min \lambda}\right) = O\left(\frac{\prod_{\delta \in I} (k_\delta - 1)}{\min\{k_\delta - 2 : \delta \in I\}}\right) = O\left(\frac{\prod_{\tau \in I_{\mathbb{R}}} k_\tau \times \prod_{\sigma \in I_{\mathbb{C}}} k_\sigma^2}{\min\{k_\delta : \delta \in I_\infty\}}\right). \quad \square$$

Appendix A

Twisted von Neumann ranks

If Γ is a countable group, let $\ell^2(\Gamma)$ be the complex Hilbert space on an orthonormal basis given by the elements of Γ . Hence, Γ acts on $\ell^2(\Gamma)$ on both sides as unitary operators. Any $r \times s$ matrix A over $\mathbb{C}[\Gamma]$ induces a bounded linear operator $\phi_\Gamma^A \ell^2(\Gamma)^s \rightarrow \ell^2(\Gamma)^r$ between Hilbert spaces by left multiplication. The *von Neumann rank* of A is defined as

$$\mathrm{rk}_\Gamma = \dim_\Gamma \overline{\mathrm{im} \phi_\Gamma^A} = s - \dim_\Gamma \ker \phi_\Gamma^A,$$

where $\dim_\Gamma$ is the *von Neumann dimension*¹ of Γ . If Γ is not countable, then for every matrix A over $\mathbb{C}[\Gamma]$ we define $\mathrm{rk}_\Gamma A$ to be $\mathrm{rk}_{\Gamma'} A$ where Γ' is any finitely generated subgroup of Γ containing the elements that appear in the entries of A . This does not depend on the choice of Γ' .²

We recall that if Γ is finite, then $\ell^2(\Gamma) = \mathbb{C}[\Gamma]$, a Hilbert Γ -module V is simply a $\mathbb{C}[\Gamma]$ -module and we have that

$$\dim_\Gamma V = \frac{\dim_{\mathbb{C}} V}{|\Gamma|} \text{ and } \mathrm{rk}_\Gamma A = \frac{\mathrm{rk}_{\mathbb{C}} A}{|\Gamma|}.$$

Recall the Lück approximation Theorem 3.1.16: Let $\Gamma_1 > \Gamma_2 > \dots$ be a chain of normal subgroups of Γ with finite index such that $\bigcap \Gamma_i = \{1\}$, and for each matrix A over $\mathbb{C}[\Gamma]$ let A_i denote its image over $\mathbb{C}[\Gamma/\Gamma_i]$. Then we have that

$$\lim_{i \rightarrow \infty} \mathrm{rk}_{\Gamma/\Gamma_i} A_i = \lim_{i \rightarrow \infty} \frac{\mathrm{rk}_{\mathbb{C}} A_i}{|\Gamma/\Gamma_i|} = \mathrm{rk}_\Gamma A.$$

Let Z be a finite central subgroup of Γ . For any character $\chi: Z \rightarrow \mathbb{C}^\times$, we consider the trace $\mathrm{tr}_\chi: \Gamma \rightarrow \mathbb{C}$ on Γ given by

$$\mathrm{tr}_\chi(g) = \begin{cases} \chi(g), & \text{if } g \in Z, \\ 0, & \text{otherwise.} \end{cases}$$

¹Lück, *L2-invariants*, Def. 1.10.

²Ibid., Thm. 6.29.

Linearly extending this formula to $\mathbb{C}[\Gamma]$ turns it into a tracial $*$ -algebra, to which we will associate a Hilbert space $\ell^2(\Gamma)_\chi$ and a cyclic representation $\mathbb{C}[\Gamma] \rightarrow \mathcal{B}(\ell^2(\Gamma)_\chi)$ to its algebra of bounded operators. Recall that a trace tr is *faithful* if $\text{tr}(x^*x) = 0$ implies $x = 0$. The set

$$N = \ker(\text{tr}_\chi) = \{x \in \mathbb{C}[\Gamma] : \text{tr}_\chi(x^*x) = 0\}$$

is a two-sided ideal of $\mathbb{C}[\Gamma]$ and tr_χ induces a faithful trace on $\mathbb{C}[\Gamma]/N$. We will make use of the Gelfand-Neumann-Segal (or GNS for short) construction, given by the following proposition.³

Proposition A.o.1. *Let $\mathcal{A}$ be a $*$ -algebra with a faithful trace tr . Assume that $\mathcal{A}$ is generated by unitary elements, i.e. the elements $x \in \mathcal{A}$ such that $x^*x = xx^* = 1$. Then, up to unitary equivalence, there exists a unique Hilbert space $\mathcal{H}$ and two commuting representations $\rho, \lambda : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H})$ with a cyclic generating vector $e \in \mathcal{H}$ such that*

$$\langle \lambda(x)(e), e \rangle = \langle \rho(x)(e), e \rangle = \text{tr}(x)$$

for every $x \in \mathcal{A}$.

In the previous proposition, the Hilbert space $\mathcal{H}$ is the Hilbert completion of $\mathcal{A}$ with respect to the inner product

$$\langle x, y \rangle = \text{tr}(y^*x),$$

the cyclic generating vector is the identity element $e \in \mathcal{A}$ and the representations are the extensions of the right and left regular action of $\mathcal{A}$ on itself. Before continuing with this construction, we list some properties of tr_χ .

Lemma A.o.2. *Let Γ, Z, χ and tr_χ be as before.*

(a) *The kernel N of the trace tr_χ is the two-sided ideal of $\mathbb{C}[\Gamma]$ generated by the central elements $\{z - \chi(z) : z \in Z\}$.*

(b) *Let T be a transversal for Γ/Z . Then, the image $T + N$ of T in the quotient $\mathbb{C}[\Gamma]_\chi = \mathbb{C}[\Gamma]/N$ is a basis consisting of unitary elements.*

Proof. Fix a transversal T for Γ/Z . It is clear that the image of T is a basis for

$$\mathbb{C}[\Gamma]_\chi = \mathbb{C}[\Gamma] / \mathbb{C}[\Gamma]\{z - \chi(z) : z \in Z\}.$$

We want to prove that $\mathbb{C}[\Gamma]_\chi = \mathbb{C}[\Gamma]/N$. Observe that N contains every element of the form $z - \chi(z)$, and therefore the surjection $\mathbb{C}[\Gamma] \rightarrow \mathbb{C}[\Gamma]_\chi$ factors through $\mathbb{C}[\Gamma]/N$.

³Alexandru Nica and Roland Speicher. *Lectures on the Combinatorics of Free Probability*. Vol. 335. London Mathematical Society Lecture Note Series. Cambridge: Cambridge University Press, 2006. ISBN: 9780521858526. DOI: 10.1017/CB09780511735127, Prop. 7.2 and 7.6.

Let $x = \sum_{t \in T} \sum_{z \in Z} a_{tz} tz$ be an element of $\mathbb{C}[\Gamma]$ and $\bar{x} = \sum_{t \in T} \sum_{z \in Z} a_{tz} t\chi(z)$ be its image in $\mathbb{C}[\Gamma]_\chi$. Since T is a basis of $\mathbb{C}[\Gamma]_\chi$, we get that $\bar{x}$ is zero if and only if

$$\sum_{t \in T} \left\| \sum_{z \in Z} a_{tz} \chi(z) \right\|^2 = \sum_{t \in T} \sum_{z, w \in Z} \overline{a_{tw}} a_{tz} \chi(w^{-1}z) = 0.$$

Writing $x^* = \sum_{s \in T} \sum_{w \in Z} \overline{a_{sw}} s^{-1} w^{-1}$, we get that

$$\begin{aligned} \text{tr}_\chi(x^* x) &= \text{tr}_\chi \left(\sum_{t, s \in T} \sum_{z, w \in Z} \overline{a_{sw}} a_{tz} s^{-1} t w^{-1} z \right) \\ &= \text{tr}_\chi \left(\sum_{t \in T} \sum_{z, w \in Z} \overline{a_{tw}} a_{tz} w^{-1} z \right), \text{ since } \text{tr}_\chi(s^{-1} t w^{-1} z) \neq 0 \iff s = t, \\ &= \sum_{t \in T} \sum_{z, w \in Z} \overline{a_{tw}} a_{tz} \chi(w^{-1}z). \end{aligned}$$

Hence, $\bar{x} = 0$ if and only if $\text{tr}_\chi(x^* x) = 0$, showing that the surjection $\mathbb{C}[\Gamma]/N \rightarrow \mathbb{C}[\Gamma]_\chi$ is an isomorphism and proving (a). For (b), it remains to check that the elements $t + N$ for $t \in T$ are unitary, which is immediate since $(t + N)^* = t^{-1} + N$. $\square$

Hence, by Proposition A.o.1 we can associate to $\mathbb{C}[\Gamma]_\chi$ two commuting representations as bounded operators of a Hilbert space $\ell^2(\Gamma)_\chi$, which are unique up to unitary equivalence. Let $\mathcal{N}(\Gamma)_\chi$ be the weak closure of the left regular representation $\lambda(\mathbb{C}[\Gamma]_\chi) \leq \mathcal{B}(\ell^2(\Gamma)_\chi)$. This is a von Neumann algebra, that is, a weakly closed subset of $\mathcal{B}(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. We call $\ell^2(\Gamma)_\chi$ the *twisted ℓ^2 -completion* of $\mathbb{C}[\Gamma]$ and $\mathcal{N}(\Gamma)_\chi$ the *twisted group von Neumann algebra*.

Observe that the trace tr_χ extends to $\mathcal{N}(\Gamma)_\chi$ by setting

$$\text{tr}_\chi(A) = \langle A(1), 1 \rangle$$

for any operator $\mathcal{N}(\Gamma)_\chi$. Any closed right Γ -invariant subspace $M \leq \ell^2(\Gamma)_\chi$ comes with an orthogonal projection $P_M: \ell^2(\Gamma)_\chi \rightarrow M$, which is the unique element in $\mathcal{N}(\Gamma)_\chi$ such that $P_M^2 = P_M = P_M^*$ and $\text{im}(P_M) = M$. We define the *twisted von Neumann dimension* of M to be

$$\dim_\Gamma^\chi(M) = \text{tr}_\chi(P_M).$$

For a fixed $A \in \mathcal{N}(\Gamma)_\chi$, we define its *twisted von Neumann rank* as

$$\text{rk}_\Gamma^\chi(A) = \dim_\Gamma^\chi \overline{\text{im}(A)} = 1 - \dim_\Gamma^\chi \ker(A).$$

Note that the twisted von Neumann dimension extends naturally to closed right Γ -invariant subspaces $M \leq \ell^2(\Gamma)_\chi^n$ by setting

$$\dim_\Gamma^\chi(M) = \text{tr}_\chi(P_M) = \sum_{i=1}^n \langle P_M(e_i), e_i \rangle,$$

where $P_M \in \text{Mat}_{n \times n}(\mathcal{N}(\Gamma)_\chi)$ is the unique element such that $P_M^2 = P_M = P_M^*$ and each e_i is the element of $\ell^2(\Gamma)_\chi$ containing 1 in the i -th coordinate and zero on the remaining ones. Hence, the twisted von Neumann rank is also defined for matrices over $\mathcal{N}(\Gamma)_\chi$.

If Γ is not countable, we can still define the twisted von Neumann rank for matrices A over $\mathbb{C}[\Gamma]$. Indeed, let Γ' be any countable subgroup of Γ containing the elements in the entries of A , $Z' = Z \cap \Gamma'$ and χ' be the restriction of χ to Z' . Then, we define $\text{rk}_\Gamma^\chi(A)$ to be $\text{rk}_{\Gamma'}^{\chi'}(A)$. The following lemma implies that $\text{rk}_\Gamma^\chi(A)$ is well-defined and does not depend on the choice of Γ' .

Lemma A.o.3. *Let Γ' , Z' and χ' be as above. Fix a subgroup $\Gamma'' \leq \Gamma'$ and let $Z'' = Z' \cap \Gamma''$ and χ'' be the restriction of χ' to Z'' . The inclusion $\Gamma'' \rightarrow \Gamma'$ induces a functor I from the category of closed $\mathcal{N}(\Gamma'')_{\chi''}$ -submodules of $\ell^2(\Gamma'')_{\chi''}^n$ to the category of closed $\mathcal{N}(\Gamma')_{\chi'}$ -submodules of $\ell^2(\Gamma')_{\chi'}^n$ for any n . Moreover, we have that*

$$\dim_{\Gamma''}^{\chi''} M = \dim_{\Gamma'}^{\chi'} I(M).$$

Proof. First, observe that the Hilbert completion of the pre-Hilbert structure on

$$\ell^2(\Gamma'')_{\chi''} \otimes_{\mathbb{C}[\Gamma'']_{\chi''}} \mathbb{C}[\Gamma']_{\chi'} \simeq_{\mathbb{C}} \bigoplus_{\Gamma'/\Gamma''Z'} \ell^2(\Gamma'')_{\chi''}$$

is $\mathcal{N}(\Gamma')_{\chi'}$ -isometrically isomorphic to $\ell^2(\Gamma')_{\chi'}$. Indeed, if T'' is a transversal for Γ''/Z'' and X is a transversal for $\Gamma'/\Gamma''Z'$, then $T' = T''X$ is a transversal for Γ'/Z' and the map

$$\bigoplus_{\Gamma'/\Gamma''Z'} \ell^2(\Gamma'')_{\chi''} \rightarrow \ell^2(\Gamma')_{\chi'}$$

given by

$$(\alpha_x)_{x \in X} \mapsto \sum_{x \in X} \alpha_x x$$

is injective, $\mathbb{C}[\Gamma']_{\chi'}$ equivariant and has dense image. Hence, if M is a closed $\mathcal{N}(\Gamma'')_{\chi''}$ -submodule of $\ell^2(\Gamma'')_{\chi''}^n$, then the Hilbert completion $I(M)$ of $M \otimes_{\mathbb{C}[\Gamma'']_{\chi''}} \mathbb{C}[\Gamma']_{\chi'}$ is a closed $\mathcal{N}(\Gamma')_{\chi'}$ -submodule of $\ell^2(\Gamma')_{\chi'}^n$.

In particular, I also induces an homomorphism of $*$ -algebras $I: \mathcal{N}(\Gamma'')_{\chi''} \rightarrow \mathcal{N}(\Gamma')_{\chi'}$. We will now show that, for any $A \in \mathcal{N}(\Gamma'')_{\chi''}$, we have

$$\text{tr}_{\chi''}(A) = \text{tr}_{\chi'}(I(A)).$$

By continuity and additivity of the trace, it suffices to check this condition for $A = g \in \Gamma''$. This follows immediately from the definition of the action of $I(g)$ on $\bigoplus_{\Gamma'/\Gamma''Z'} \ell^2(\Gamma'')_{\chi''}$ and hence on its Hilbert completion.

Now, for $M \leq \ell^2(\Gamma'')_{\chi''}$ we have that $\dim_{\Gamma''}^{\chi''} M = \text{tr}_{\chi''}(P_M)$ where $P_M \in \text{Mat}_{n \times n}(\mathcal{N}(\Gamma'')_{\chi''})$ is the unique element satisfying $P_M^2 = P_M = P_M^*$ and $\text{im}(P_M) = M$. Since $I(P_M)$ satisfies $I(P_M)^2 = I(P_M) = I(P_M)^*$ and $\text{im}(P_M) = I(M)$, we must have $P_{I(M)} = I(P_M)$ and therefore

$$\dim_{\Gamma'}^{\chi'} I(M) = \text{tr}_{\chi'}(P_{I(M)}) = \text{tr}_{\chi'}(I(P_M)) = \text{tr}_{\chi''}(P_M) = \dim_{\Gamma''}^{\chi''}(M). \quad \square$$

Remark A.o.4. If $Z = \{1\}$, then tr_χ is simply the characteristic function $\text{tr}_\chi(g) = \delta_{1,g}$ and $\ell^2(\Gamma)_\chi \simeq \ell^2(\Gamma)$. Hence, in this case the twisted von Neumann rank rk_Γ^χ coincides with the classical von Neumann rank rk_Γ . Even if $Z \neq \{1\}$, if Γ' is a subgroup of Γ that intersects Z trivially and A is a matrix over $\mathbb{C}[\Gamma']$, then we have that

$$\text{rk}_\Gamma^\chi(A) = \text{rk}_{\Gamma'}(A)$$

by Lemma A.o.3. This is particularly the case whenever Γ' is torsion-free.

When Γ has torsion central elements, then distinct central characters lead to distinct rank functions. This is why the hypothesis on central characters is essential in Conjectures 6.2.1 and 6.2.2. As an example, consider $\Gamma = Z$ the cyclic group of order two inside $\text{SL}_2(\mathbb{C})$ generated by $g = -\text{Id}$ and the element $g - 1$ inside $\mathbb{C}[\Gamma]$. Then, for the trivial character $1(g) = 1$, we have that $\text{rk}_\Gamma^1(g-1) = 0$, while for the non-trivial character $-1(g) = -1$ we have that $\text{rk}_\Gamma^{-1}(g-1) = 1$. This reflects on the cohomology of Γ with coefficients in $\text{Sym}^k(\mathbb{C}^2)$ by the appearance of two clearly converging subsequences

$$\frac{\dim_{\mathbb{C}} H^0(\Gamma, \text{Sym}^k(\mathbb{C}^2))}{\dim_{\mathbb{C}} \text{Sym}^k(\mathbb{C}^2)} = \begin{cases} 1, & \text{if } k \text{ is even,} \\ 0, & \text{if } k \text{ is odd,} \end{cases}$$

where the parity of k determines whether Γ acts on $\text{Sym}^k(\mathbb{C}^2)$ through 1 or -1 . ■

We recall that a von Neumann algebra whose center is exactly $\mathbb{C}$ is called a *factor*. It is a well known fact that the group von Neumann algebra $\mathcal{N}(\Gamma)$ is a factor if and only if Γ is an ICC group, i.e. every the conjugacy class of any non-trivial element is infinite. In particular, Γ/Z is ICC if and only if $\mathcal{N}(\Gamma)_1$ is a factor. We show that one direction still holds for all twisted group von Neumann algebras.

Lemma A.o.5. *If Γ/Z is ICC then $\mathcal{N}(\Gamma)_\chi$ is a factor.*

Proof. After modding out Γ by the kernel of χ , we may assume that χ is injective. We will construct a special transversal for Γ/Z . First, let T_0 be a set of representatives on Γ for the conjugacy classes of Γ/Z . For any $t \in T_0$, let S_t be a set of left coset representatives of the stabilizer

$$\text{Stab}_\Gamma(tZ) = \{b \in \Gamma : btb^{-1} \in tZ\}$$

of the coset tZ under the conjugation action of Γ . Observe that the centralizer $C_\Gamma(t)$ of t is contained in $\text{Stab}_\Gamma(tZ)$. Then, the set

$$T = \{sts^{-1} : t \in T_0, s \in S_t\}$$

is a complete set of representatives for the quotient Γ/Z containing 1.

Fix an element $A \in \mathcal{N}(\Gamma)_\chi$. When we restrict A to the dense subset $\mathbb{C}[\Gamma]_\chi$, it acts as left multiplication by $x = A(1) \in \ell^2(\Gamma)_\chi$. Hence, it suffices to show that the only elements in $\ell^2(\Gamma)_\chi$ fixed under the conjugation action of Γ are the constant multiples of the identity.

Write a fixed element $x \in \ell^2(\Gamma)_\chi$ uniquely as

$$x = \sum_{t \in T_0} \sum_{s \in S_t} a_{sts^{-1}} sts^{-1}$$

with $a_{sts^{-1}} \in \mathbb{C}$. If h is any element of Γ , then for every $t \in T_0$ and every $s \in S_t$ there are unique elements $r = r(h, t, s) \in S_t$ and $u = u(h, t, s) \in Z$ such that

$$hsts^{-1}h^{-1} = rtr^{-1}u.$$

If for some h, t, s the element u is non-trivial, then $r^{-1}hs$ is an element of $\text{Stab}_\Gamma(tZ)$ that does not belong to $C_\Gamma(t)$. We will show that if x is central then $a_{sts^{-1}} = 0$ for $t \neq 1$ arbitrary. Fixing one such t , there are two cases to consider:

Case 1. There exists an element $h \in \text{Stab}_\Gamma(tZ) \setminus C_\Gamma(t)$.

In this case, we have that $tht^{-1} = tz$ for some non-trivial $z \in Z$. This also implies that for any $\tilde{s} \in S_t$ the element $\tilde{s}h\tilde{s}^{-1}$ belongs to $\text{Stab}_\Gamma(\tilde{s}t\tilde{s}^{-1}Z) \setminus C_\Gamma(\tilde{s}t\tilde{s}^{-1})$. Moreover, if for some $s \in S_t$ we have

$$(\tilde{s}h\tilde{s}^{-1})(sts^{-1})(\tilde{s}h^{-1}\tilde{s}^{-1}) = \tilde{s}t\tilde{s}^{-1}z'$$

for some $z' \in Z$, then $h\tilde{s}^{-1}s \in \text{Stab}_\Gamma(tZ)$ and therefore $s = \tilde{s}$ by our definition of S_t . We now compute

$$\begin{aligned} (\tilde{s}h\tilde{s}^{-1})x(\tilde{s}h^{-1}\tilde{s}^{-1}) &= \sum_{s \in S_t} a_{sts^{-1}}(\tilde{s}h\tilde{s}^{-1})(sts^{-1})(\tilde{s}h^{-1}\tilde{s}^{-1}) + \sum_{t \neq t' \in T_0} \dots \\ &= a_{\tilde{s}t\tilde{s}^{-1}}\chi(z)\tilde{s}t\tilde{s}^{-1} + \sum_{\tilde{s} \neq \tilde{r} \in S_t} \tilde{r}t\tilde{r}^{-1}\chi(\tilde{u}) + \sum_{t \neq t' \in T_0} \dots \end{aligned}$$

Thus, if $(\tilde{s}h\tilde{s}^{-1})x(\tilde{s}h^{-1}\tilde{s}^{-1}) = x$, we have that $a_{\tilde{s}t\tilde{s}^{-1}}\chi(z) = a_{\tilde{s}t\tilde{s}^{-1}}$, which is only possible if $a_{\tilde{s}t\tilde{s}^{-1}} = 0$. Since the choice of $\tilde{s}$ was arbitrary, we are done with this case.

Case 2. We have $\text{Stab}_\Gamma(tZ) = C_\Gamma(t)$.

In particular, for any $s \in S_t$, we also have

$$\text{Stab}_\Gamma(sts^{-1}Z) = {}_s\text{Stab}_\Gamma(tZ)s^{-1} = {}_sC_\Gamma(t)s^{-1} = C_\Gamma(sts^{-1}).$$

This shows that for every $h \in \Gamma$ and every $s \in S_t$ there is a unique $r = r(h, s) \in S_t$ such that $hsts^{-1}h^{-1} = rtr^{-1}$. Therefore, for h fixed, the map $s \mapsto r(h, s)$ is a bijection in S_t . Take any $h \in \Gamma$ and consider the element

$$\begin{aligned} hxb^{-1} &= \sum_{s \in S_t} a_{sts^{-1}}hsts^{-1}h^{-1} + \sum_{t \neq t' \in T_0} \dots \\ &= \sum_{s \in S_t} a_{r(b^{-1},s)tr(b^{-1},s)^{-1}}sts^{-1} + \sum_{t \neq t' \in T_0} \dots \end{aligned}$$

If $hxb^{-1} = x$, then by our choice of T_0 for every $s \in S_t$ we must have $a_{r(b^{-1},s)tr(b^{-1},s)^{-1}} = a_{sts}$. This implies that the coefficients a_{sts} are constant in s . Since $t \neq 1$, its conjugacy class is infinite, and so we must have infinitely many repetitions of the coefficient a_t in x . As the coefficients of x must form a square-summable sequence in $\mathbb{C}$, this is only possible if $a_{sts} = 0$ for all $s \in S_t$. $\square$

Remark A.o.6. The “only if” direction of the previous lemma is false in the twisted setting. For example, if

$$\Gamma = Q_8 = \langle e, i, j, k | e^2 = 1, i^2 = j^2 = k^2 = ijk = e \rangle$$

denotes the quaternion group, then its center $Z = Z(\Gamma) = \langle e \rangle$ is cyclic of order two. Now, for the character $\chi: Z \rightarrow \mathbb{C}^\times$ given by $\chi(e) = -1$, one gets that $\mathbb{C}[\Gamma]_\chi$ is a four dimensional quaternion algebra over $\mathbb{C}$, and in particular a central simple algebra. The finite groups Γ for which there exists a central character $\chi: Z(\Gamma) \rightarrow \mathbb{C}^\times$ such that $Z(\mathbb{C}[\Gamma]_\chi) = \mathbb{C}$ are called *groups of central type*. $\blacksquare$

A $\ast$ -algebra is called $\ast$ -regular if it is a von Neumann regular ring and its involution $\ast$ is proper, i.e. $x^\ast x = 0$ if and only if $x = 0$. In the case of the twisted von Neumann algebra $\mathcal{N}(\Gamma)_\chi$, it was shown by Holger Reich⁴ that the set $S \subset \mathcal{N}(\Gamma)$ of non-zero divisors satisfies the right Ore condition, and thus we can form the Ore localization $\mathcal{U}(\Gamma)_\chi = \mathcal{N}(\Gamma)_\chi S^{-1}$. The following lemma explains the properties of $\mathcal{U}(\Gamma)_\chi$.

Lemma A.o.7. *The following assertions are true:*

- (a) *The map $\iota: \mathcal{N}(\Gamma)_\chi \rightarrow \mathcal{U}(\Gamma)_\chi$ is injective.*
- (b) *The involution $\ast$, the twisted von Neumann dimension $\dim_\Gamma^\chi$ and the twisted von Neumann rank rk_Γ^χ extend from $\mathcal{N}(\Gamma)_\chi$ to $\mathcal{U}(\Gamma)_\chi$.*
- (c) *The ring $\mathcal{U}(\Gamma)_\chi$ is $\ast$ -regular.*
- (d) *If $\mathcal{N}(\Gamma)_\chi$ is a factor, then the center of $\mathcal{U}(\Gamma)_\chi$ is $Z(\mathcal{N}(\Gamma)_\chi) \simeq \mathbb{C}$, which equals the centralizer of the image of Γ in $\mathcal{U}(\Gamma)_\chi$.*

Proof. For (a), note that the kernel of $\mathcal{N}(\Gamma)_\chi \rightarrow \mathcal{U}(\Gamma)_\chi$ consists of the elements $x \in \mathcal{N}(\Gamma)_\chi$ such that $xy = 0$ for some $y \in S$. However, the elements of S are not zero divisors. Assertions (b) and (c) can be found in Reich’s thesis,⁵ and for (d) we point the reader to Zhengwei Liu’s 2012 paper.⁶ $\square$

⁴Holger Reich. “Group von Neumann Algebras and Related Algebras”. PhD thesis. Göttingen, Germany: Universität Göttingen, 1998, Prop. 2.8.

⁵Ibid., Prop. 2.10, Note 2.11 and Prop. 3.9.

⁶Zhengwei Liu. “A double commutant theorem for Murray–von Neumann algebras”. In: *Proceedings of the National Academy of Sciences* 109.20 (2012), pp. 7676–7681. DOI: 10.1073/pnas.1119679109, Prop. 30.

The ring $\mathcal{U}(\Gamma)_\chi$ can be identified⁷ with the ring of unbounded operators affiliated to $\mathcal{N}(\Gamma)_\chi$, although we will not need this description. Observe that if Γ' is a subgroup of Γ and χ' is the restriction of χ to $Z' = Z \cap \Gamma'$, then the inclusion map $\mathbb{C}[\Gamma']_{\chi'} \rightarrow \mathbb{C}[\Gamma]_\chi$ induces an embedding $\mathcal{U}(\Gamma')_{\chi'} \rightarrow \mathcal{U}(\Gamma)_\chi$ by the universal property of the Ore localization.

We recall that a ring R' is called a *crossed product* of a ring R and a group G if R can be identified with a subring of R' and there is a map $\mu: G \rightarrow (R')^\times$ satisfying the following properties:

(CrPri) R' is a free left R -module on the set $\mu(G)$;

(CrPr2) for all $g \in G$, R is closed under conjugation by $\mu(g)$; and

(CrPr3) for every pair of elements $g, g' \in G$, the element

$$\tau(g, g') = \mu(g)\mu(g')\mu(gg')^{-1}$$

belongs to $R^\times$.

In particular, μ induces a function $\bar{\mu}: G \rightarrow \text{Aut}(R)$ that satisfies the following:

(CrPri') for all g, g' and g'' in G , we have that

$$\tau(g, g')\tau(gg', g'') = \bar{\mu}(g)(\tau(g', g''))\tau(g, g'g''),$$

in $R^\times$; and

(CrPr2') For all $g, g' \in G$ and $r \in R$, we have that

$$\tau(g, g')\bar{\mu}(gg')(r)\tau(g, g')^{-1} = \bar{\mu}(g)(\bar{\mu}(g')(r)).$$

Conversely, given $\bar{\mu}: G \rightarrow \text{Aut}(R)$ and $\tau: G \times G \rightarrow R^\times$ satisfying (CrPri') and (CrPr2'), we can construct a ring $R * G$ with underlying set the free left R -module on the set G and multiplication obtained by linearly extending

$$(rg) \cdot (r'g') = (r\mu(g)(r')\tau(g, g'))gg'$$

for $g, g' \in G$ and $r, r' \in R$. If R' is a crossed product of R and G , then there is a unique isomorphism of rings $R * G \rightarrow R'$ extending μ on G and the identity on R , where $R * G$ is constructed using the maps induced from R' . If R' is a crossed product of R and G and $\psi: R' \rightarrow \bigoplus_{g \in G} R\mu(g)$ is a left R -module isomorphism, we say that (ψ, μ, τ) is an (*internal*) *twisting data*. A pair $(\bar{\mu}, \tau)$ satisfying (CrPri') and (CrPr2') is called an (*external*) *twisting data*. The previous considerations show that there exists a bijection between internal and

⁷Reich, "Group von Neumann Algebras and Related Algebras", Sec. 2.

external twisting datum, and any twisting data determines a unique crossed product up to R -ring isomorphism.

The crossed product satisfies the following universal property: if $(\bar{\mu}, \tau)$ is a twisting data of G on R , $\varphi: R \rightarrow R'$ is a ring homomorphism and $\nu: G \rightarrow (R')^\times$ is any function such that (1) $\nu(g)\varphi(r)\nu(g)^{-1} = \varphi(\bar{\mu}(g)(r))$ and (2) $\varphi(\tau(g, g')) = \nu(g)\nu(g')\nu(gg')^{-1}$, then there exist a unique ring homomorphism $R * G \rightarrow R'$ extending φ and ν .

Suppose now that R is contained in a larger ring R' and $(\bar{\mu}, \tau)$ is a twisting data of G on R . If there exists a function $\bar{\mu}': G \rightarrow \text{Aut}(R')$ such that (a) for every $g \in G$ and $r \in R$ we have $\bar{\mu}'(g)(r) = \bar{\mu}(g)(r)$ and (b) the pair $(\bar{\mu}', \tau)$ satisfies (CrPrz') on R' , then it also satisfies (CrPri'). Then, get a unique injective ring map $R * G \rightarrow R' * G$ extending the identity on both R and G where the crossed products are constructed using their respective twisting data. In this case, we say that the crossed product structure on R *extends* to a crossed product structure on R' .

We will now explore how the rings $\mathbb{C}[\Gamma]_\chi$, $\mathcal{N}(\Gamma)_\chi$ and $\mathcal{U}(\Gamma)_\chi$ decomposed as crossed products over finite quotients of Γ . Let Γ' be a finite index normal subgroup of Γ , and if $\chi: Z \rightarrow \mathbb{C}^\times$ denotes the central character, we let $Z' = Z \cap \Gamma'$ and χ' be the restriction of χ to Z' . By taking $\Gamma_\chi = \Gamma/Z$ and $\Gamma'_{\chi'} = \Gamma'/Z'$, we have an embedding $\Gamma'_{\chi'} \leq \Gamma_\chi$ and the quotient is isomorphic to $\Gamma/(\Gamma'Z)$.

Lemma A.o.8. *The following are true:*

(a) *The conjugation action of Γ on Γ' induces a twisting data $\bar{\mu}: \Gamma_\chi/\Gamma'_{\chi'} \rightarrow \text{Aut}(\mathbb{C}[\Gamma']_{\chi'})$ and $\tau: (\Gamma_\chi/\Gamma'_{\chi'})^2 \rightarrow \mathbb{C}[\Gamma']_{\chi'}^\times$ and $\mathbb{C}[\Gamma]_\chi \simeq \mathbb{C}[\Gamma']_{\chi'} * (\Gamma_\chi/\Gamma'_{\chi'})$.*

(b) *The crossed product structure on $\mathbb{C}[\Gamma']_{\chi'}$ extends to a crossed product structure on $\mathcal{N}(\Gamma')_{\chi'}$ and $\mathcal{U}(\Gamma')_{\chi'}$.*

(c) *The subring of $\mathcal{U}(\Gamma)_\chi$ generated by $\mathcal{U}(\Gamma')_{\chi'}$ and Γ is isomorphic to the crossed product $\mathcal{U}(\Gamma')_{\chi'} * (\Gamma_\chi/\Gamma'_{\chi'})$.*

Proof. (a) Fix a set of representatives T' for Γ'/Z' and X for $\Gamma/\Gamma'Z$ so that $T = T'X$ is a transversal for Γ/Z and the projection $\Gamma \rightarrow \Gamma_\chi/\Gamma'_{\chi'}$ is bijective when restricted to X . For each $g \in \Gamma_\chi/\Gamma'_{\chi'}$, let x_g be its representative in X . For any pair g and g' in $\Gamma_\chi/\Gamma'_{\chi'}$ there is a unique element $u = u(g, g') \in \Gamma'Z$ such that $x_g x_{g'} = x_{gg'} u$ in Γ . Since $\Gamma'_{\chi'}$ is isomorphic to $\Gamma'Z/Z$, one can also find unique elements $h = h(g, g') \in T_0$ and $z = z(g, g') \in Z$ such that $u = zh$.

Define $\bar{\mu}(g) \in \text{Aut}(\mathbb{C}[\Gamma']_{\chi'})$ to be conjugation by x_g . Set $\tau(g, g') = \chi(z)h$, where z and h are as above. A straightforward computation shows that $(\bar{\mu}, \tau)$ satisfies (CrPri') and (CrPrz'), and so we can build a crossed product $\mathbb{C}[\Gamma']_{\chi'} * \Gamma_\chi/\Gamma'_{\chi'}$. The injective map $\mathbb{C}[\Gamma']_{\chi'} \rightarrow \mathbb{C}[\Gamma]_\chi$ and the function $\nu: \Gamma_\chi/\Gamma'_{\chi'} \rightarrow \mathbb{C}[\Gamma]_\chi^\times$ given by $\nu(g)x_g$ induce by the universal property of the crossed product a ring homomorphism $\mathbb{C}[\Gamma']_{\chi'} * \Gamma_\chi/\Gamma'_{\chi'} \rightarrow \mathbb{C}[\Gamma]_\chi$ which is bijective in view of part (b) of Lemma A.o.2.

(b) Since the image of x_g in both $\mathcal{N}(\Gamma)_\chi$ and $\mathcal{U}(\Gamma)_\chi$ is a unit that normalizes Γ' , one only needs to show that $(\bar{u}, \tau)$ satisfies (CrPr2') on $\mathcal{N}(\Gamma')_{\chi'}$ and $\mathcal{U}(\Gamma')_{\chi'}$ respectively. For $\mathcal{N}(\Gamma')_{\chi'}$, it suffices to show that the actions of both sides of (CrPr2'), seen as operators in $\ell^2(\Gamma')_{\chi'}$, coincide on the dense subset $\mathbb{C}[\Gamma']_{\chi'}$. In fact, it suffices to check that they send the identity element to the same element, which is immediate from the definitions. For $\mathcal{U}(\Gamma')_{\chi'}$, observe that both sides of (CrPr2') define for $r \in \mathcal{U}(\Gamma')_{\chi'}$ automorphisms of $\mathcal{U}(\Gamma')_{\chi'}$ that coincide in the image of $\mathcal{N}(\Gamma')_{\chi'}$, so they must be equal.

(c) This subring coincides with the image of the induced map $\varphi: \mathcal{U}(\Gamma')_{\chi'} * \Gamma_\chi / \Gamma'_{\chi'}$, so it suffices to show that φ is injective. This is equivalent to the statement that the elements of X are linearly independent over $\mathcal{N}(\Gamma')_{\chi'}$ as operators on $\ell^2(\Gamma)_\chi$. From the isomorphism

$$\ell^2(\Gamma)_\chi \simeq \bigoplus_{x \in X} \ell^2(\Gamma')_{\chi'}$$

of $\mathcal{N}(\Gamma')_{\chi'}$ -modules where X acts by permuting the entries and multiplying by u , the desired independence follows. $\square$

We recall that in a $*$ -regular ring R' , for every element $r \in R'$ there are unique⁸ projections e and f such that $R'e = R'r$ and $fR' = rR'$. Moreover, there exists⁹ a unique element $r^{[-1]}$, called the *relative inverse* of r , such that $e = r^{[-1]}r$ and $f = rr^{[-1]}$.

If R is a subring of a $*$ -regular ring R' , then there exists¹⁰ a unique minimal $*$ -regular subring $\mathcal{R}(R, R')$ of R' containing R , called the *$*$ -regular closure* of R in R' . This ring is constructed inductively,¹¹ by taking $\mathcal{R}_0(R, R') = R$ and $\mathcal{R}_i(R, R')$ the subring of R' generated by $\mathcal{R}_{i-1}(R, R')$ and all the relative inverses of xx^* for $x \in \mathcal{R}_i(R, R')$.

Lemma A.o.9. *Let S be a subring of $\mathbb{C}$ and Γ' be a finite index normal subgroup of Γ . Then, we also get a decomposition $S[\Gamma]_\chi \simeq S[\Gamma']_{\chi'} * (\Gamma_\chi / \Gamma'_{\chi'})$, the crossed product structure on $S[\Gamma']_{\chi'}$ extends to*

$$\mathcal{R}(\Gamma')_{\chi'}^S = \mathcal{R}(S[\Gamma']_{\chi'}, \mathcal{U}(\Gamma')_{\chi'})$$

*and the subring of $\mathcal{R}(\Gamma)_\chi^S$ generated by $\mathcal{R}(\Gamma')_{\chi'}^S$ and Γ is isomorphic to $\mathcal{R}(\Gamma')_{\chi'}^S * (\Gamma_\chi / \Gamma'_{\chi'})$.*

Proof. The decomposition of $S[\Gamma]_\chi$ as a crossed product is obtained in the same way as the decomposition of $\mathbb{C}[\Gamma]_\chi$ as a crossed product in part (a) of Lemma A.o.8.

Assume now that we have already extended the twisting data for the ring $S[\Gamma']_{\chi'}$ all the way to $\mathcal{R}_i(S[\Gamma']_{\chi'}, \mathcal{U}(\Gamma')_{\chi'})$. To show that it extends to $\mathcal{R}_{i+1}(S[\Gamma']_{\chi'}, \mathcal{U}(\Gamma')_{\chi'})$ all we must show

⁸Jaikin-Zapirain, “The base change in the Atiyah and the Lück approximation conjectures”, Prop. 3.2(3).

⁹Ibid., Prop. 3.2(4).

¹⁰Ibid., Sec. 3.4.

¹¹Pere Ara and Kenneth R. Goodearl. “The Realization Problem for Some Wild Monoids and the Atiyah Problem”. In: *Transactions of the American Mathematical Society* 369.8 (2017), pp. 5665–5710. DOI: 10.1090/tran/6793, Prop. 6.2.

is that this ring is invariant under conjugation by $x \in X$, where X is the chosen transversal in Γ of $\Gamma_\chi/\Gamma'_{\chi'}$, since it will inherit (CrPr2') from the crossed product decomposition of $\mathcal{U}(\Gamma')_{\chi'} * (\Gamma_\chi/\Gamma'_{\chi'})$.

Hence, let $r \in \mathcal{R}_i(S[\Gamma']_{\chi'}, \mathcal{U}(\Gamma')_{\chi'})$ be any self-adjoint element and $e, f \in \mathcal{U}(\Gamma')_{\chi'}$ be the projections such that $\mathcal{U}(\Gamma')_{\chi'} r = \mathcal{U}(\Gamma')_{\chi'} e$ and $r \mathcal{U}(\Gamma')_{\chi'} = f \mathcal{U}(\Gamma')_{\chi'}$. Then, for any $x \in X$, we have that xex^{-1} and xfx^{-1} must be the unique projections such that

$$\mathcal{U}(\Gamma')_{\chi'}(xrx^{-1}) = x(\mathcal{U}(\Gamma')_{\chi'} r)x^{-1} = x(\mathcal{U}(\Gamma')_{\chi'} e)x^{-1} = \mathcal{U}(\Gamma')_{\chi'}(xex^{-1}),$$

$$(xrx^{-1})\mathcal{U}(\Gamma')_{\chi'} = x(r\mathcal{U}(\Gamma')_{\chi'})x^{-1} = x(f\mathcal{U}(\Gamma')_{\chi'})x^{-1} = (xfx^{-1})\mathcal{U}(\Gamma')_{\chi'}.$$

This shows that $xr^{[-1]}x^{-1}$ must be the relative inverse of xrx^{-1} . Hence, we have that

$$xr^{[-1]}x^{-1} = (xrx^{-1})^{[-1]} \in \mathcal{R}_{i+1}(S[\Gamma']_{\chi'}, \mathcal{U}(\Gamma')_{\chi'})$$

as desired.

From the universal property of the crossed product, we get a surjective map

$$\mathcal{R}(\Gamma')_{\chi'}^S * \Gamma_\chi/\Gamma'_{\chi'} \rightarrow \langle \mathcal{R}(\Gamma')_{\chi'}^S, S[\Gamma]_{\chi} \rangle$$

which is the restriction of the injective map $\mathcal{U}(\Gamma')_{\chi'} * \Gamma_\chi/\Gamma'_{\chi'} \rightarrow \mathcal{U}(\Gamma)_\chi$, and so the last claim follows. $\square$

Remark A.o.10. Observe that if $\mathcal{N}(\Gamma)_\chi$ is a factor, then by part (d) of Lemma A.o.7 the center of $\mathcal{R}(\Gamma)_\chi^S$ is contained in the center of $\mathcal{U}(\Gamma)_\chi$. Since $\mathcal{R}(\Gamma)_\chi^S$ is $*$ -regular, this is a subfield of $\mathbb{C}$. $\blacksquare$

We can now describe how we are going to make use of Lemma A.o.5. We keep some notation from Section 6.2: $G(\mathbb{C})$ denotes a semisimple complex Lie group, $\chi: Z(G(\mathbb{C})) \rightarrow \mathbb{C}^\times$ a central character, $\mathfrak{g}$ the complex Lie algebra of $G(\mathbb{C})$ and $P(\mathfrak{g})_\chi$ the set of highest weights λ of the irreducible complex representations V_λ of $G(\mathbb{C})$ onto which $Z(G(\mathbb{C}))$ acts through χ .

Assume now that Γ is a Zariski-dense subgroup of $G(\mathbb{C})$ containing $Z = Z(G(\mathbb{C}))$, so that Γ/Z is a Zariski-dense subgroup of $G(\mathbb{C})/Z$. Let Γ' be a finite index subgroup of Γ and set $Z' = Z \cap \Gamma'$ and χ' the restriction of χ to Z' . Recall that the representations V_λ induce Sylvester matrix rank functions rk_λ on $S[\Gamma]$ and $S[\Gamma']$ for any subring S of $\mathbb{C}$. The twisted von Neumann ranks rk_Γ^χ and $\text{rk}_{\Gamma'}^{\chi'}$ are also Sylvester matrix rank functions on $S[\Gamma]$ and $S[\Gamma']$ respectively. The main goal of this appendix is to show:

Theorem A.o.11. *Let $\Gamma, \Gamma', \chi, \chi'$ and V_λ be as above. If rk_λ converges to $\text{rk}_{\Gamma'}^{\chi'}$ in $\mathbb{P}(S[\Gamma'])$ along the filter of cofinite subsets, then rk_λ converges to rk_Γ^χ in $\mathbb{P}(S[\Gamma])$ along the filter of cofinite subsets.*

Proof. Let ω be a non-principal ultrafilter on $P(\mathfrak{g})_\lambda$, and consider the ultraproduct of tracial $*$ -regular rings

$$\mathcal{D} = \prod_{\omega} \text{End}_{\mathbb{C}}(V_\lambda) .$$

We refer the reader to Jaikin-Zapirain's paper¹² for the construction of this ultraproduct. The maps $S[\Gamma'] \rightarrow \text{End}_{\mathbb{C}}(V_\lambda)$ induce a ring homomorphism $S[\Gamma'] \rightarrow \mathcal{D}$ whose $*$ -regular closure we will denote by $\mathcal{S}$. The fact that

$$\lim_{\omega} \text{rk}_{\lambda} = \text{rk}_{\Gamma'}^{\chi'}$$

on $\mathbb{P}(S[\Gamma'])$ imply¹³ that we have a rank preserving homomorphism $\varphi: \mathcal{R}(\Gamma')_{\chi'}^S \rightarrow \mathcal{S}$ such that the diagram

$$\begin{array}{ccc} \mathcal{R}(\Gamma')_{\chi'}^S & \xrightarrow{\varphi} & \mathcal{S} \\ \uparrow & \nearrow & \\ S[\Gamma'] & & \end{array}$$

commutes.

The maps $S[\Gamma] \rightarrow \text{End}_{\mathbb{C}}(V_\lambda)$ also induce a ring homomorphism $S[\Gamma] \rightarrow \mathcal{D}$ extending the map from $S[\Gamma']$. By the universal property of the crossed product, we get a homomorphism

$$\tilde{\varphi}: \mathcal{R}(\Gamma)_{\chi}^S \simeq \mathcal{R}(\Gamma')_{\chi'}^S * \Gamma_{\chi} / \Gamma_{\chi'} \rightarrow \mathcal{D}$$

that extends φ . By Lemma A.0.5 and Remark A.0.10, the center of $\mathcal{R}(\Gamma)_{\chi}^S$ is a subfield of $\mathbb{C}$. It was shown by Jaikin¹⁴ (see also Ken Goodearl's book¹⁵) that in this case $\mathcal{R}(\Gamma)_{\chi}^S$ admits at most one Sylvester matrix rank function. Since both $\text{rk}_{\Gamma}^{\chi}$ and $\lim_{\omega} \text{rk}_{\lambda}$ define rank functions on $\mathcal{R}(\Gamma)_{\chi}^S$ they must coincide. As the choice of ω was arbitrary, the result follows.¹⁶ $\square$

We finish this chapter with a proposition that allows us to deduce information about the twisted rank of central quotients.

Proposition A.0.12. *Let Γ, Z and χ be as above, choose $Z' \leq \ker \chi$ and define $\Gamma' = \Gamma / Z'$ with the induced character $\chi': Z / Z' \rightarrow \mathbb{C}^{\times}$. Let $s: \Gamma' \rightarrow \Gamma$ be a section such that $s(1) = 1$. Then, for any matrix A over $\mathbb{C}[\Gamma']$, if we denote by $s(A)$ the induced matrix over $\mathbb{C}[\Gamma]$ given by applying the section s to A entrywise, we have*

$$\text{rk}_{\Gamma}^{\chi}(s(A)) = \text{rk}_{\Gamma'}^{\chi'}(A) .$$

¹²Jaikin-Zapirain, "The base change in the Atiyah and the Lück approximation conjectures", Sec. 6.7.

¹³Ibid., Thm. 6.3.

¹⁴Ibid., Prop. 5.7.

¹⁵K. R. Goodearl. *Von Neumann Regular Rings*. Vol. 4. Monographs and Studies in Mathematics. London: Pitman, 1979. ISBN: 9780273084006, Thm. 19.13 and 19.14.

¹⁶Bourbaki, *Elements of Mathematics*, Prop. I.6.7.

Proof. Let tr_χ and $\text{tr}_{\chi'}$ be the respective traces on $\mathbb{C}[\Gamma]$ and $\mathbb{C}[\Gamma']$, $\pi: \mathbb{C}[\Gamma] \rightarrow \mathbb{C}[\Gamma']$ be the projection map and $s: \mathbb{C}[\Gamma'] \rightarrow \mathbb{C}[\Gamma]$ be the injective linear extension of s to a map of $\mathbb{C}$ -vector spaces. We claim that for any $x \in \mathbb{C}[\Gamma]$, we have

$$\text{tr}_\chi(x) = \text{tr}_{\chi'}(\pi(x)) .$$

Indeed, since tr is a linear map, it suffices to check this equality for $x \in \Gamma$. If $x \notin Z$, then $\pi(x) \notin Z/Z'$ since $Z = \pi^{-1}(Z/Z')$ and both sides of the equation are zero. If $x \in Z$, then since $\chi = \chi' \circ \pi$ we have:

$$\text{tr}_\chi(x) = \chi(x) = \chi'(\pi(x)) = \text{tr}_{\chi'}(\pi(x)) .$$

Hence, the faithful tracial $*$ -algebras $\mathbb{C}[\Gamma]_\chi$ and $\mathbb{C}[\Gamma']_{\chi'}$ are isomorphic under the map induced by π . In particular, the canonical projection map $\mathbb{C}[\Gamma'] \rightarrow \mathbb{C}[\Gamma']_{\chi'}$ coincides with the composition

$$\mathbb{C}[\Gamma'] \xrightarrow{s} \mathbb{C}[\Gamma] \rightarrow \mathbb{C}[\Gamma]_\chi \xrightarrow{\pi} \mathbb{C}[\Gamma']_{\chi'} .$$

The isomorphism $\pi: \mathbb{C}[\Gamma]_\chi \rightarrow \mathbb{C}[\Gamma']_{\chi'}$ extends to an isomorphism $\pi: \mathcal{N}(\Gamma)_\chi \rightarrow \mathcal{N}(\Gamma')_{\chi'}$ of tracial von Neumann algebras such that the diagram

$$\begin{array}{ccc} \mathbb{C}[\Gamma] & \longrightarrow & \mathcal{N}(\Gamma)_\chi \\ \uparrow s & & \downarrow \pi \\ \mathbb{C}[\Gamma'] & \longrightarrow & \mathcal{N}(\Gamma')_{\chi'} \end{array}$$

commutes, even though s is not a ring homomorphism. This yields the desired equality. $\square$

Conclusion

The main results obtained in this thesis are the following:

Theorem 2.3.17. *Let G be a finitely presented pro- p group and R a commutative pro- p local ring. Then, if $R[[G]]$ is a Sylvester domain, one of the following holds:*

(a) $\text{gl.dim}(R) = 2$. Then G is trivial. If moreover R is Noetherian, then it must be either a power series ring $\mathbb{F}_q[[x, y]]$ in two variables over a finite field or an Eisenstein extension of the ring of power series $\mathcal{O}_K[[x]]$ in one variable over the ring of integers of a finite extension K of $\mathbb{Q}_p$ with residue field $R/J(R)$.

(b) $\text{gl.dim}(R) = 1$. Then G is a finitely generated free pro- p group and R is either a power series ring $\mathbb{F}_q[[x]]$ over a finite field or the ring of integers $\mathcal{O}_K$ of a finite extension K of $\mathbb{Q}_p$.

(c) $\text{gl.dim}(R) = 0$. Then $R = \mathbb{F}_q$ is a finite field and $\text{cd}_p G \leq 2$.

Theorem 2.4.1. *If S is profinite and Δ is a collection of square matrices over S , then $\Delta^{-1}S$ admits a Hausdorff ring topology such that the induced map $S \rightarrow \Delta^{-1}S$ is continuous.*

Theorem 3.2.5. *Let G be a pro- p group, $(\mathbb{Q}, w)$ a division ring with a non-archimedean discrete valuation $w: \mathbb{Q} \rightarrow \{p^k: k \in \mathbb{Z}\} \cup \{0\}$ and $\varphi: \mathbb{F}_p[[G]] \rightarrow \mathbb{Q}$ a continuous embedding with respect to the topology on $\mathbb{Q}$ induced by w . Then*

$$\varphi^\# \text{rk}_{\mathbb{Q}} \leq \text{rk}_G$$

as rank functions on $\mathbb{F}_p[[G]]$. In particular, if $\varphi^\# \text{rk}_{\mathbb{Q}} = \text{irk}_{\mathbb{F}_p[[G]]}$, we have $\text{irk}_{\mathbb{F}_p[[G]]} = \text{rk}_G$.

Theorem 4.3.7. *Let G be a torsion-free finitely generated pro- p group. Then, the following are equivalent:*

(A) $\mathbb{F}_q[[G]]$ is a Sylvester domain for any finite field $\mathbb{F}_q$ of characteristic p .

(B) $\mathbb{F}_q[[U]]$ is a Sylvester domain for every open subgroup U of G and any finite field $\mathbb{F}_q$ of characteristic p .

(C) $\mathbb{F}_q[[U]]$ is a Sylvester domain for some open subgroup U of G and some finite field $\mathbb{F}_q$ of characteristic p .

Moreover, in case one of the above holds and U is an open normal subgroup of G with a universal embedding $\mathbb{F}_p[[U]] \rightarrow \mathbb{Q}$, then $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}) * G/U$ is a division ring and induced map $\mathbb{F}_q[[G]] \rightarrow (\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}) * G/U$ is also a universal embedding.

Theorem 4.6.6. *Let G be a finitely generated torsion-free virtually free-by- $\mathbb{Z}_p$ pro- p group. Then $\mathbb{F}_q[[G]]$ is a Sylvester domain and $\text{rk}_G = \text{irk}_{\mathbb{F}_q[[G]]}$.*

Corollary 5.4.8. *Let $G = \prod_{i=1}^k \text{SL}_2$, p and odd prime and G_1 be the first congruence subgroup of $G(\mathbb{Z}_p)$. For almost all primes p , we have for all matrices A over $\mathbb{Z}_p[[G_1]]$ that*

$$|\text{rk}_{\mathbb{Q}}(A) - \text{rk}_{\lambda}(A)| = O\left(\frac{1}{\min \lambda}\right),$$

where $\mathbb{Q}$ is the Ore division ring of fractions of $\mathbb{Z}_p[[G_1]]$. In particular,

$$\lim_{\min \lambda \rightarrow \infty} \text{rk}_{\lambda} = \text{rk}_{\mathbb{Q}}.$$

Theorem 6.1.5. *Let G be a free-by- $\mathbb{Z}_p$ pro- p group and Γ a finitely generated abstract subgroup of G . Let $G > U_1 > U_2 > \dots$ be a chain of open normal subgroups of G with trivial intersection and define $\Gamma_i = \Gamma \cap U_i$. Then, for any matrix A over $\mathbb{F}_q[\Gamma]$ we have*

$$\lim_{i \rightarrow \infty} \text{rk}_{\Gamma/\Gamma_i}(A) = \text{rk}_{\mathbb{Q}}(A)$$

where $\mathbb{Q}$ is the universal division ring of fractions of $\mathbb{F}_q[\Gamma]$.

Corollary 6.2.7. *Let $G(\mathbb{C})$ be a connected semisimple complex Lie group such that its Lie algebra $\mathfrak{g}$ is isomorphic to $\prod_{i=1}^k \mathfrak{sl}_2(\mathbb{C})$ and Γ be a finitely generated subgroup of $G(\mathbb{C})$. Then, for any central character χ and each finitely generated subring S_0 of $\mathbb{C}$, Conjecture 6.2.1 holds over S_0 for some finite index subgroup of Γ and Conjecture 6.2.2 holds over $\mathbb{C}$ for Γ .*

Corollary 6.3.2. *Let M be a hyperbolic 3-manifold with Euler characteristic $E(M)$ such that its fundamental group $\pi_1(M)$ is finitely generated. Then:*

$$\beta_i^{(2)}(M) = \begin{cases} 0, & \text{if } i \neq 1, \\ -E(M), & \text{if } i = 1. \end{cases}$$

Theorem 6.4.3. *If F is not totally real, then*

$$\dim_{\mathbb{C}} S_k(\Gamma) = O\left(\frac{\prod_{\tau \in I_{\mathbb{R}}} k_{\tau} \times \prod_{\sigma \in I_{\mathbb{C}}} k_{\sigma}^2}{\min\{k_{\delta} : \delta \in I_{\infty}\}}\right).$$

Conclusión

Los principales resultados obtenidos en esta tesis son los siguientes:

Teorema 2.3.17. *Sea G un grupo pro- p finitamente presentado y R un anillo local pro- p conmutativo. Si $R[[G]]$ es un dominio de Sylvester, entonces se da uno de los siguientes casos:*

(a) $\text{gl.dim}(R) = 2$. Entonces G es trivial. Además, si R es noetheriano, entonces debe ser o bien un anillo de series formales $\mathbb{F}_q[[x, y]]$ en dos variables sobre un cuerpo finito, o bien una extensión de Eisenstein del anillo de series formales $\mathbb{O}_K[[x]]$ en una variable sobre el anillo de enteros de una extensión finita K de $\mathbb{Q}_p$ cuyo cuerpo residual es $R/J(R)$.

(b) $\text{gl.dim}(R) = 1$. Entonces G es un grupo pro- p libre finitamente generado y R es o bien un anillo de series formales $\mathbb{F}_q[[x]]$ sobre un cuerpo finito, o bien el anillo de enteros $\mathbb{O}_K$ de una extensión finita K de $\mathbb{Q}_p$.

(c) $\text{gl.dim}(R) = 0$. Entonces $R = \mathbb{F}_q$ es un cuerpo finito y $\text{cd } G \leq 2$.

Teorema 2.4.1. *Si S es un anillo profinito y Δ es una colección de matrices cuadradas sobre S , entonces $\Delta^{-1}S$ admite una topología de anillo Hausdorff tal que la aplicación inducida $S \rightarrow \Delta^{-1}S$ es continua.*

Teorema 3.2.5. *Sea G un grupo pro- p , $(\mathbb{Q}, w)$ un anillo de división con una valoración discreta no arquimediana*

$$w : \mathbb{Q} \longrightarrow \{p^k : k \in \mathbb{Z}\} \cup \{0\},$$

y sea $\varphi : \mathbb{F}_p[[G]] \rightarrow \mathbb{Q}$ un monomorfismo continuo con respecto a la topología en $\mathbb{Q}$ inducida por w . Entonces

$$\varphi^\# \text{rk}_{\mathbb{Q}} \leq \text{rk}_G$$

como funciones de rango sobre $\mathbb{F}_p[[G]]$. En particular, si $\varphi^\# \text{irk}_{\mathbb{Q}} = \text{irk}_{\mathbb{F}_p[[G]]}$, se tiene $\text{irk}_{\mathbb{F}_p[[G]]} = \text{rk}_G$.

Teorema 4.3.7. *Sea G un grupo pro- p sin torsión finitamente generado. Entonces son equivalentes:*

(A) $\mathbb{F}_q[[G]]$ es un dominio de Sylvester para todo cuerpo finito $\mathbb{F}_q$ de característica p .

(B) $\mathbb{F}_q[[U]]$ es un dominio de Sylvester para todo subgrupo abierto U de G y todo cuerpo finito $\mathbb{F}_q$ de característica p .

(C) $\mathbb{F}_q[[U]]$ es un dominio de Sylvester para algún subgrupo abierto U de G y algún cuerpo finito $\mathbb{F}_q$ de característica p .

Además, en el caso en que se cumpla alguna de las condiciones anteriores y U sea un subgrupo abierto normal de G con un encaje universal $F_p[[U]] \rightarrow \mathbb{Q}$, el anillo $(\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}) * G/U$ es un anillo de división y la aplicación inducida

$$\mathbb{F}_q[[G]] \rightarrow (\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{Q}) * G/U$$

también es un encaje universal.

Teorema 4.6.6. Sea G un grupo pro- p sin torsión, finitamente generado y virtualmente libre-por- $\mathbb{Z}_p$. Entonces $\mathbb{F}_q[[G]]$ es un dominio de Sylvester y

$$\mathrm{rk}_G = \mathrm{irk}_{\mathbb{F}_q[[G]]}.$$

Corolario 5.4.8. Sea $G = \prod_{i=1}^k \mathrm{SL}_2$, p un primo impar y G_1 el primer subgrupo de congruencia de $G(\mathbb{Z}_p)$. Para casi todos los primos p , se tiene que para toda matriz A sobre $\mathbb{Z}_p[[G_1]]$,

$$|\mathrm{rk}_{\mathbb{Q}}(A) - \mathrm{rk}_{\lambda}(A)| = O\left(\frac{1}{\min \lambda}\right),$$

donde $\mathbb{Q}$ es el anillo de fracciones de Ore de $\mathbb{Z}_p[[G_1]]$. En particular,

$$\lim_{\min \lambda \rightarrow \infty} \mathrm{rk}_{\lambda} = \mathrm{rk}_{\mathbb{Q}}.$$

Teorema 6.1.5. Sea G un grupo pro- p libre-por- $\mathbb{Z}_p$ y Γ un subgrupo abstracto finitamente generado de G . Sea

$$G > U_1 > U_2 > \dots$$

una sucesión descendente de subgrupos abiertos normales de G con intersección trivial y definamos $\Gamma_i = \Gamma \cap U_i$. Entonces, para cualquier matriz A sobre $\mathbb{F}_q[\Gamma]$ se tiene

$$\lim_{i \rightarrow \infty} \mathrm{rk}_{\Gamma/\Gamma_i}(A) = \mathrm{rk}_{\mathbb{Q}}(A),$$

donde $\mathbb{Q}$ es el anillo de división universal de $\mathbb{F}_q[\Gamma]$.

Corolario 6.2.7. Sea $G(\mathbb{C})$ un grupo de Lie complejo semisimple conexo cuya álgebra de Lie $\mathfrak{g}$ es isomorfa a $\prod_{i=1}^k \mathfrak{sl}_2(\mathbb{C})$, y sea Γ un subgrupo finitamente generado de $G(\mathbb{C})$. Entonces, para cualquier carácter central χ y todo subanillo finitamente generado $S_0 \subset \mathbb{C}$, la Conjetura 6.2.1 se

verifica sobre S_0 para algún subgrupo de índice finito de Γ , y la Conjetura 6.2.2 se verifica sobre $\mathbb{C}$ para Γ .

Corolario 6.3.2. Sea M una 3-variedad hiperbólica con característica de Euler $E(M)$ tal que su grupo fundamental $\pi_1(M)$ es finitamente generado. Entonces

$$b_i^{(2)}(M) = \begin{cases} 0, & \text{si } i \neq 1, \\ -E(M), & \text{si } i = 1. \end{cases}$$

Teorema 6.4.3. Si F no es totalmente real, entonces

$$\dim_{\mathbb{C}} S_k(\Gamma) = O\left(\frac{\prod_{\tau \in I_{\mathbb{R}}} k_{\tau} \times \prod_{\sigma \in I_{\mathbb{C}}} k_{\sigma}^2}{\min\{k_{\delta} : \delta \in I_{\infty}\}}\right).$$

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